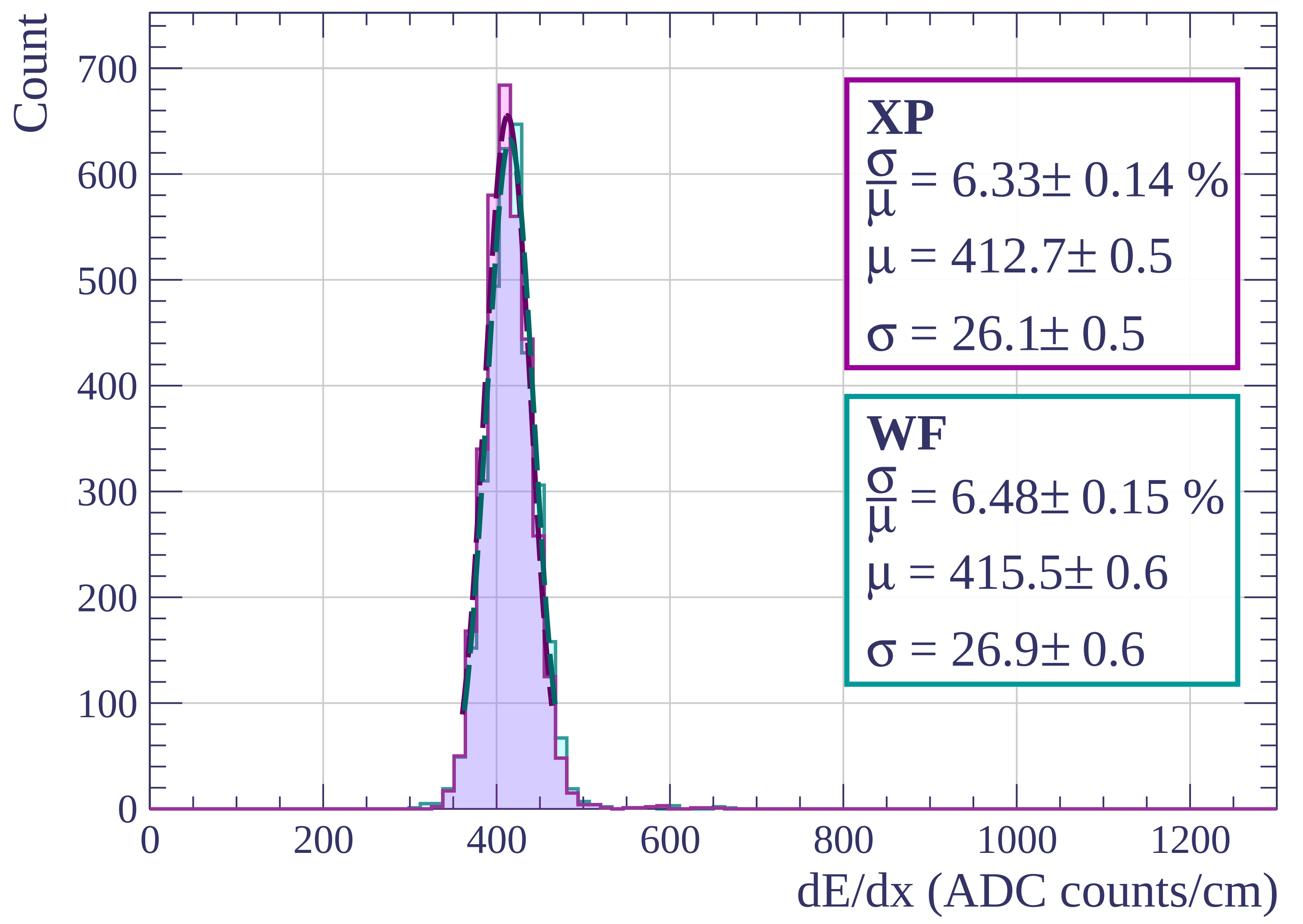
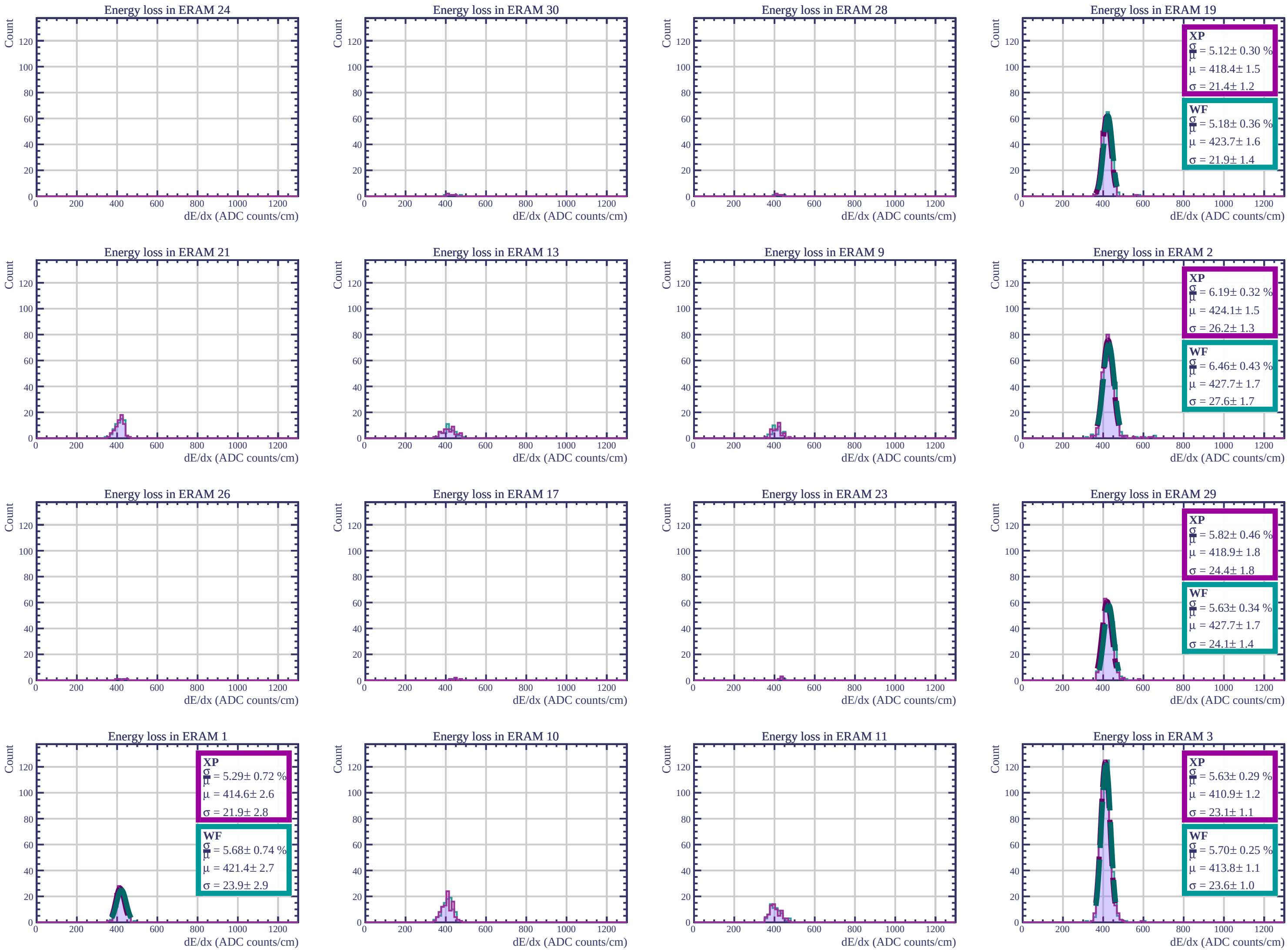
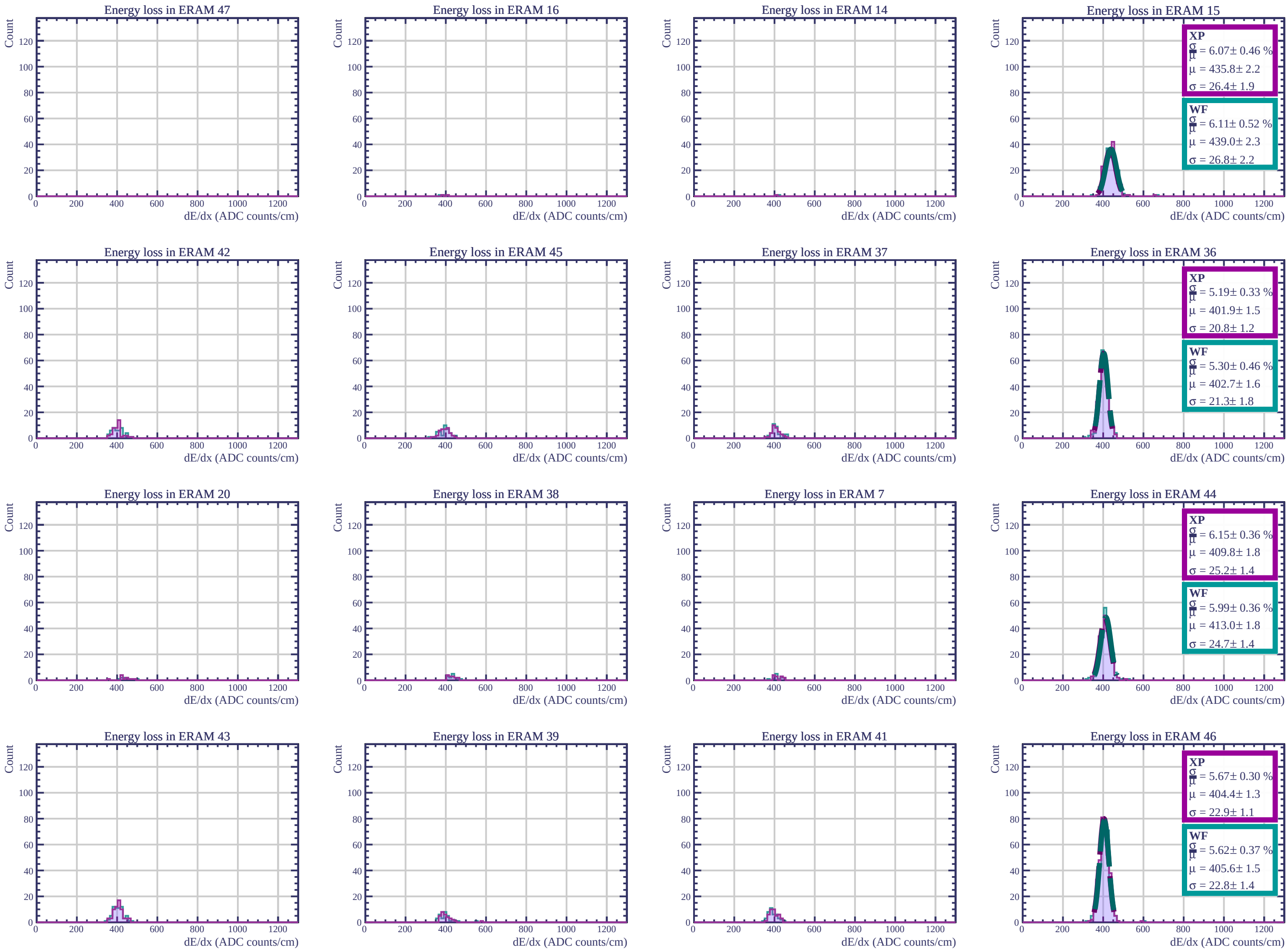
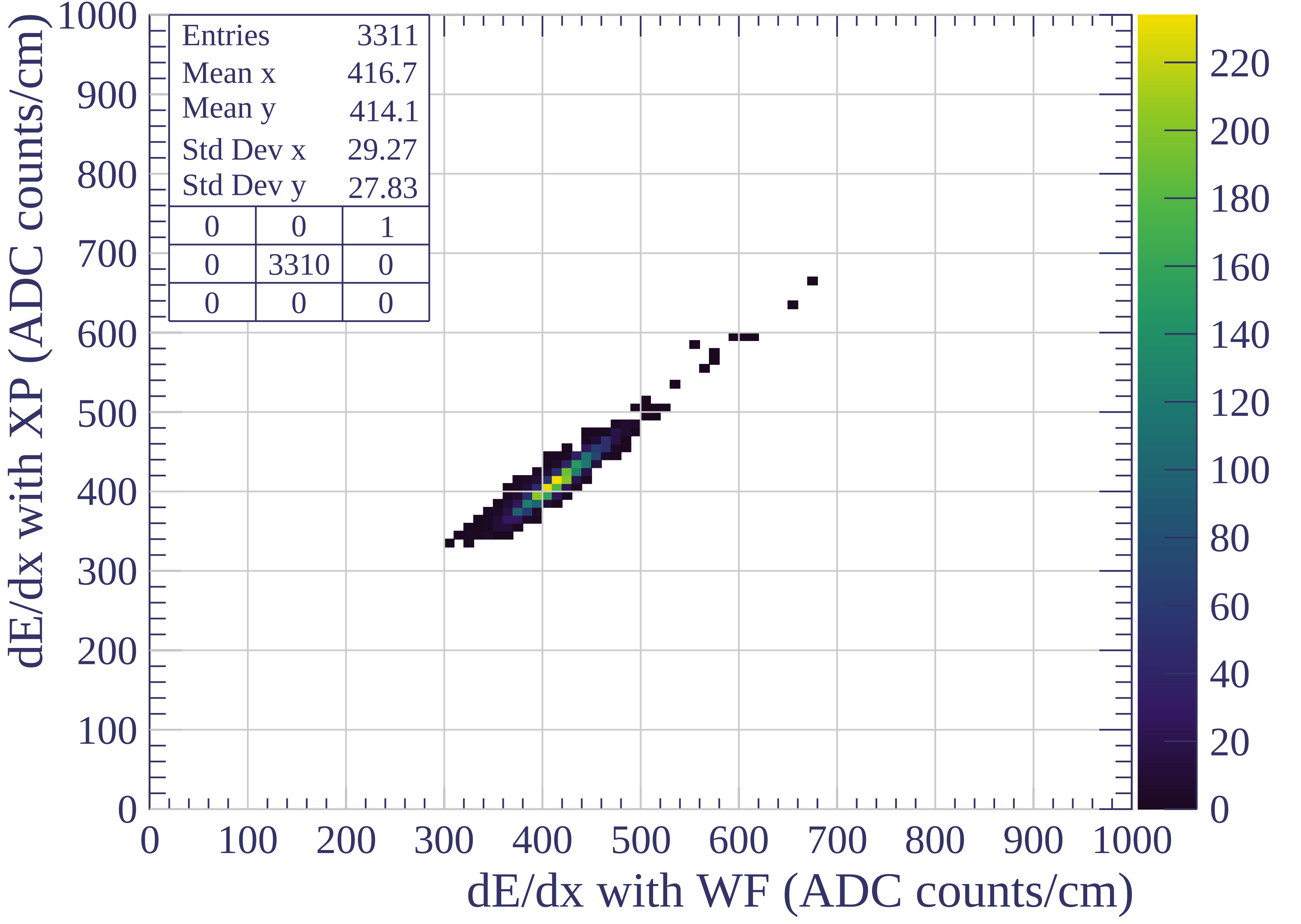

Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.33± 0.14 %
μ = 412.7± 0.5
σ = 26.1± 0.5
WF
$\frac{\sigma}{\mu}$ = 6.48± 0.15 %
μ = 415.5± 0.6
σ = 26.9± 0.6



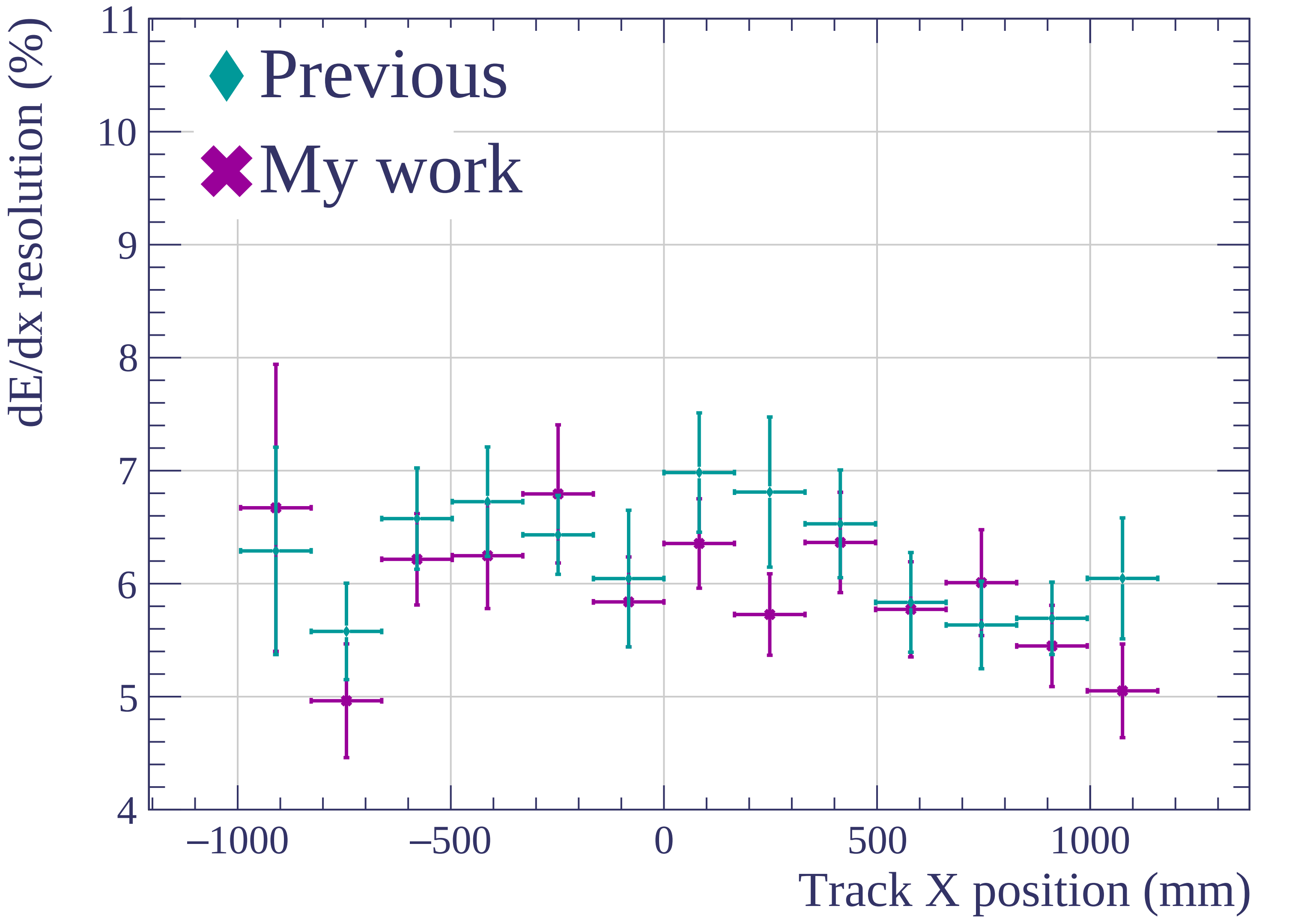
Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
XP
σ/μ = 5.12± 0.30 %
μ = 418.4± 1.5
σ = 21.4± 1.2
WF
σ/μ = 5.18± 0.36 %
μ = 423.7± 1.6
σ = 21.9± 1.4
Energy loss in ERAM 21
Energy loss in ERAM 13
Energy loss in ERAM 9
Energy loss in ERAM 2
XP
σ/μ = 6.19± 0.32 %
μ = 424.1± 1.5
σ = 26.2± 1.3
WF
σ/μ = 6.46± 0.43 %
μ = 427.7± 1.7
σ = 27.6± 1.7
Energy loss in ERAM 26
Energy loss in ERAM 17
Energy loss in ERAM 23
Energy loss in ERAM 29
XP
σ/μ = 5.82± 0.46 %
μ = 418.9± 1.8
σ = 24.4± 1.8
WF
σ/μ = 5.63± 0.34 %
μ = 427.7± 1.7
σ = 24.1± 1.4
Energy loss in ERAM 1
XP
σ/μ = 5.29± 0.72 %
μ = 414.6± 2.6
σ = 21.9± 2.8
WF
σ/μ = 5.68± 0.74 %
μ = 421.4± 2.7
σ = 23.9± 2.9
Energy loss in ERAM 10
Energy loss in ERAM 11
Energy loss in ERAM 3
XP
σ/μ = 5.63± 0.29 %
μ = 410.9± 1.2
σ = 23.1± 1.1
WF
σ/μ = 5.70± 0.25 %
μ = 413.8± 1.1
σ = 23.6± 1.0
Count
dE/dx (ADC counts/cm)

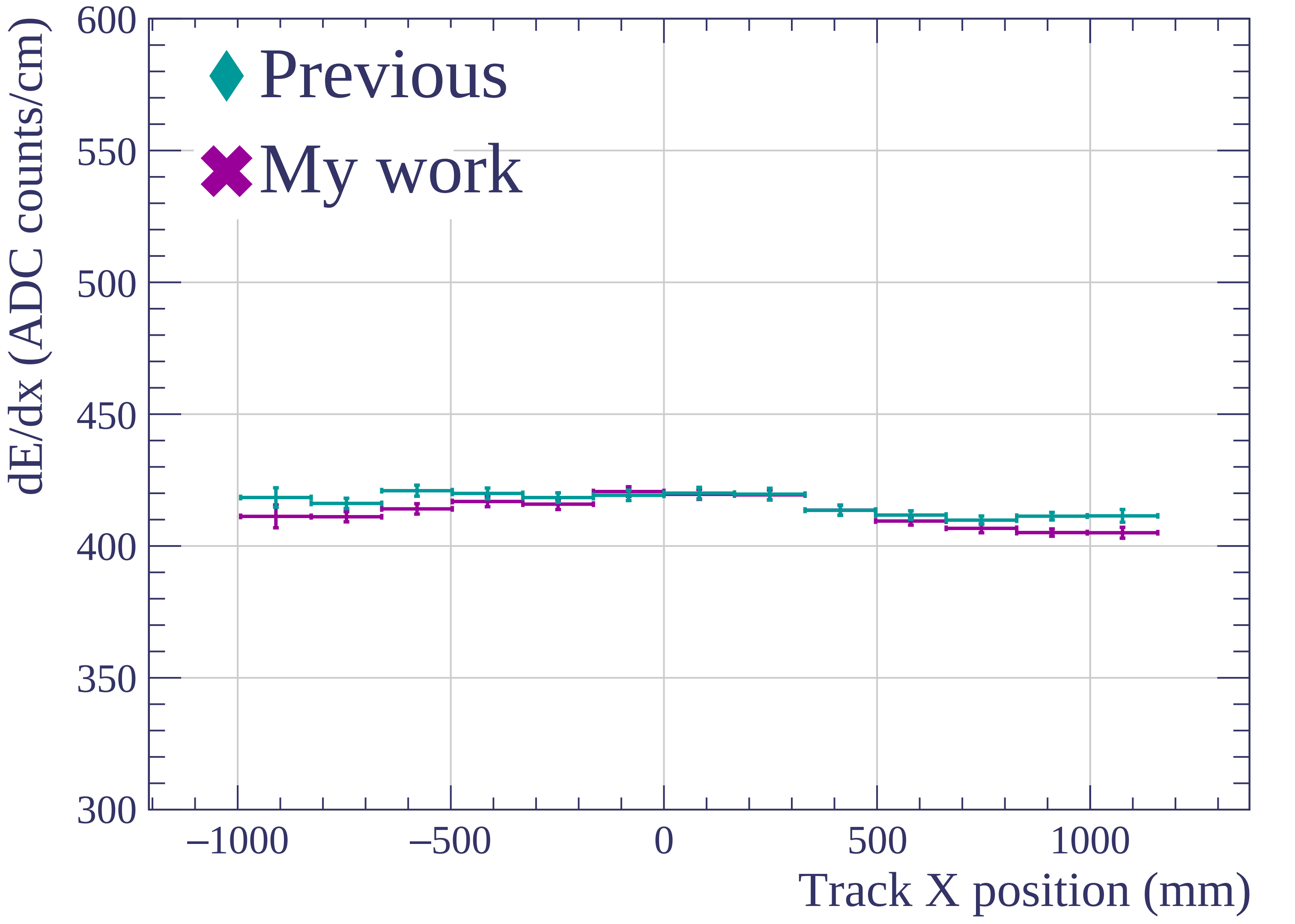


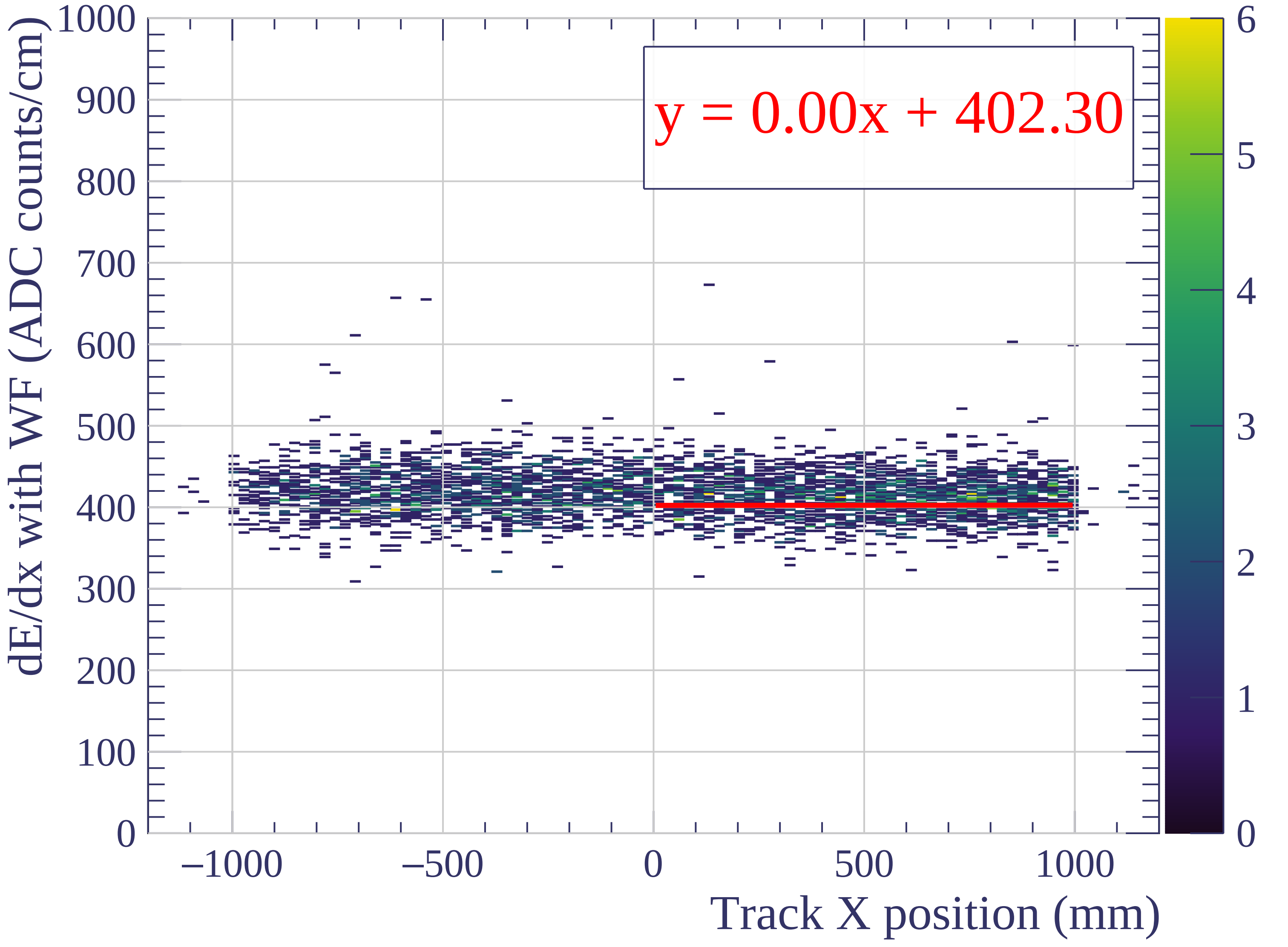
Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
XP
σ/μ = 6.07± 0.46 %
μ = 435.8± 2.2
σ = 26.4± 1.9
WF
σ/μ = 6.11± 0.52 %
μ = 439.0± 2.3
σ = 26.8± 2.2
Energy loss in ERAM 42
Energy loss in ERAM 45
Energy loss in ERAM 37
Energy loss in ERAM 36
XP
σ/μ = 5.19± 0.33 %
μ = 401.9± 1.5
σ = 20.8± 1.2
WF
σ/μ = 5.30± 0.46 %
μ = 402.7± 1.6
σ = 21.3± 1.8
Energy loss in ERAM 20
Energy loss in ERAM 38
Energy loss in ERAM 7
Energy loss in ERAM 44
XP
σ/μ = 6.15± 0.36 %
μ = 409.8± 1.8
σ = 25.2± 1.4
WF
σ/μ = 5.99± 0.36 %
μ = 413.0± 1.8
σ = 24.7± 1.4
Energy loss in ERAM 43
Energy loss in ERAM 39
Energy loss in ERAM 41
Energy loss in ERAM 46
XP
σ/μ = 5.67± 0.30 %
μ = 404.4± 1.3
σ = 22.9± 1.1
WF
σ/μ = 5.62± 0.37 %
μ = 405.6± 1.5
σ = 22.8± 1.4
Count
dE/dx (ADC counts/cm)



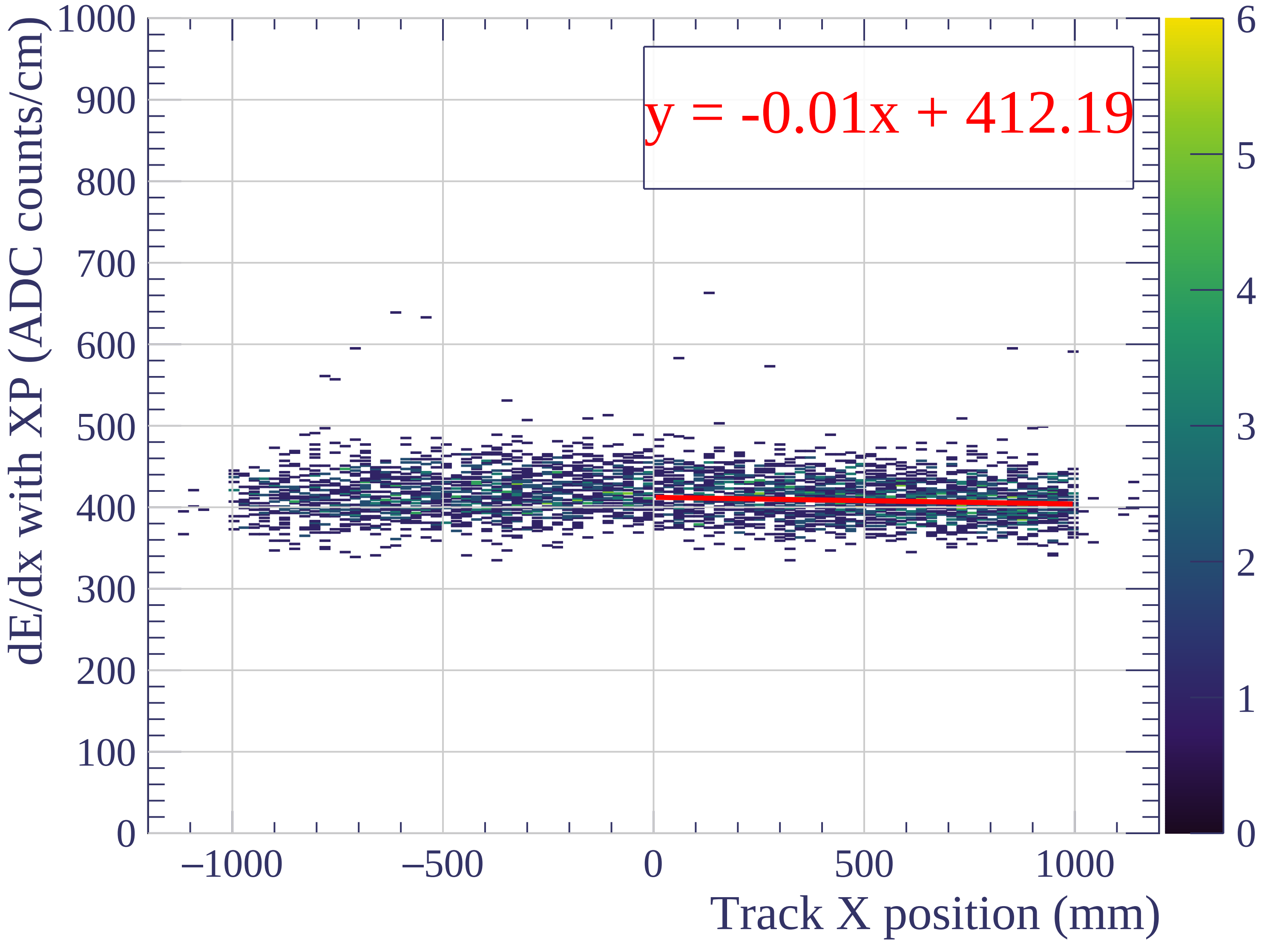
Entries 3311
Mean x 416.7
Mean y 414.1
Std Dev x 29.27
Std Dev y 27.83
0 0 1
0 3310 0
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
20
40
60
80
100
120
140
160
180
200
220


Previous
My work
dE/dx resolution (%)
Track X position (mm)
11
10
9
8
7
6
5
4
−1000
−500
0
500
1000

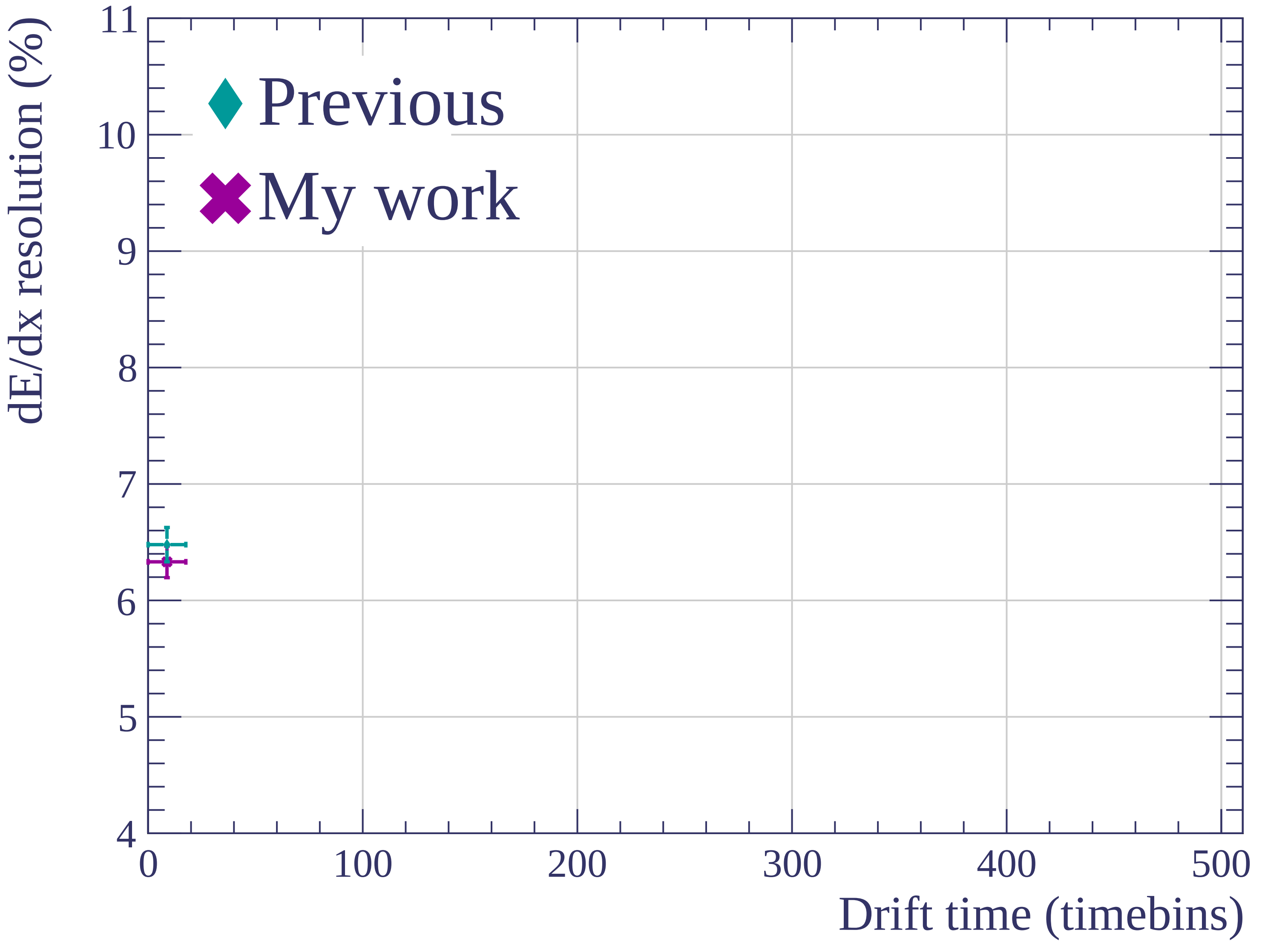

Previous
My work
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
Track X position (mm)



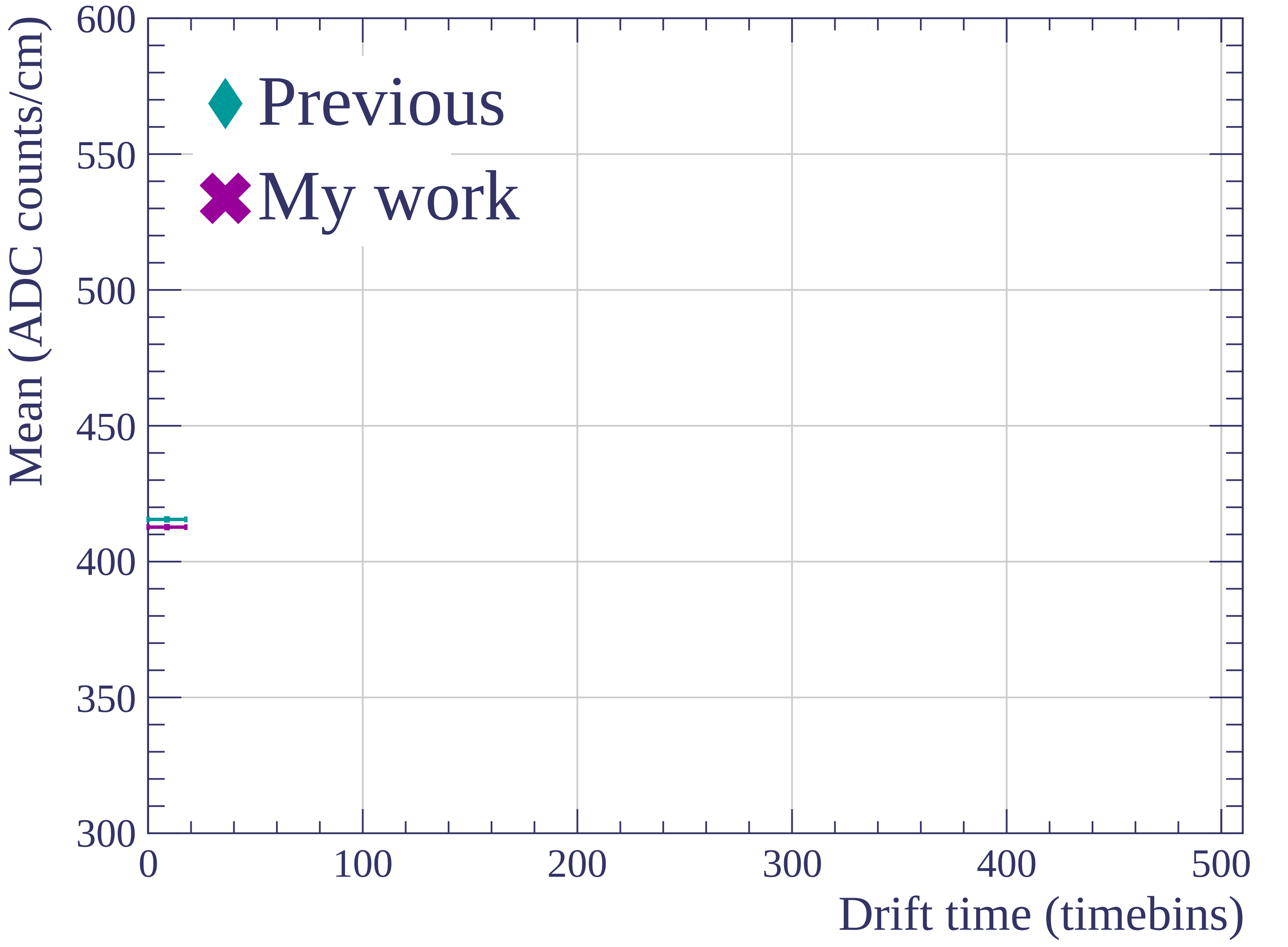
y = 0.00x + 402.30
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
1
2
3
4
5
6



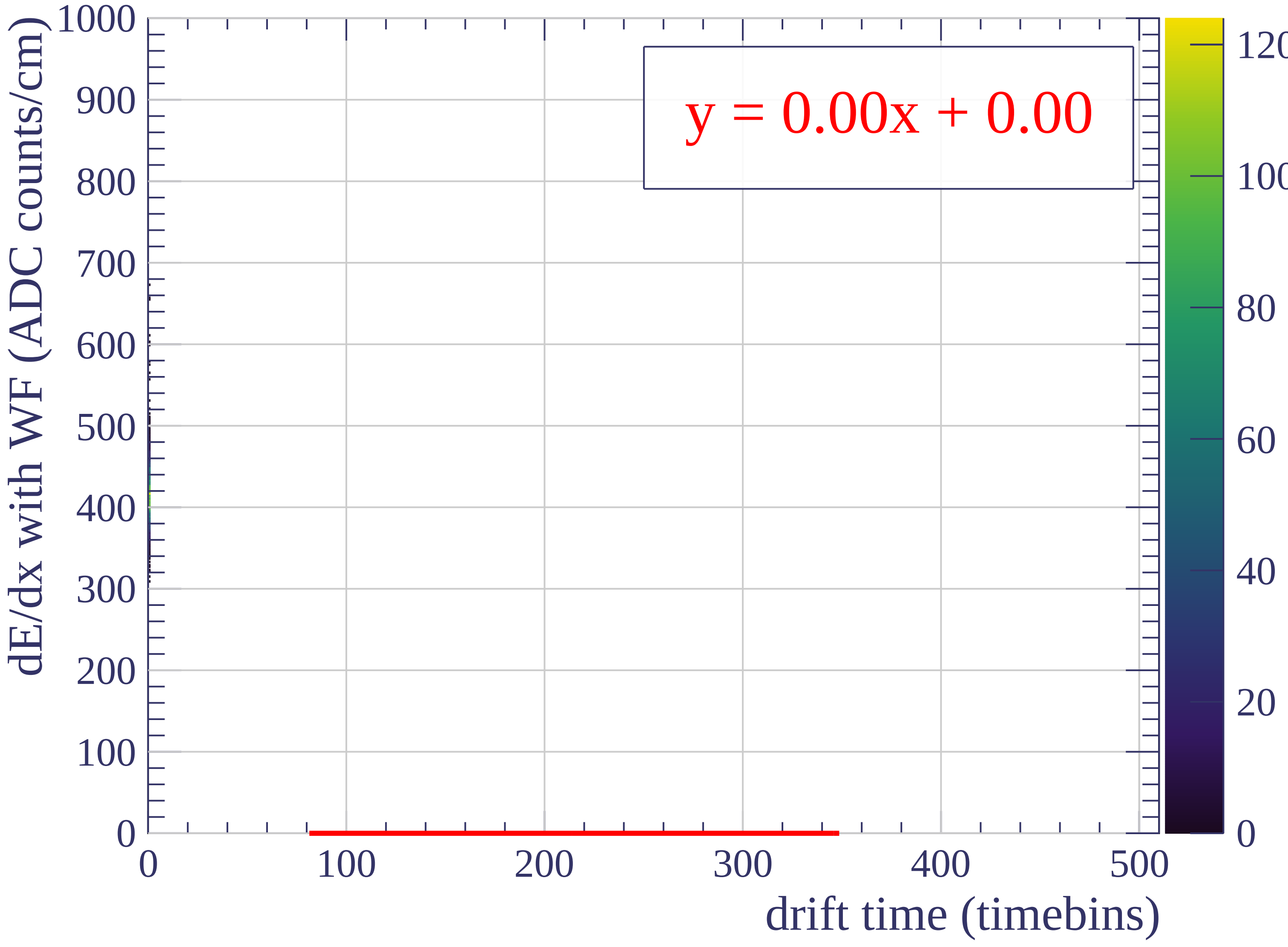
y = -0.01x + 412.19
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
1
2
3
4
5
6

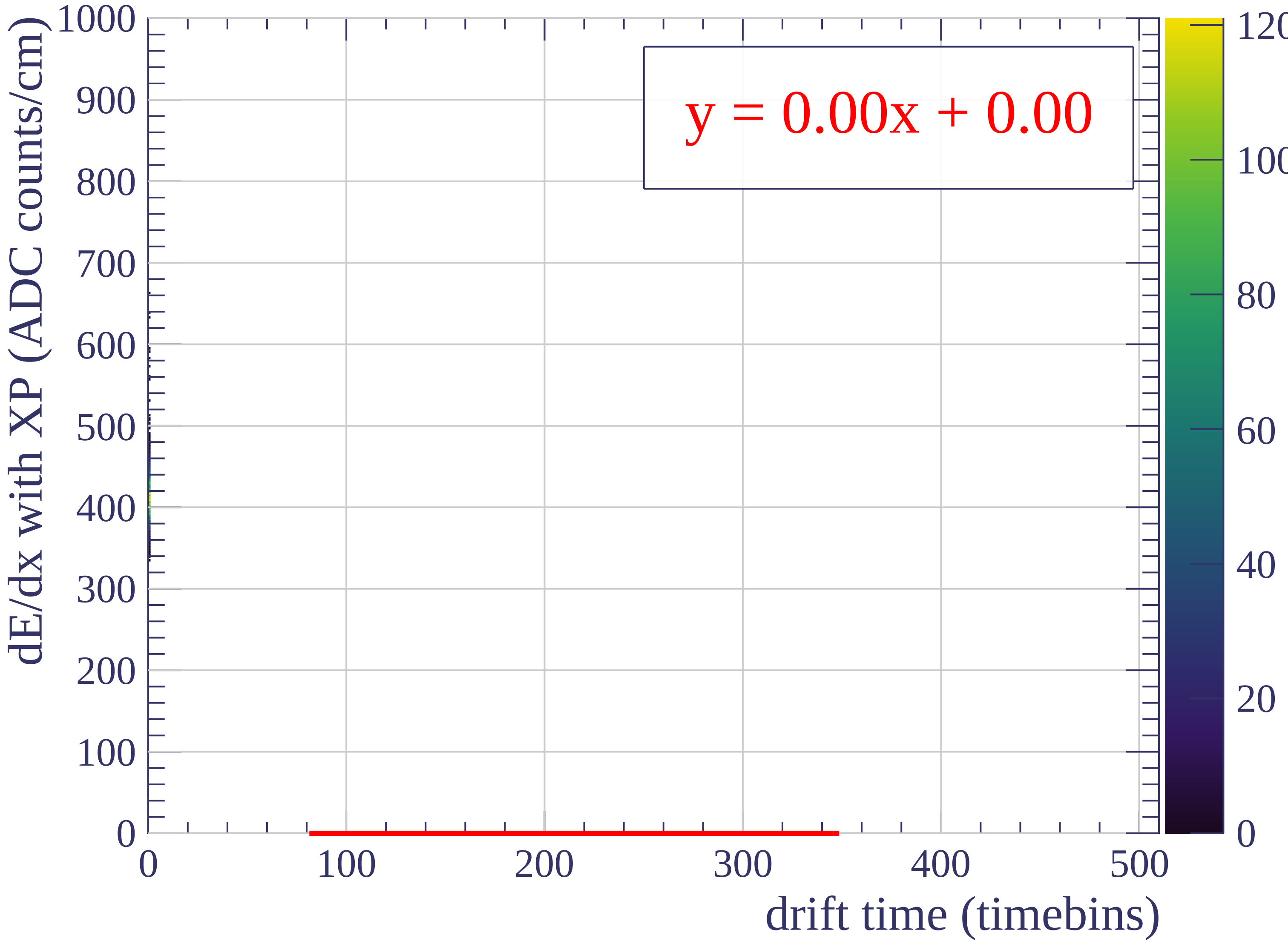


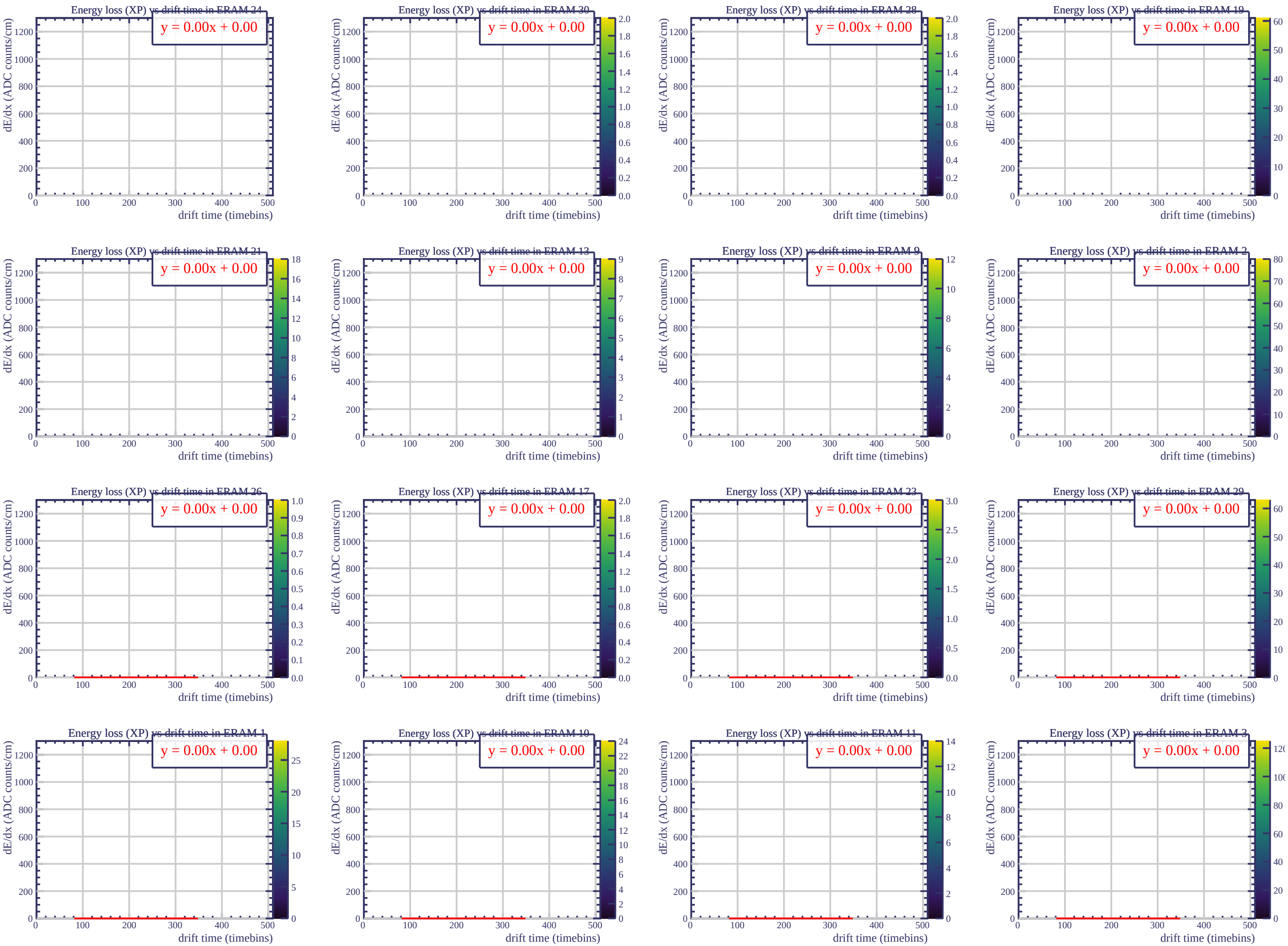
Previous
My work
dE/dx resolution (%)
Drift time (timebins)
4
5
6
7
8
9
10
11
0
100
200
300
400
500

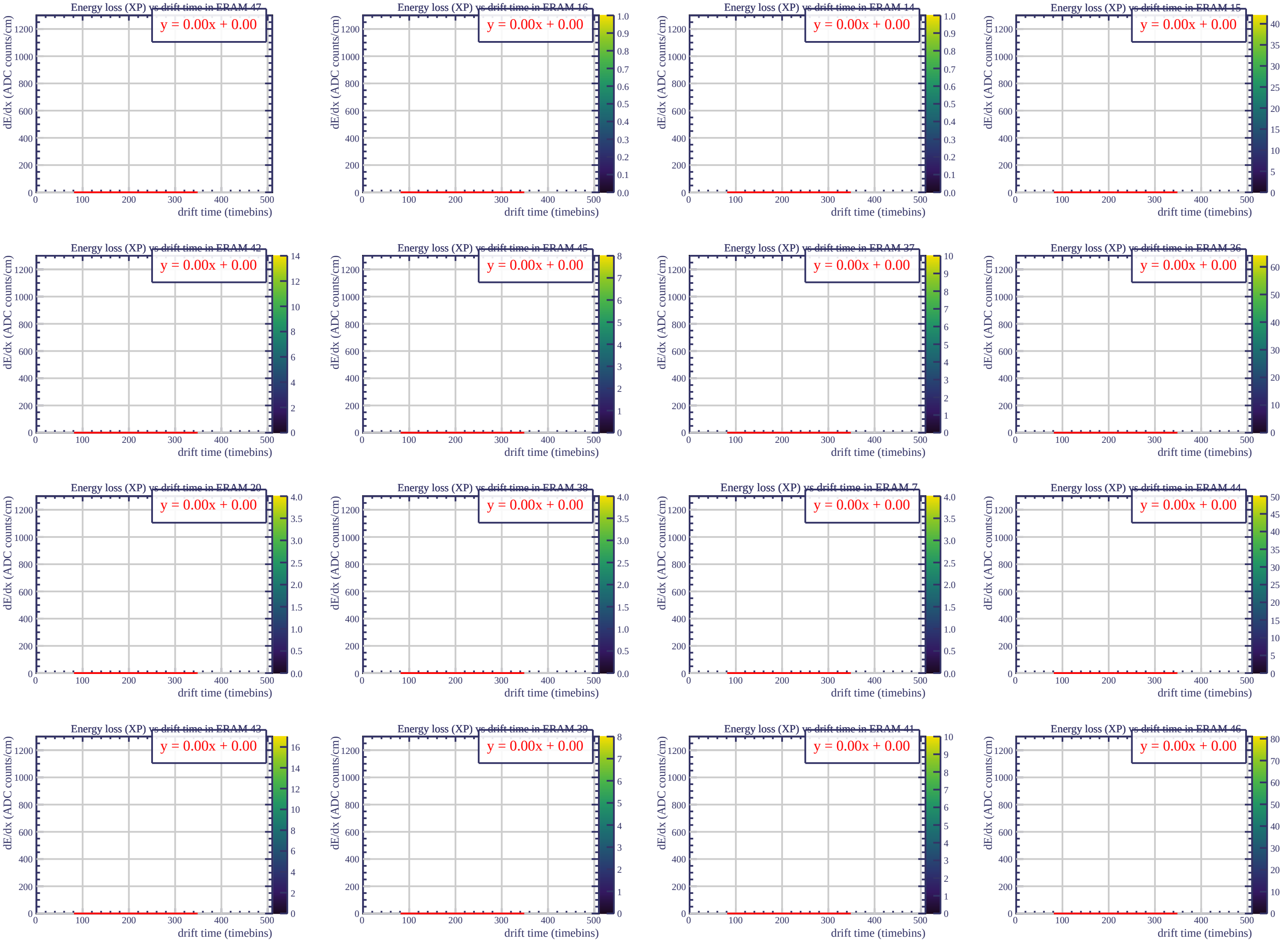


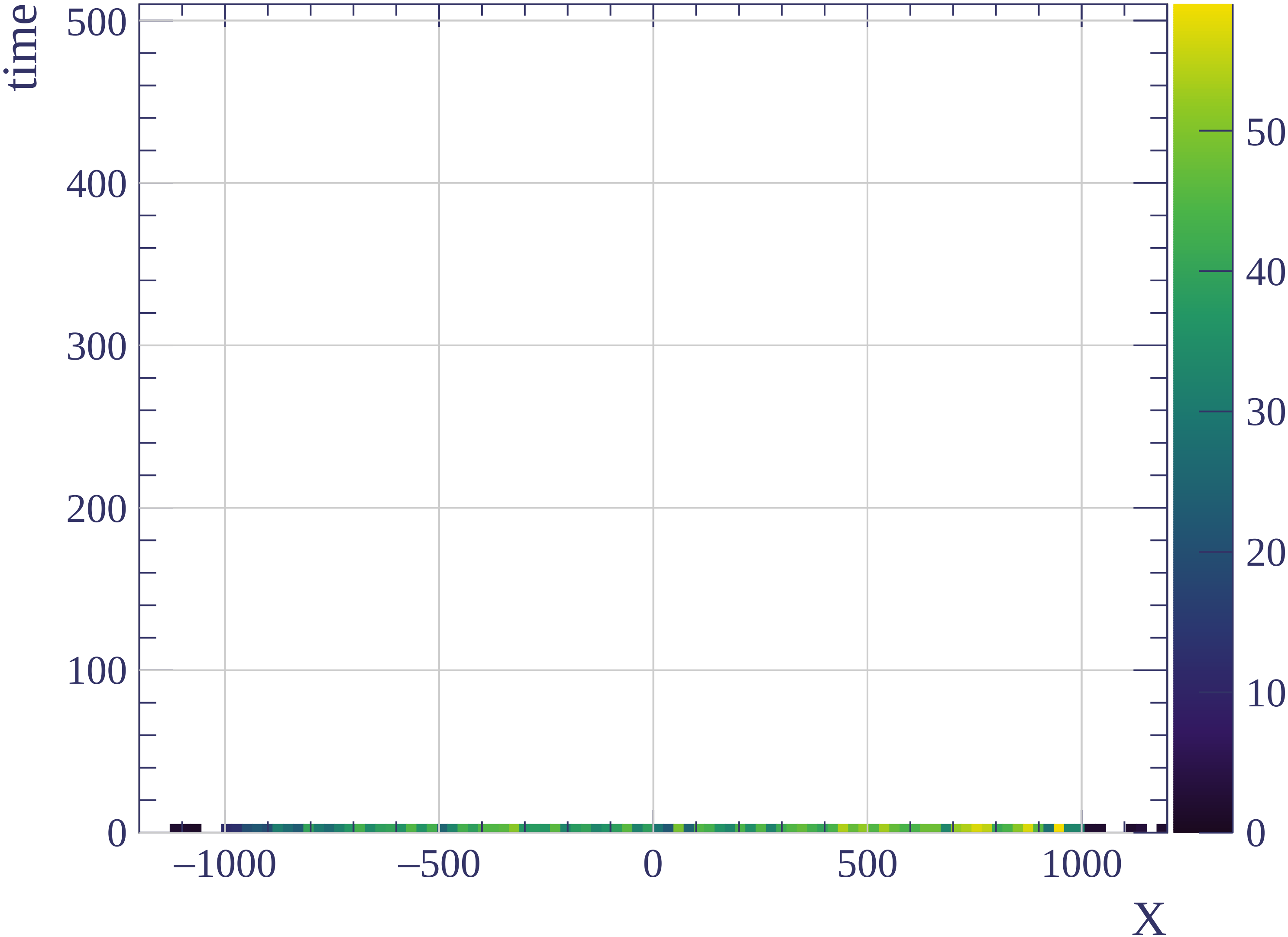
Previous
My work
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
100
200
300
400
500
Drift time (timebins)

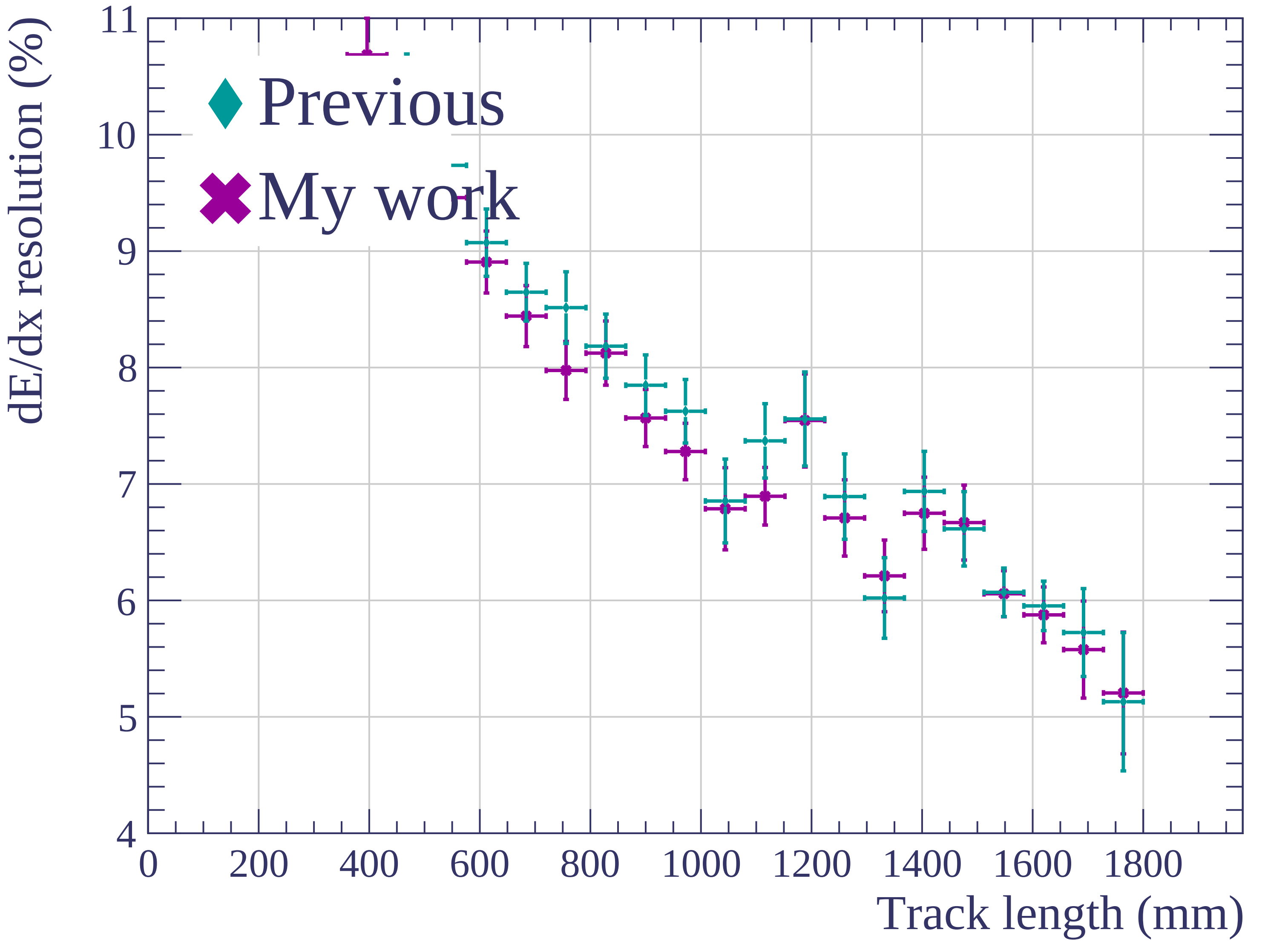

y = 0.00x + 0.00
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
20
40
60
80
100
120


y = 0.00x + 0.00
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
20
40
60
80
100
120

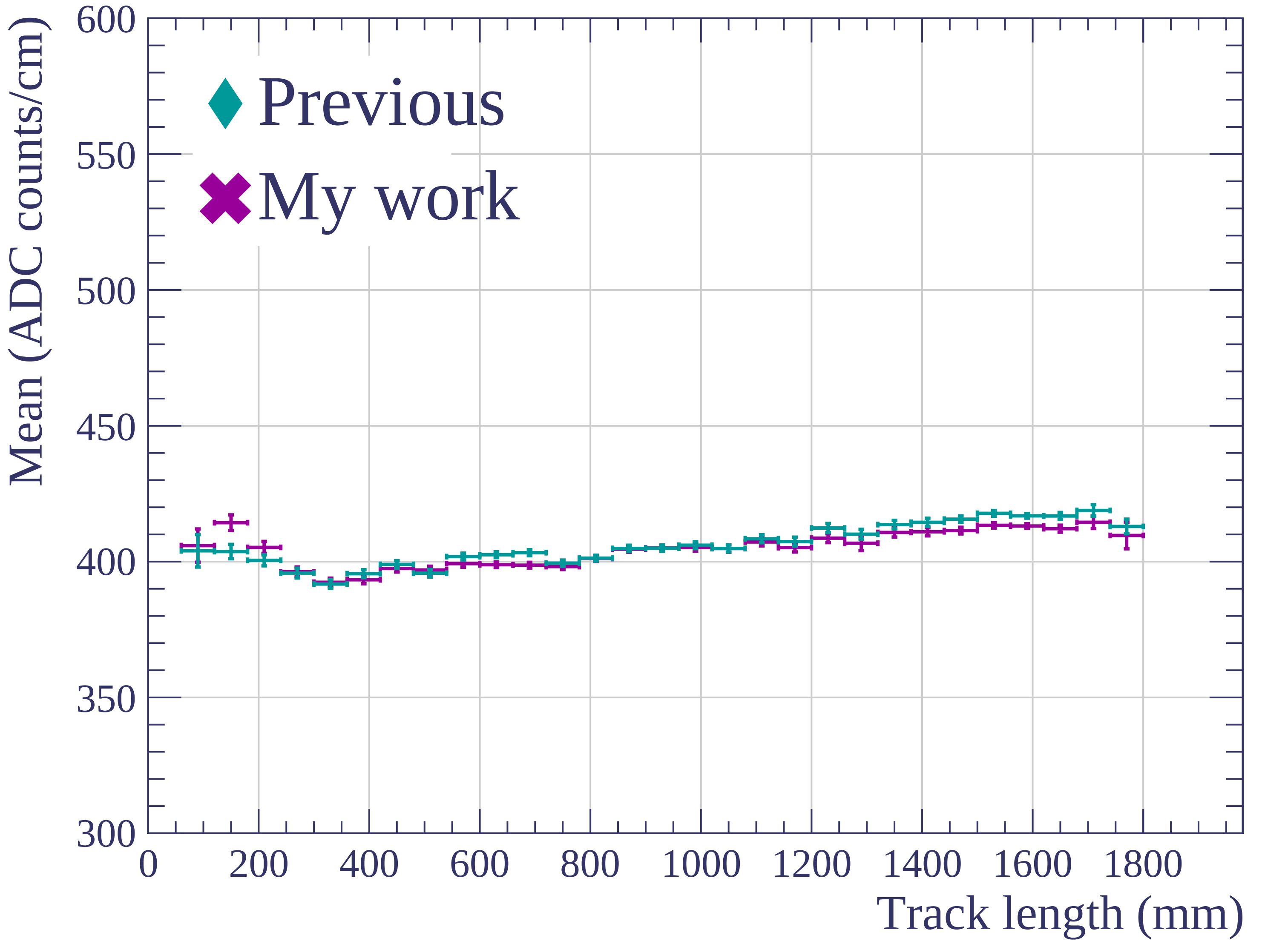

Energy loss (XP) vs drift time in ERAM 24
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 30
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 28
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 19
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 21
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 13
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 9
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 2
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 26
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 17
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 23
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 29
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 1
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 10
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 11
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 3
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

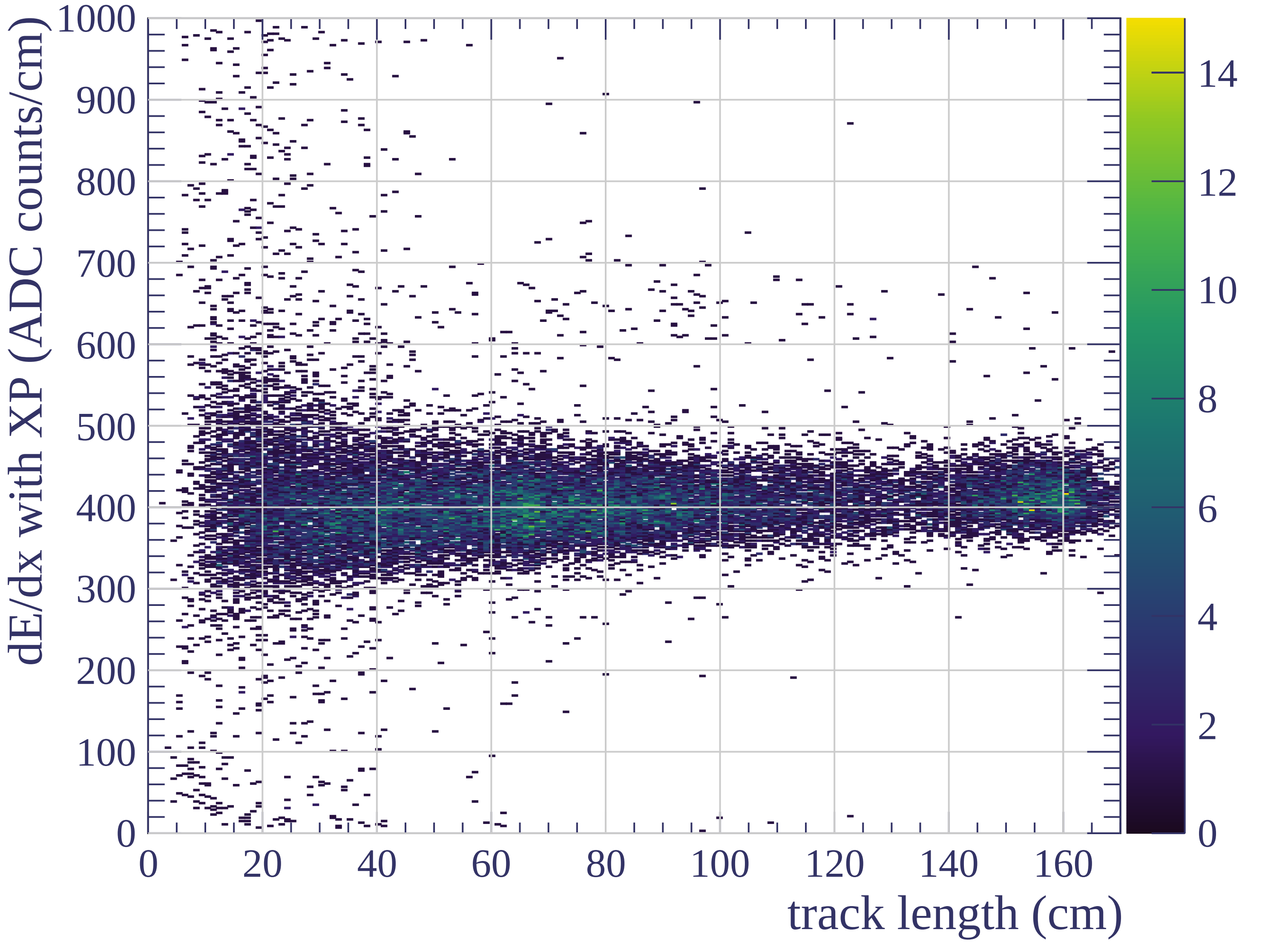

Energy loss (XP) vs drift time in ERAM 47
Energy loss (XP) vs drift time in ERAM 16
Energy loss (XP) vs drift time in ERAM 14
Energy loss (XP) vs drift time in ERAM 15
Energy loss (XP) vs drift time in ERAM 42
Energy loss (XP) vs drift time in ERAM 45
Energy loss (XP) vs drift time in ERAM 37
Energy loss (XP) vs drift time in ERAM 36
Energy loss (XP) vs drift time in ERAM 20
Energy loss (XP) vs drift time in ERAM 38
Energy loss (XP) vs drift time in ERAM 7
Energy loss (XP) vs drift time in ERAM 44
Energy loss (XP) vs drift time in ERAM 43
Energy loss (XP) vs drift time in ERAM 39
Energy loss (XP) vs drift time in ERAM 41
Energy loss (XP) vs drift time in ERAM 46
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)


time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
50
40
30
20
10
0

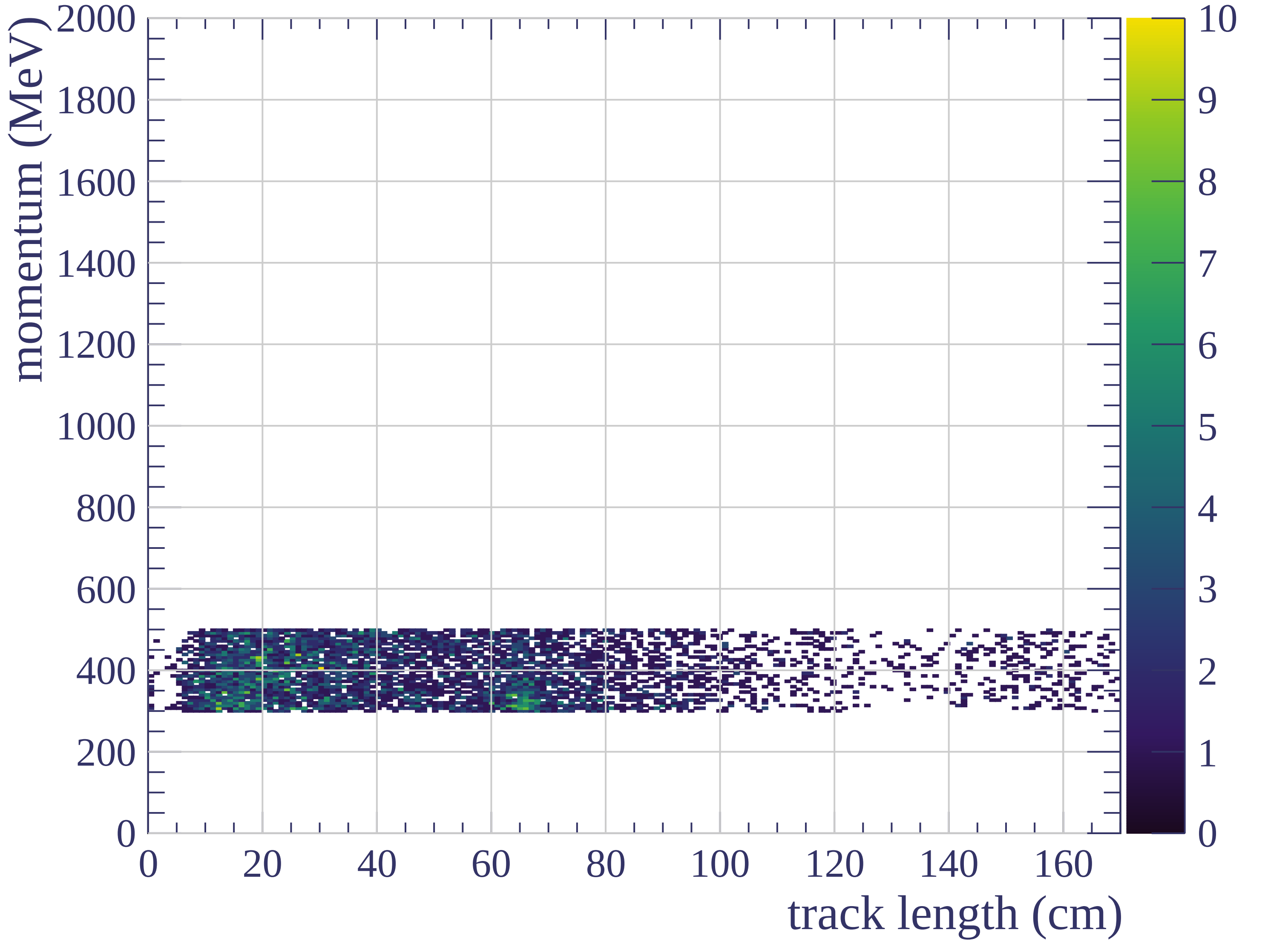


dE/dx resolution (%)
11
10
9
8
7
6
5
4
0
200
400
600
800
1000
1200
1400
1600
1800
Track length (mm)
Previous
My work

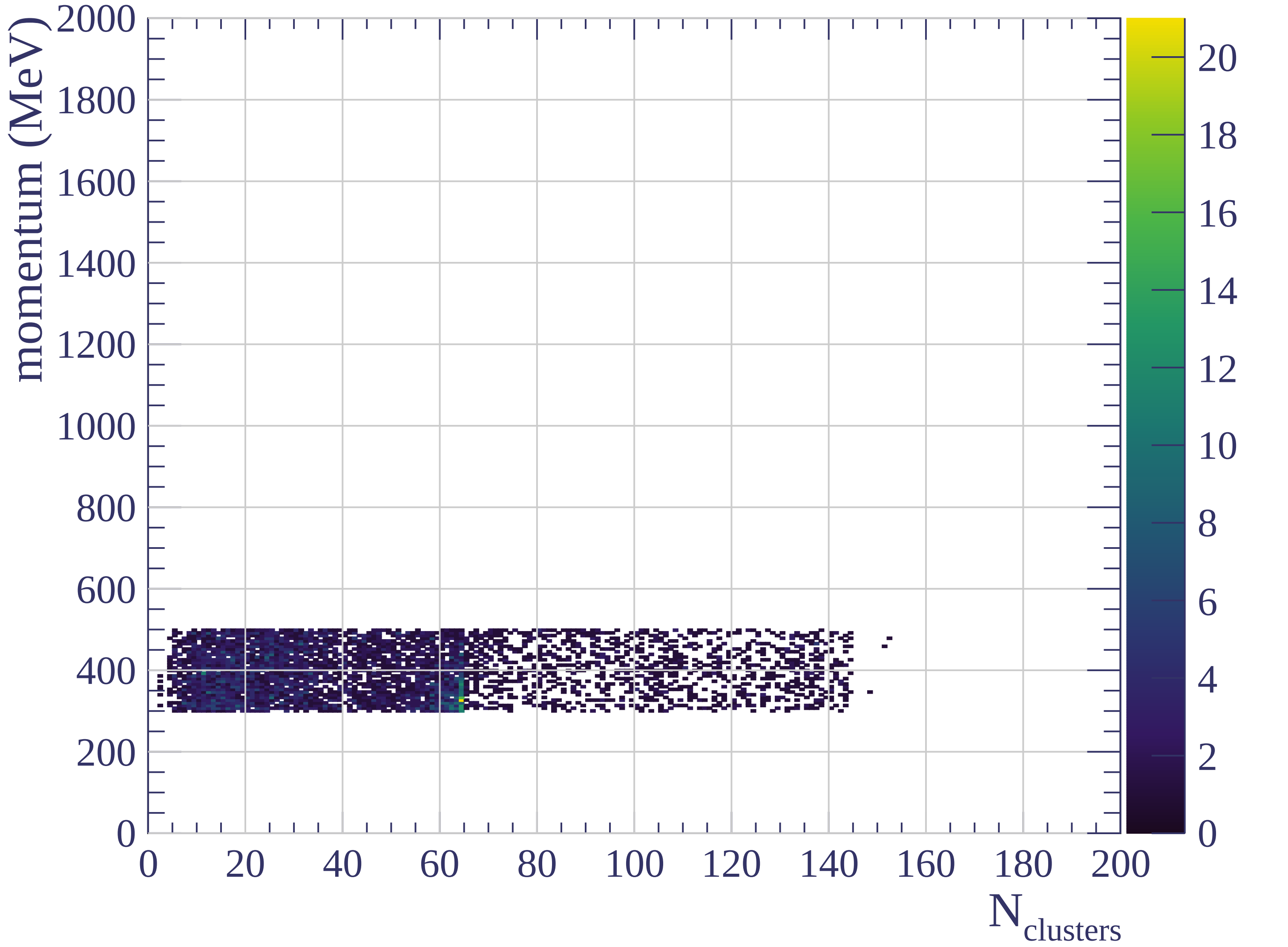

Previous
My work
Mean (ADC counts/cm)
Track length (mm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800

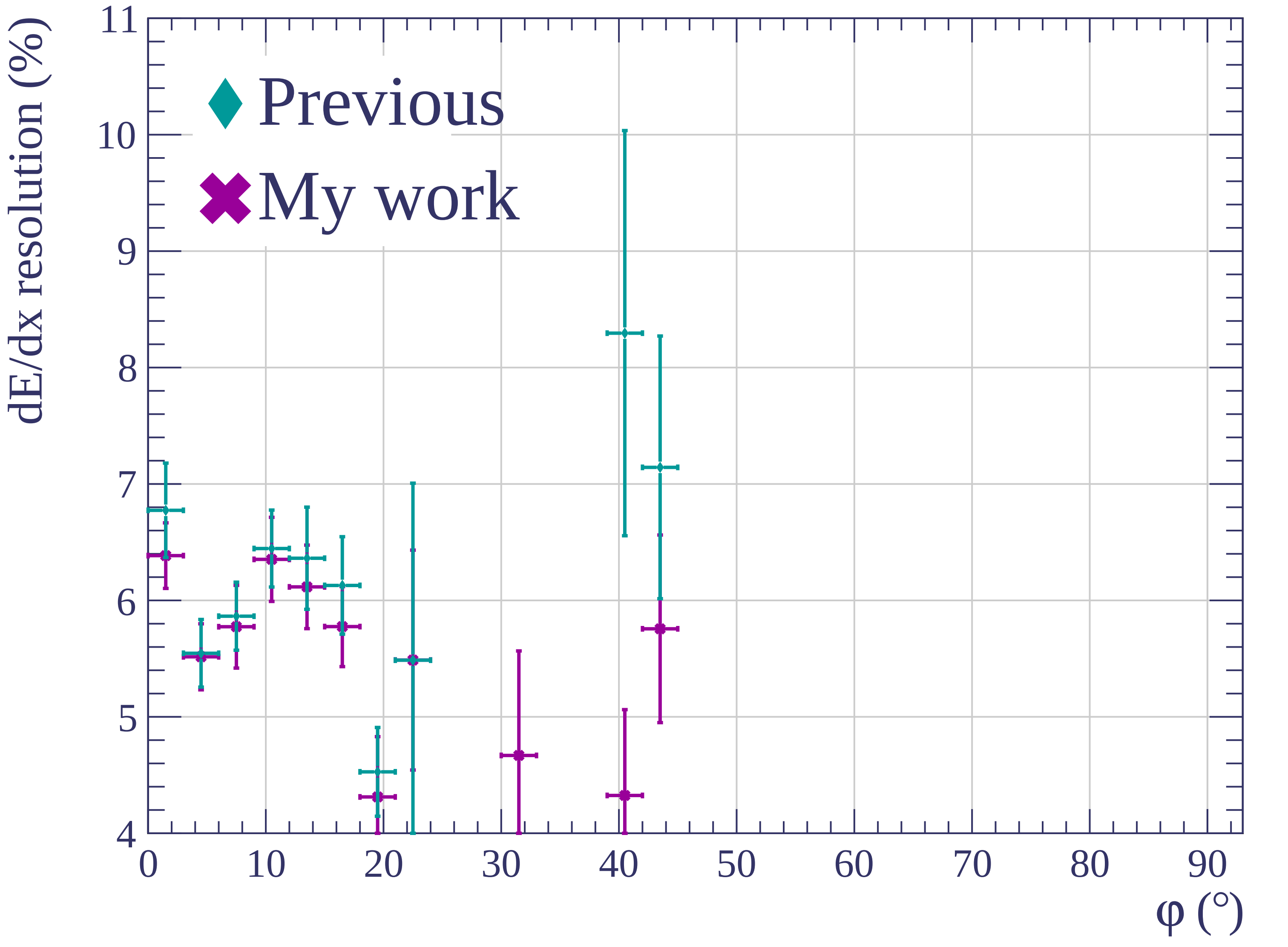


dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
2
4
6
8
10
12
14

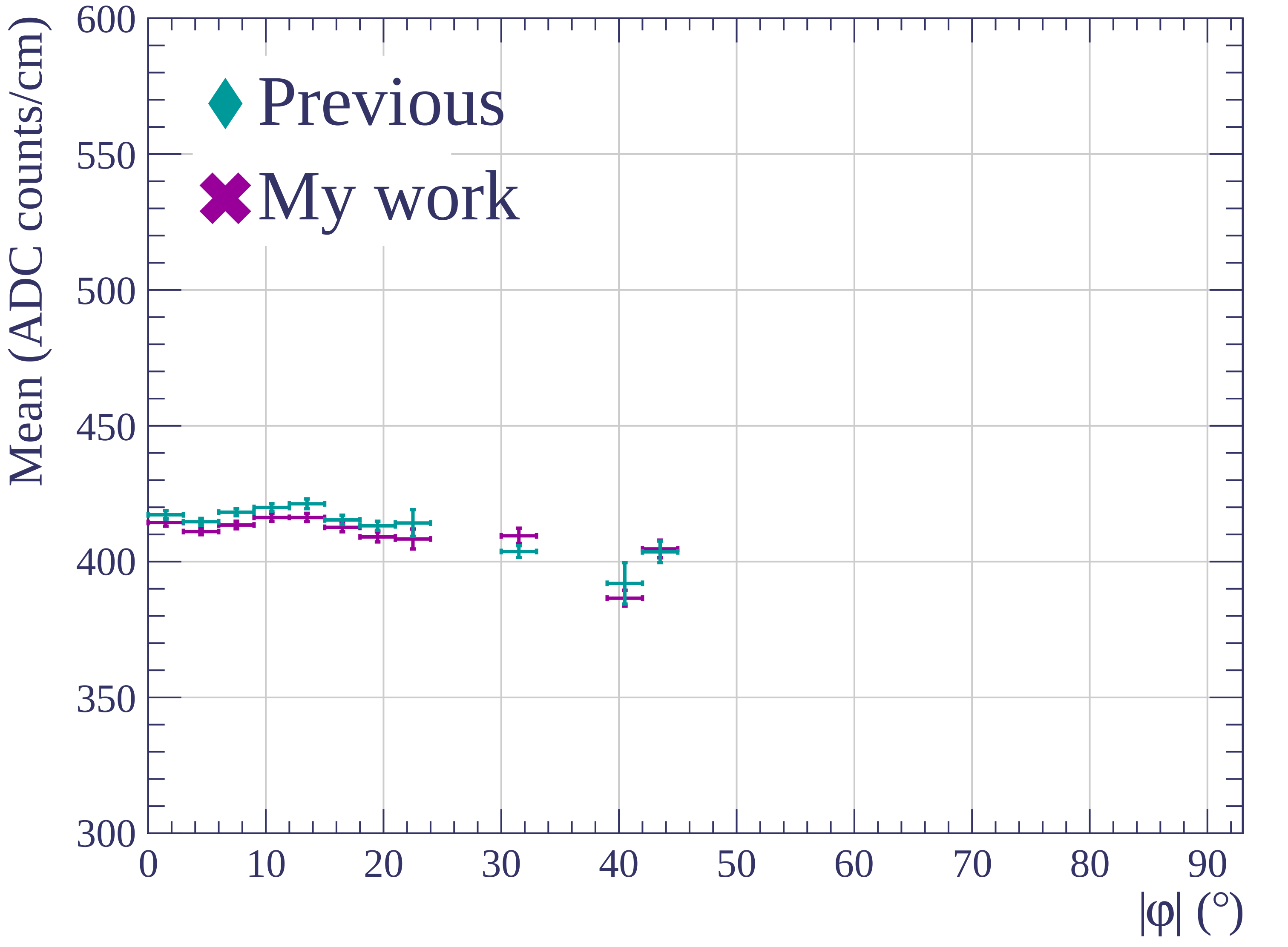


momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
track length (cm)
10
9
8
7
6
5
4
3
2
1
0

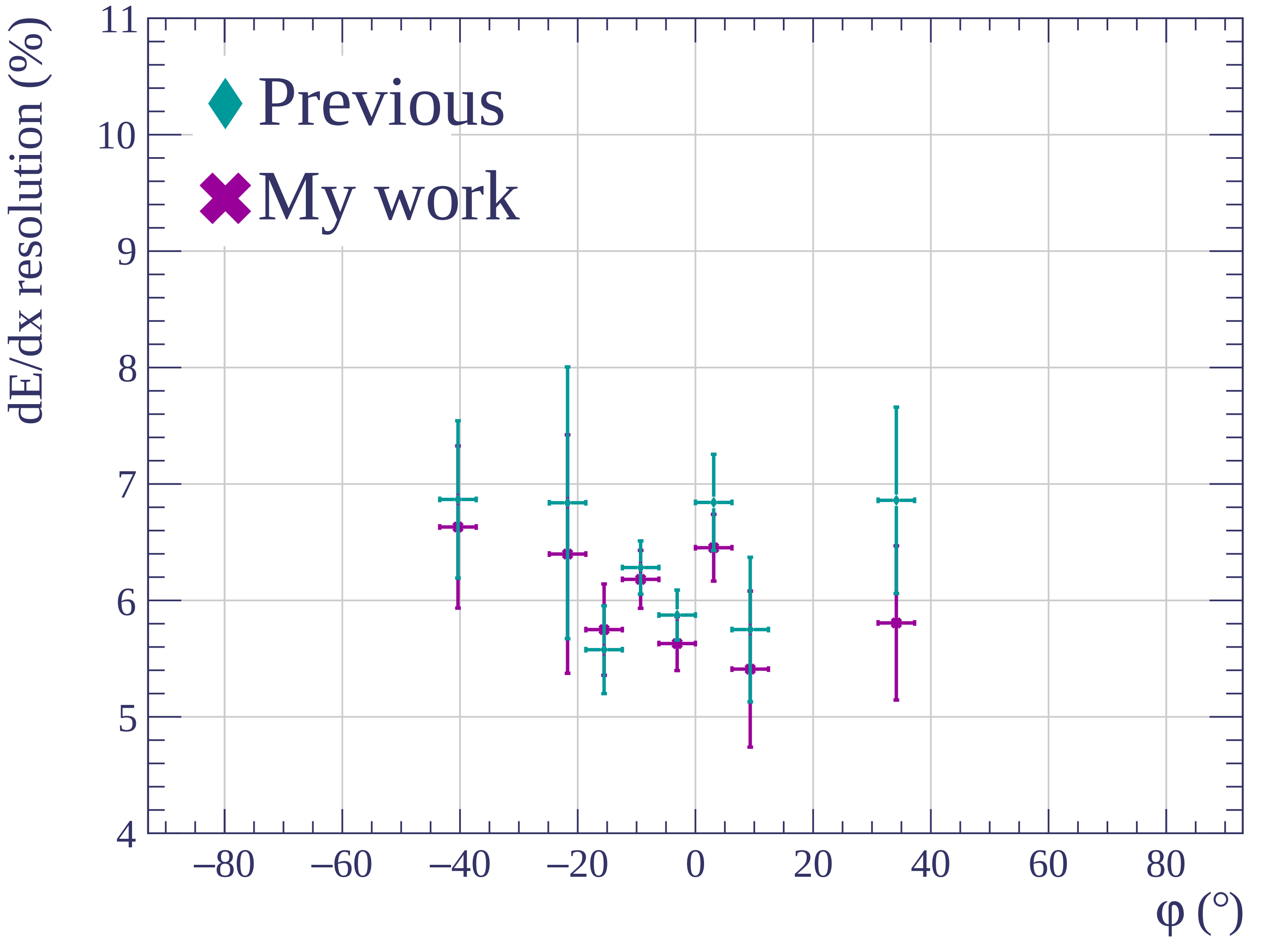

momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
180
200
$N_{clusters}$
20
18
16
14
12
10
8
6
4
2
0



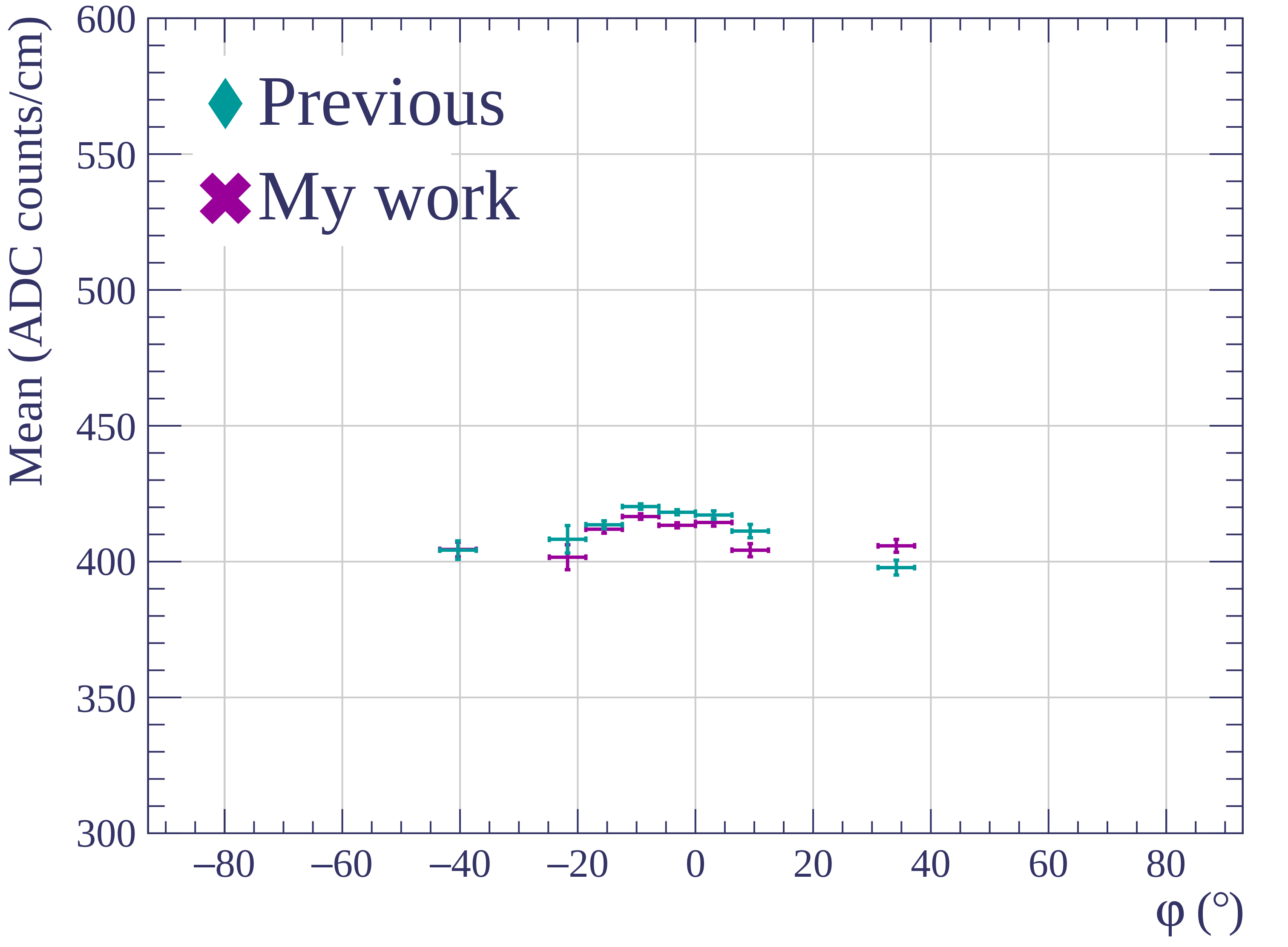
Previous
My work
dE/dx resolution (%)
φ (°)
4
5
6
7
8
9
10
11
0
10
20
30
40
50
60
70
80
90



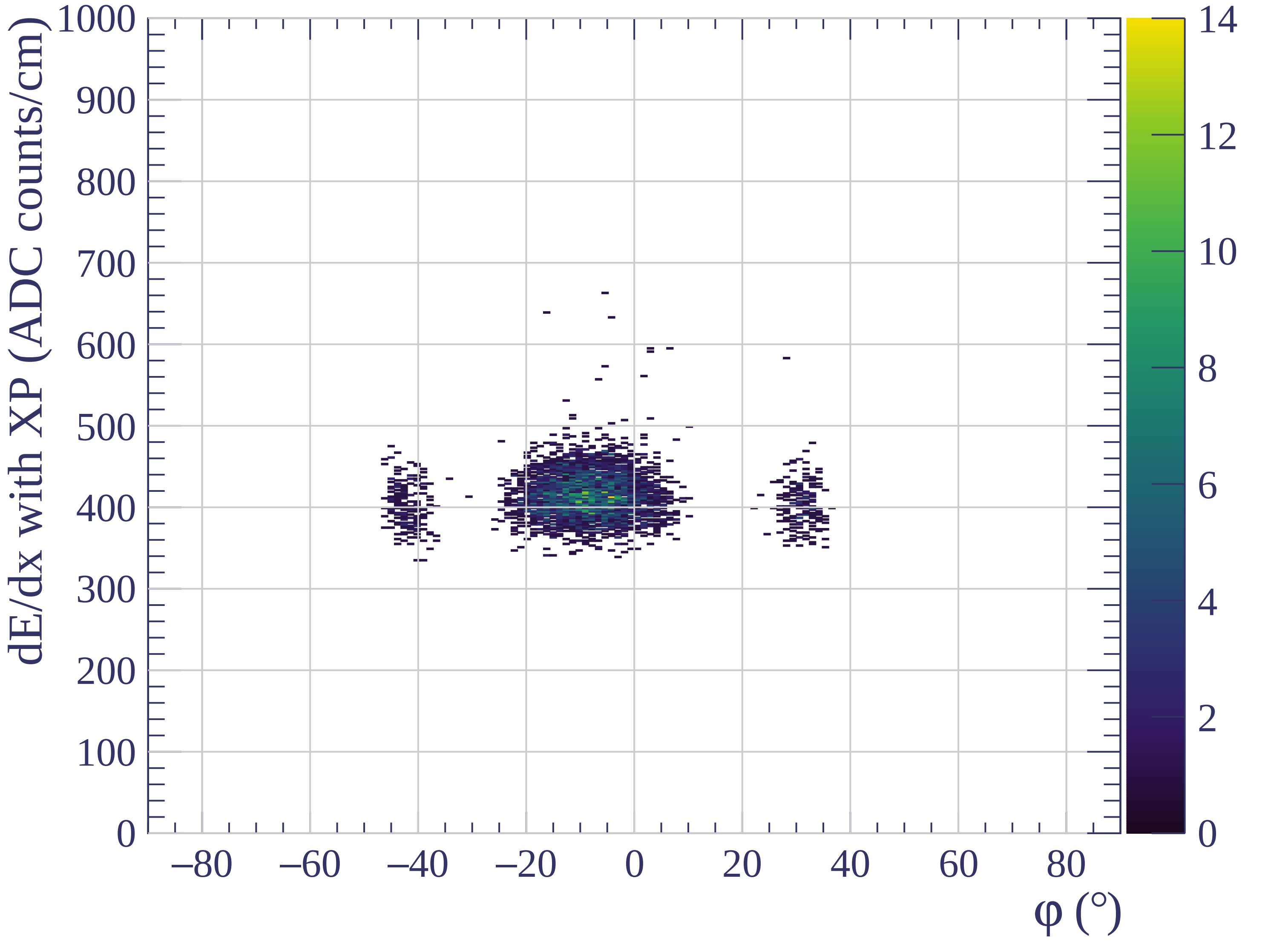
Previous
My work
Mean (ADC counts/cm)
|φ| (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90



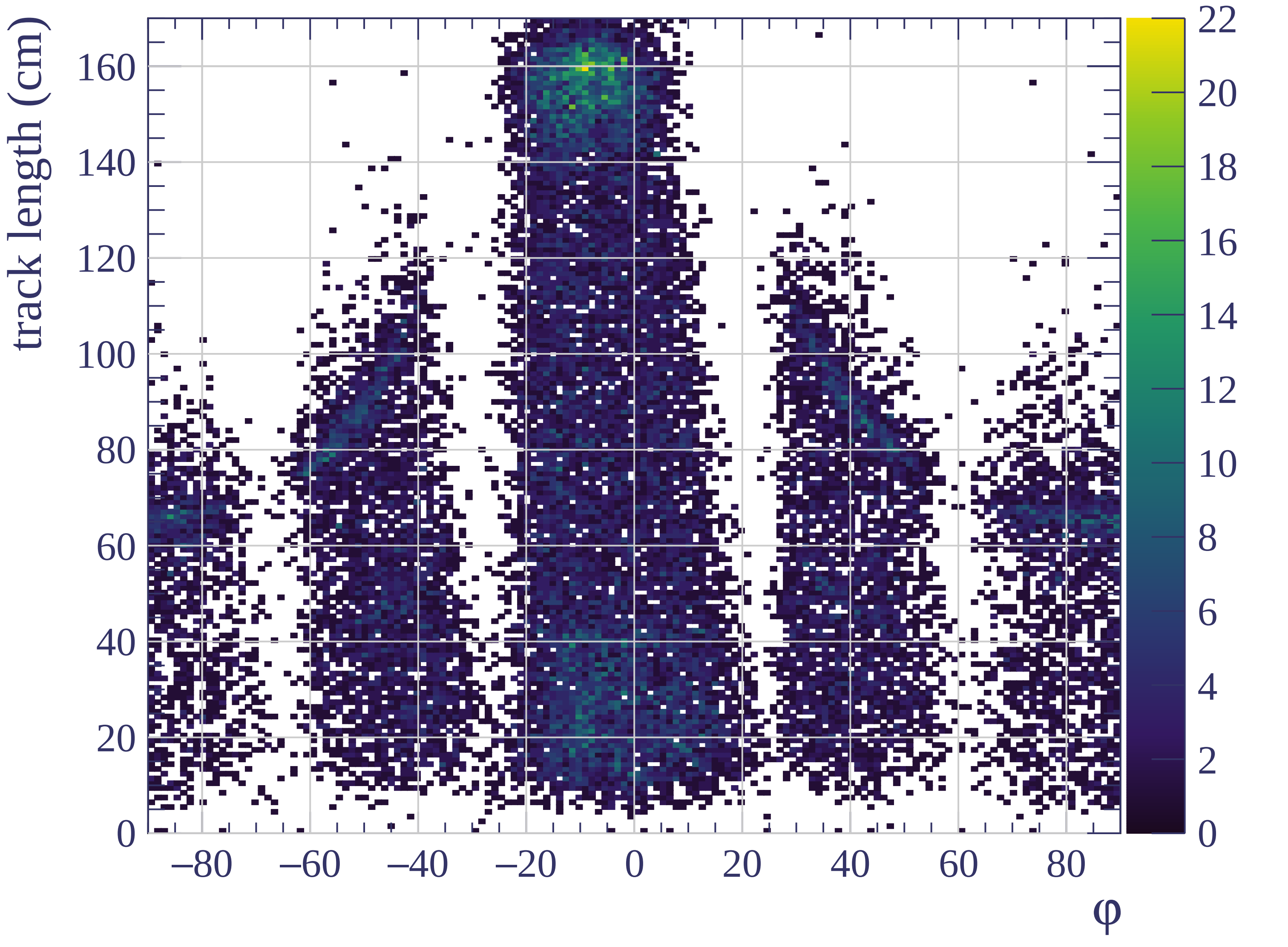
Previous
My work
dE/dx resolution (%)
φ (°)
4
5
6
7
8
9
10
11
−80
−60
−40
−20
0
20
40
60
80



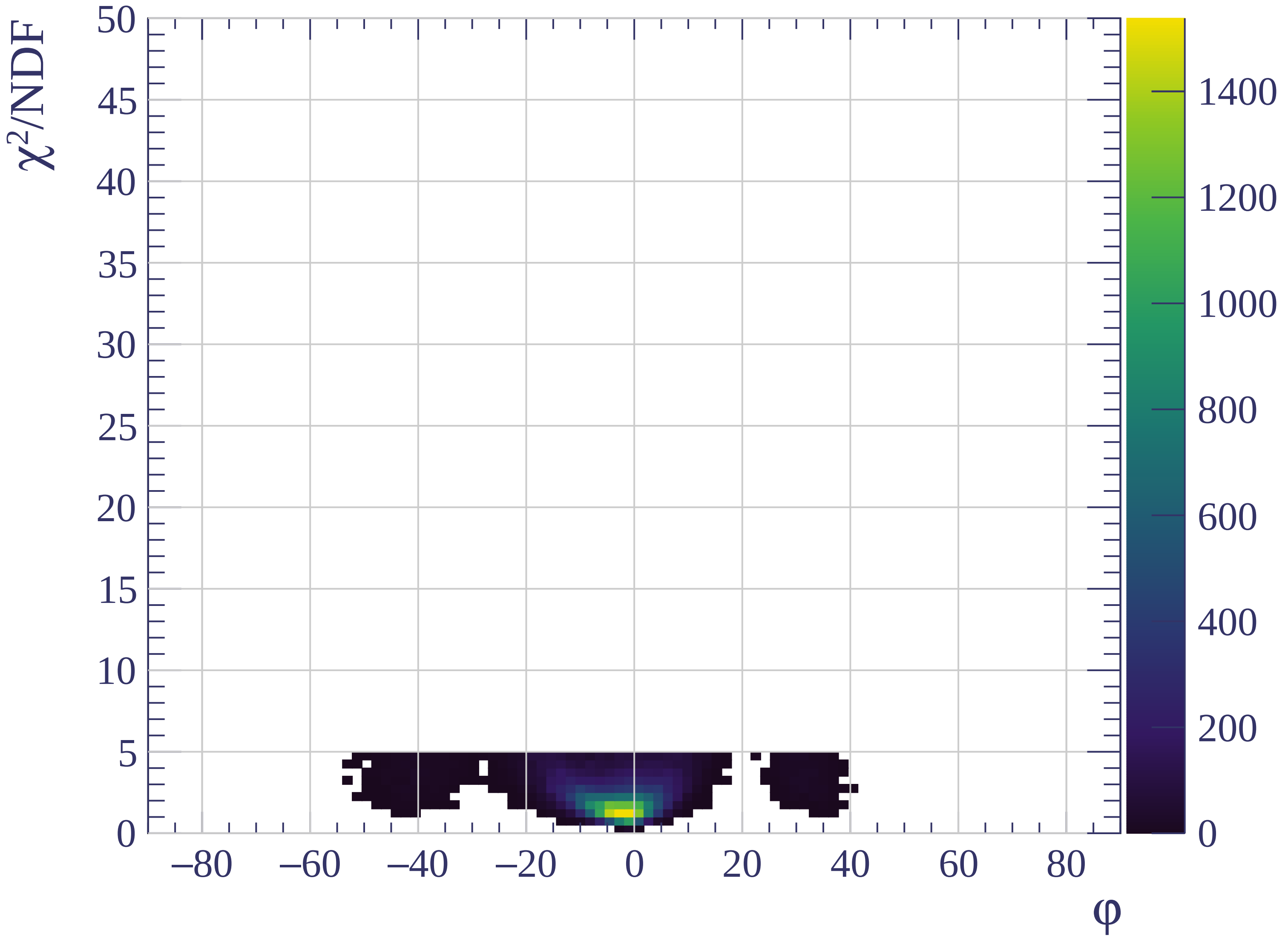
Previous
My work
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)



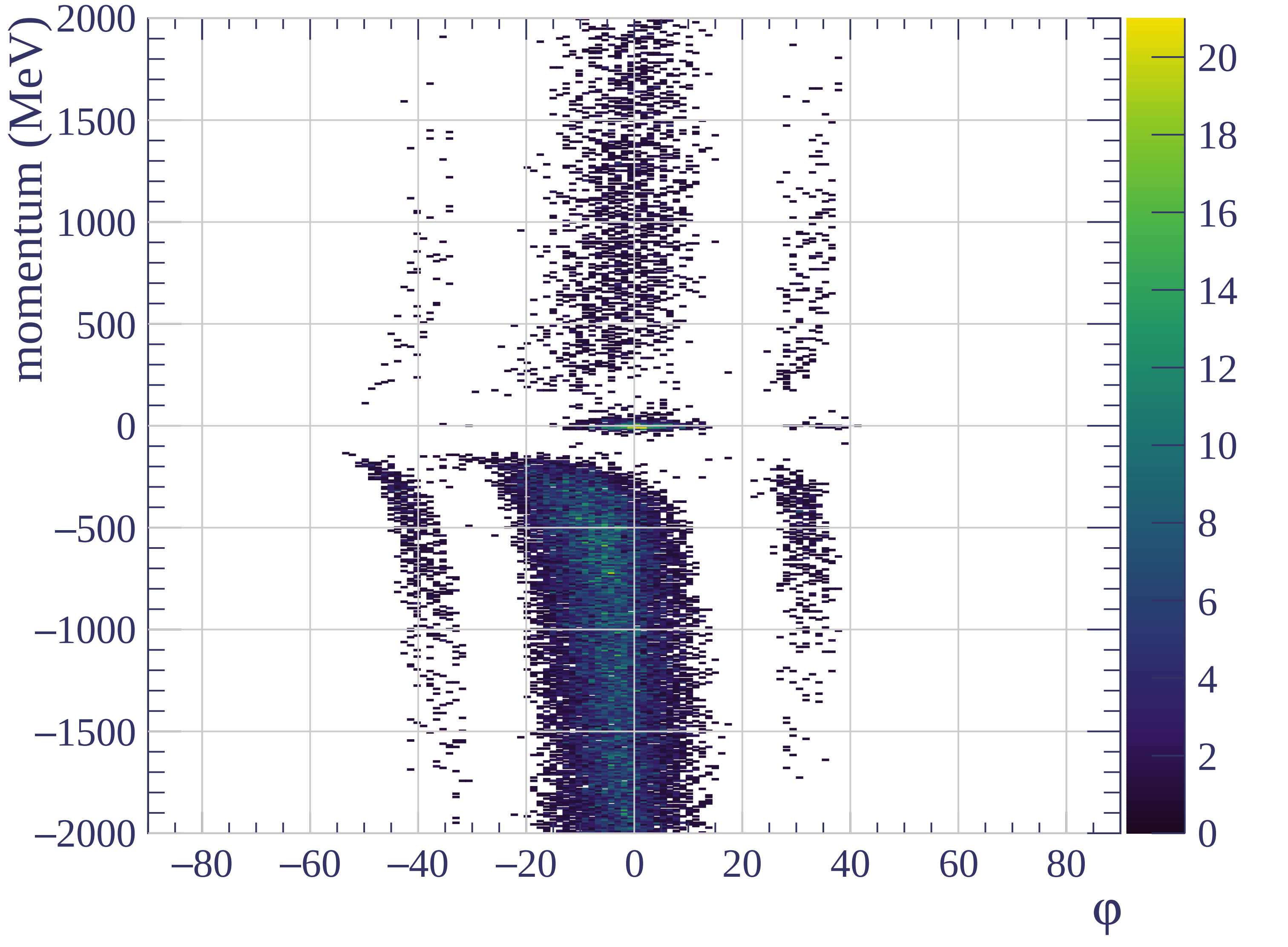
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
14
12
10
8
6
4
2
0



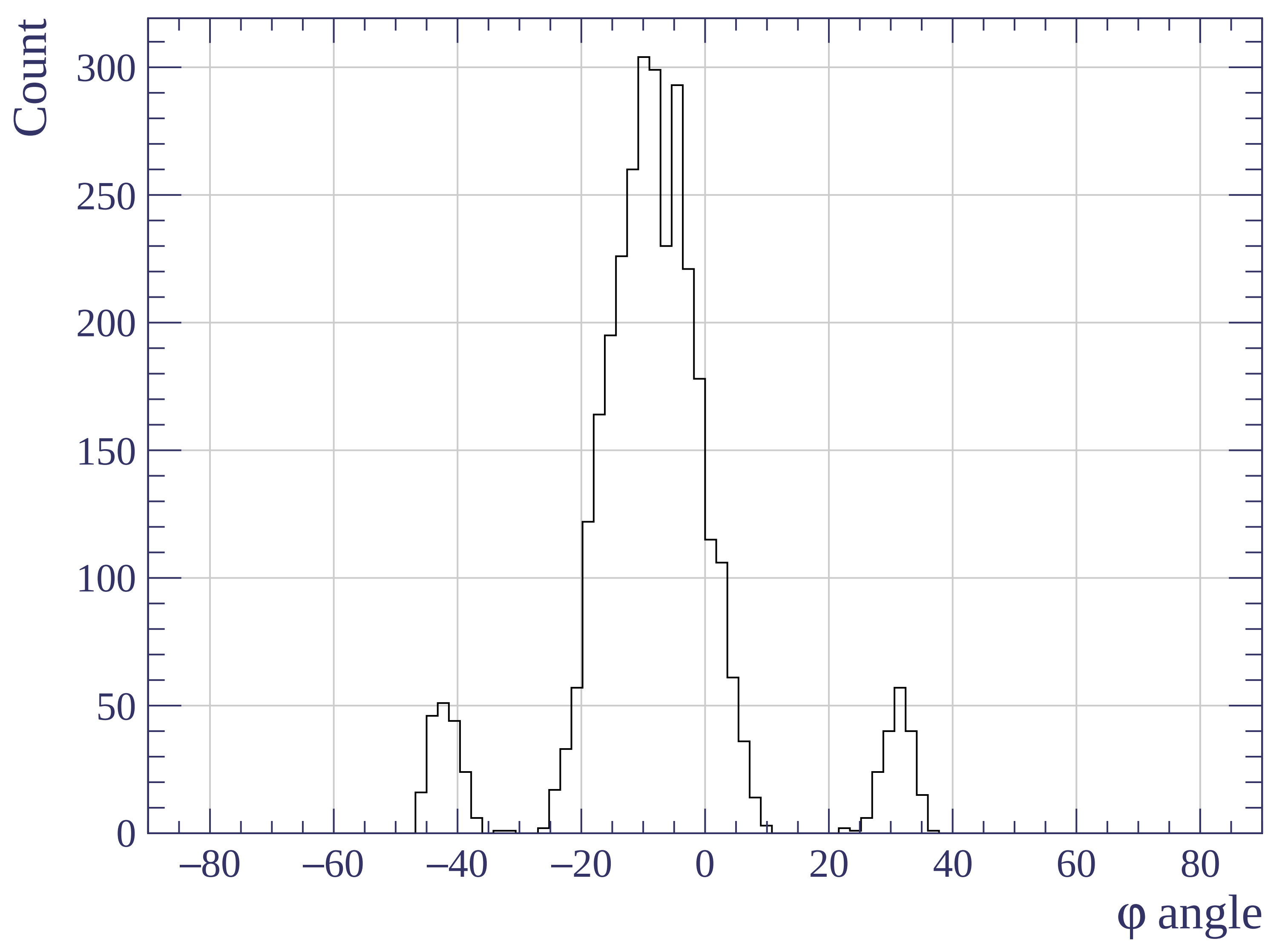
track length (cm)
φ

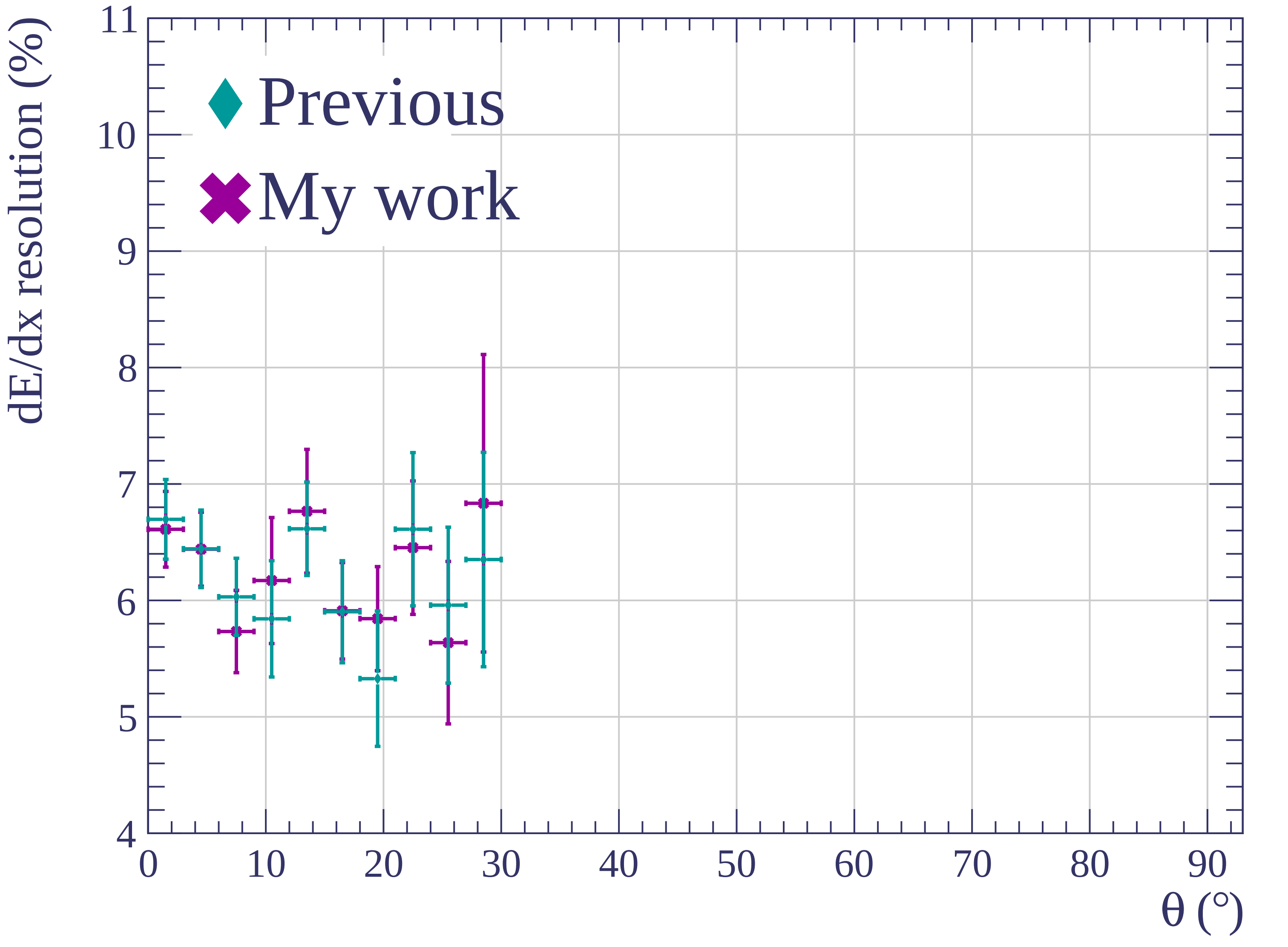


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
1400
1200
1000
800
600
400
200
0

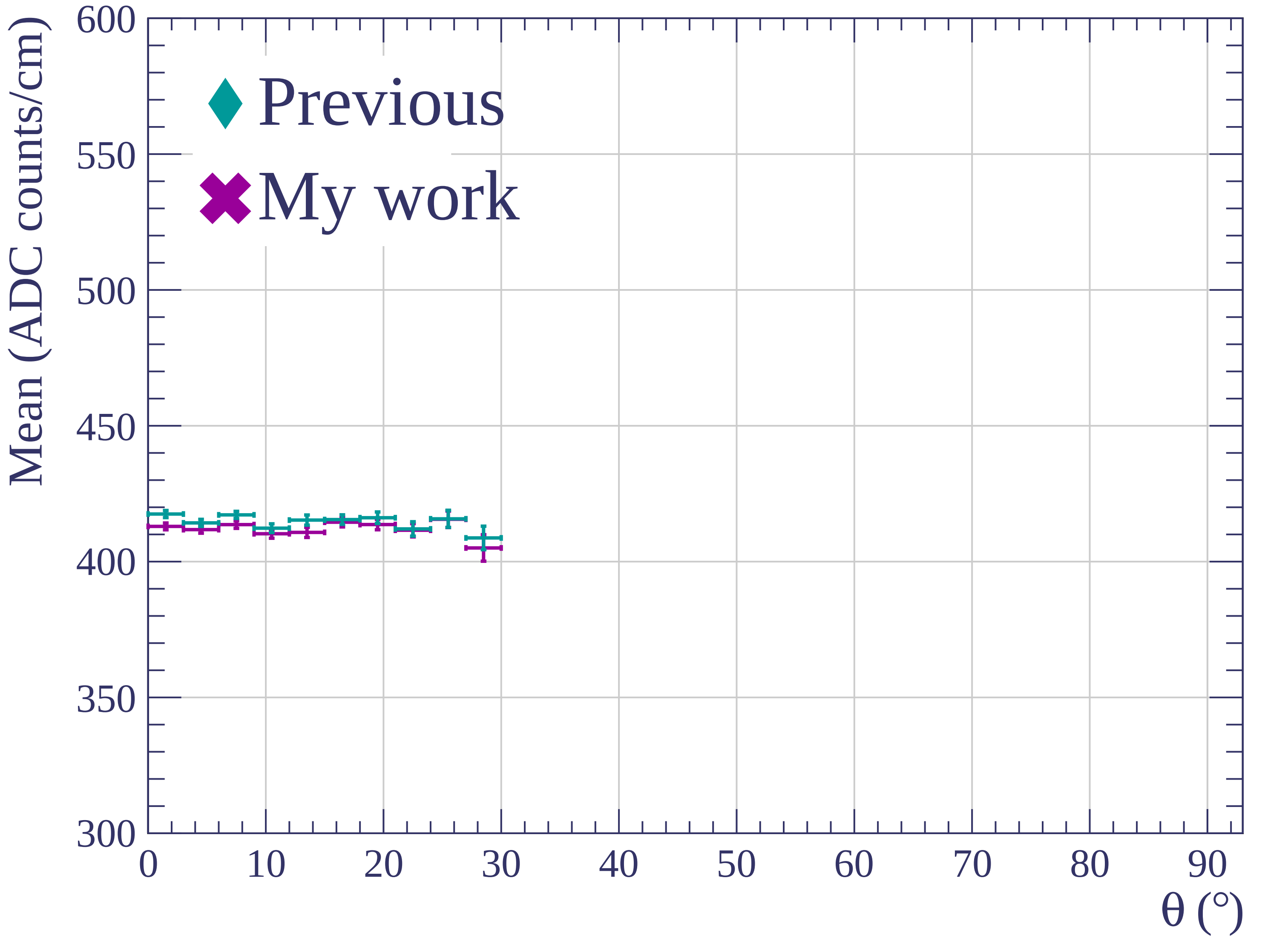


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
20
18
16
14
12
10
8
6
4
2
0

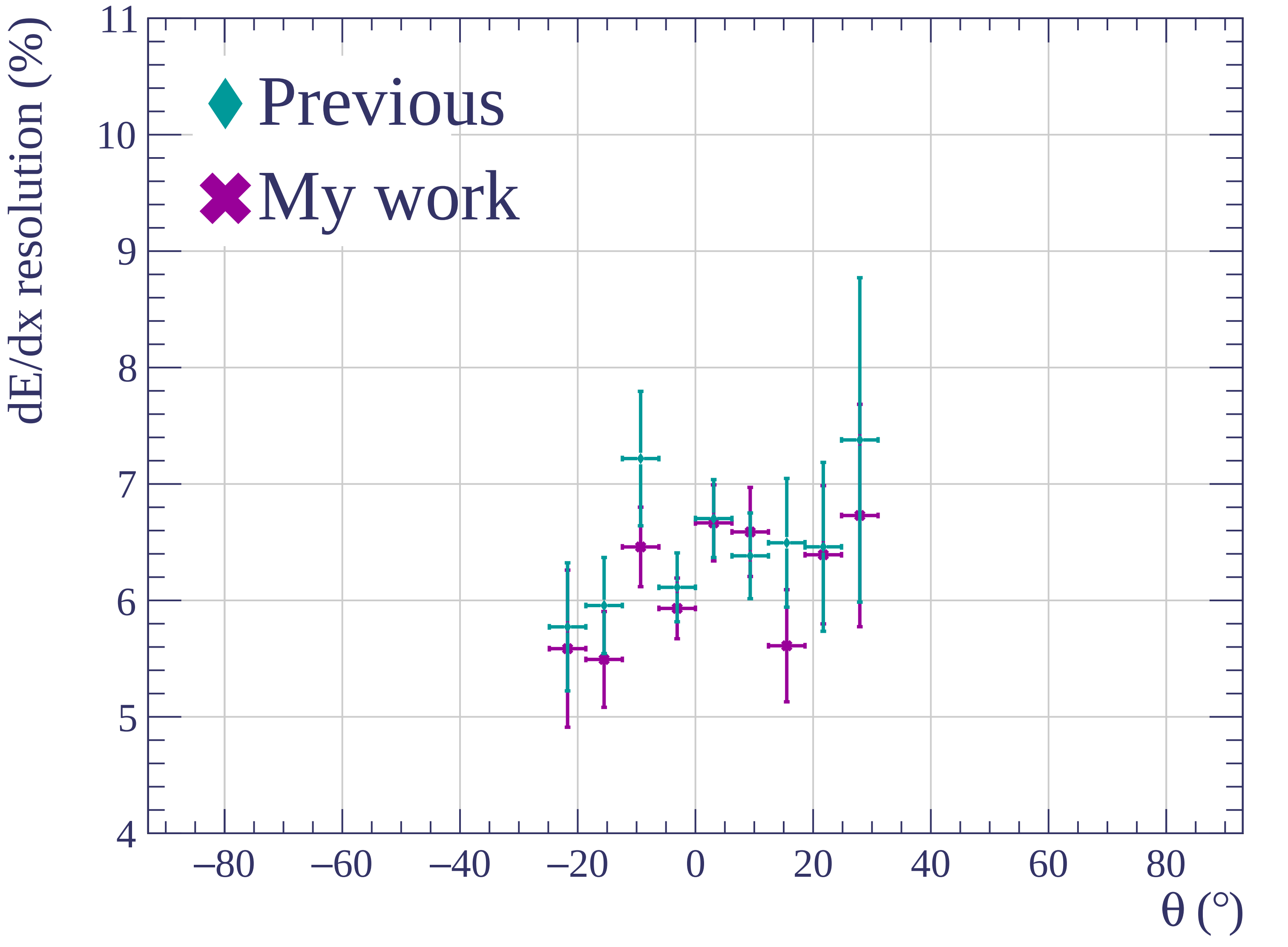

Count
300
250
200
150
100
50
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

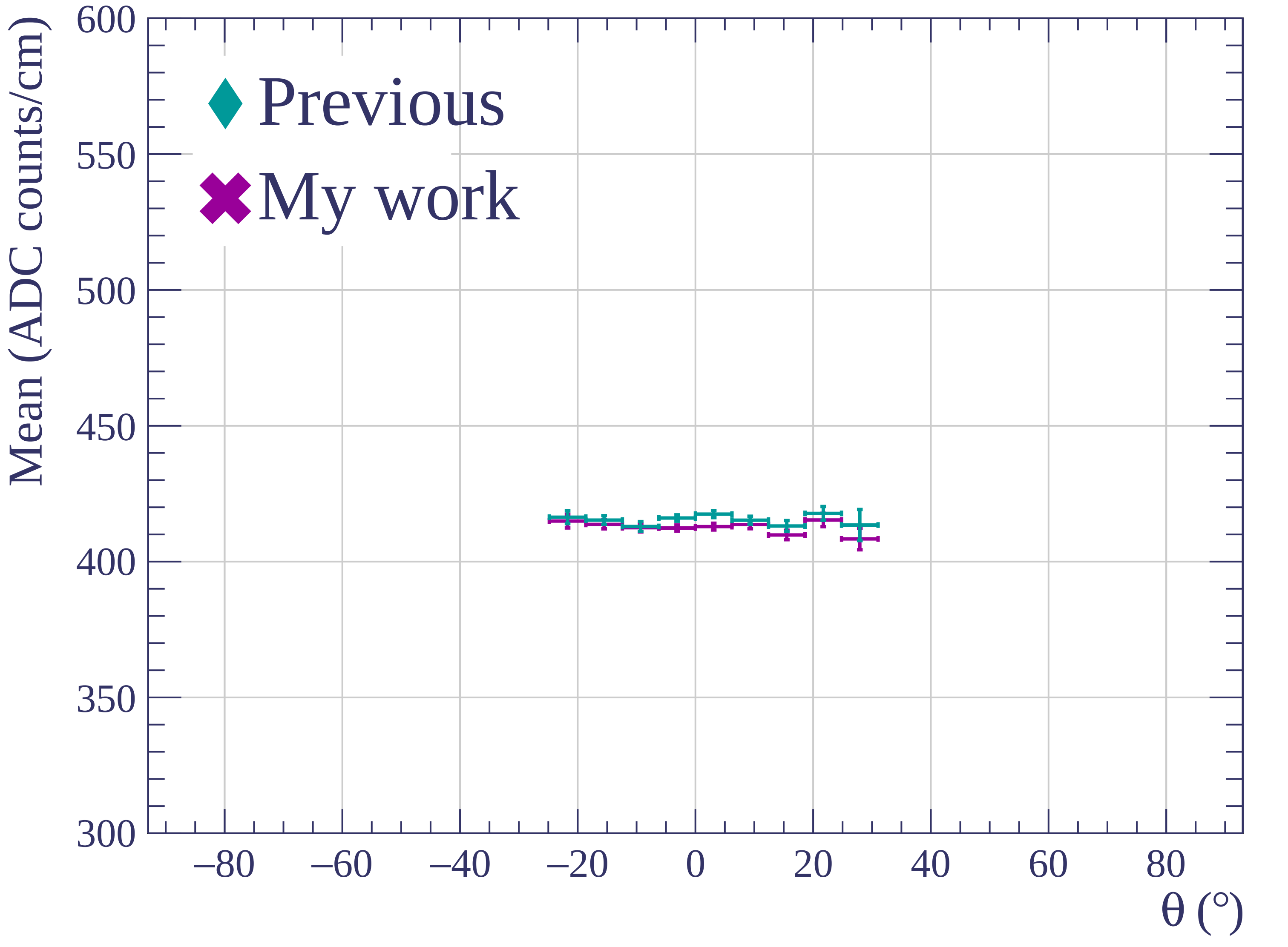


Previous
My work
dE/dx resolution (%)
θ (°)
11
10
9
8
7
6
5
4
0
10
20
30
40
50
60
70
80
90

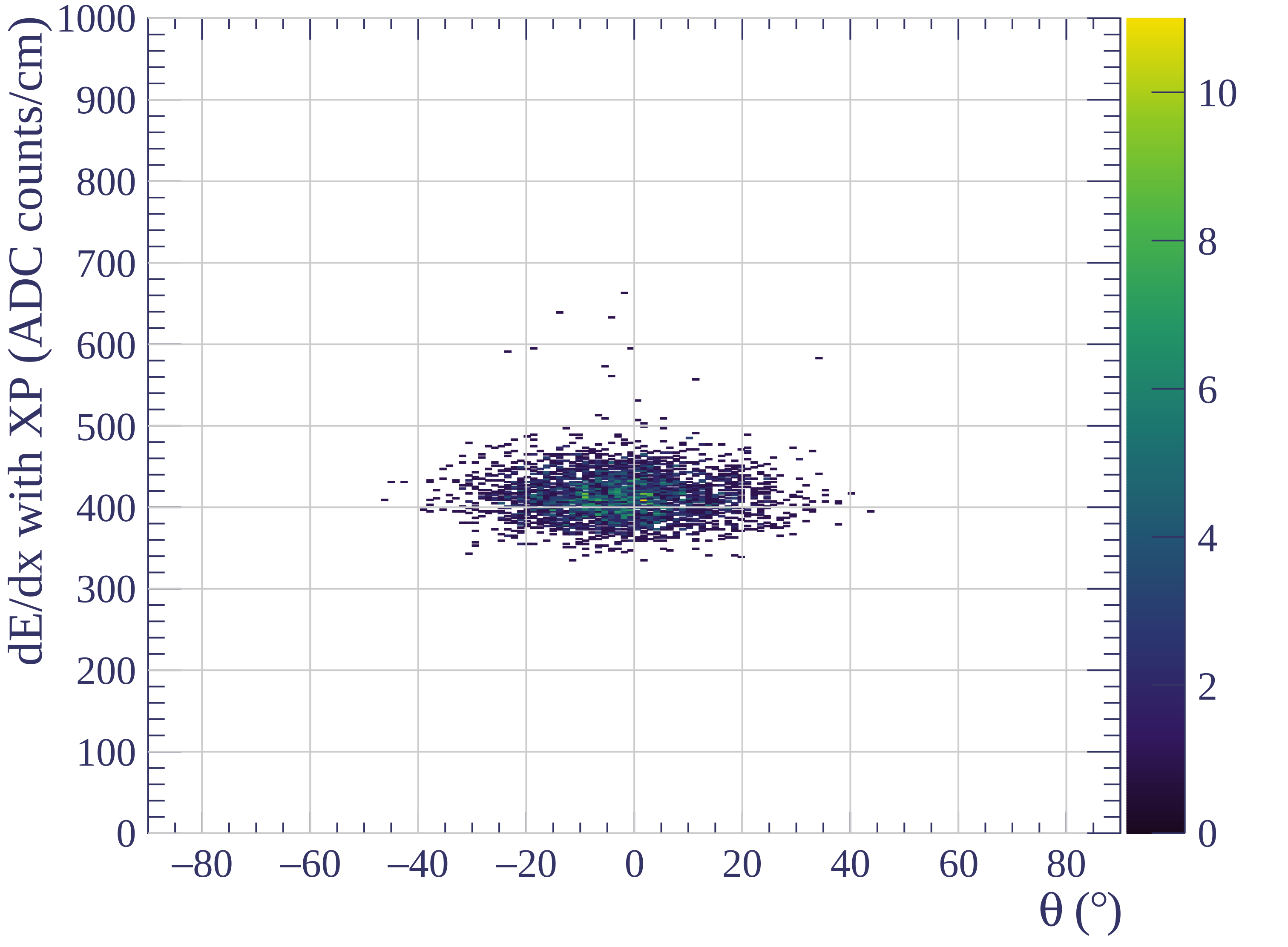


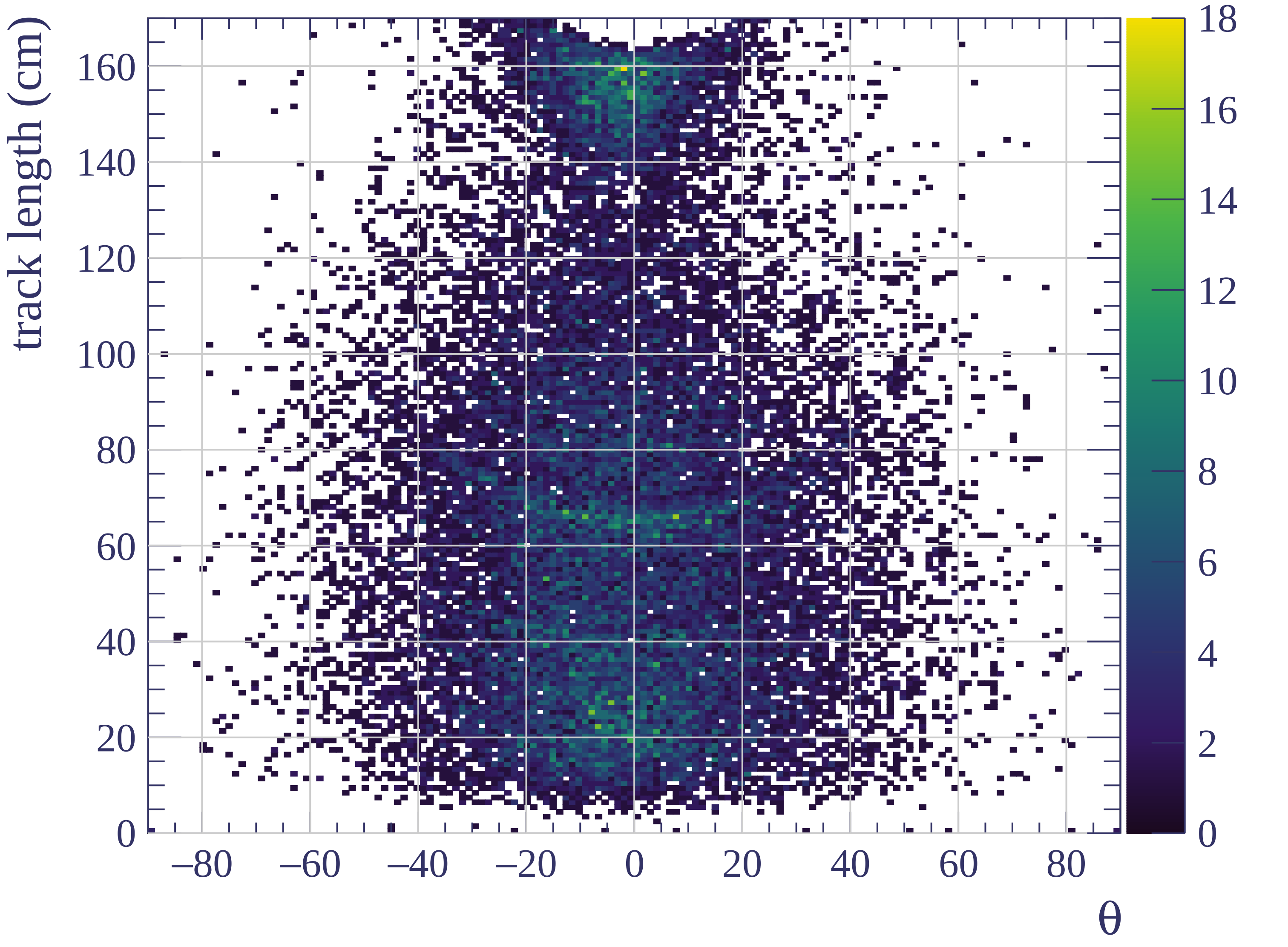
Previous
My work
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90


Previous
My work
dE/dx resolution (%)
θ (°)
4
5
6
7
8
9
10
11
−80
−60
−40
−20
0
20
40
60
80

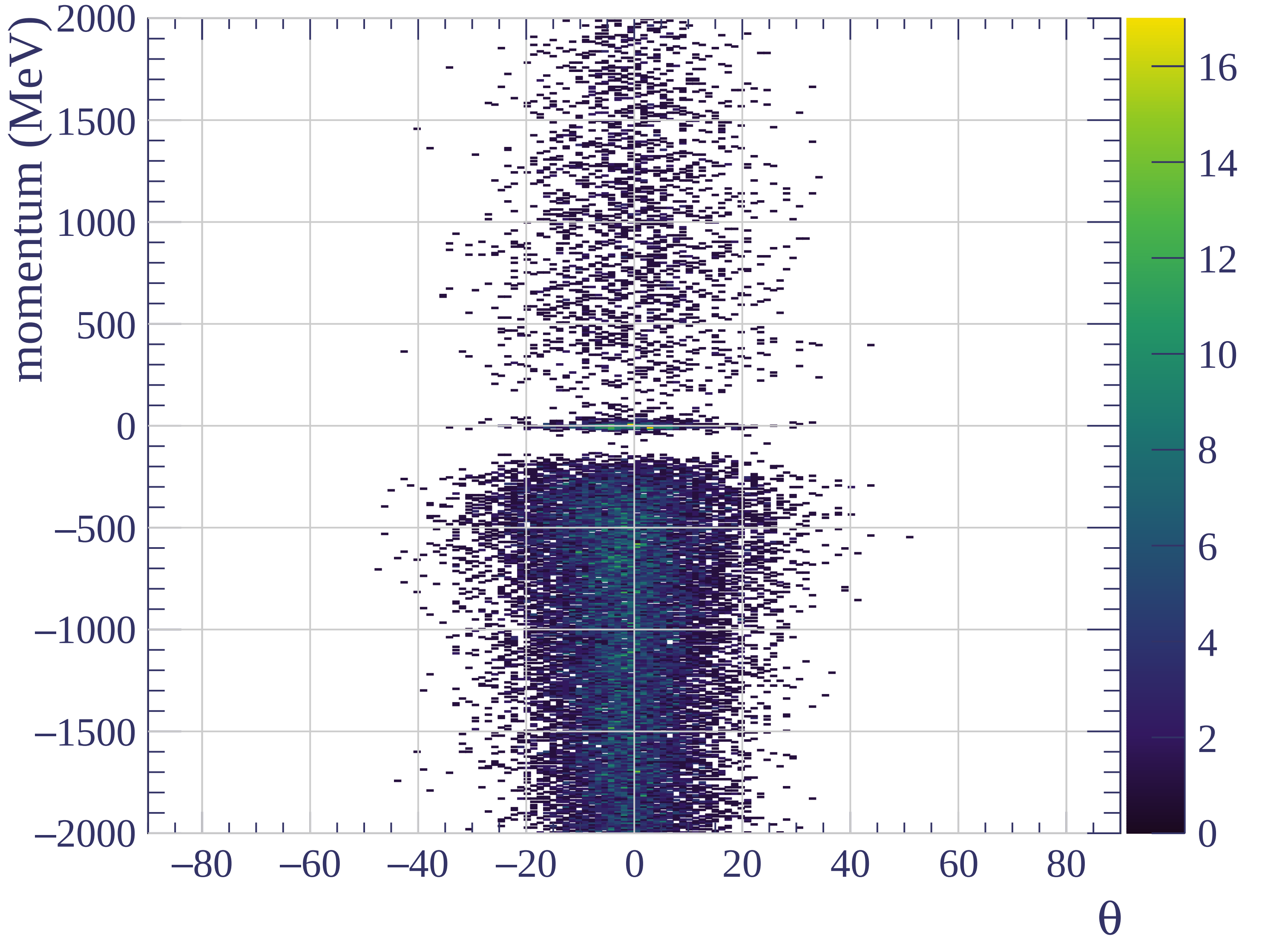


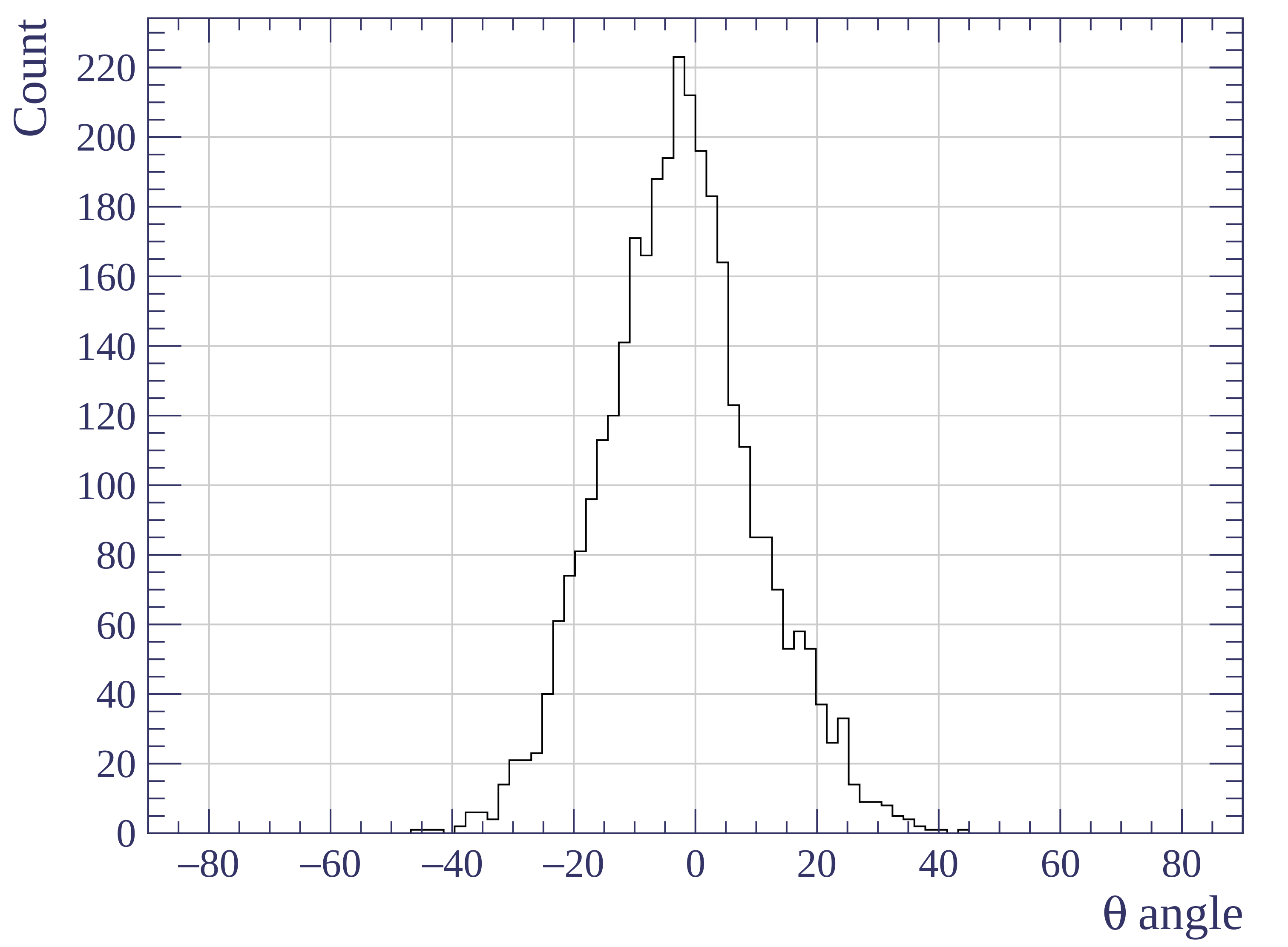
Previous
My work
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
θ (°)


dE/dx with XP (ADC counts/cm)
θ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
10
8
6
4
2
0

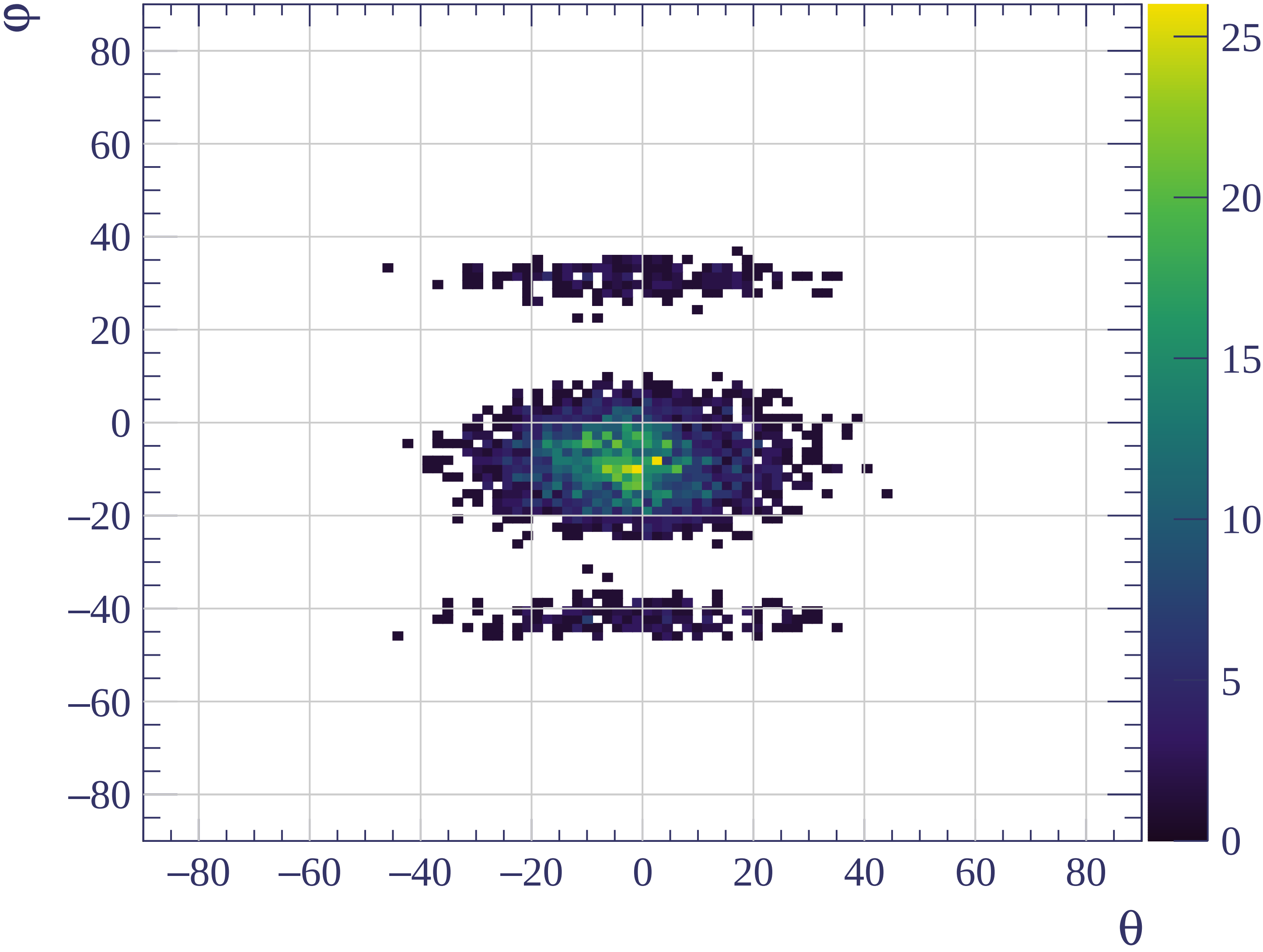


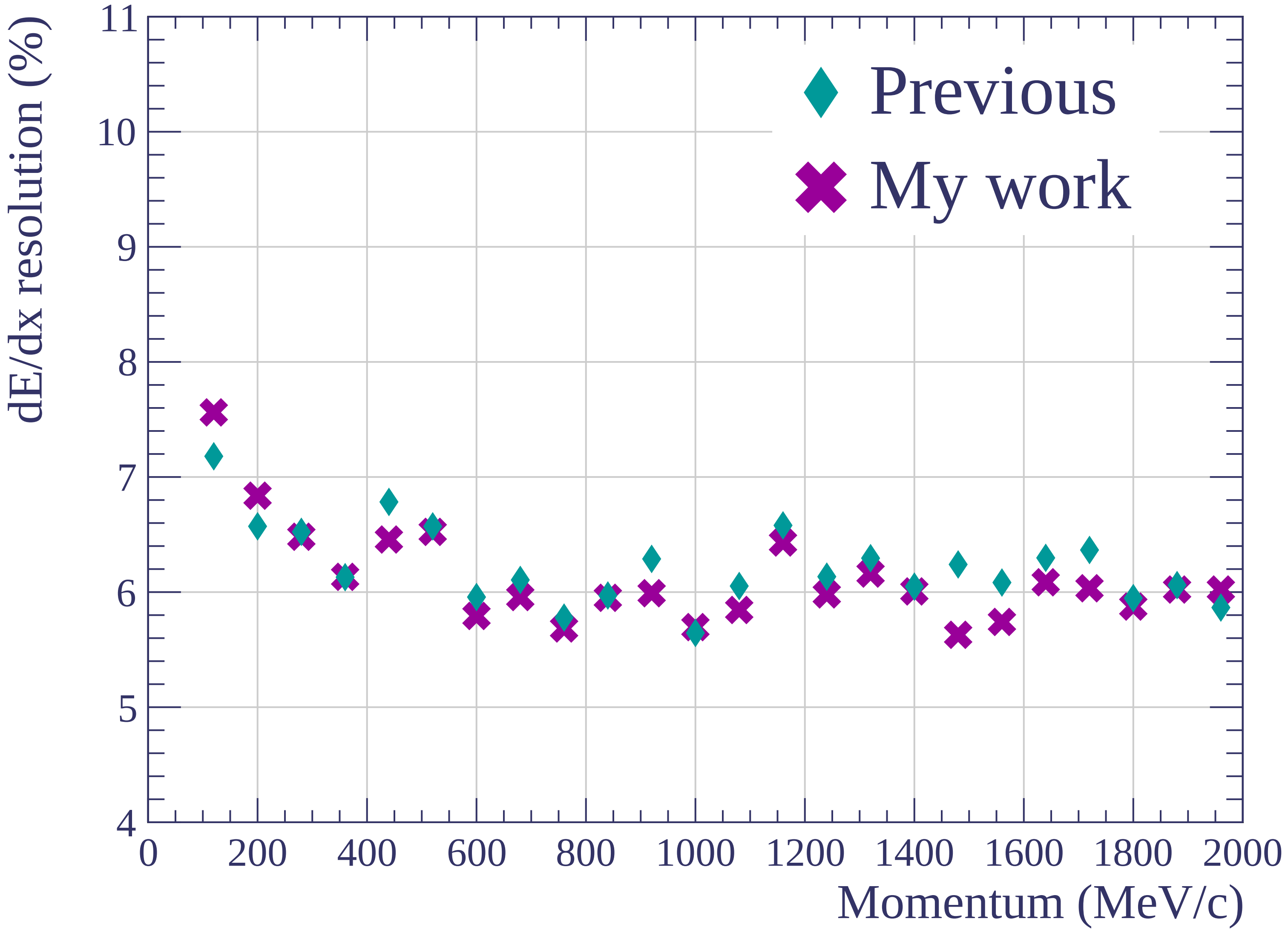
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
18
16
14
12
10
8
6
4
2
0


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
16
14
12
10
8
6
4
2
0

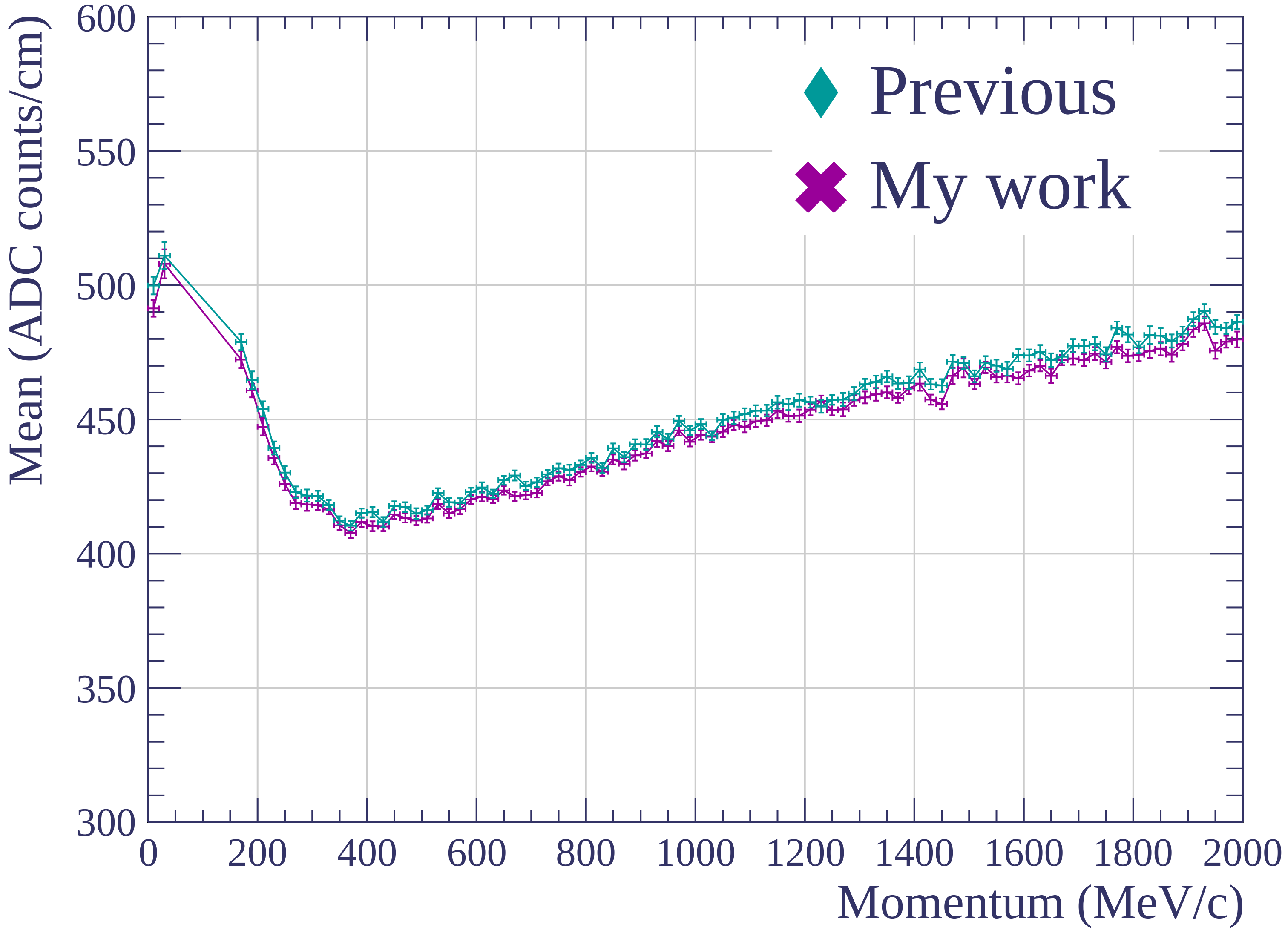


Count
0
20
40
60
80
100
120
140
160
180
200
220
−80
−60
−40
−20
0
20
40
60
80
θ angle

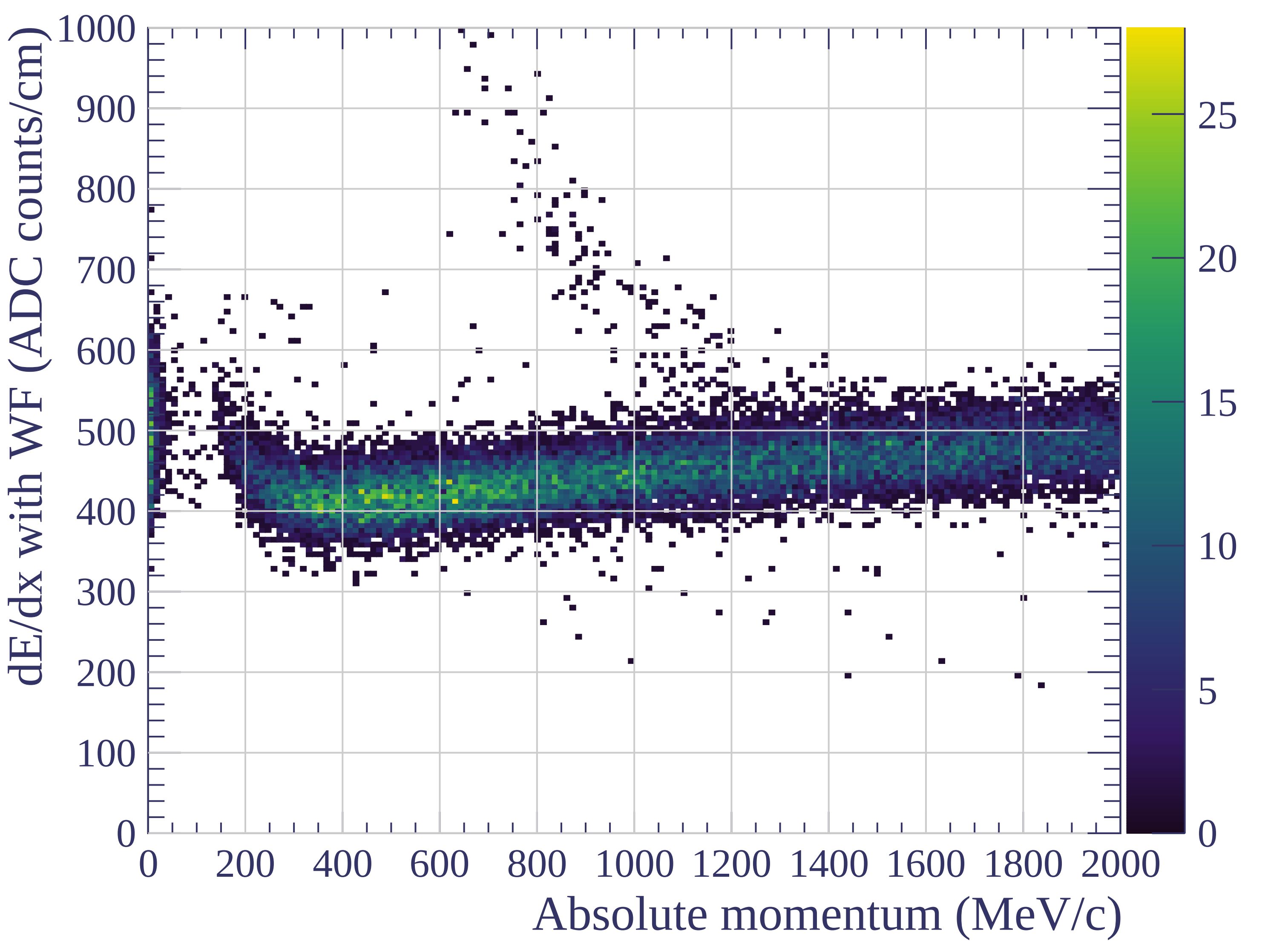

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
25
20
15
10
5
0



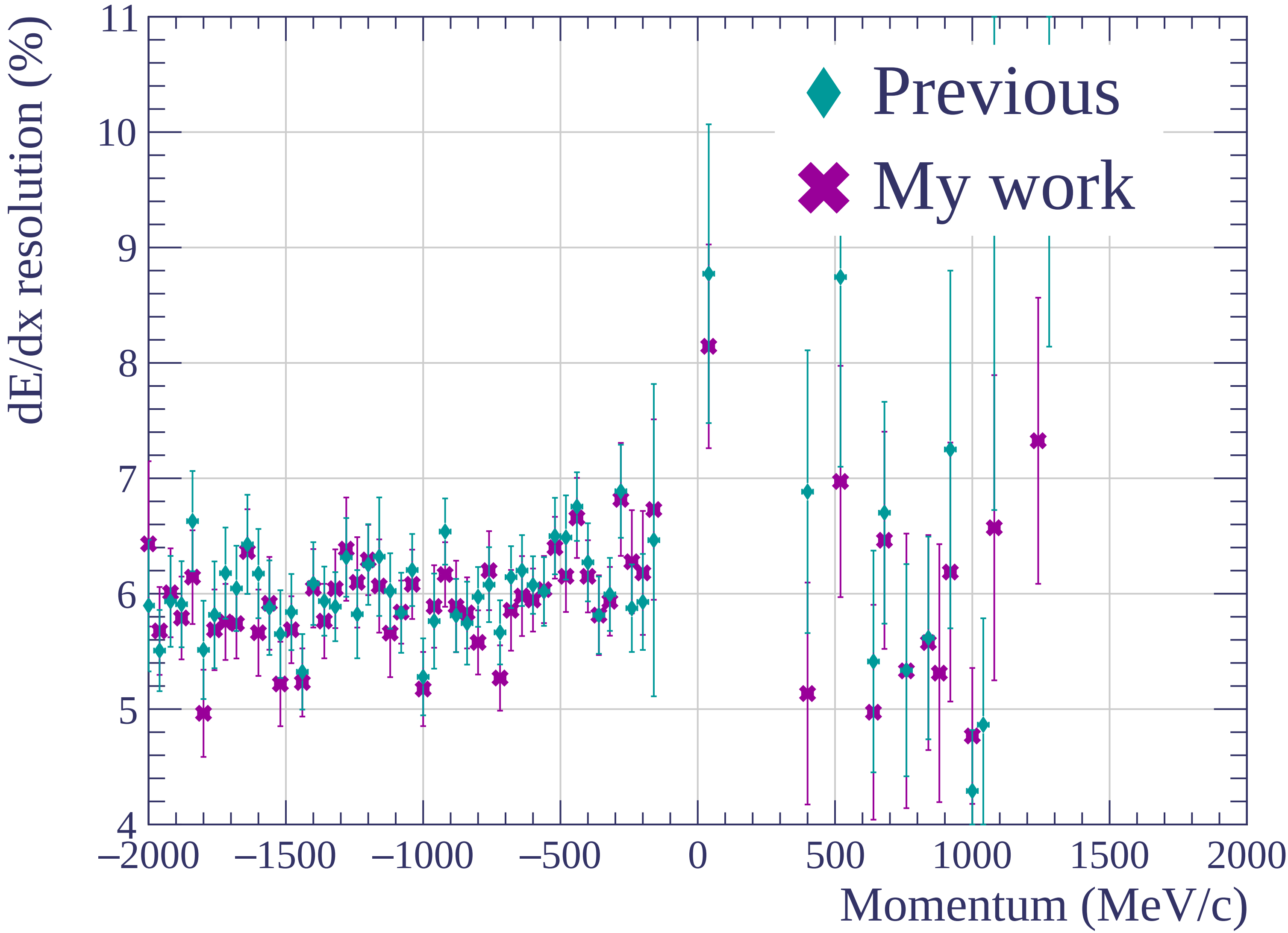
Previous
My work
dE/dx resolution (%)
Momentum (MeV/c)
4
5
6
7
8
9
10
11
0
200
400
600
800
1000
1200
1400
1600
1800
2000



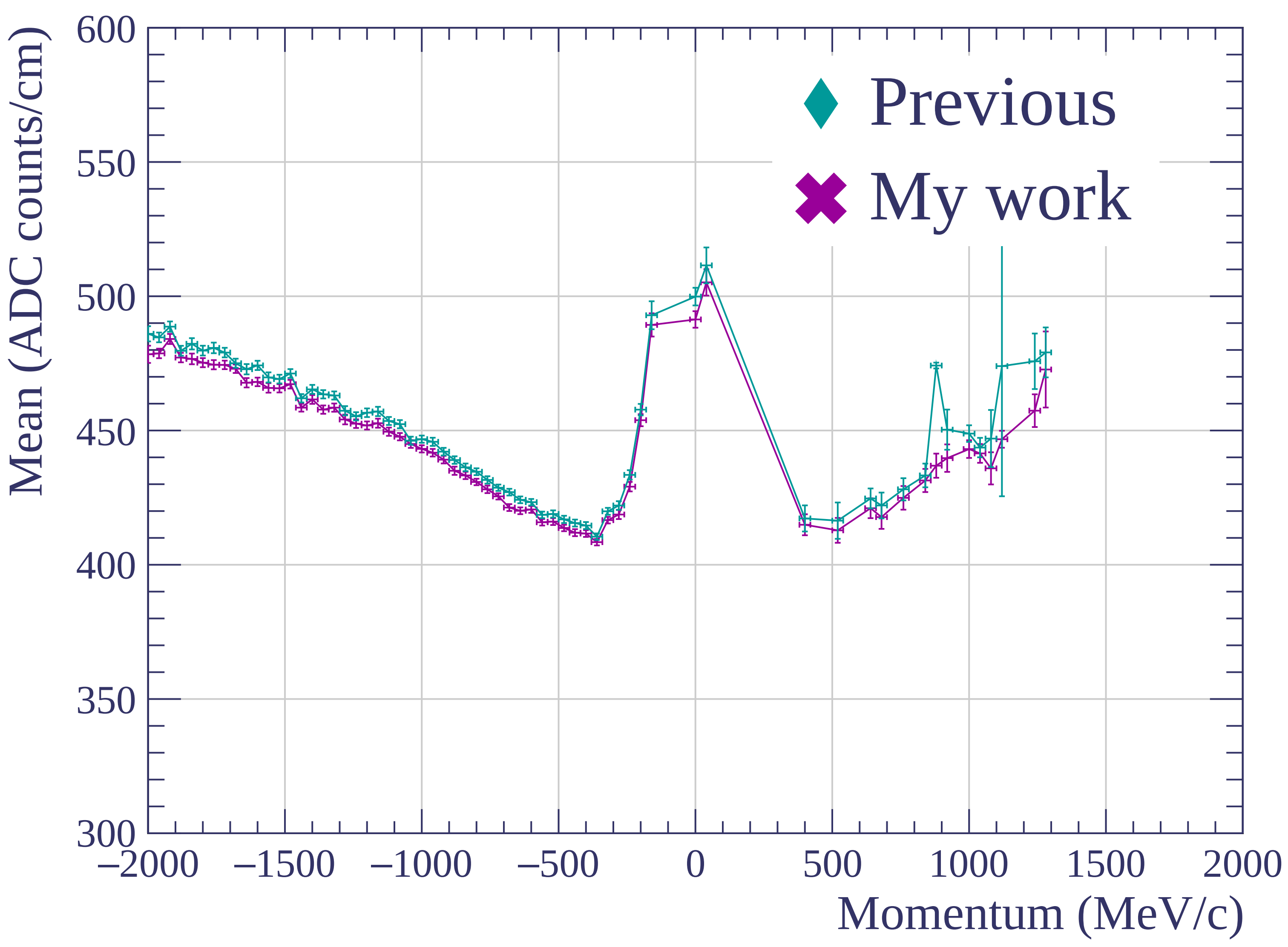
Previous
My work
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000



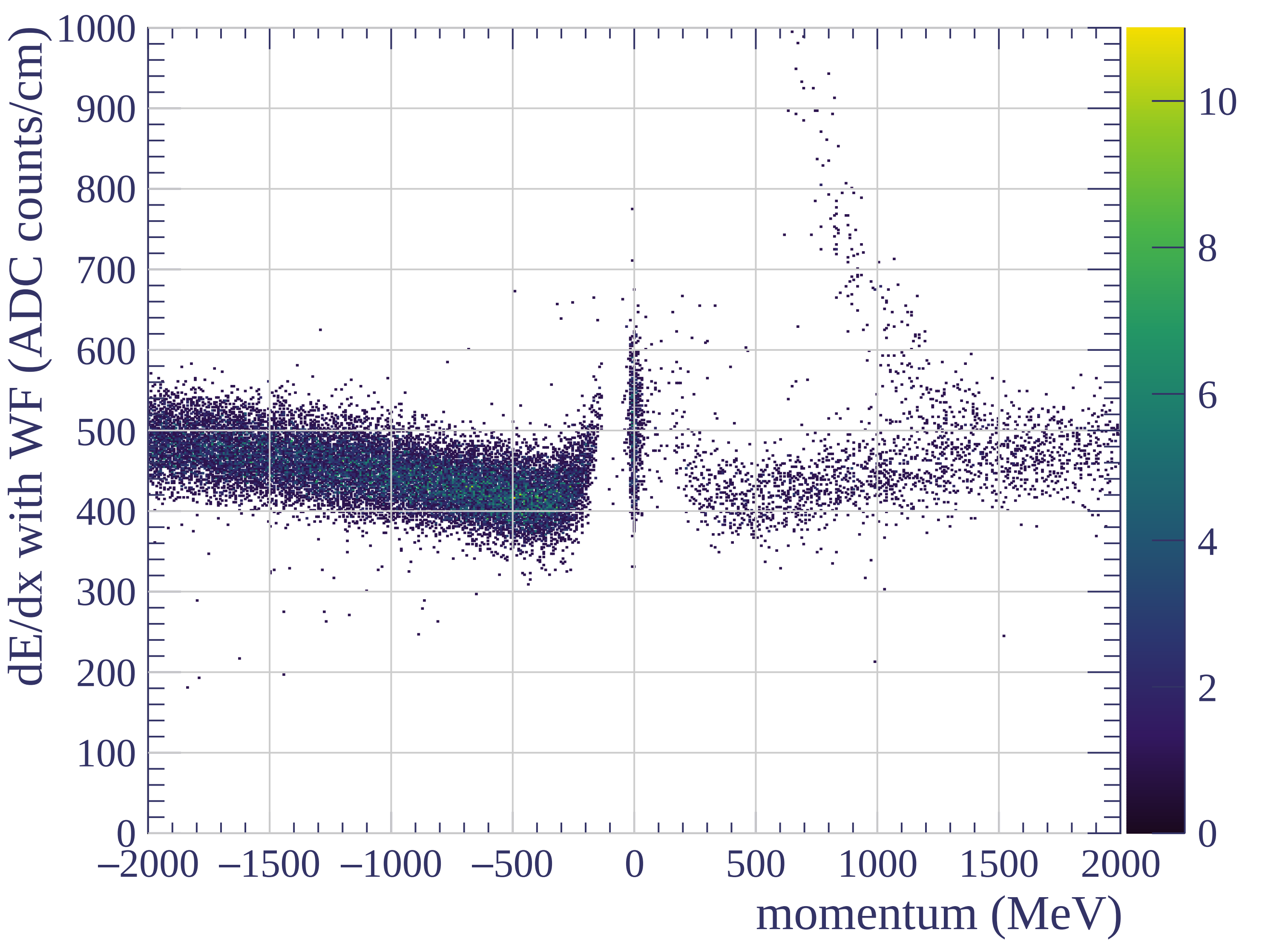
dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
5
10
15
20
25



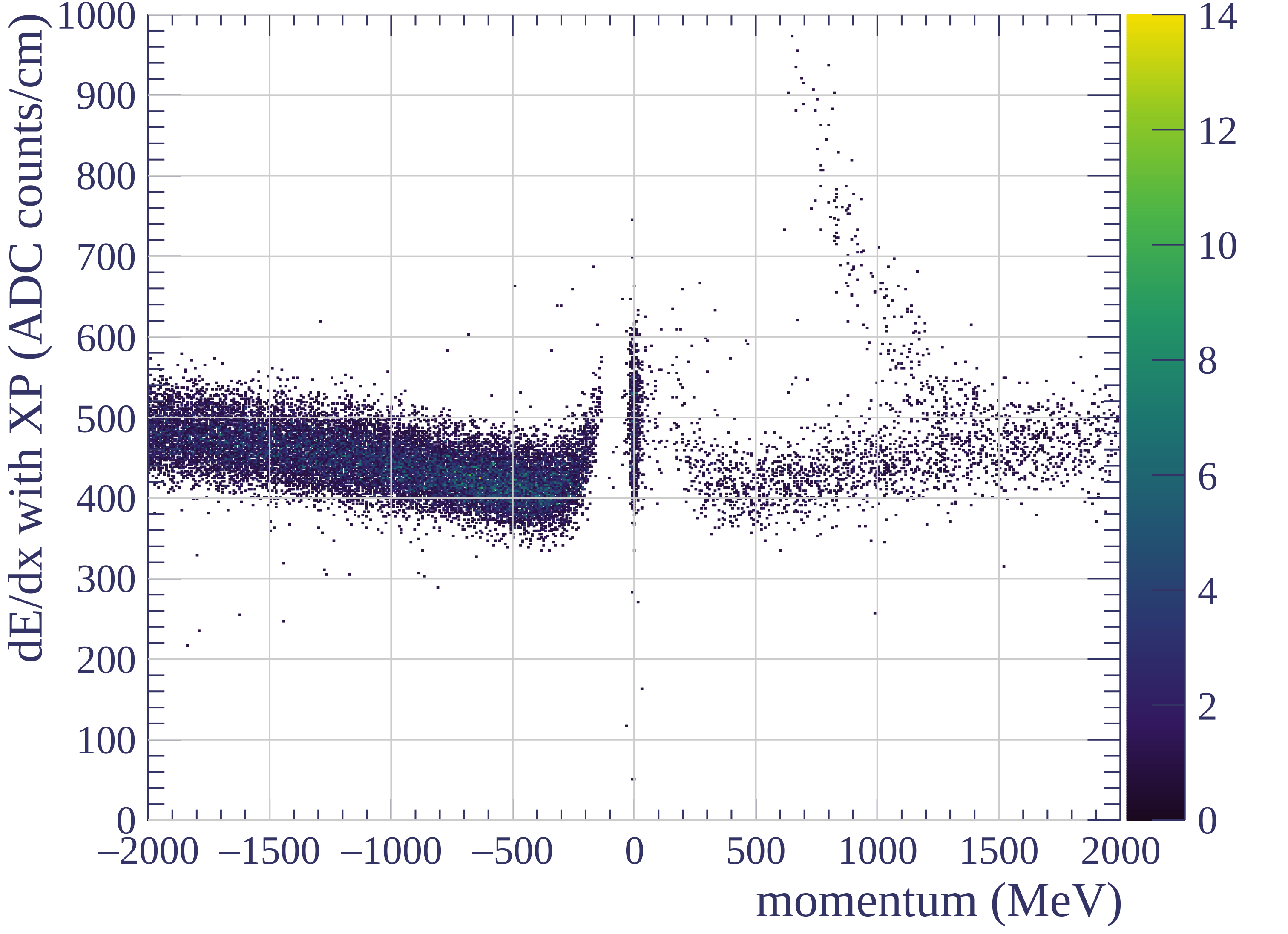
Previous
My work
dE/dx resolution (%)
Momentum (MeV/c)
4
5
6
7
8
9
10
11
−2000
−1500
−1000
−500
0
500
1000
1500
2000

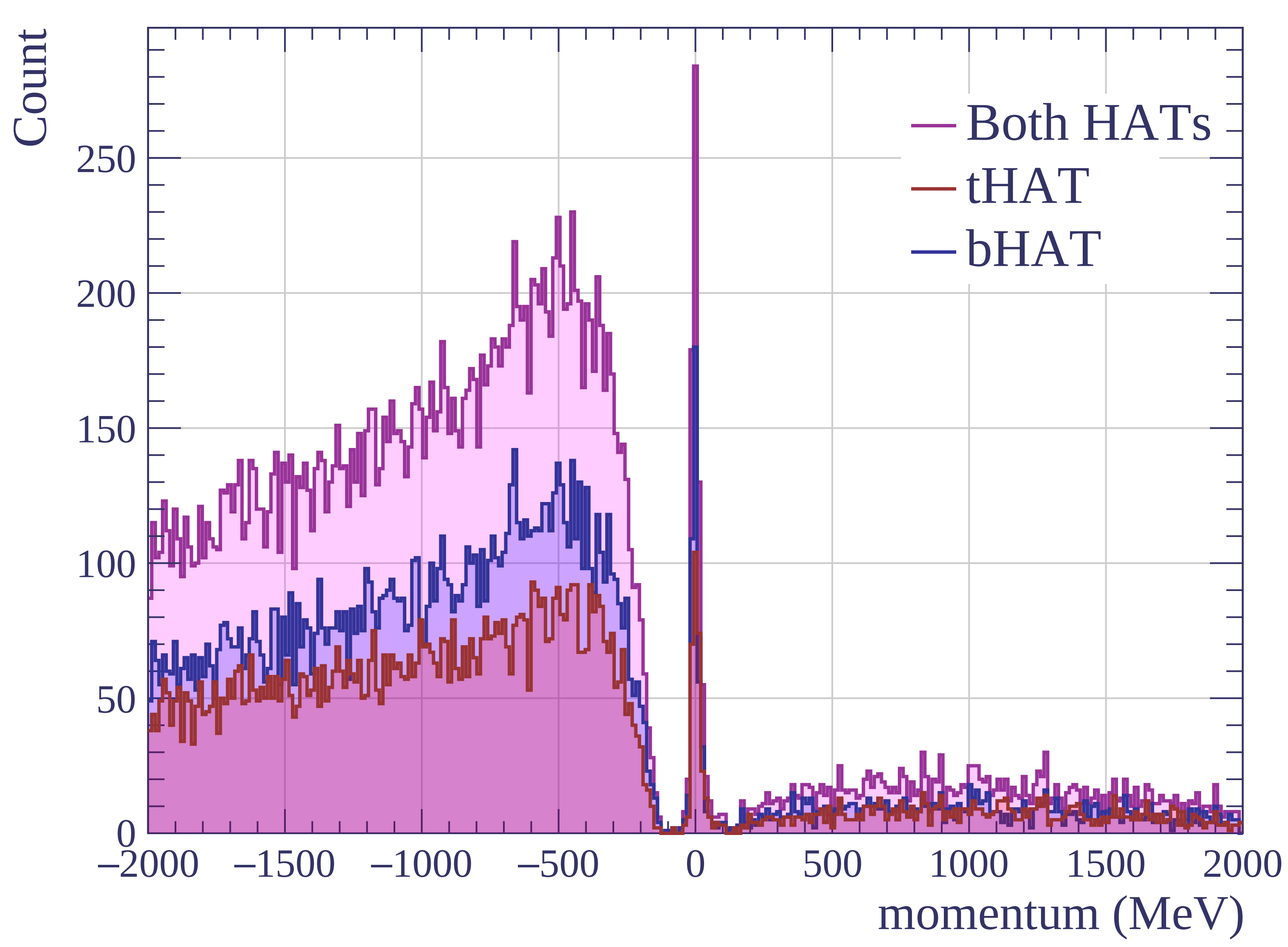


Previous
My work
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000

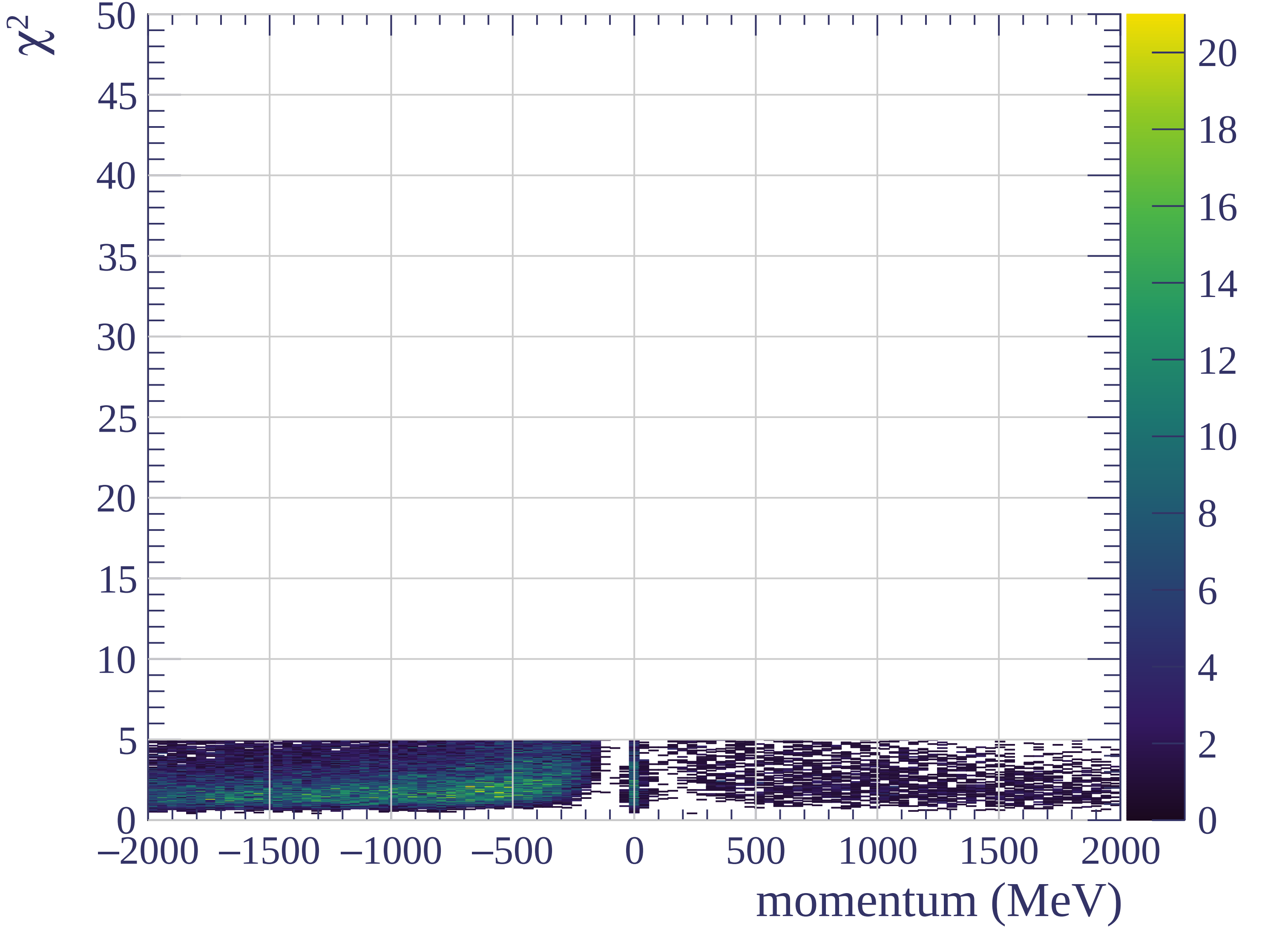


dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
10
8
6
4
2
0

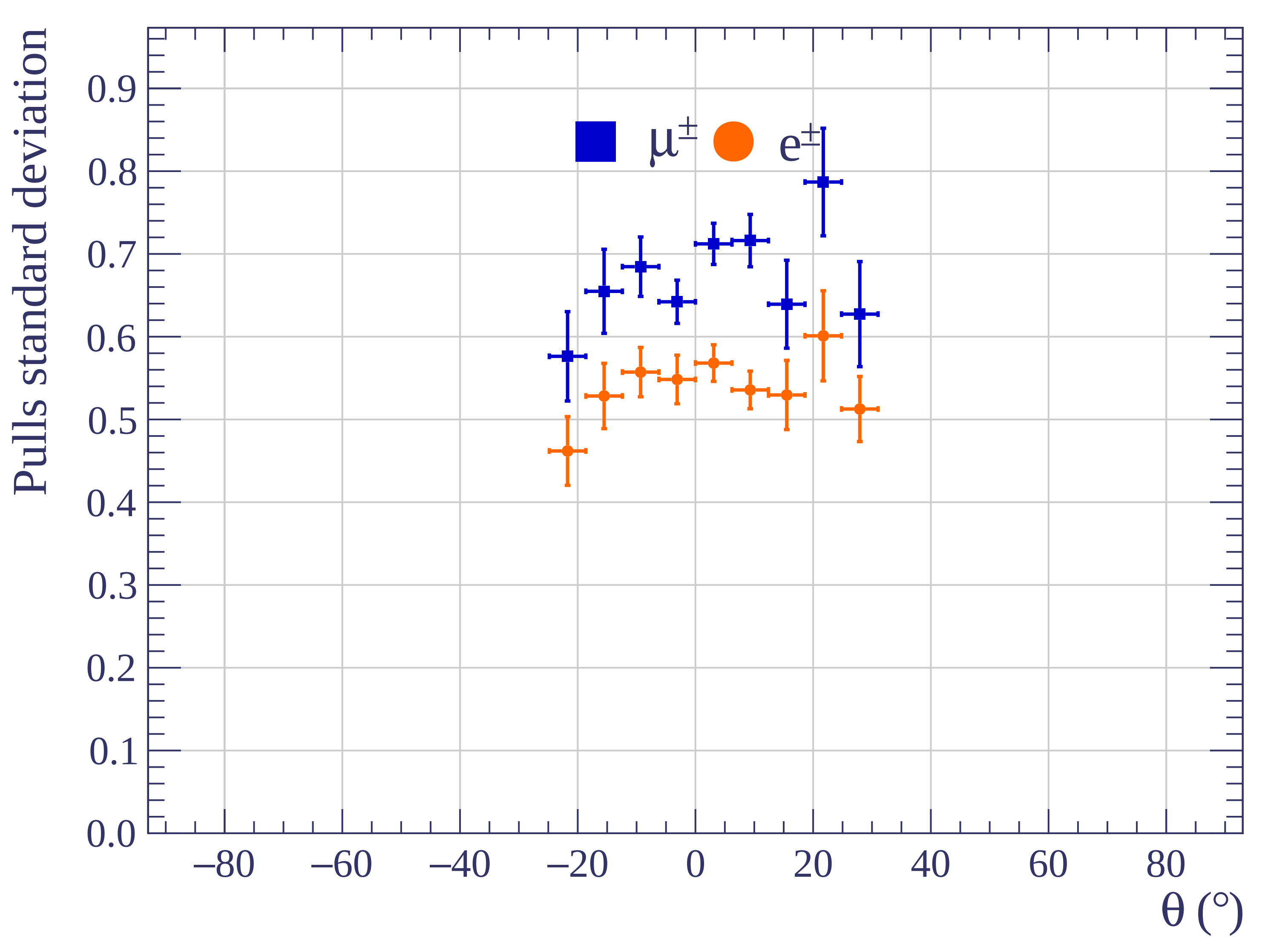

dE/dx with XP (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
14
12
10
8
6
4
2
0



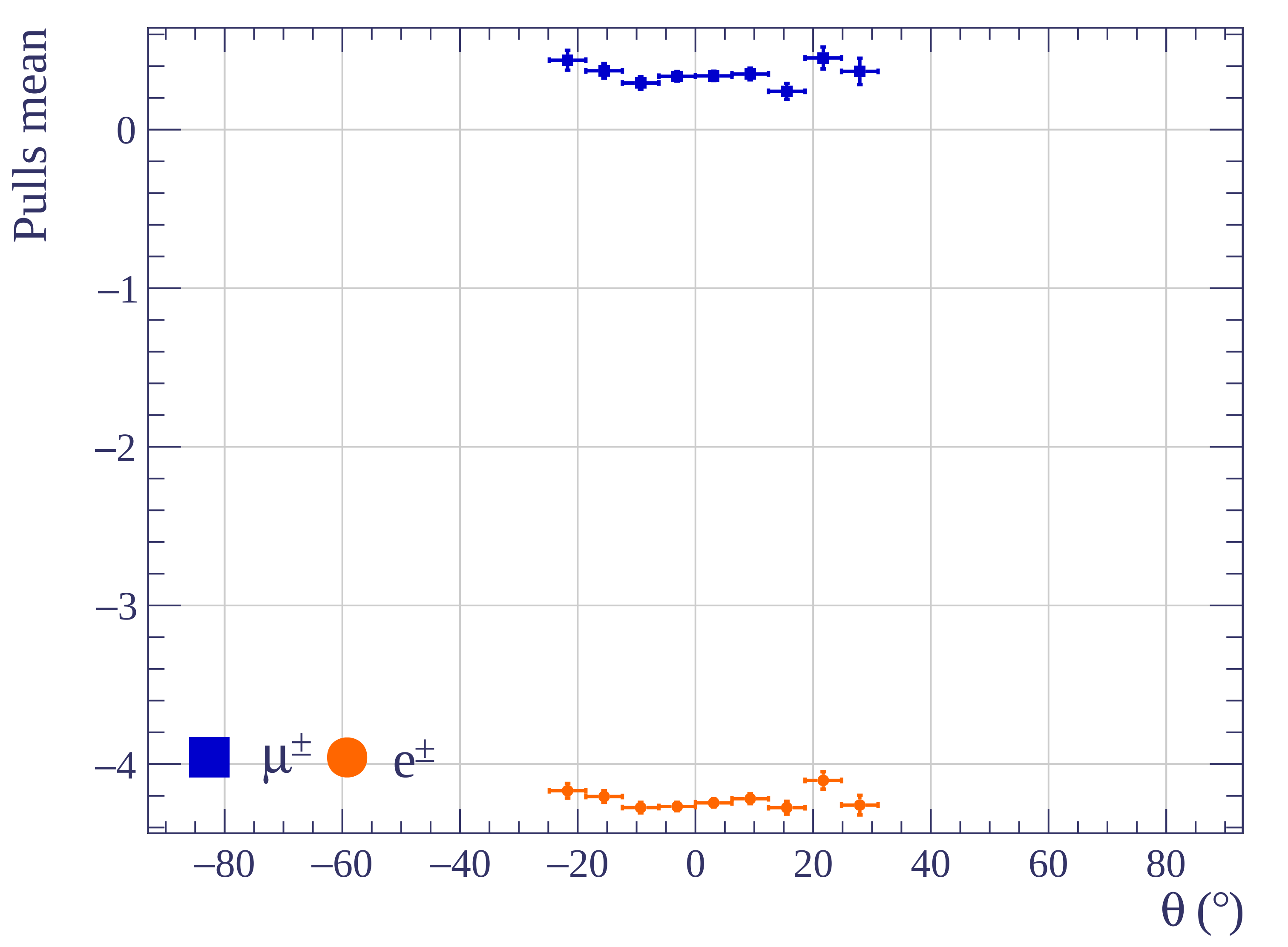
Count
250
200
150
100
50
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



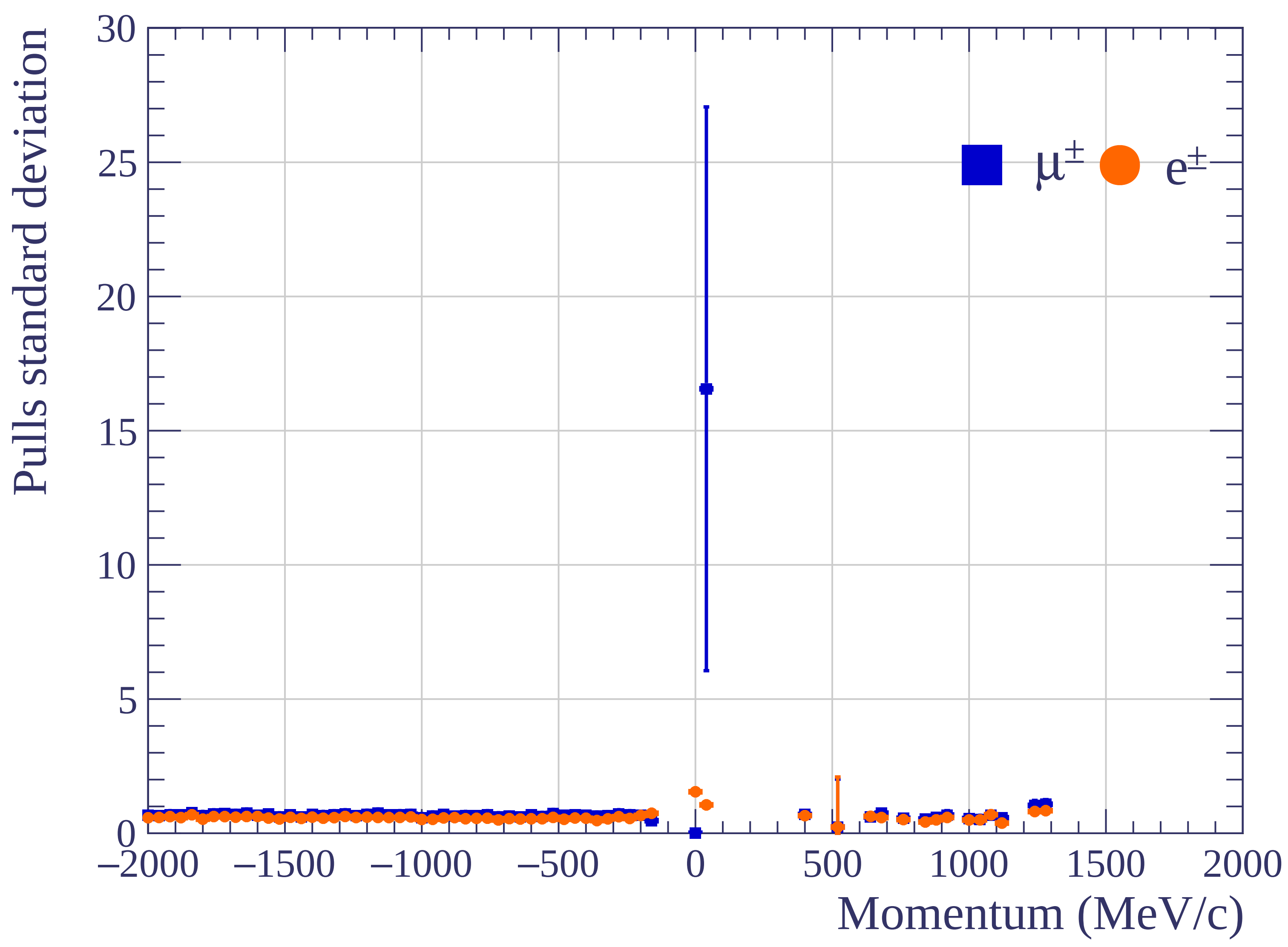
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
20
18
16
14
12
10
8
6
4
2
0



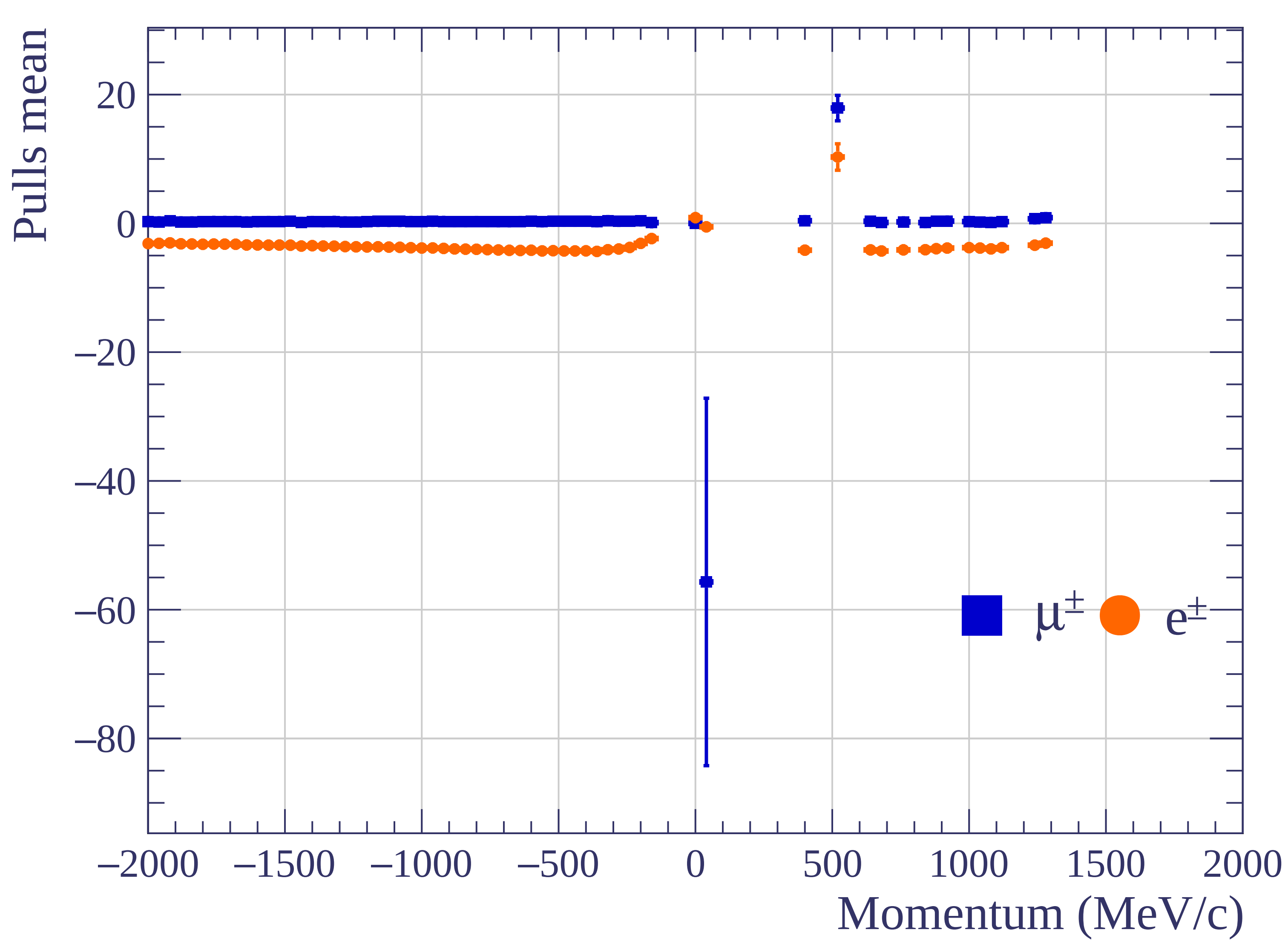
Pulls standard deviation
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



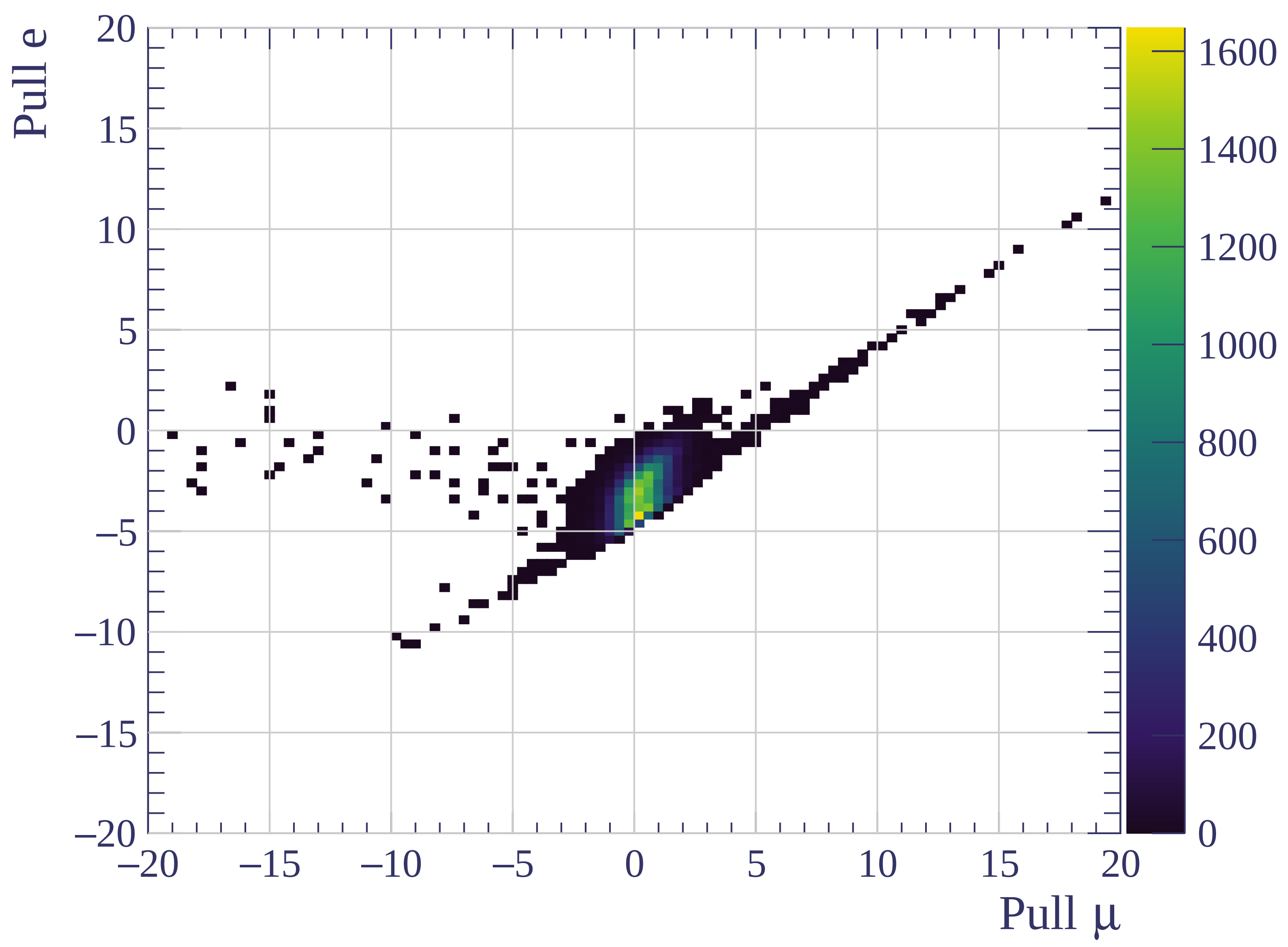
Pulls mean
0
−1
−2
−3
−4
μ±
e±
−80
−60
−40
−20
0
20
40
60
80
θ (°)

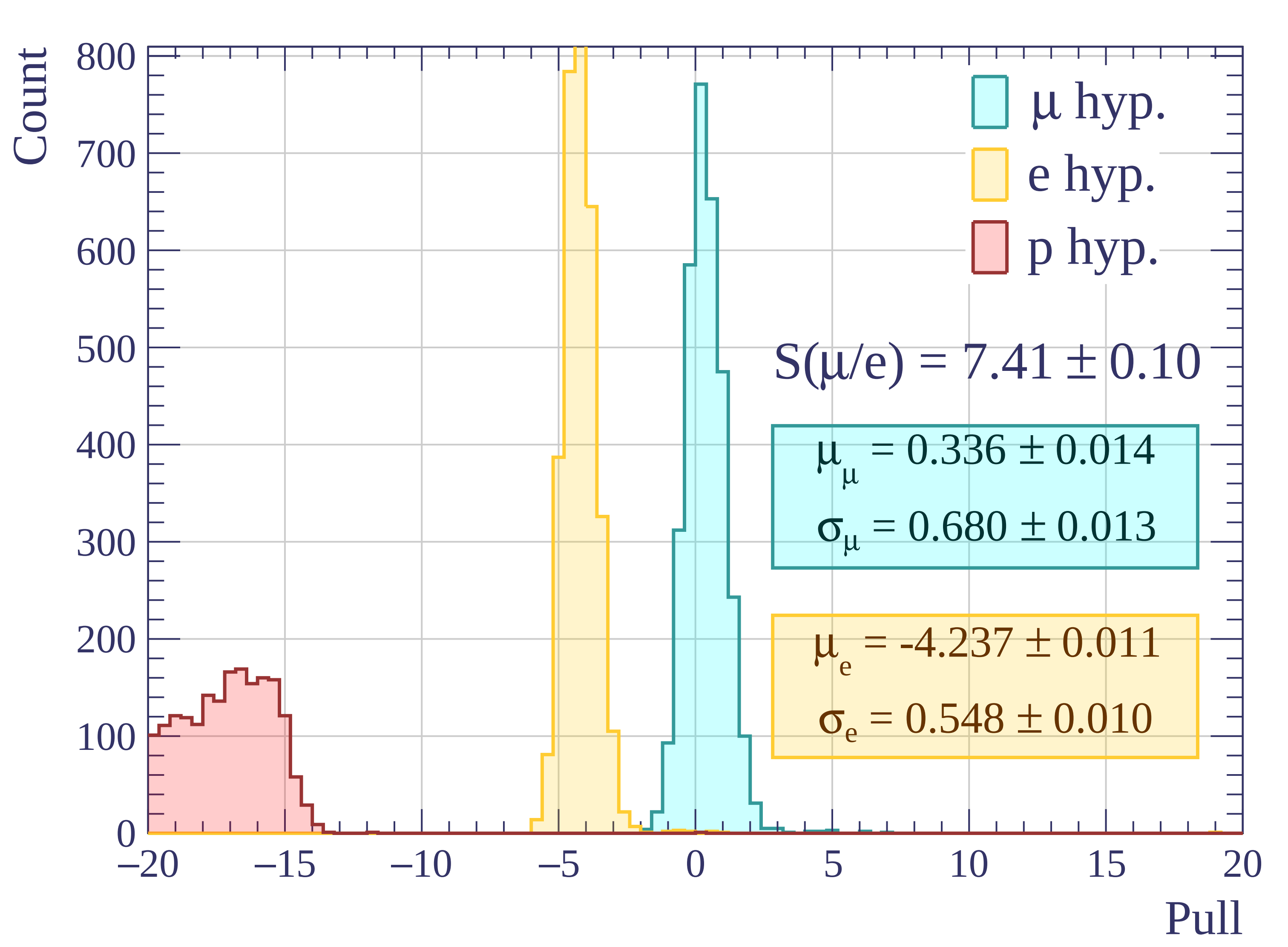


Pulls standard deviation
Momentum (MeV/c)
μ±
e±
0
5
10
15
20
25
30
−2000
−1500
−1000
−500
0
500
1000
1500
2000

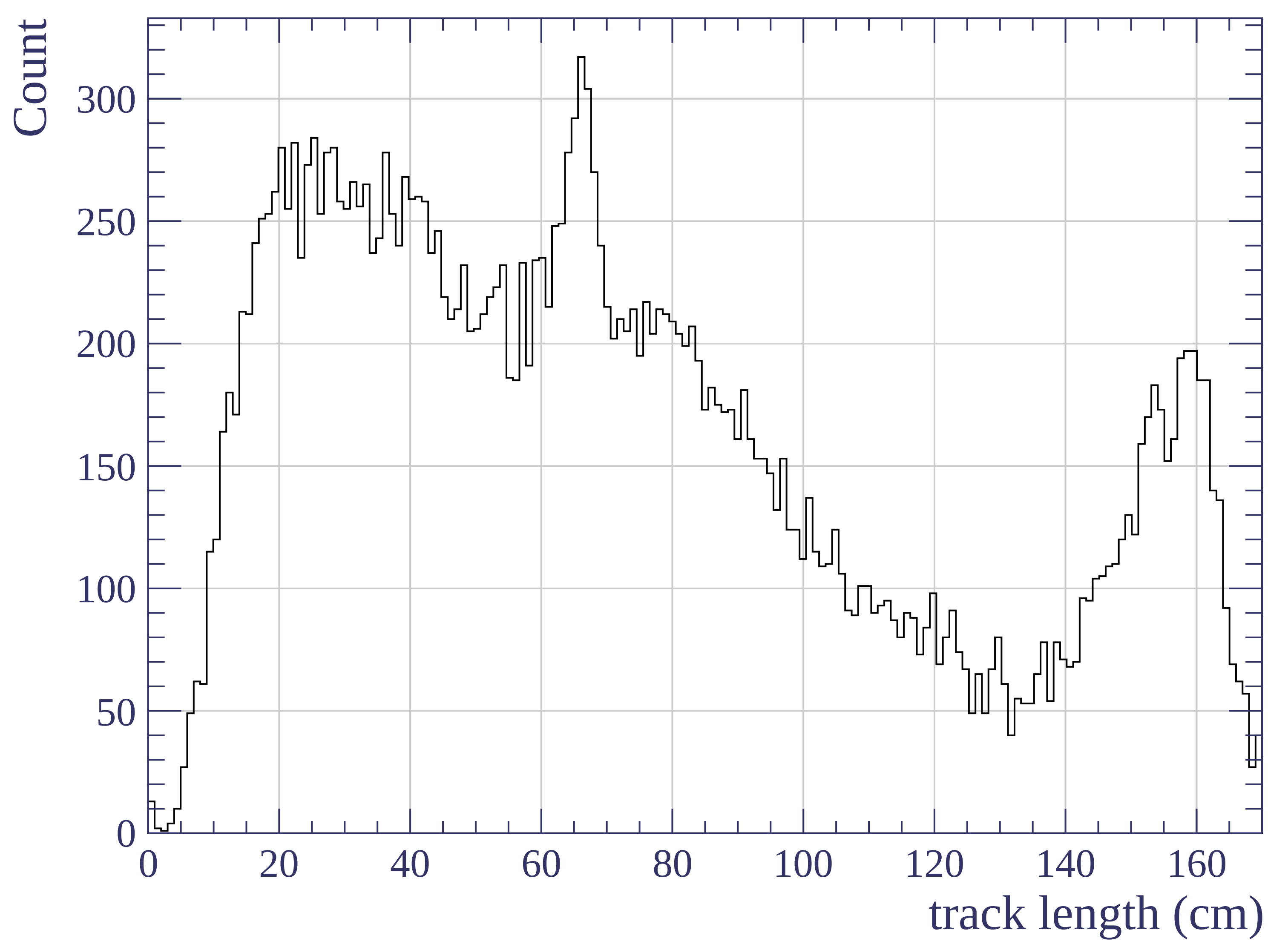


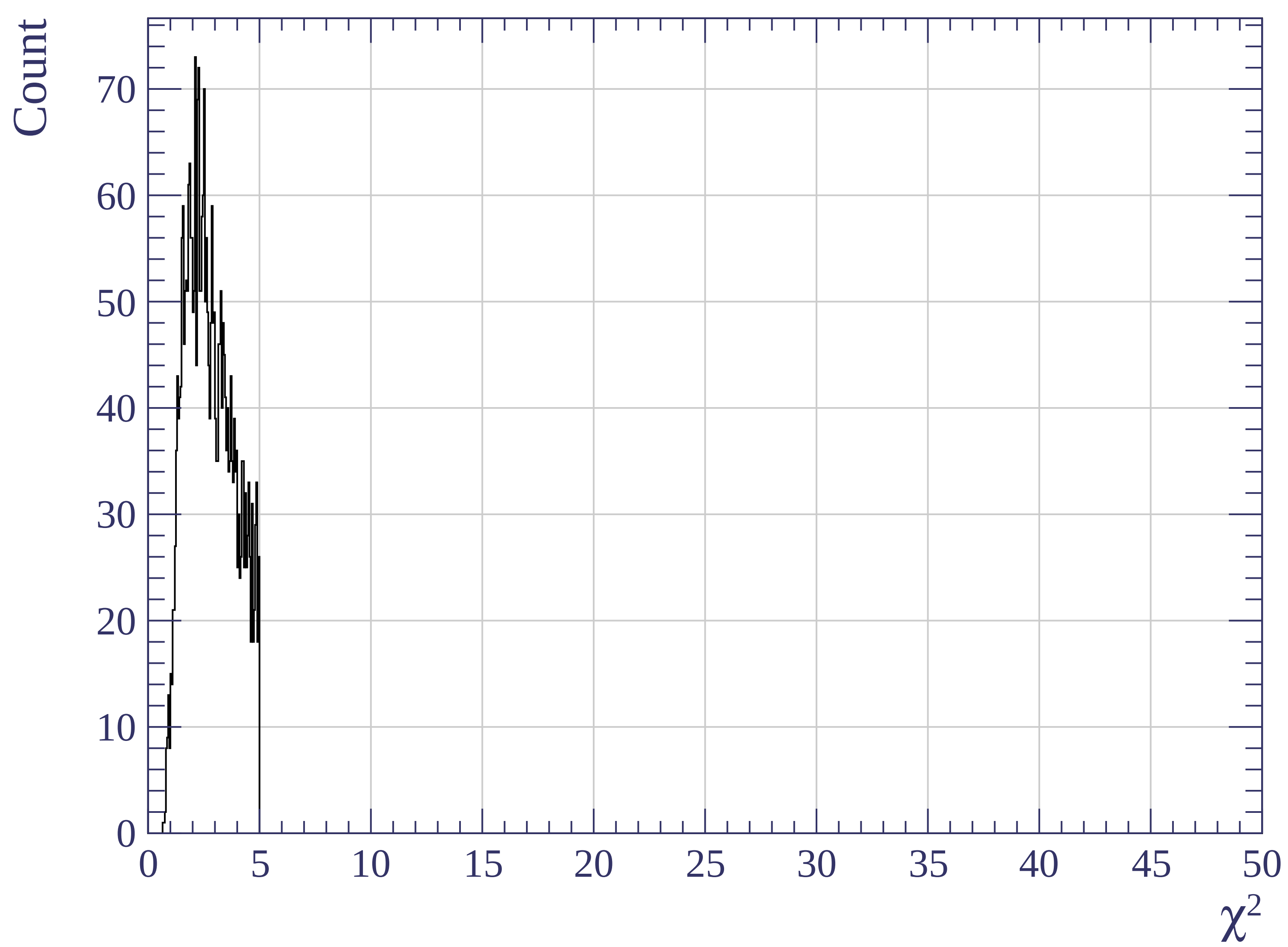
Pulls mean
20
0
−20
−40
−60
−80
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±

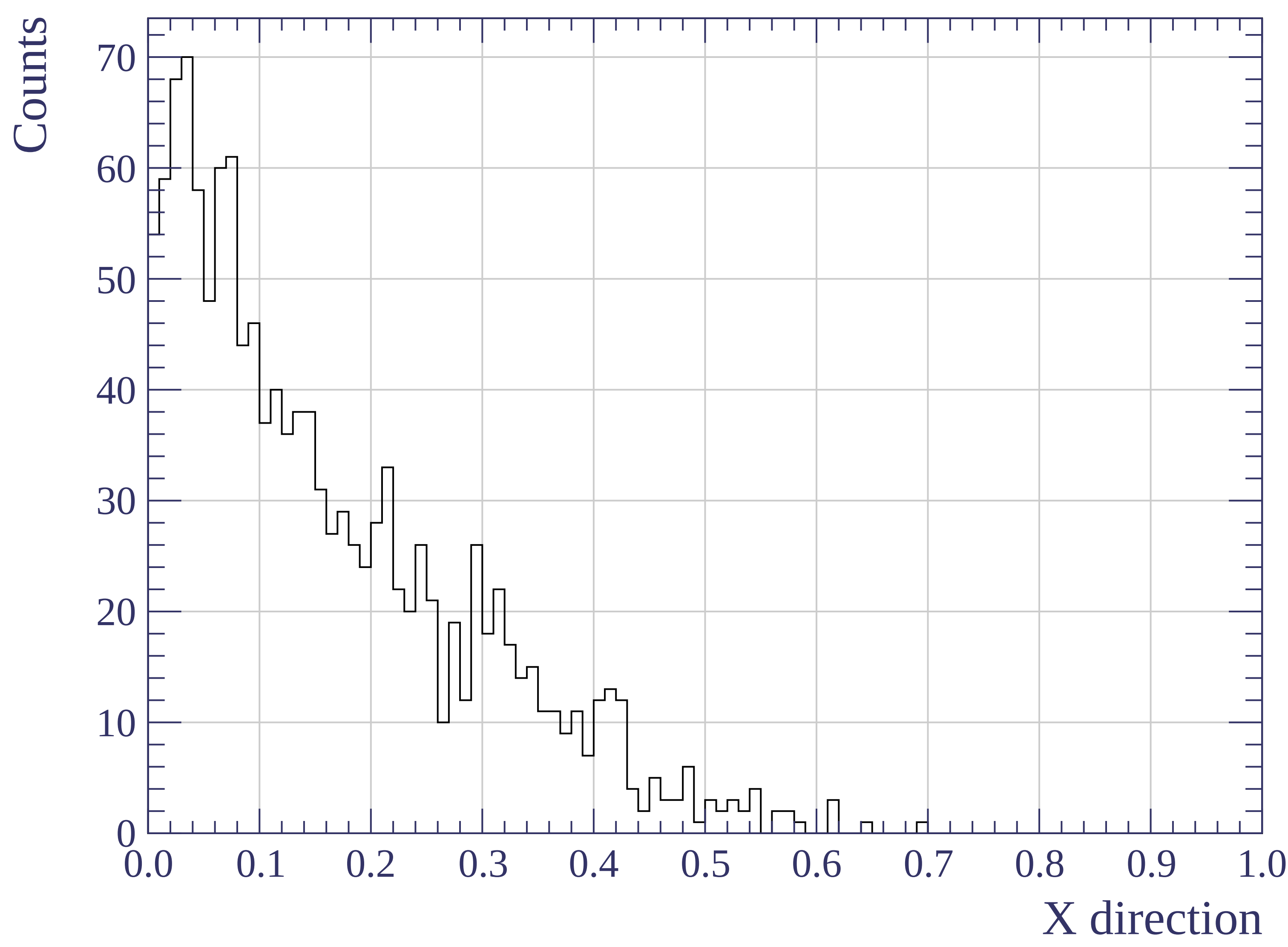

Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
1600
1400
1200
1000
800
600
400
200
0

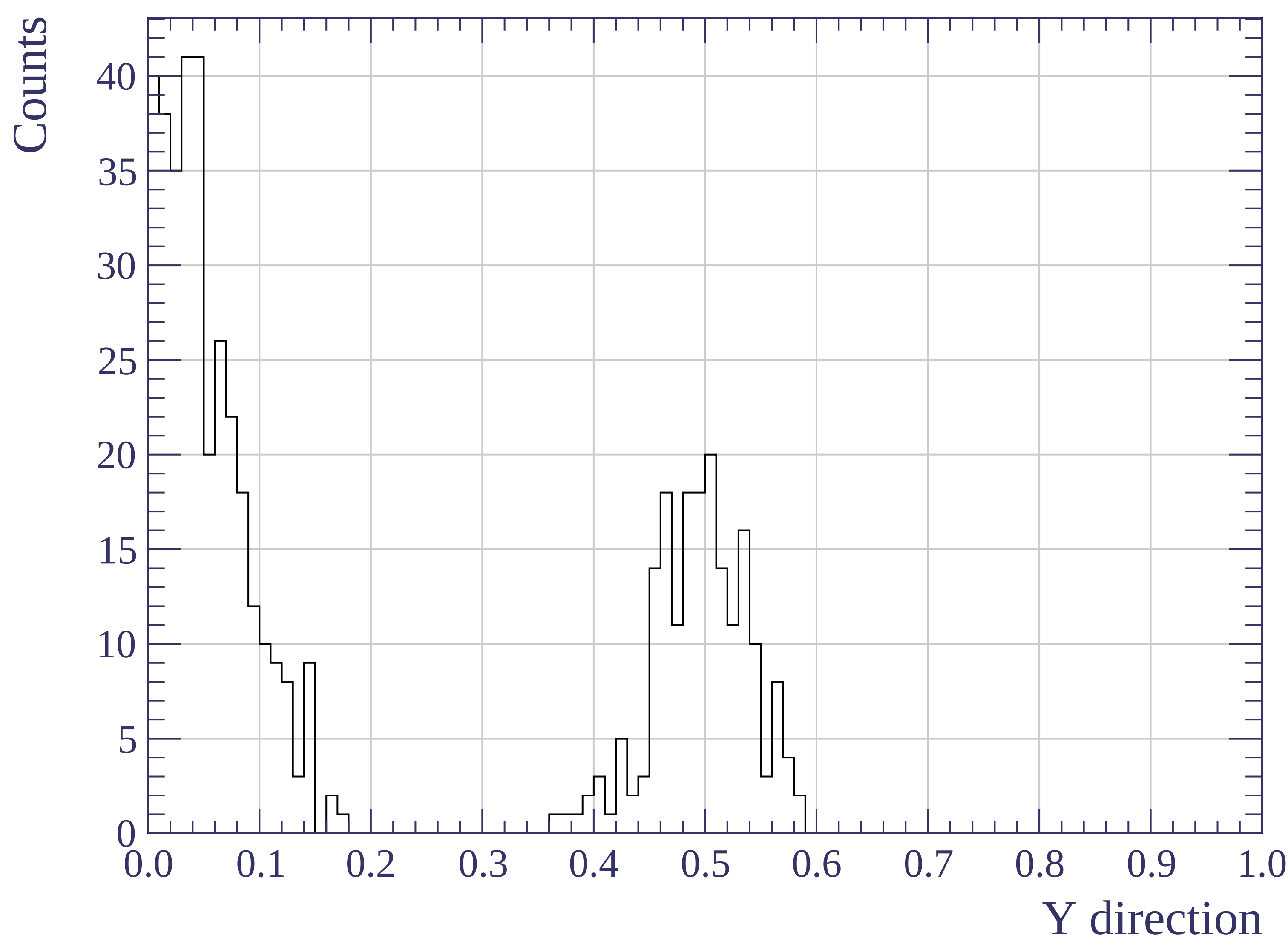


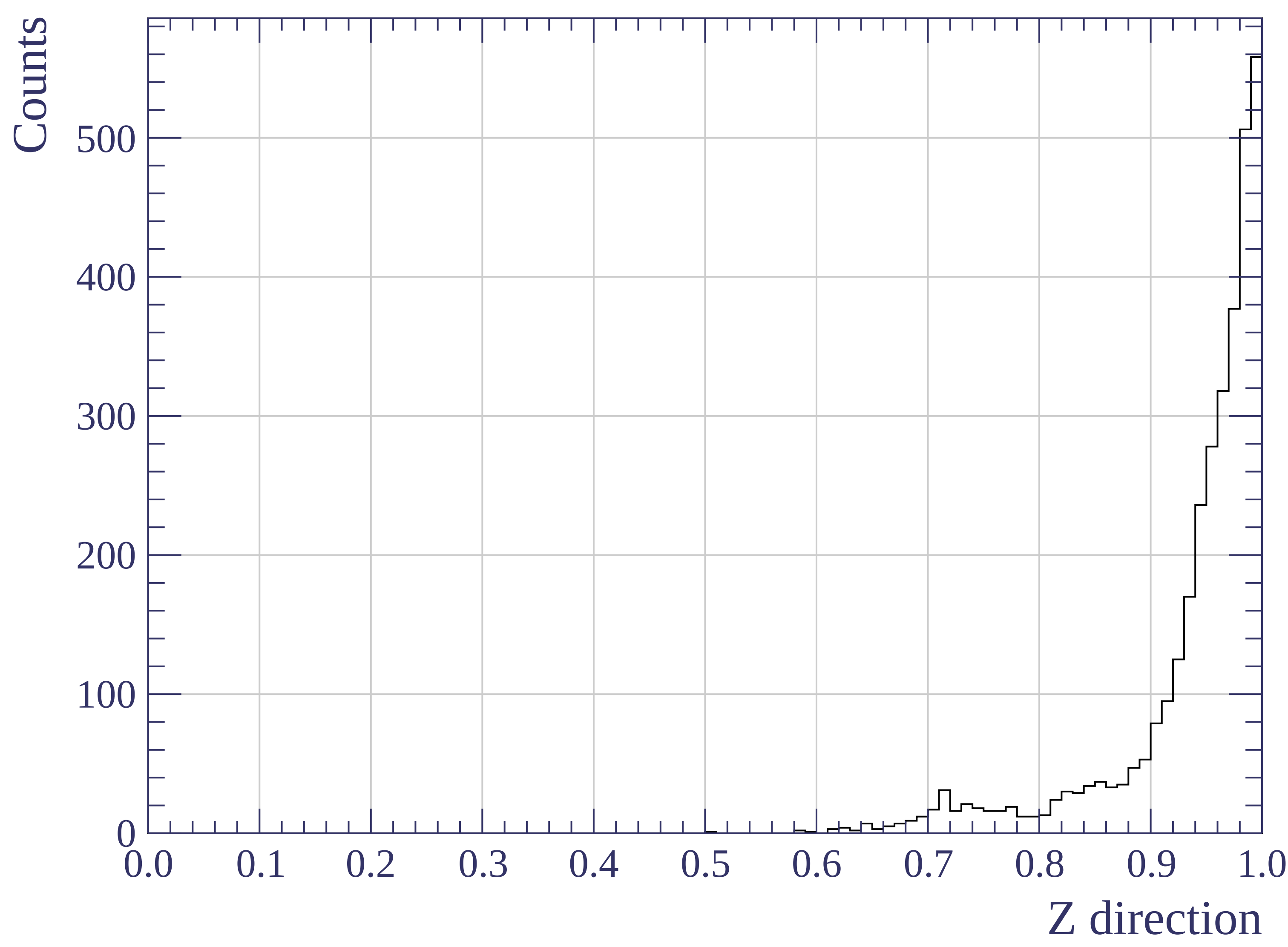
Count
800
700
600
500
400
300
200
100
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 7.41 ± 0.10
μ_{μ} = 0.336 ± 0.014
σ_{μ} = 0.680 ± 0.013
μ_{e} = -4.237 ± 0.011
σ_{e} = 0.548 ± 0.010


Count
300
250
200
150
100
50
0
0
20
40
60
80
100
120
140
160
track length (cm)

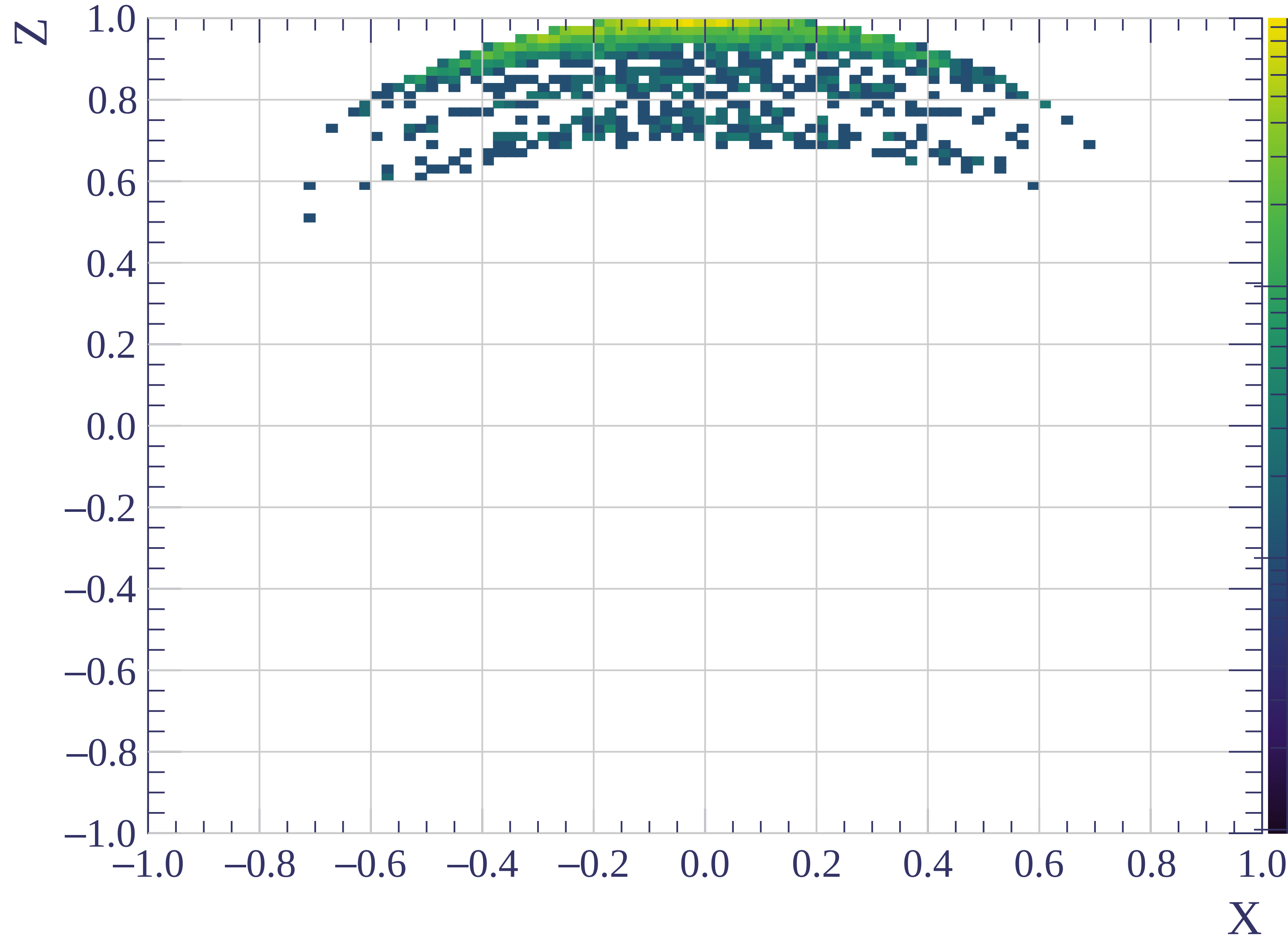

Count
0
10
20
30
40
50
60
70
0
5
10
15
20
25
30
35
40
45
50
χ^2


Counts
0
10
20
30
40
50
60
70
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction

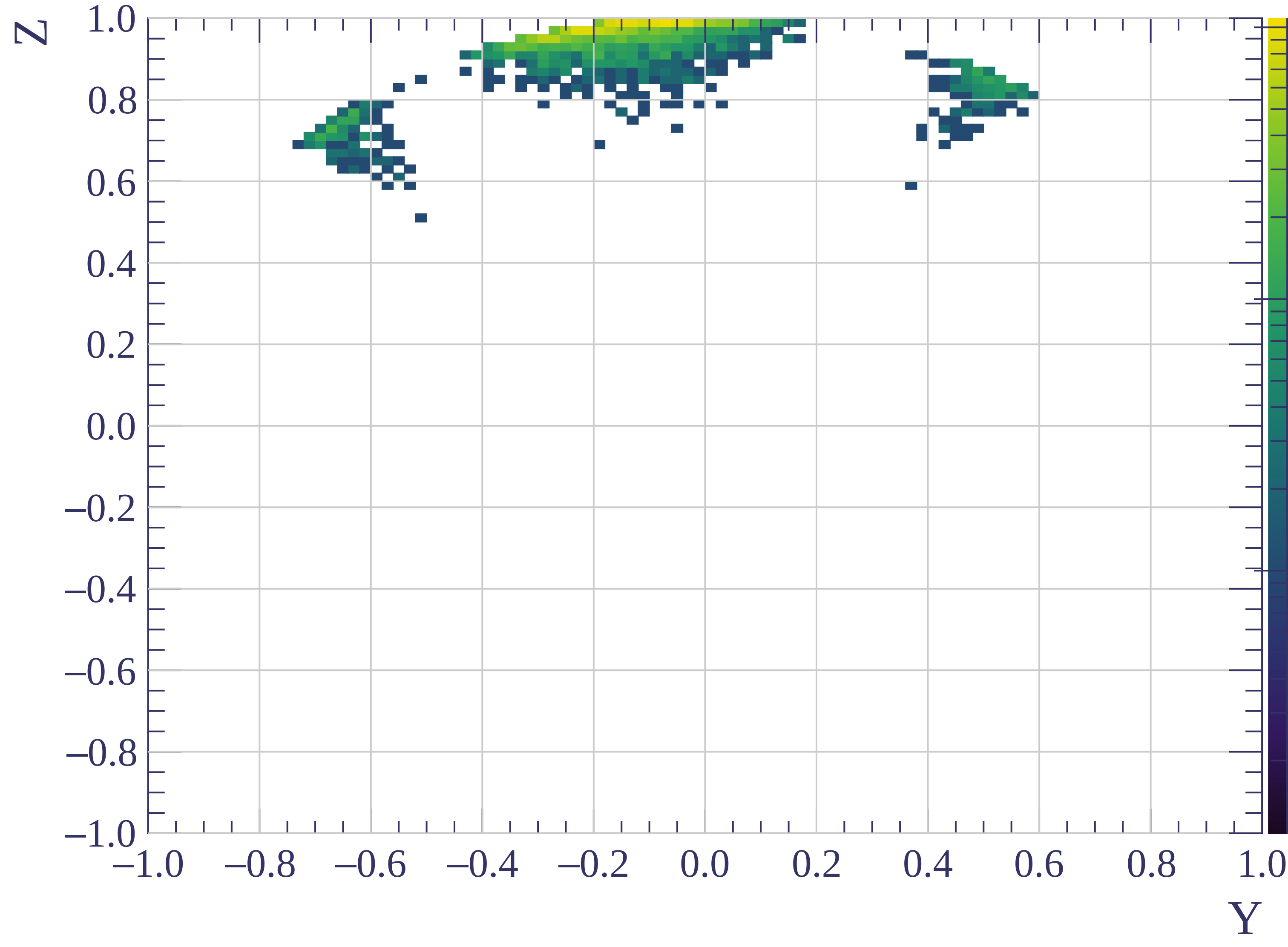

Counts
40
35
30
25
20
15
10
5
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction



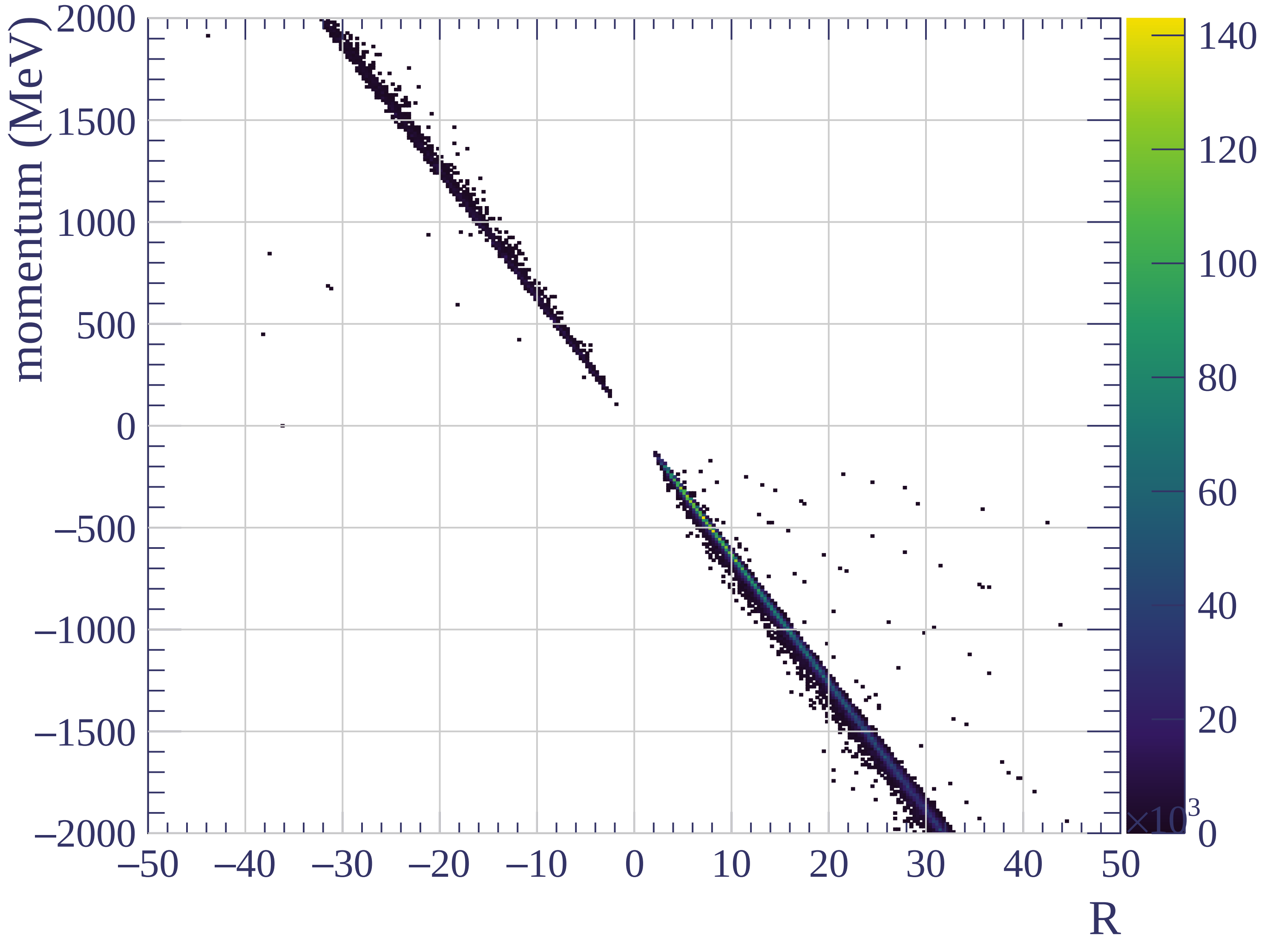
Counts
500
400
300
200
100
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction



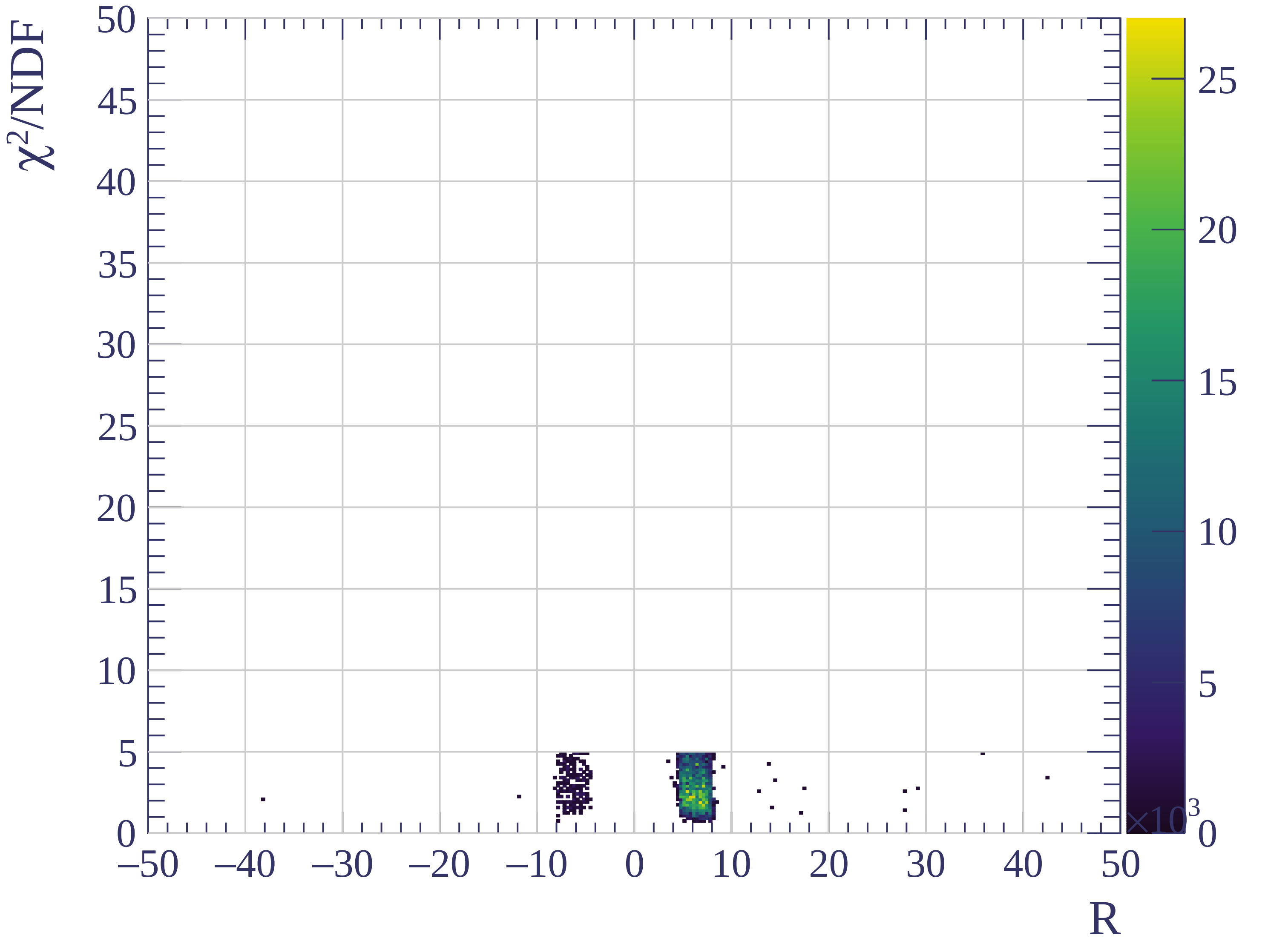
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X



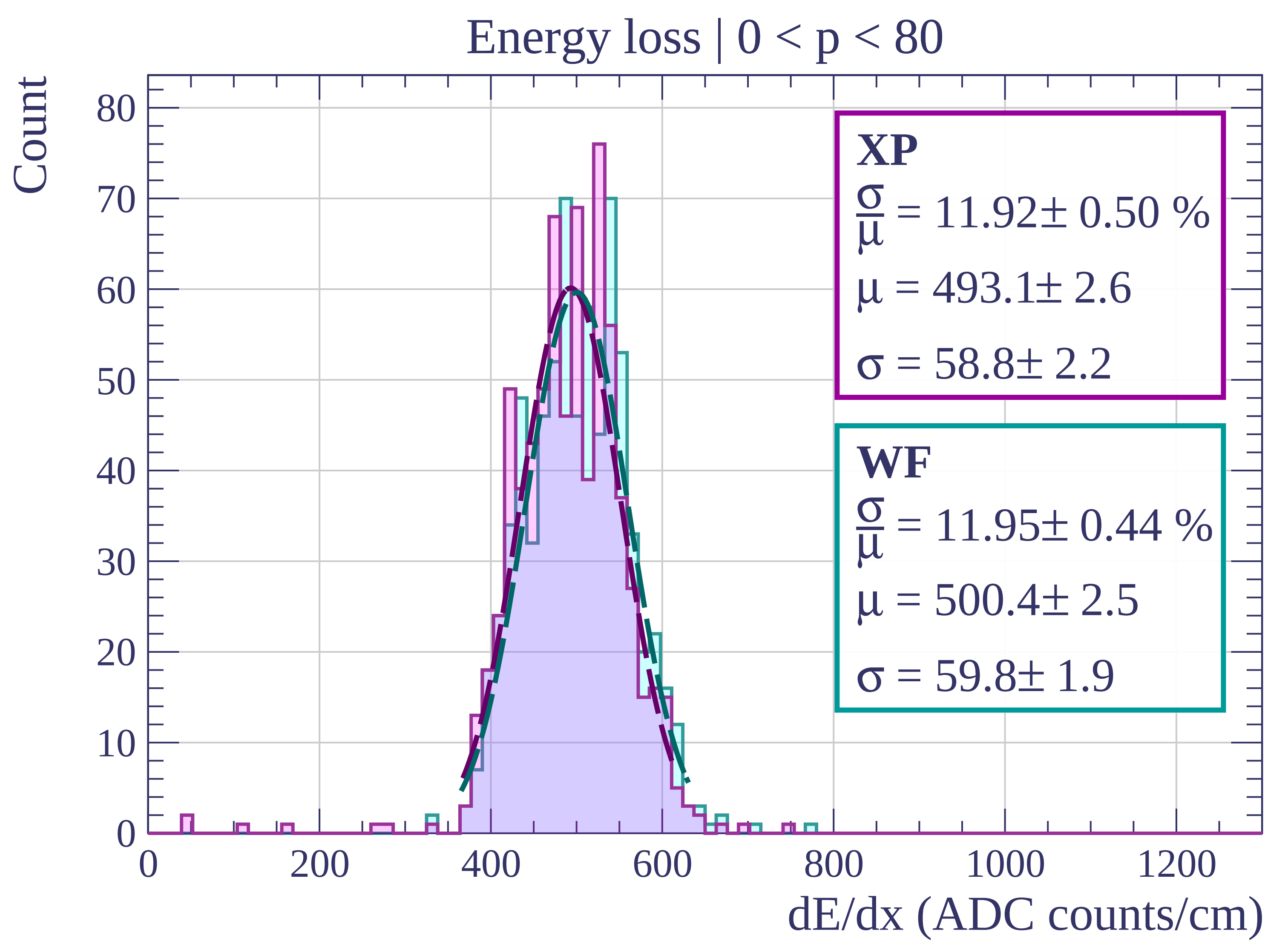
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y



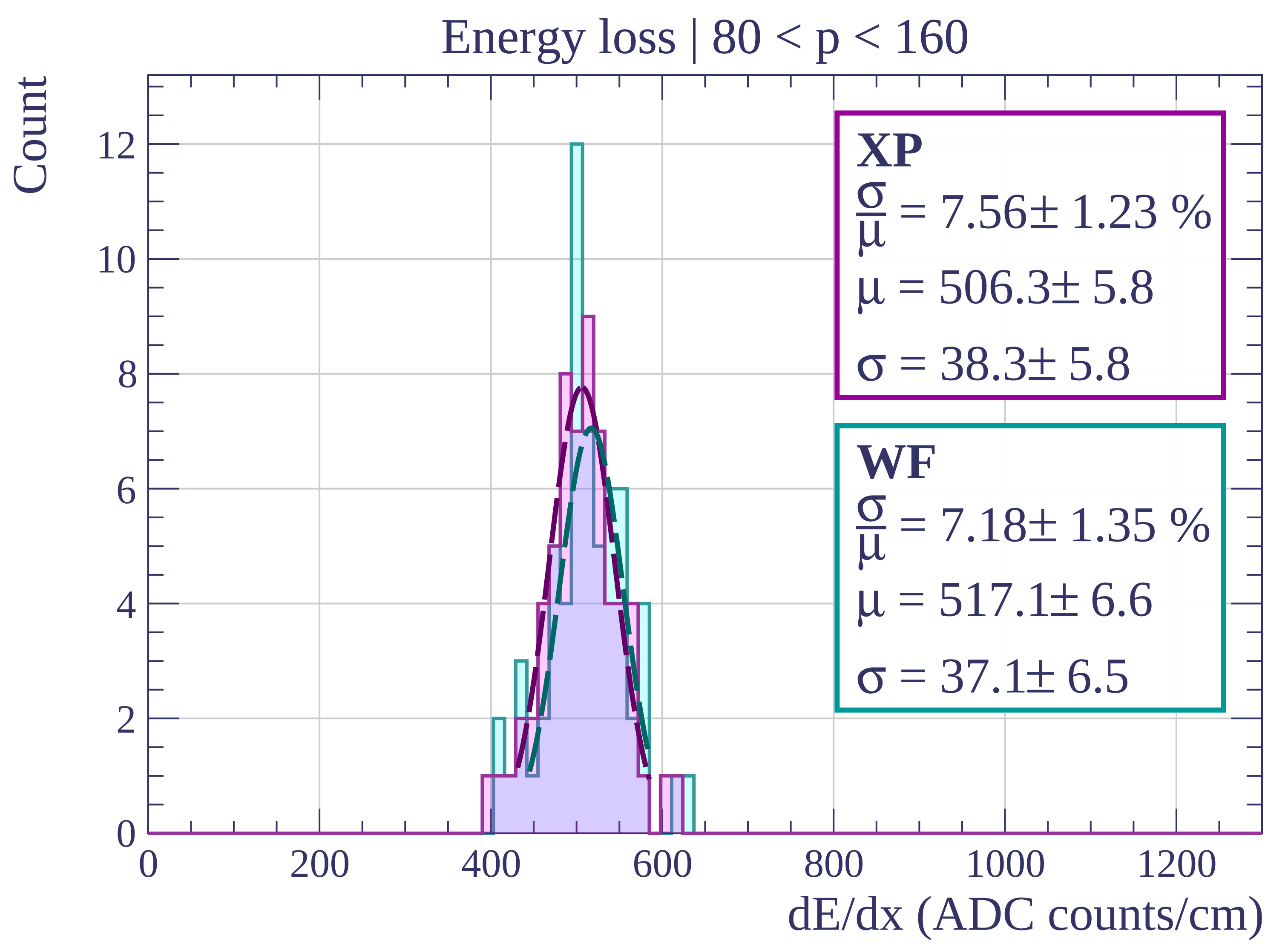
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
140
120
100
80
60
40
20
0
$\times10^3$



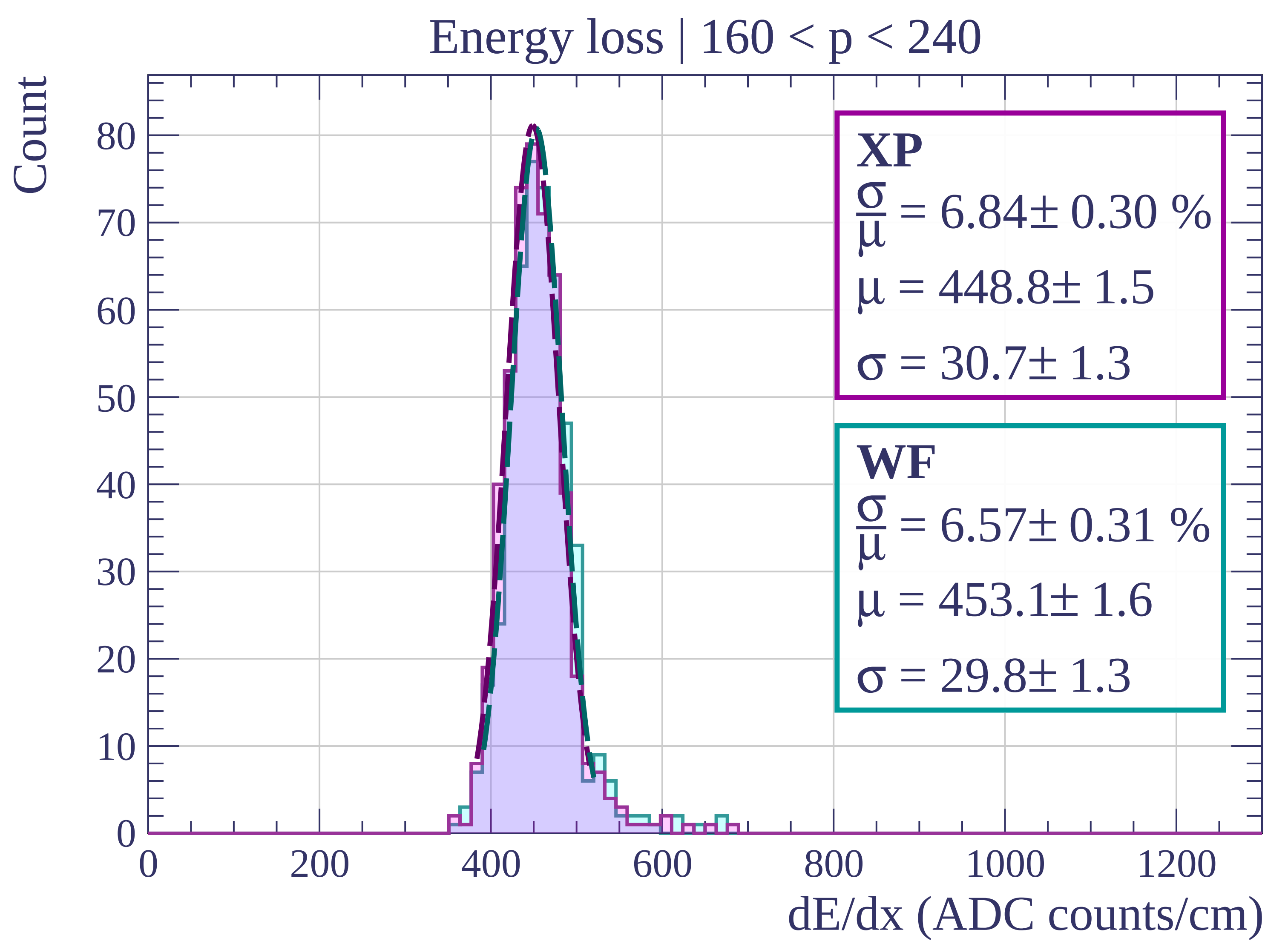
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
25
20
15
10
5
0
$\times 10^3$



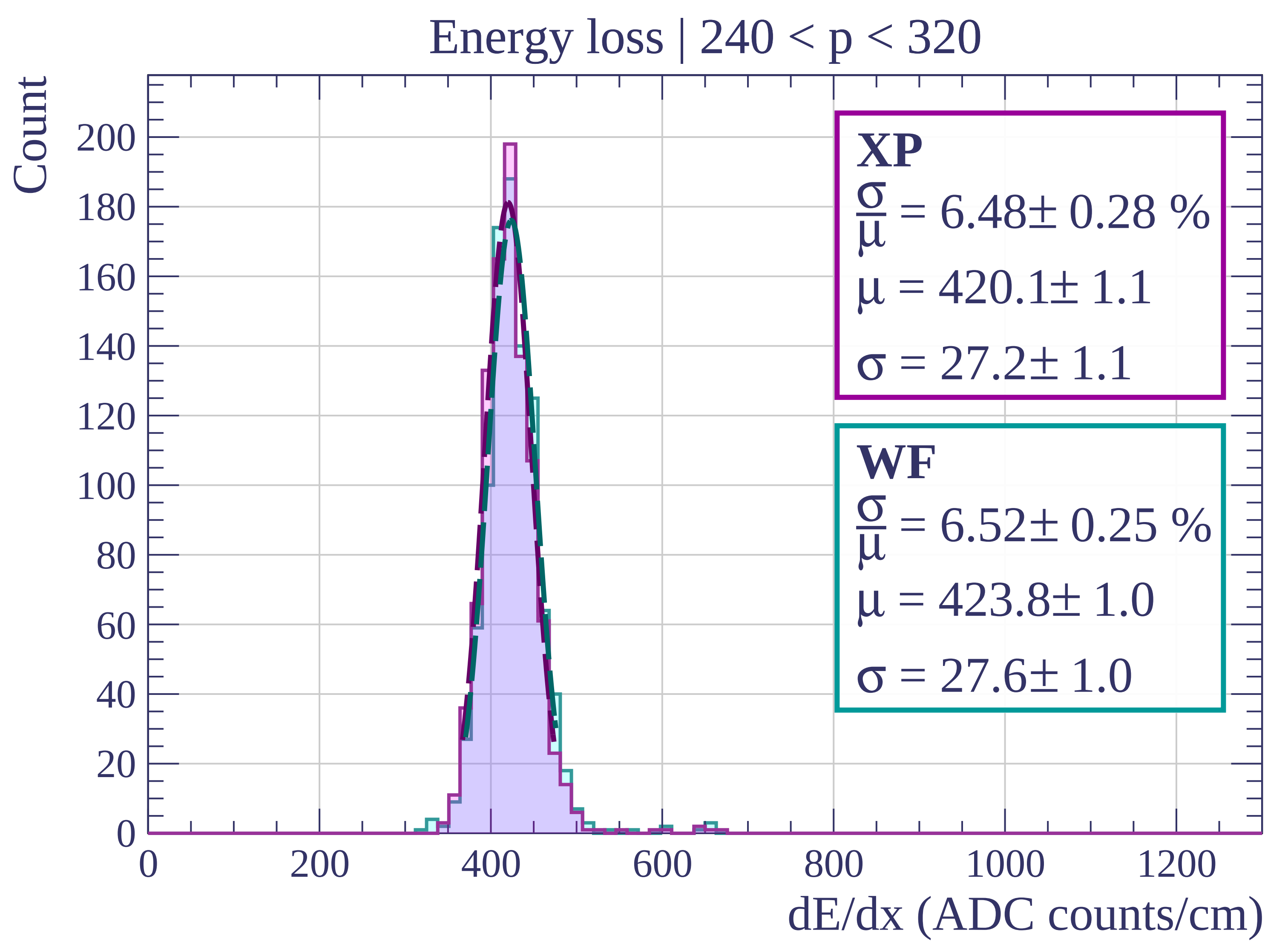
Energy loss | 0 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP
σ/μ = 11.92± 0.50 %
μ = 493.1± 2.6
σ = 58.8± 2.2
WF
σ/μ = 11.95± 0.44 %
μ = 500.4± 2.5
σ = 59.8± 1.9



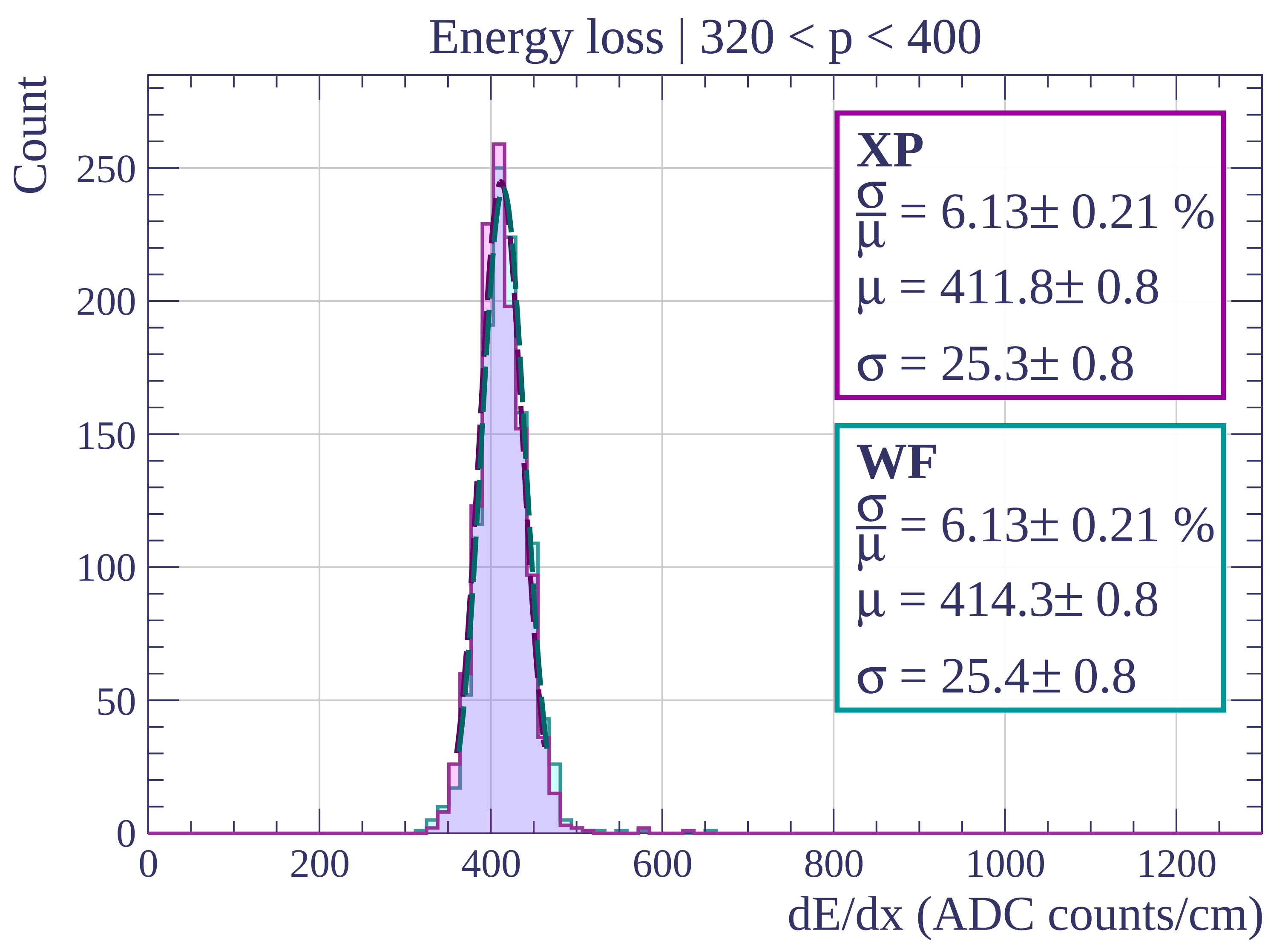
Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.56± 1.23 %
μ = 506.3± 5.8
σ = 38.3± 5.8
WF
σ/μ = 7.18± 1.35 %
μ = 517.1± 6.6
σ = 37.1± 6.5



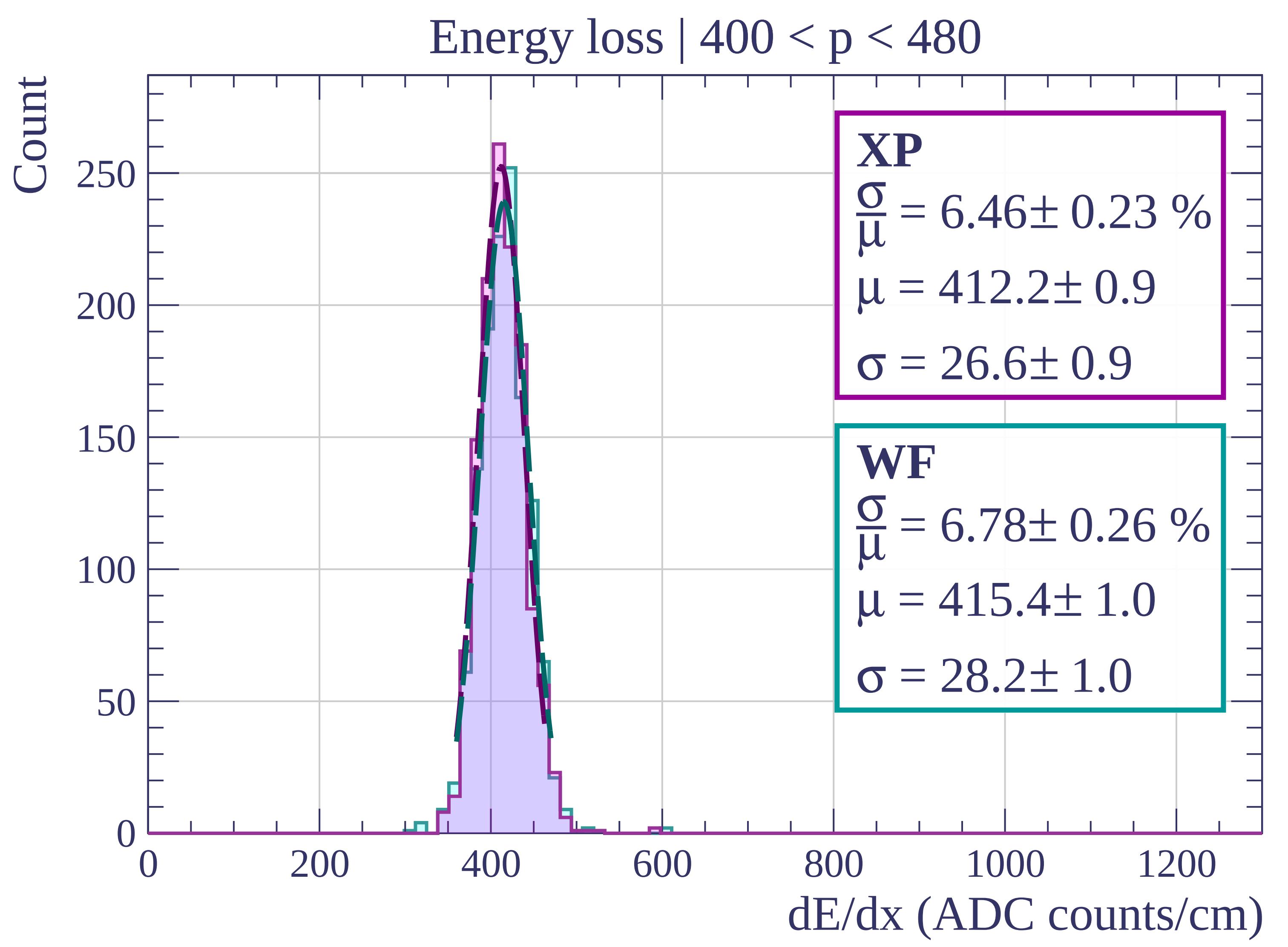
Energy loss | 160 < p < 240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP
σ/μ = 6.84± 0.30 %
μ = 448.8± 1.5
σ = 30.7± 1.3
WF
σ/μ = 6.57± 0.31 %
μ = 453.1± 1.6
σ = 29.8± 1.3



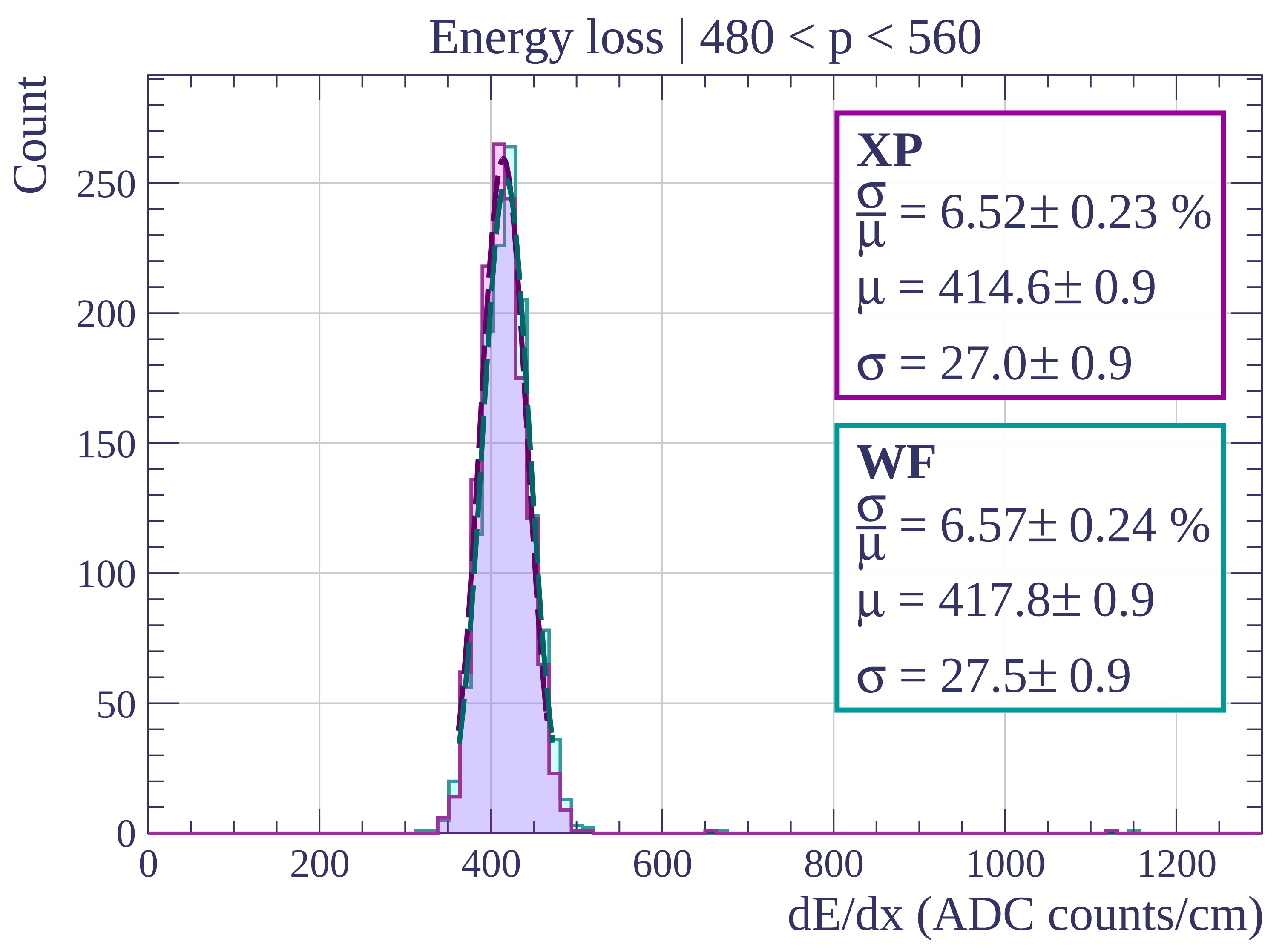
Energy loss | 240 < p < 320
Count
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.48± 0.28 %
μ = 420.1± 1.1
σ = 27.2± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.52± 0.25 %
μ = 423.8± 1.0
σ = 27.6± 1.0



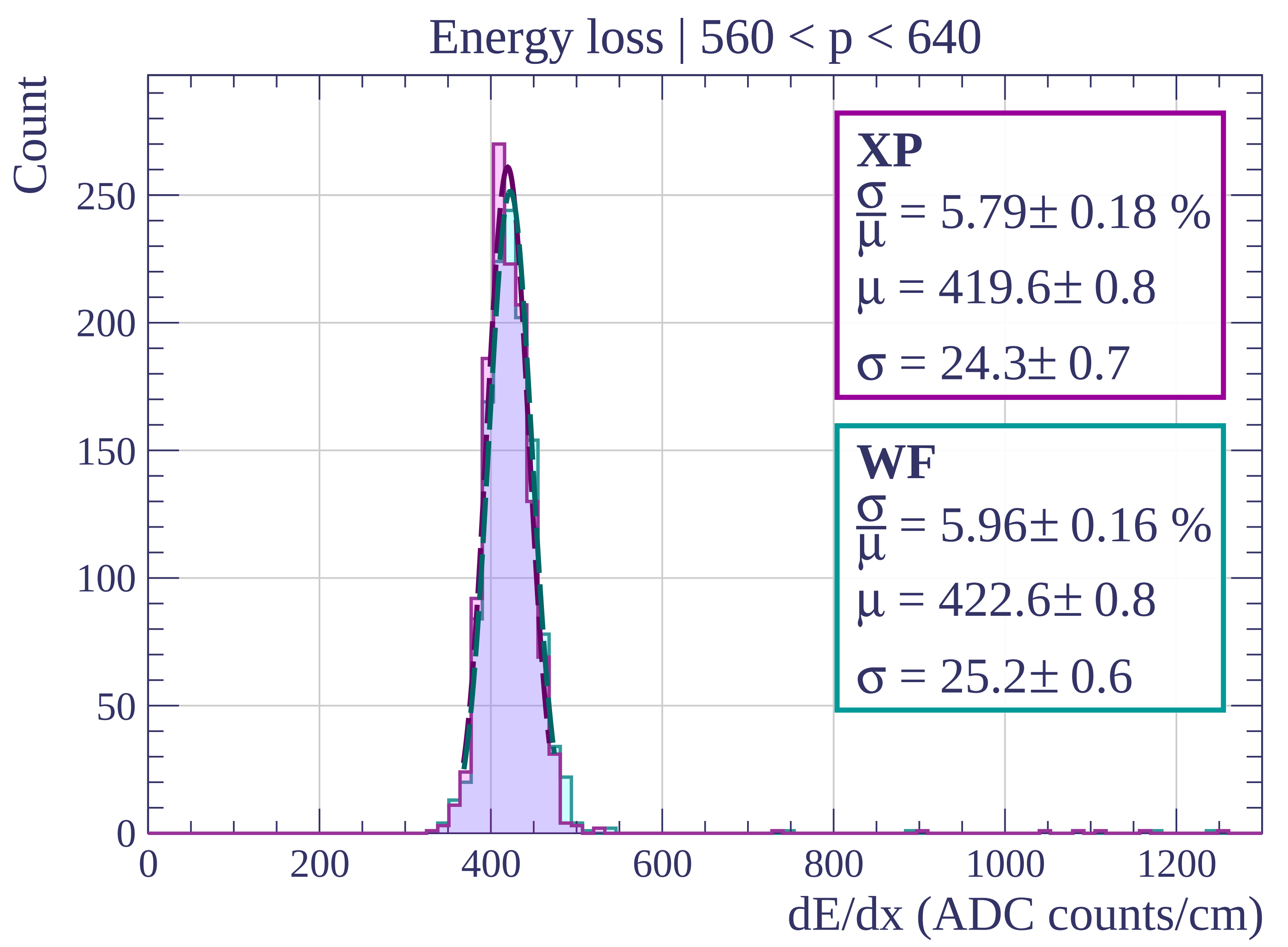
Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.13± 0.21 %
μ = 411.8± 0.8
σ = 25.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 6.13± 0.21 %
μ = 414.3± 0.8
σ = 25.4± 0.8



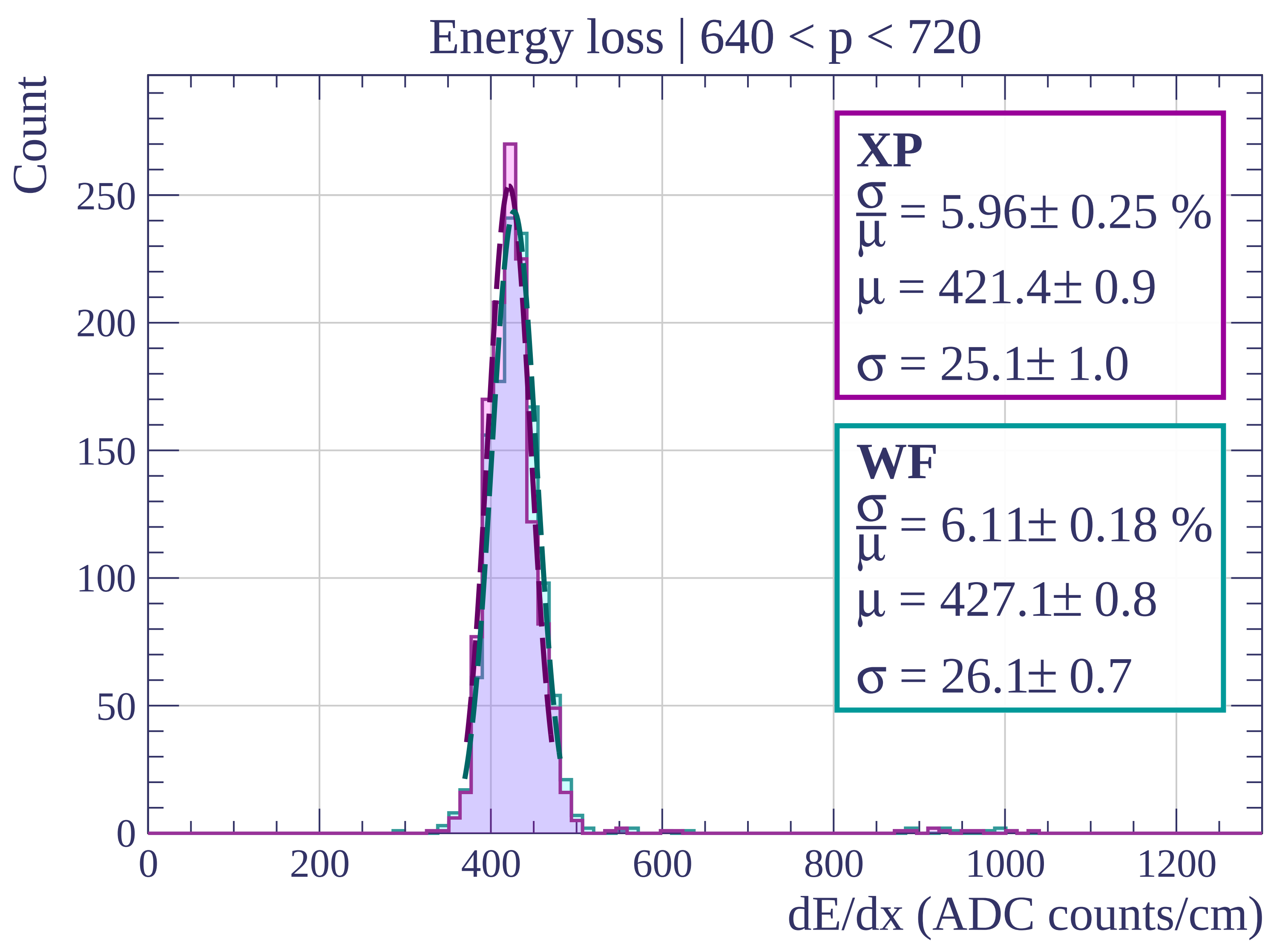
Energy loss | 400 < p < 480
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.46± 0.23 %
μ = 412.2± 0.9
σ = 26.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.78± 0.26 %
μ = 415.4± 1.0
σ = 28.2± 1.0



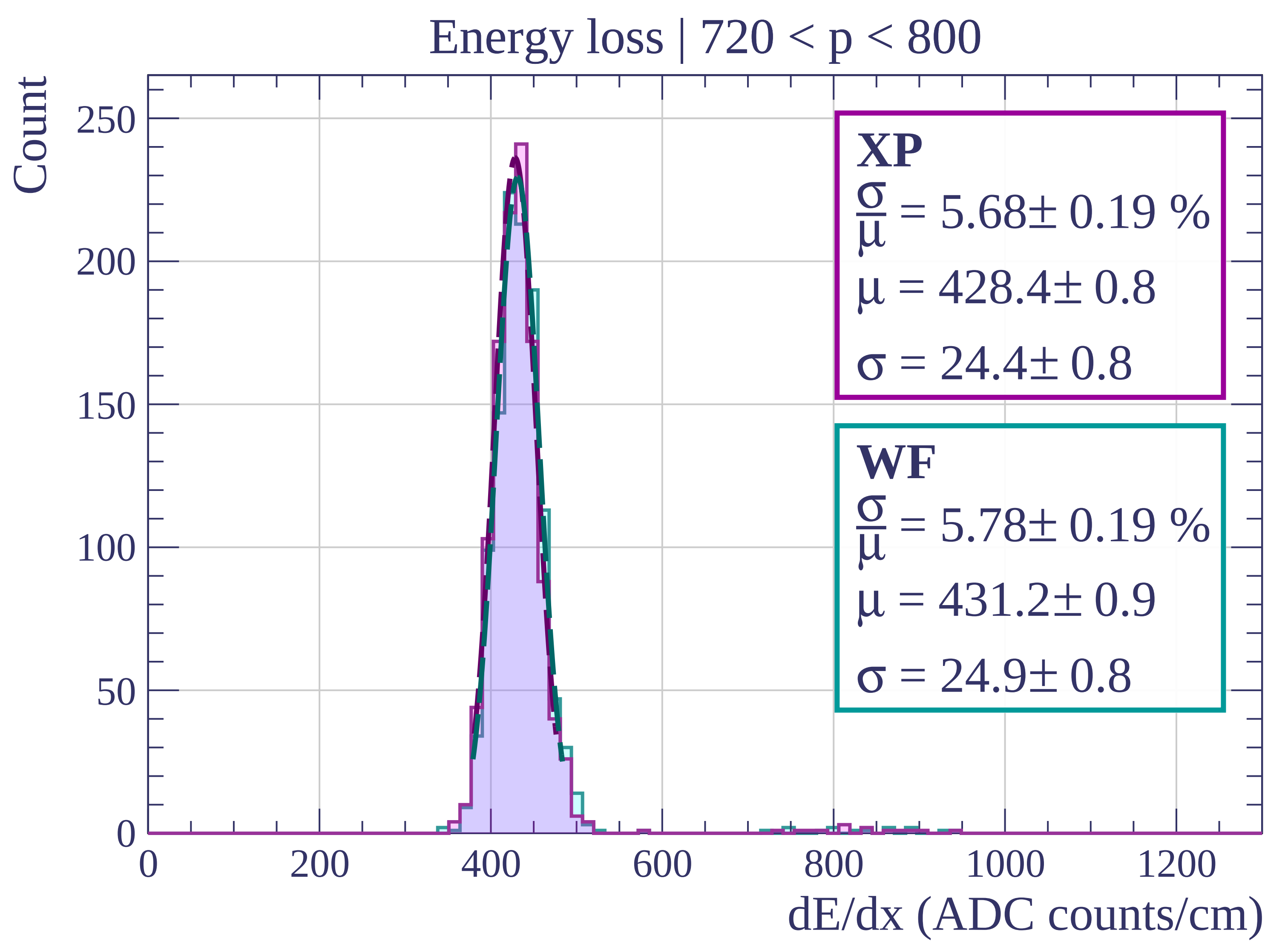
Energy loss | 480 < p < 560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 6.52± 0.23 %
μ = 414.6± 0.9
σ = 27.0± 0.9
WF
σ/μ = 6.57± 0.24 %
μ = 417.8± 0.9
σ = 27.5± 0.9



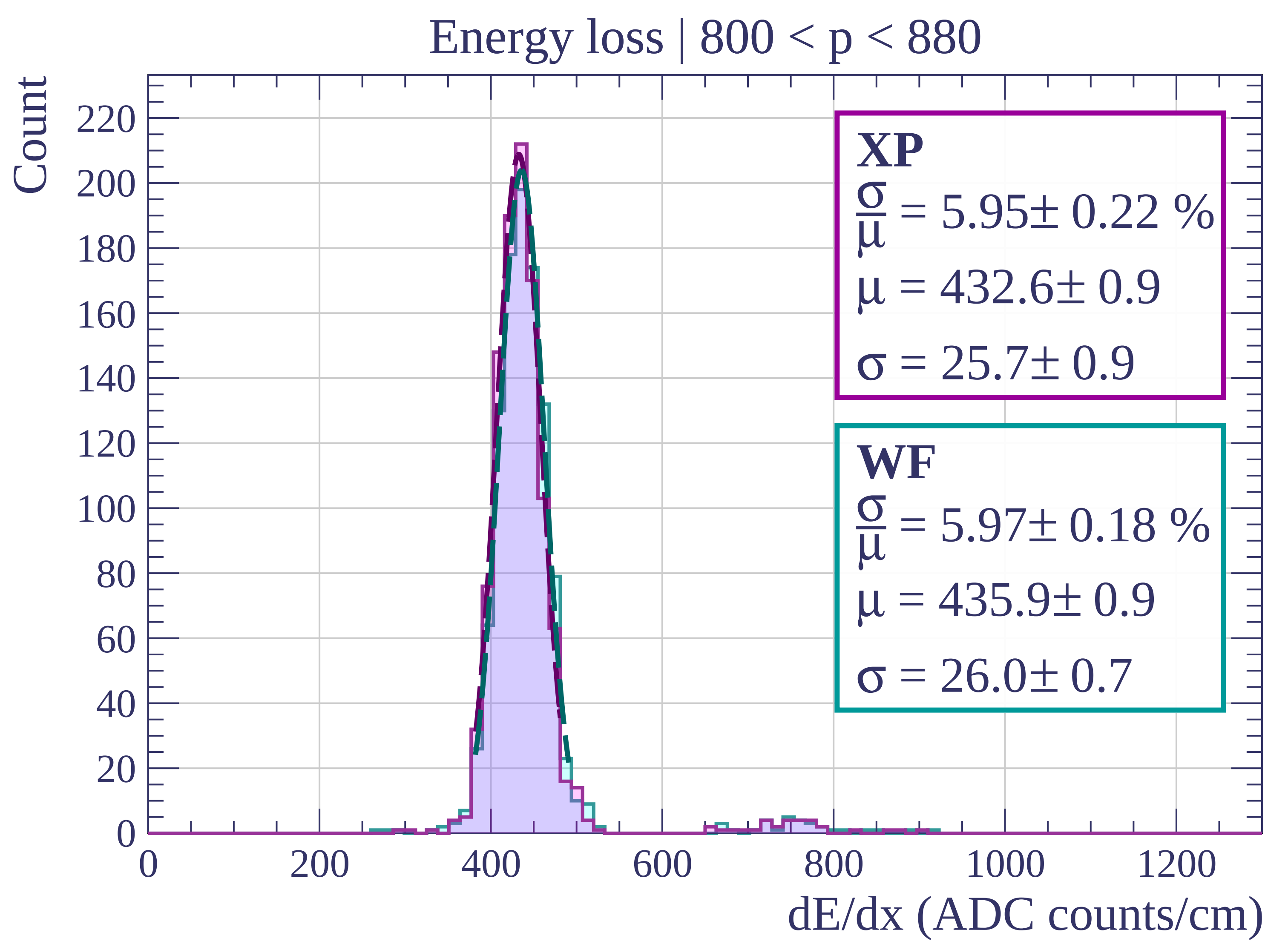
Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.79± 0.18 %
μ = 419.6± 0.8
σ = 24.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 5.96± 0.16 %
μ = 422.6± 0.8
σ = 25.2± 0.6



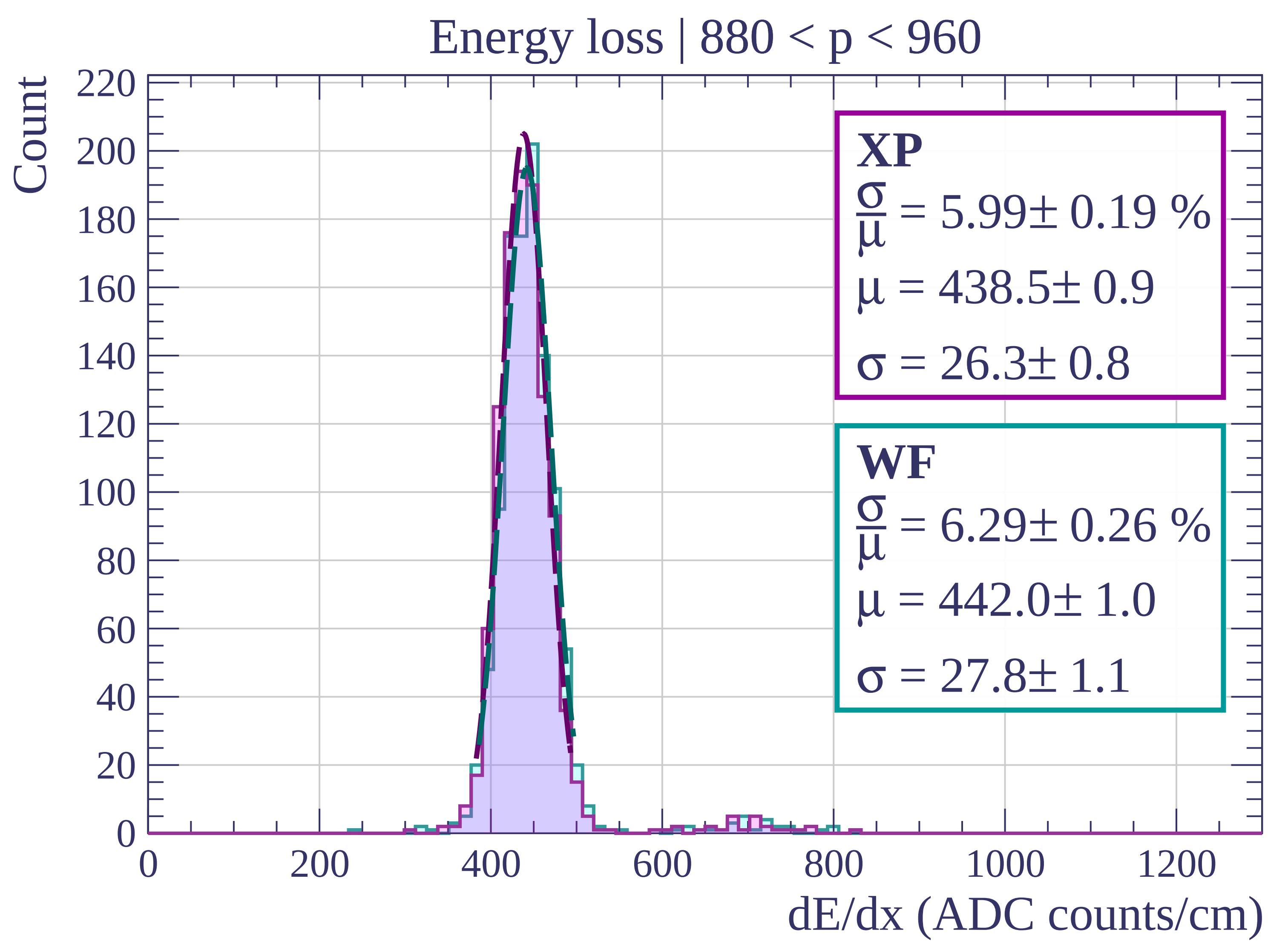
Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.96± 0.25 %
μ = 421.4± 0.9
σ = 25.1± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.11± 0.18 %
μ = 427.1± 0.8
σ = 26.1± 0.7



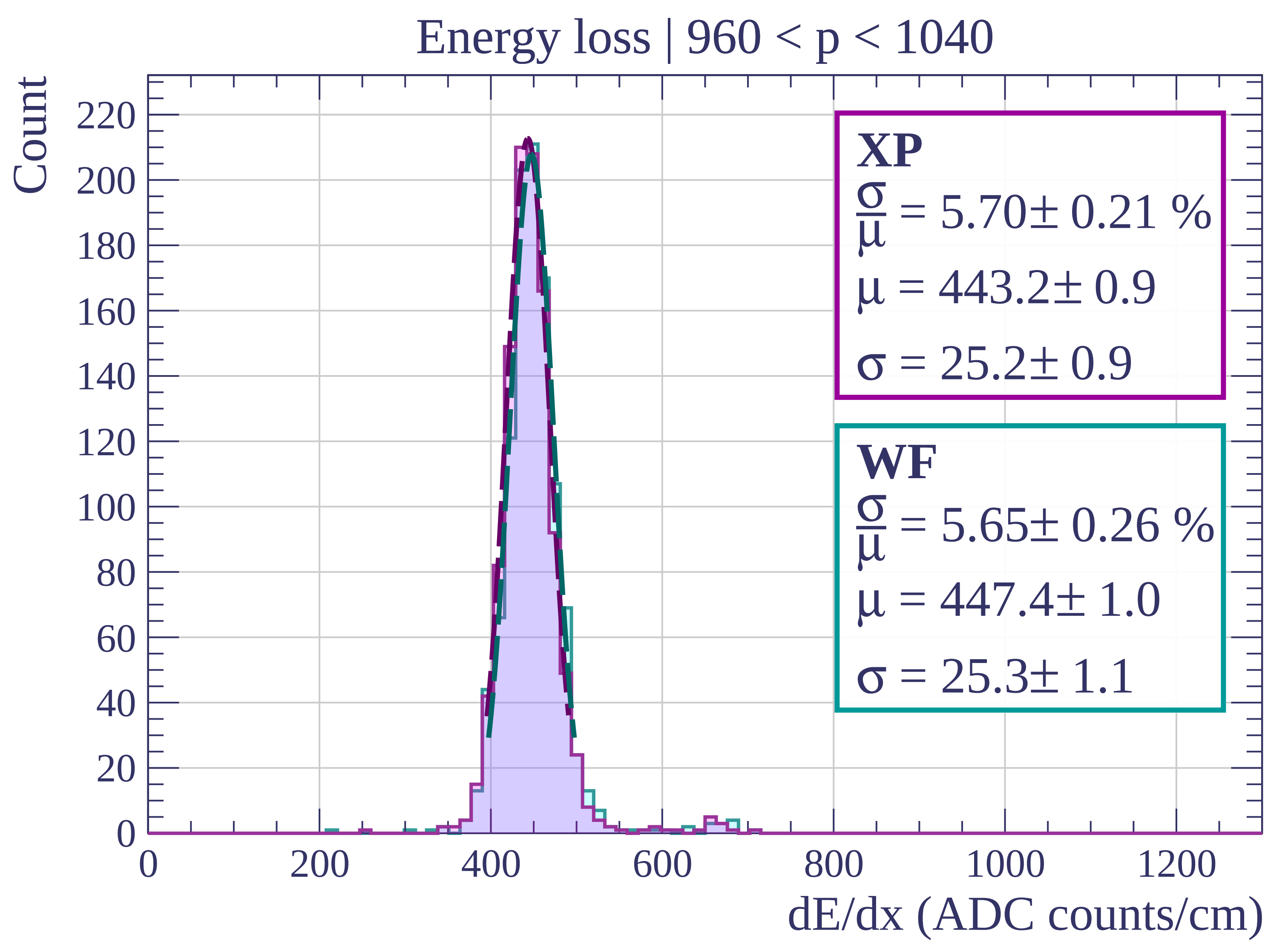
Energy loss | 720 < p < 800
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.68± 0.19 %
μ = 428.4± 0.8
σ = 24.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 5.78± 0.19 %
μ = 431.2± 0.9
σ = 24.9± 0.8



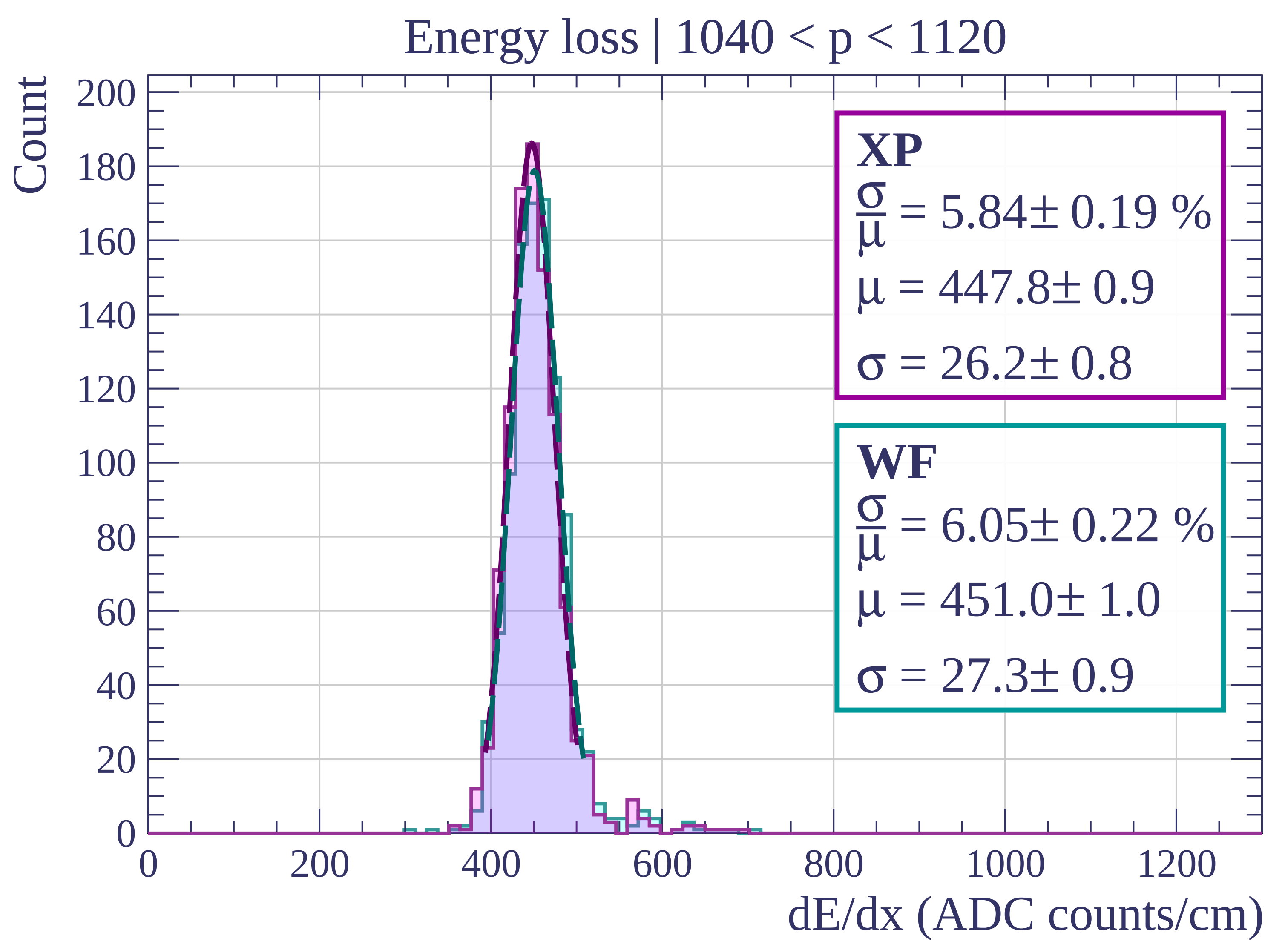
Energy loss | 800 < p < 880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.95± 0.22 %
μ = 432.6± 0.9
σ = 25.7± 0.9
WF
σ/μ = 5.97± 0.18 %
μ = 435.9± 0.9
σ = 26.0± 0.7
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200



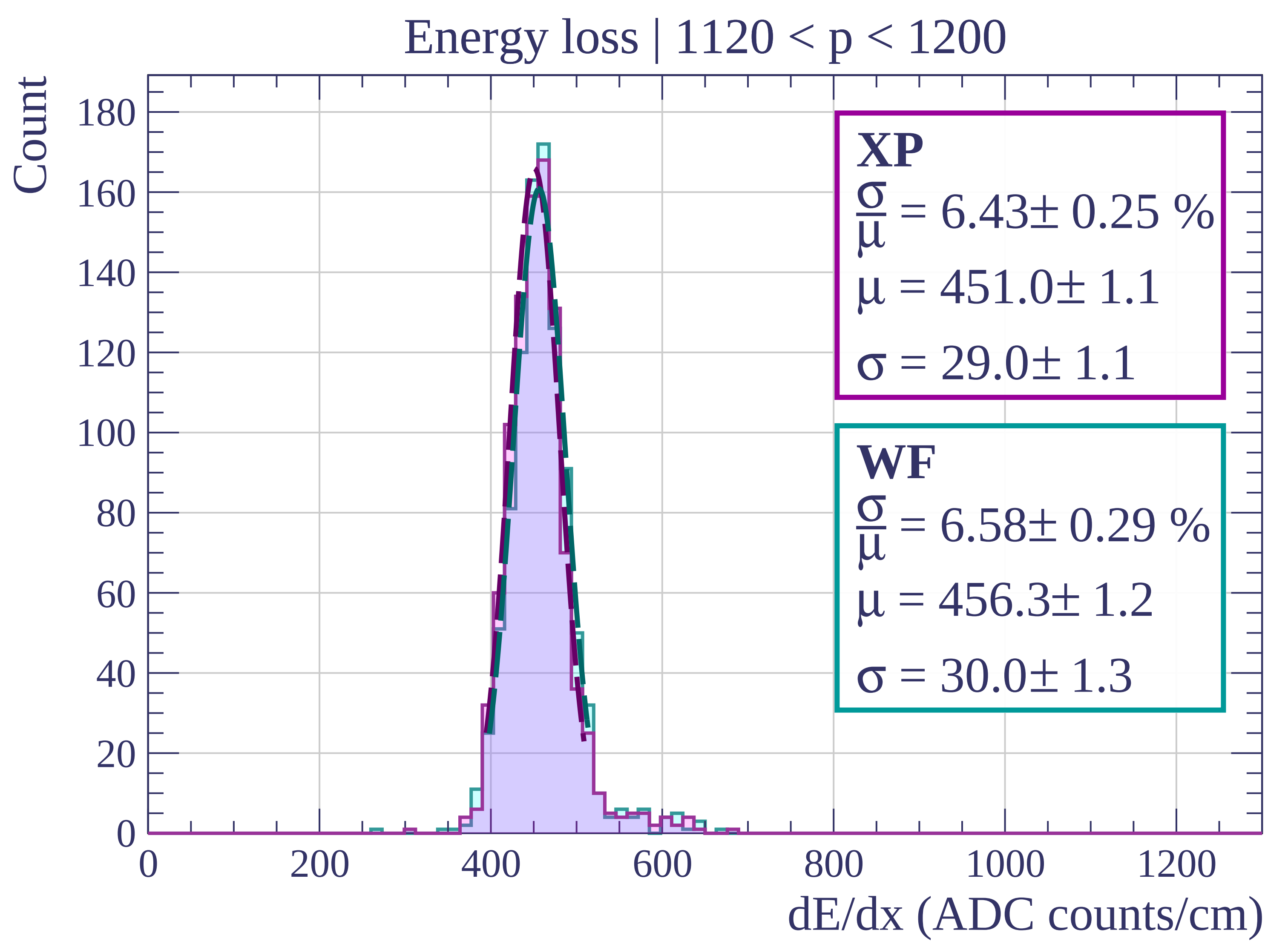
Energy loss | 880 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 5.99± 0.19 %
μ = 438.5± 0.9
σ = 26.3± 0.8
WF
σ/μ = 6.29± 0.26 %
μ = 442.0± 1.0
σ = 27.8± 1.1

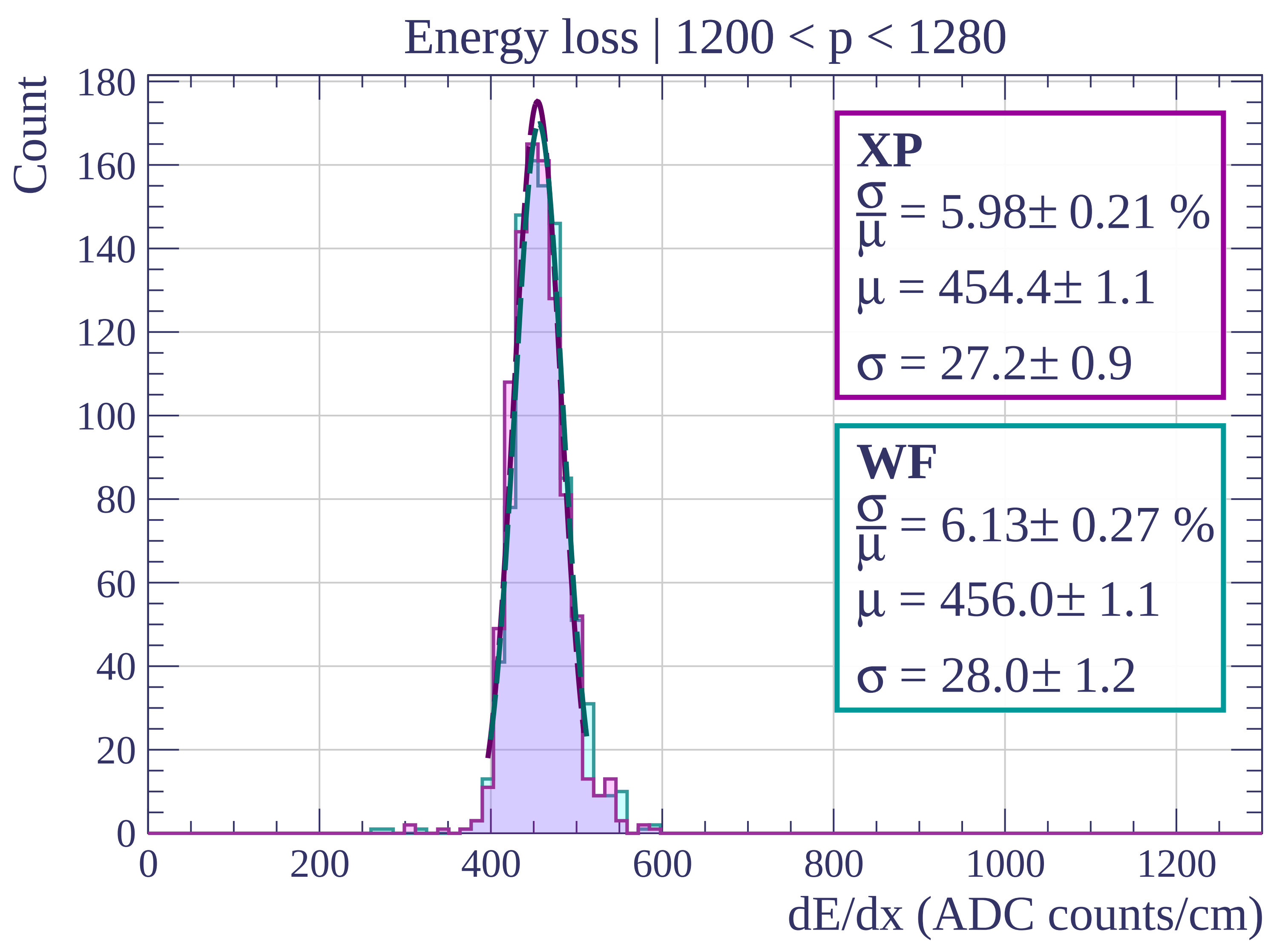


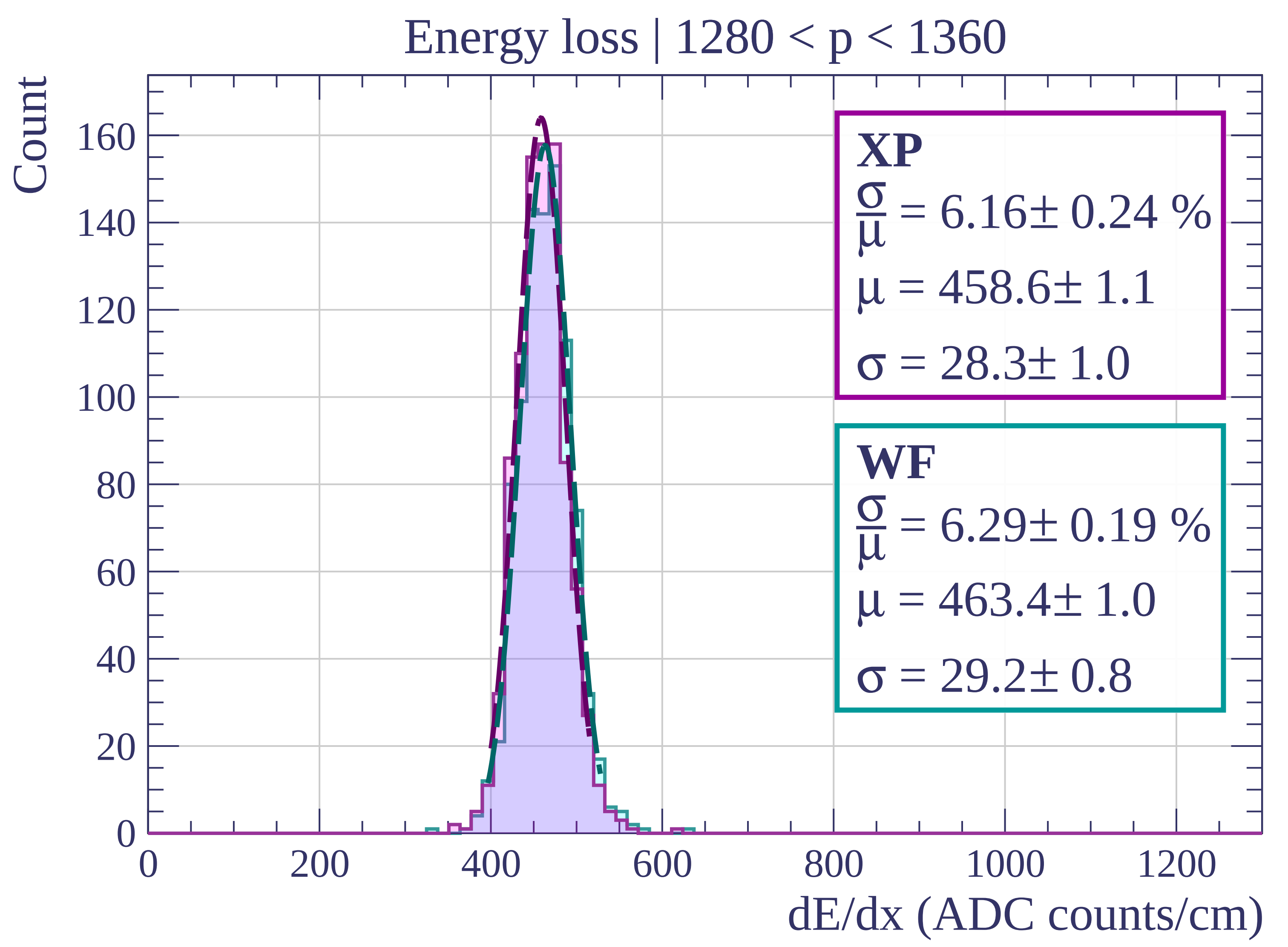
Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = 5.70± 0.21 %
μ = 443.2± 0.9
σ = 25.2± 0.9
WF
σ/μ = 5.65± 0.26 %
μ = 447.4± 1.0
σ = 25.3± 1.1



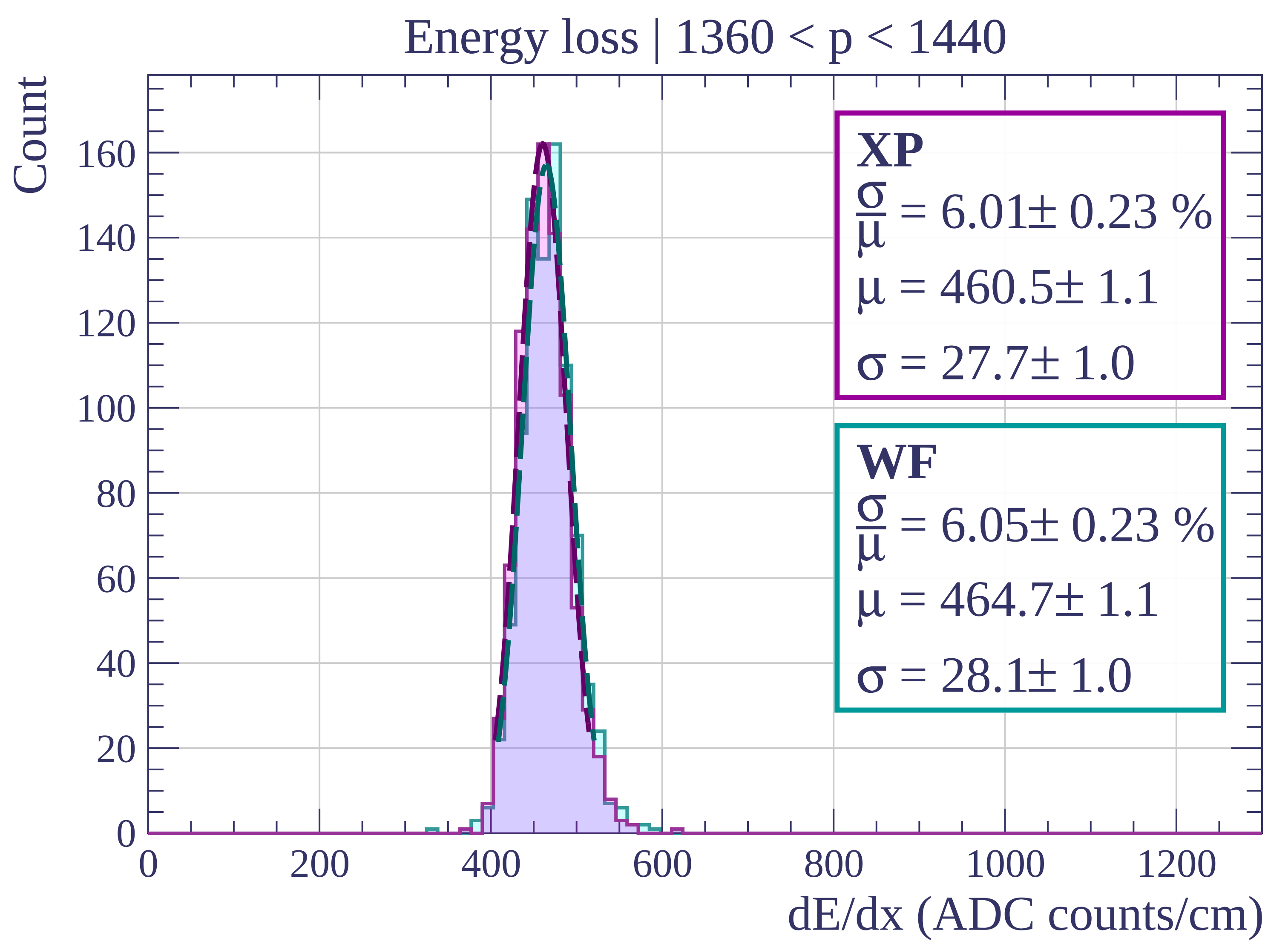
Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.84± 0.19 %
μ = 447.8± 0.9
σ = 26.2± 0.8
WF
$\frac{\sigma}{\mu}$ = 6.05± 0.22 %
μ = 451.0± 1.0
σ = 27.3± 0.9

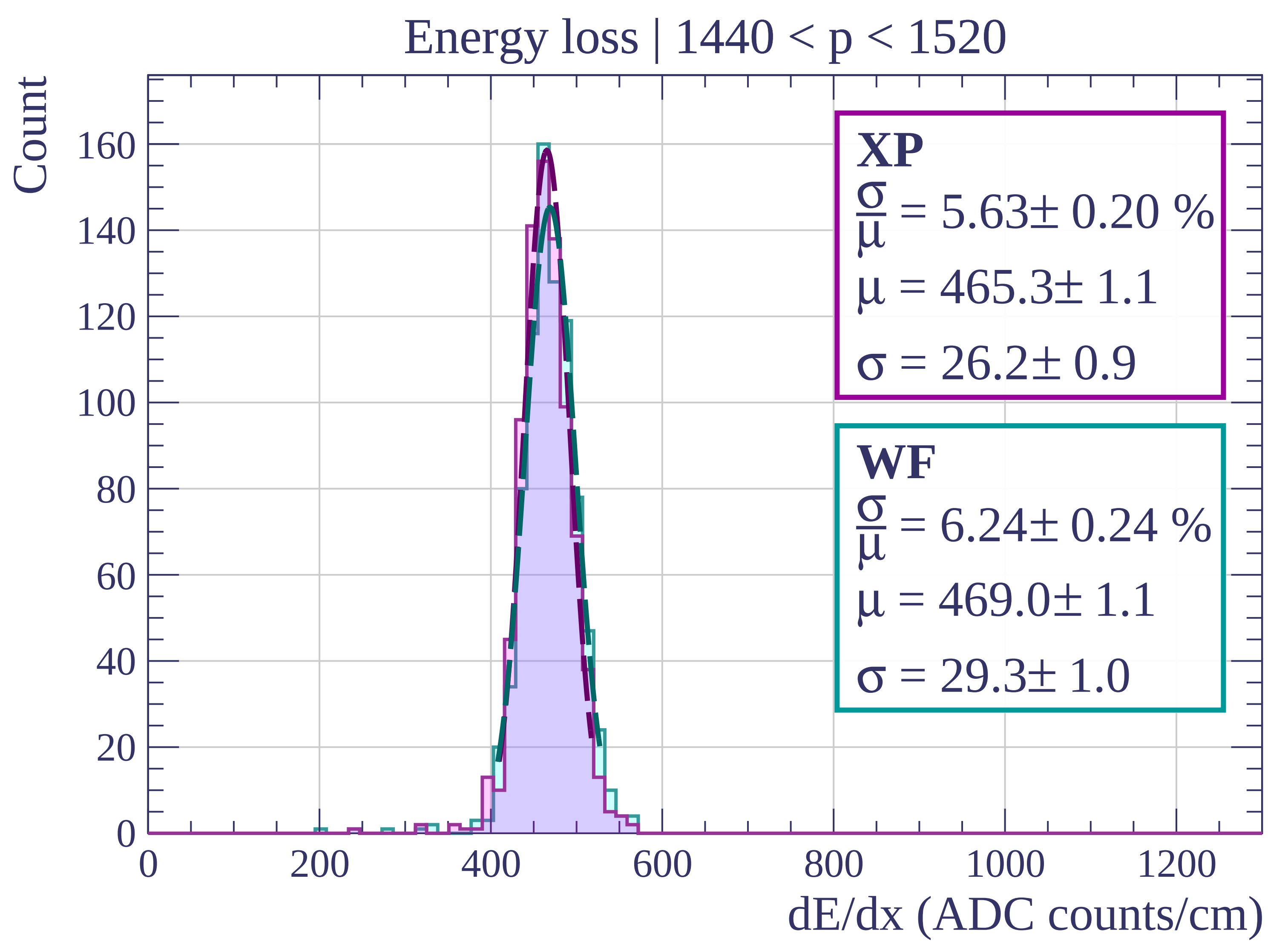

Energy loss | 1120 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 6.43± 0.25 %
μ = 451.0± 1.1
σ = 29.0± 1.1
WF
σ/μ = 6.58± 0.29 %
μ = 456.3± 1.2
σ = 30.0± 1.3


Energy loss | 1200 < p < 1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 5.98± 0.21 %
μ = 454.4± 1.1
σ = 27.2± 0.9
WF
σ/μ = 6.13± 0.27 %
μ = 456.0± 1.1
σ = 28.0± 1.2

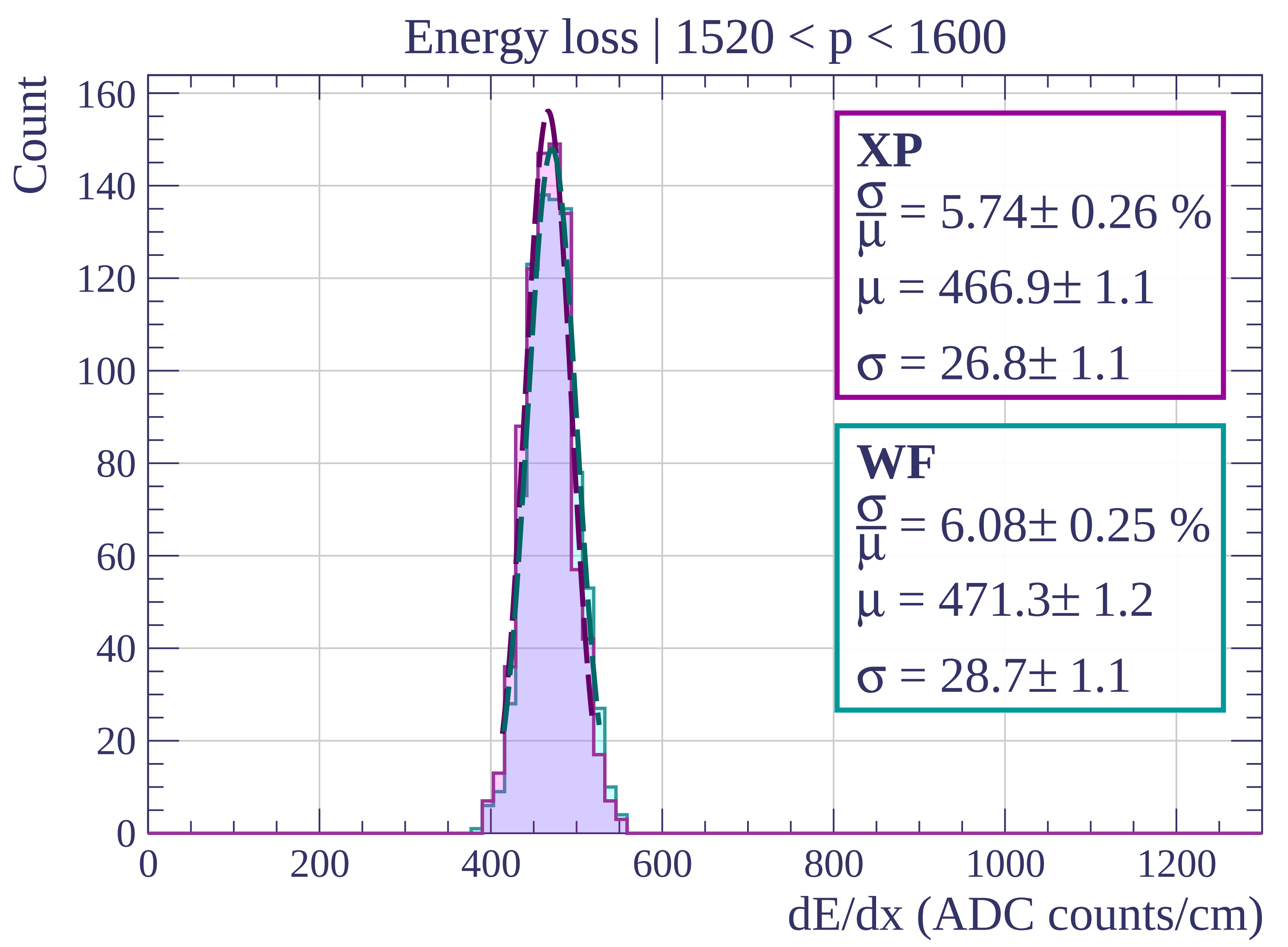


Energy loss | 1280 < p < 1360
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.16± 0.24 %
μ = 458.6± 1.1
σ = 28.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.29± 0.19 %
μ = 463.4± 1.0
σ = 29.2± 0.8

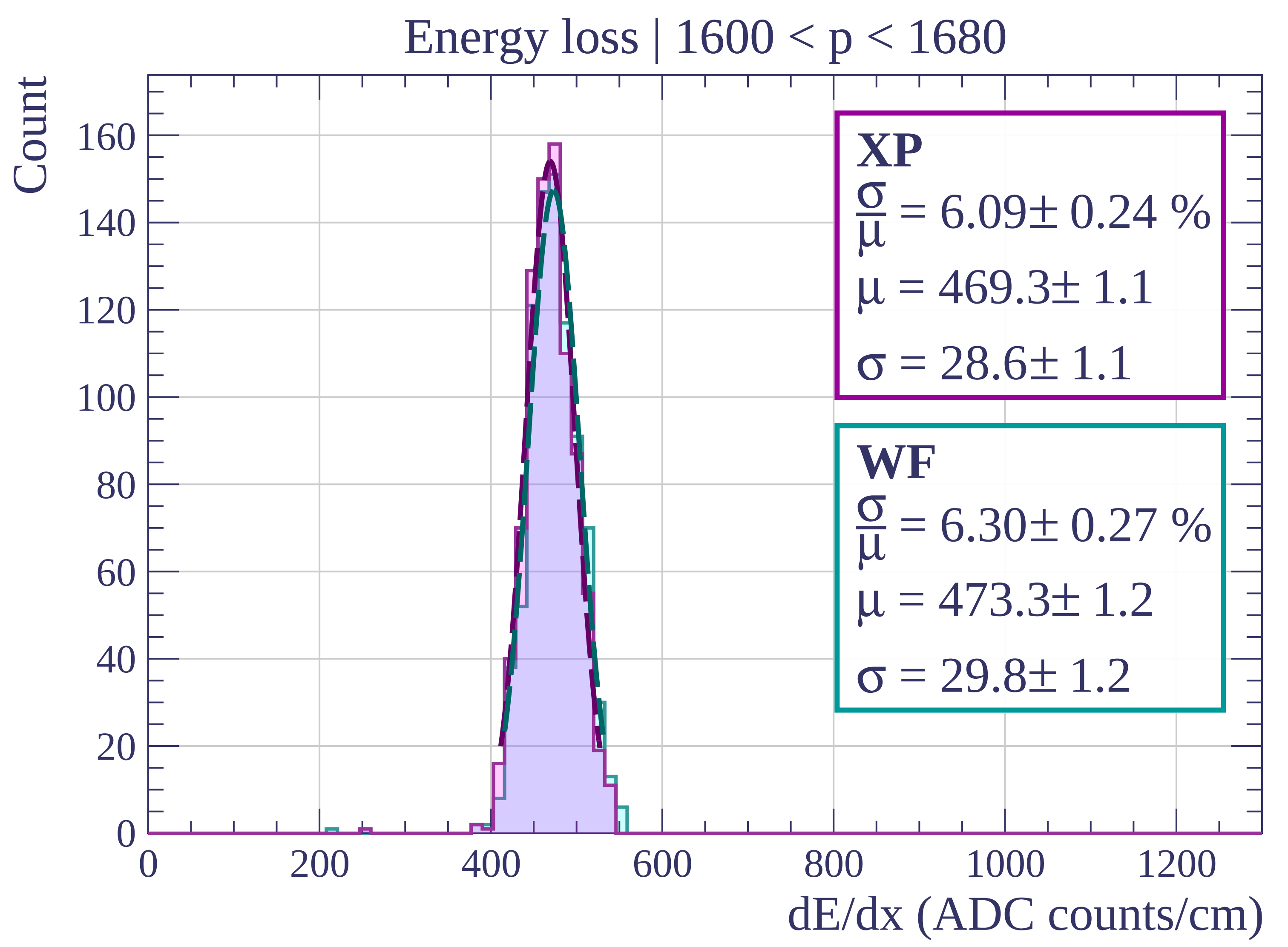

Energy loss | 1360 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
$\frac{\sigma}{\mu}$ = 6.01± 0.23 %
μ = 460.5± 1.1
σ = 27.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.05± 0.23 %
μ = 464.7± 1.1
σ = 28.1± 1.0



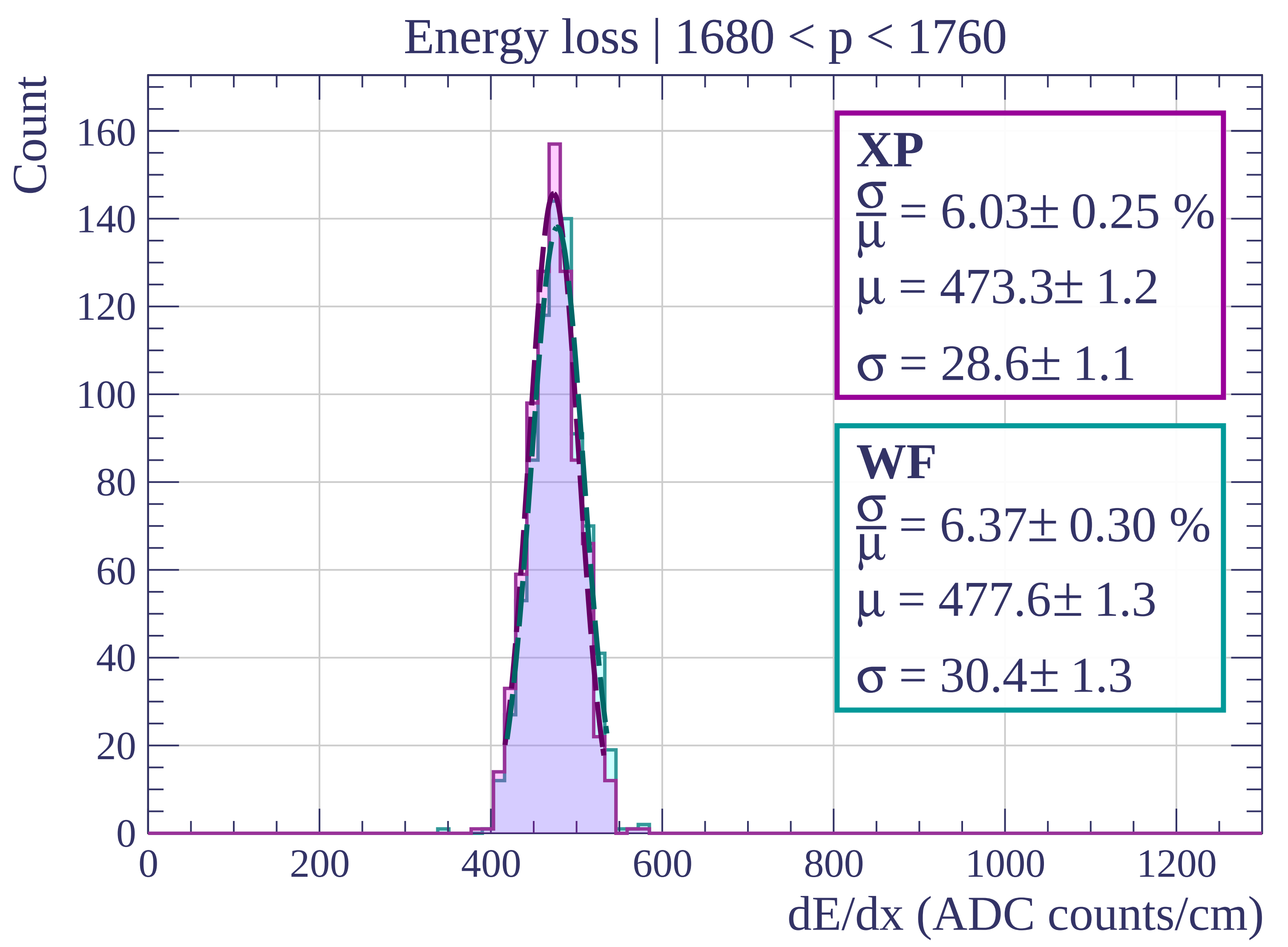
Energy loss | 1440 < p < 1520
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.63± 0.20 %
μ = 465.3± 1.1
σ = 26.2± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.24± 0.24 %
μ = 469.0± 1.1
σ = 29.3± 1.0



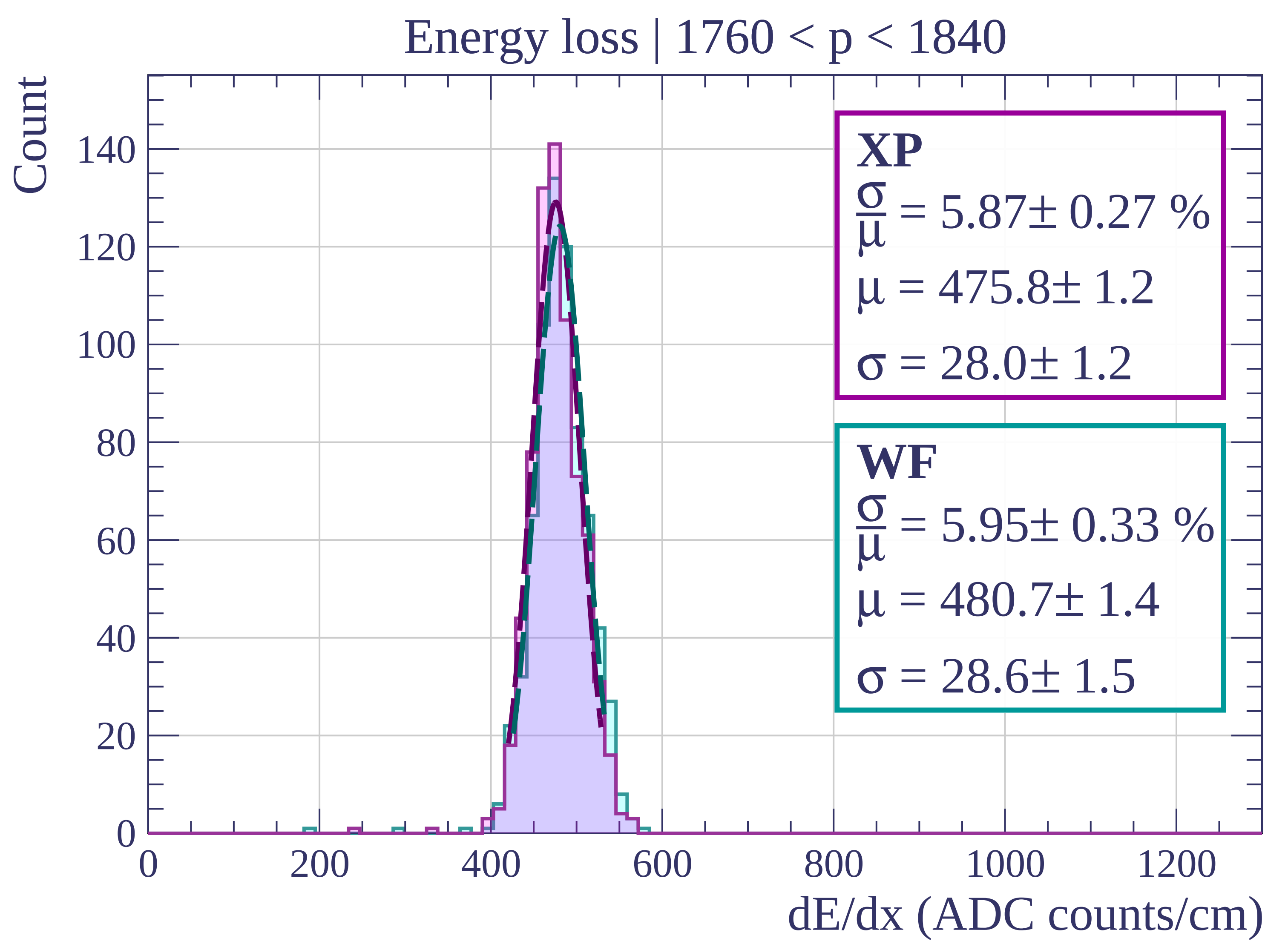
Energy loss | 1520 < p < 1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 5.74± 0.26 %
μ = 466.9± 1.1
σ = 26.8± 1.1
WF
σ/μ = 6.08± 0.25 %
μ = 471.3± 1.2
σ = 28.7± 1.1



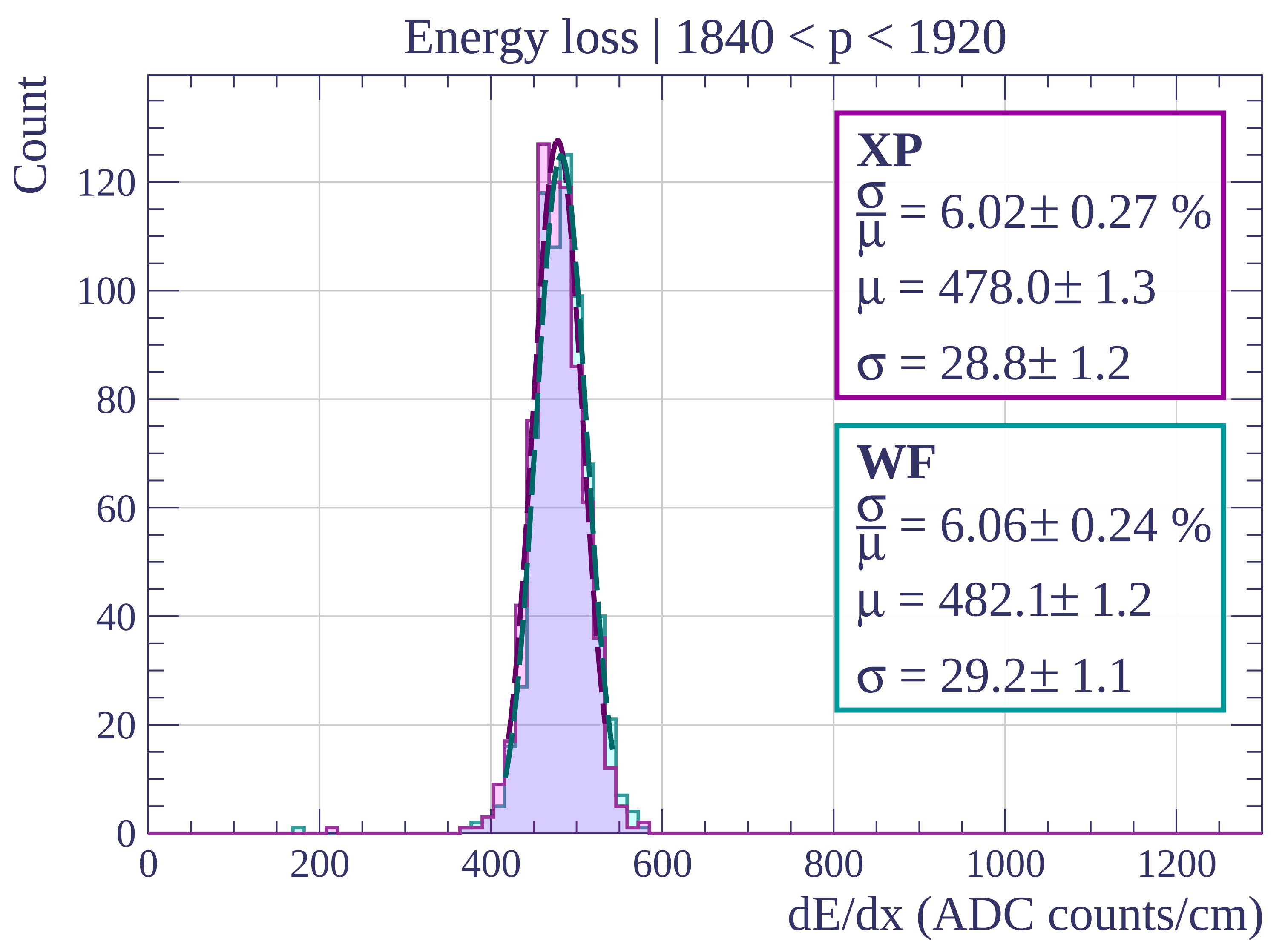
Energy loss | 1600 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
$\frac{\sigma}{\mu}$ = 6.09± 0.24 %
μ = 469.3± 1.1
σ = 28.6± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.30± 0.27 %
μ = 473.3± 1.2
σ = 29.8± 1.2



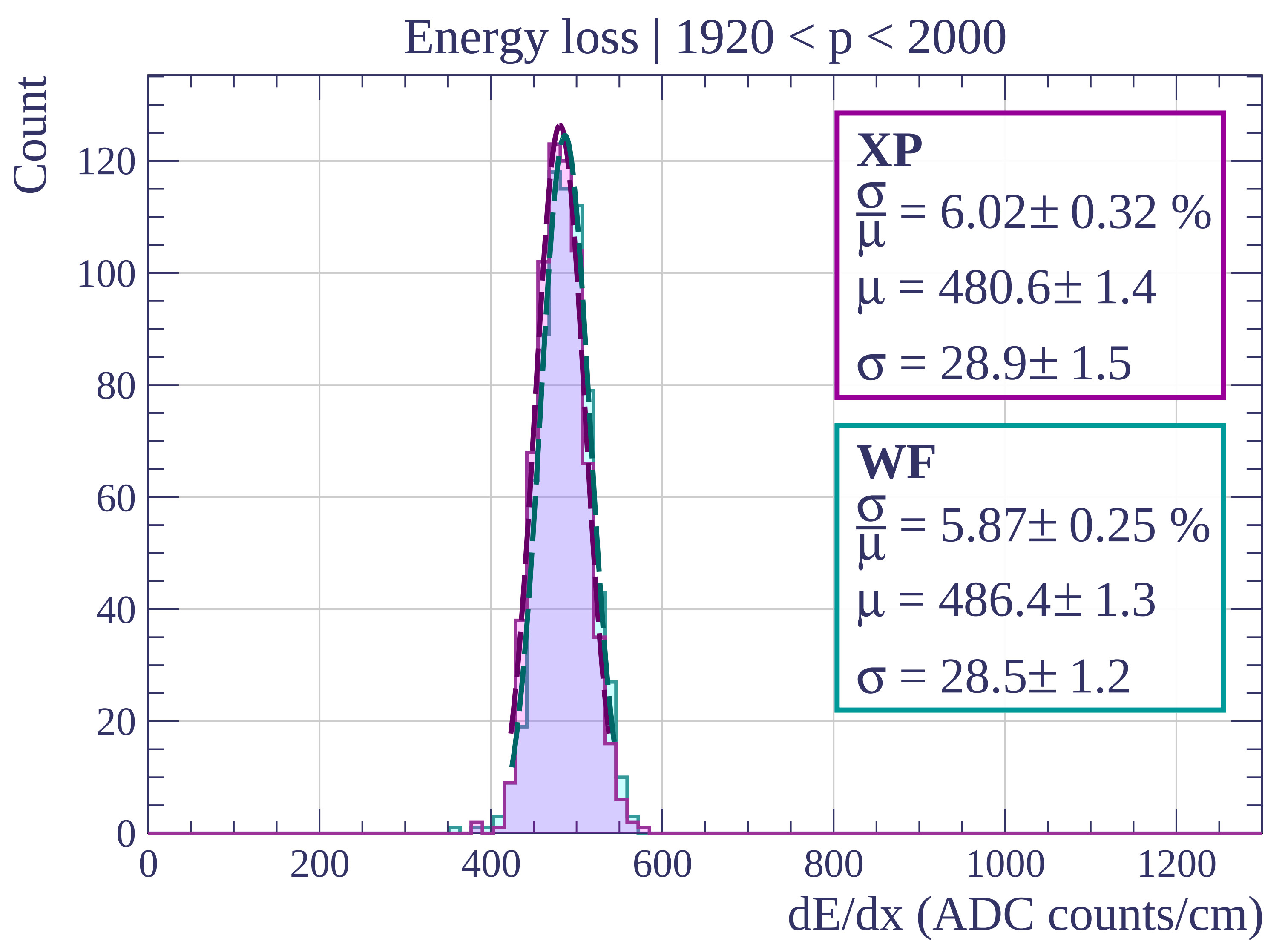
Energy loss | 1680 < p < 1760
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.03± 0.25 %
μ = 473.3± 1.2
σ = 28.6± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.37± 0.30 %
μ = 477.6± 1.3
σ = 30.4± 1.3



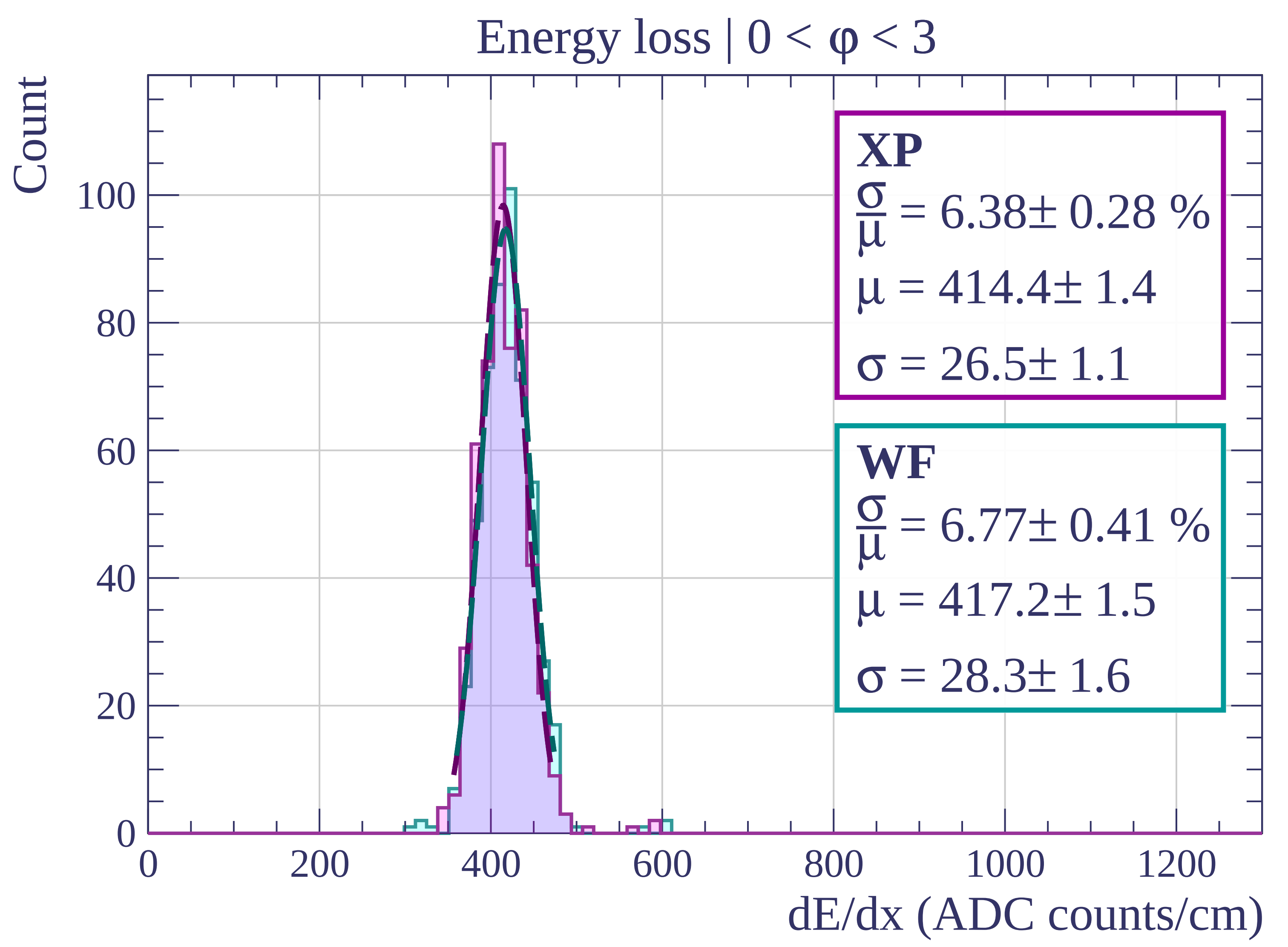
Energy loss | 1760 < p < 1840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 5.87± 0.27 %
μ = 475.8± 1.2
σ = 28.0± 1.2
WF
σ/μ = 5.95± 0.33 %
μ = 480.7± 1.4
σ = 28.6± 1.5



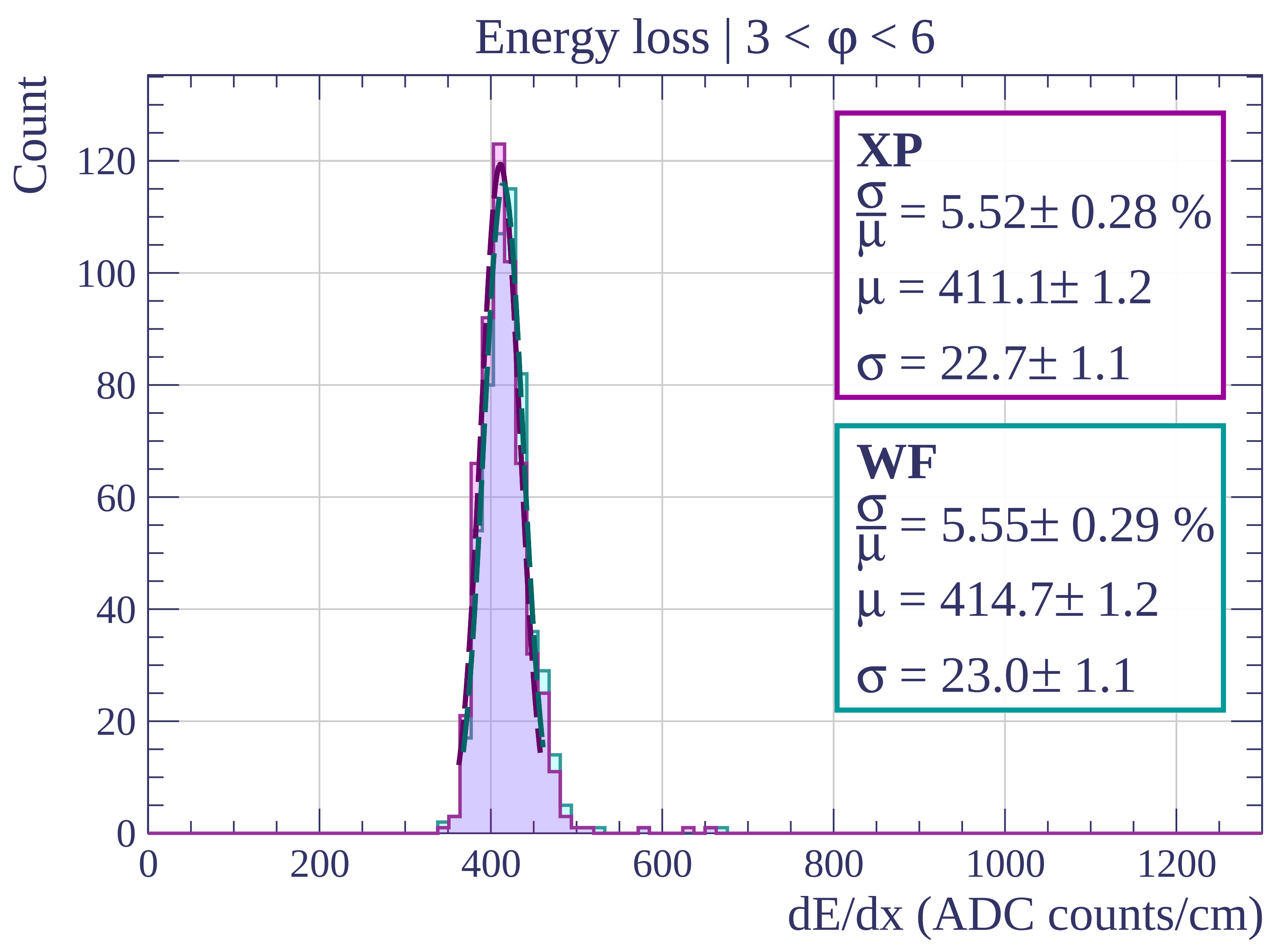
Energy loss | 1840 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP
$\frac{\sigma}{\mu}$ = 6.02± 0.27 %
μ = 478.0± 1.3
σ = 28.8± 1.2
WF
$\frac{\sigma}{\mu}$ = 6.06± 0.24 %
μ = 482.1± 1.2
σ = 29.2± 1.1

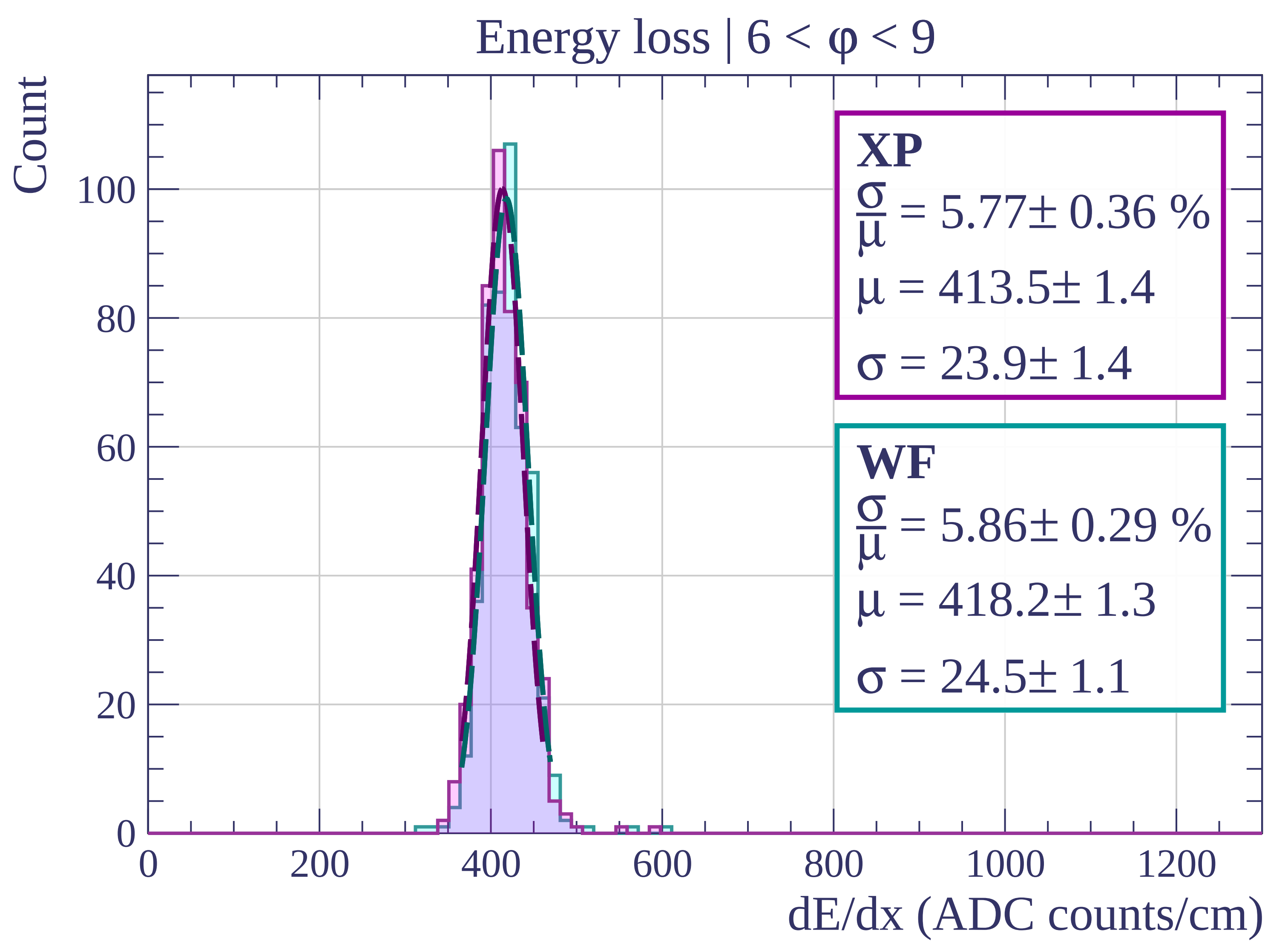


Energy loss | 1920 < p < 2000
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.02± 0.32 %
μ = 480.6± 1.4
σ = 28.9± 1.5
WF
$\frac{\sigma}{\mu}$ = 5.87± 0.25 %
μ = 486.4± 1.3
σ = 28.5± 1.2

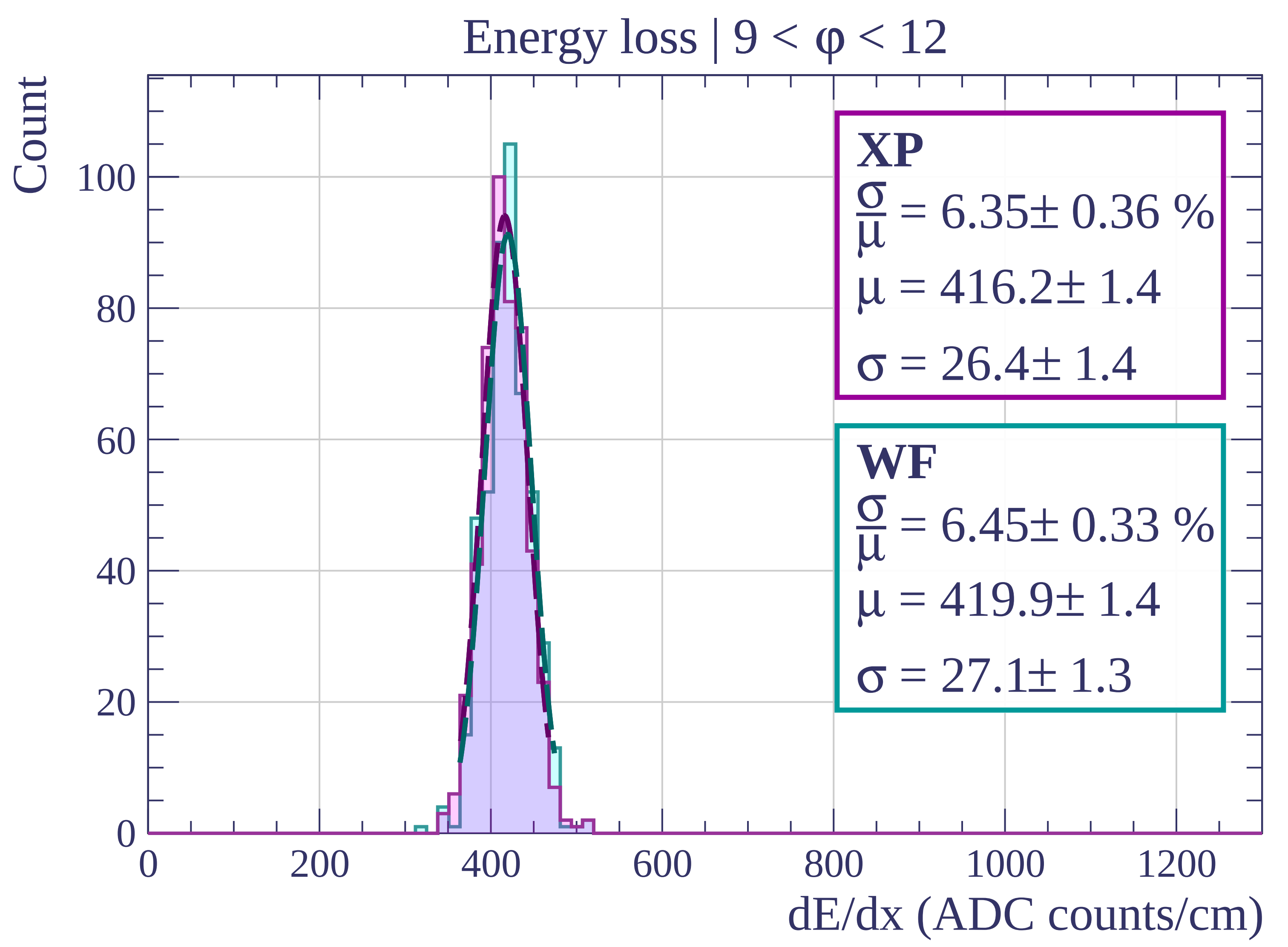


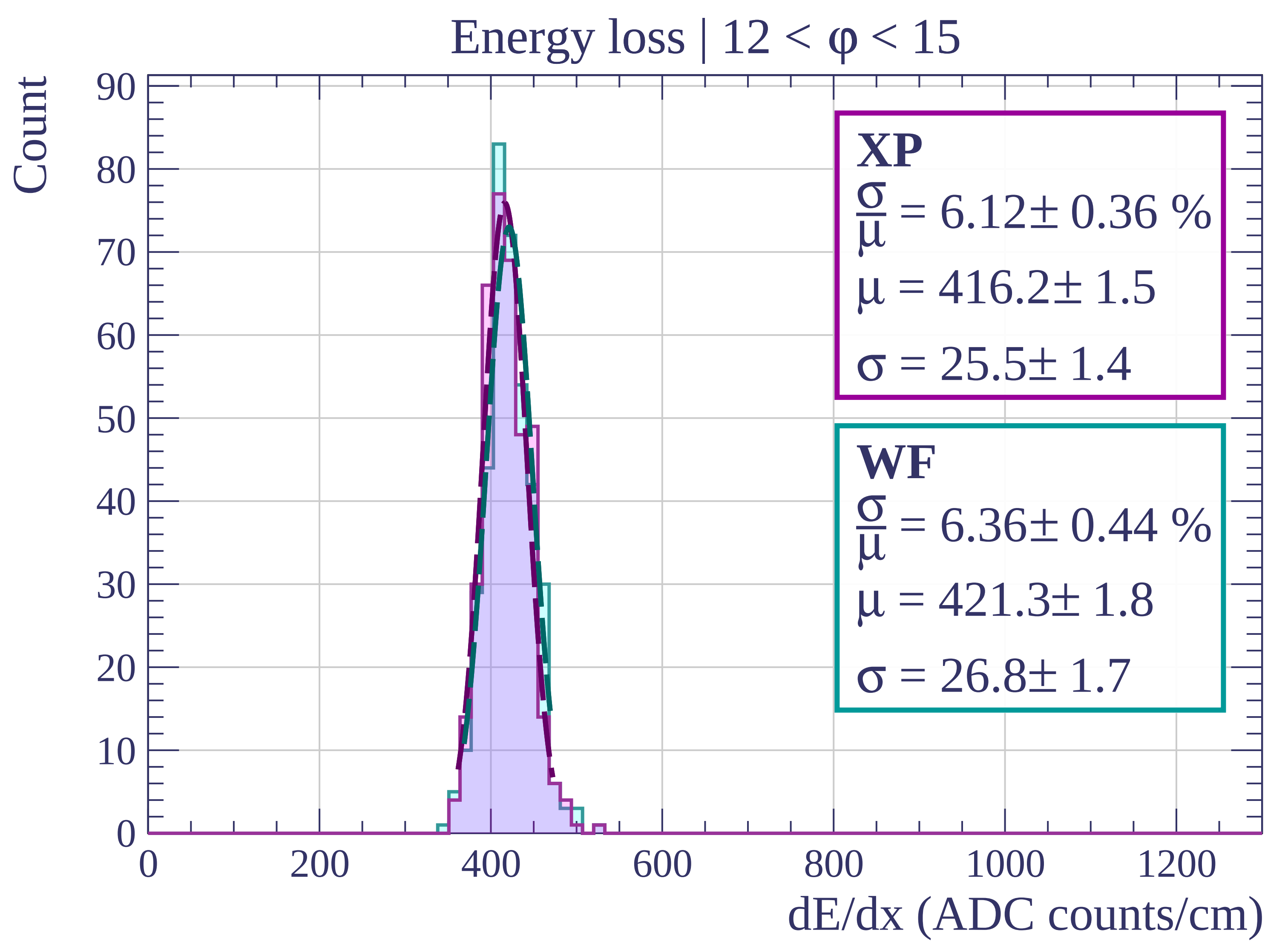
Energy loss | 0 < φ < 3
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
$\frac{\sigma}{\mu}$ = 6.38± 0.28 %
μ = 414.4± 1.4
σ = 26.5± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.77± 0.41 %
μ = 417.2± 1.5
σ = 28.3± 1.6

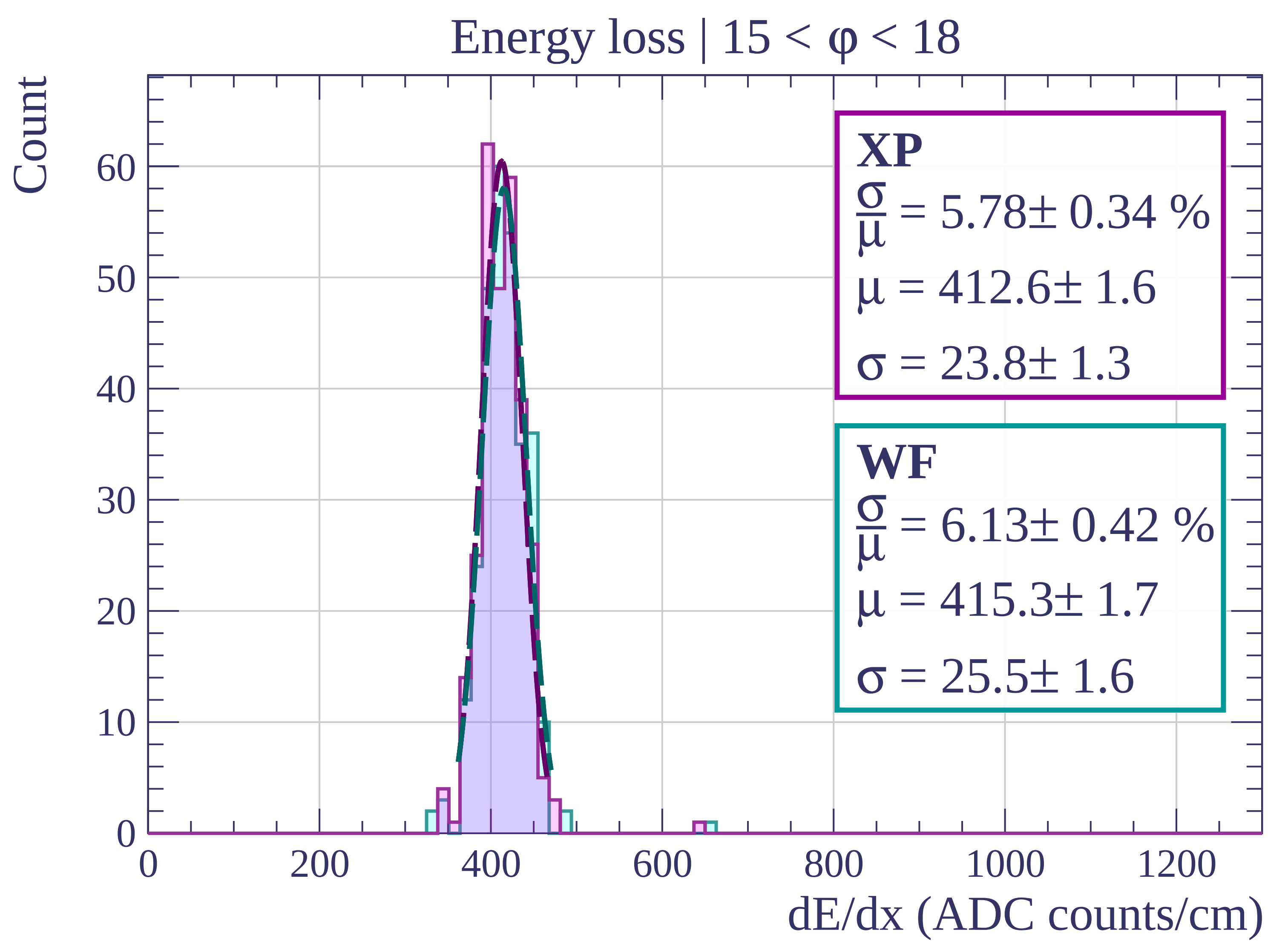

Energy loss | 3 < φ < 6
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 5.52± 0.28 %
μ = 411.1± 1.2
σ = 22.7± 1.1
WF
σ/μ = 5.55± 0.29 %
μ = 414.7± 1.2
σ = 23.0± 1.1



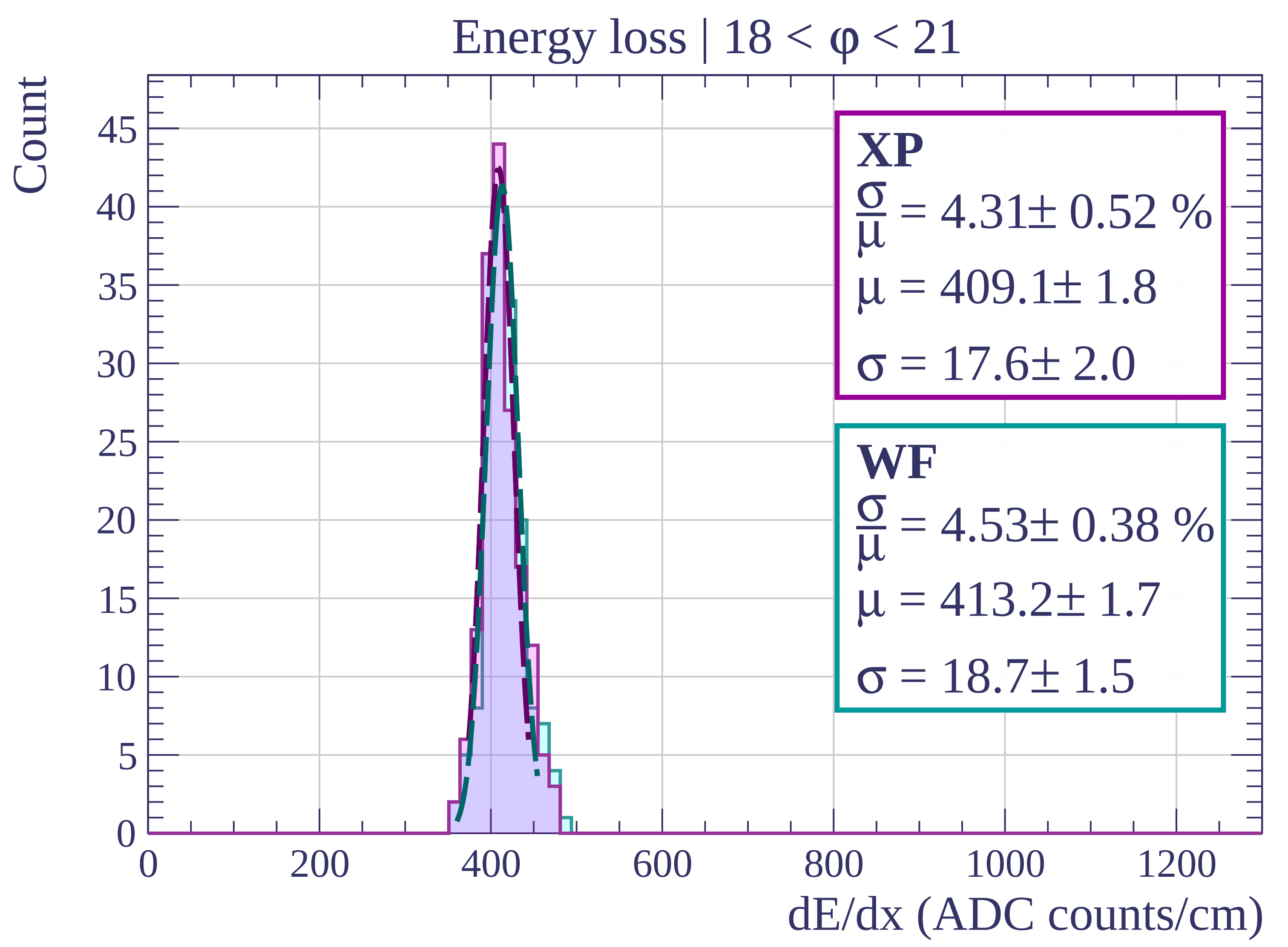
Energy loss | 6 < φ < 9
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 5.77± 0.36 %
μ = 413.5± 1.4
σ = 23.9± 1.4
WF
σ/μ = 5.86± 0.29 %
μ = 418.2± 1.3
σ = 24.5± 1.1


Energy loss | 9 < φ < 12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.35± 0.36 %
μ = 416.2± 1.4
σ = 26.4± 1.4
WF
σ/μ = 6.45± 0.33 %
μ = 419.9± 1.4
σ = 27.1± 1.3

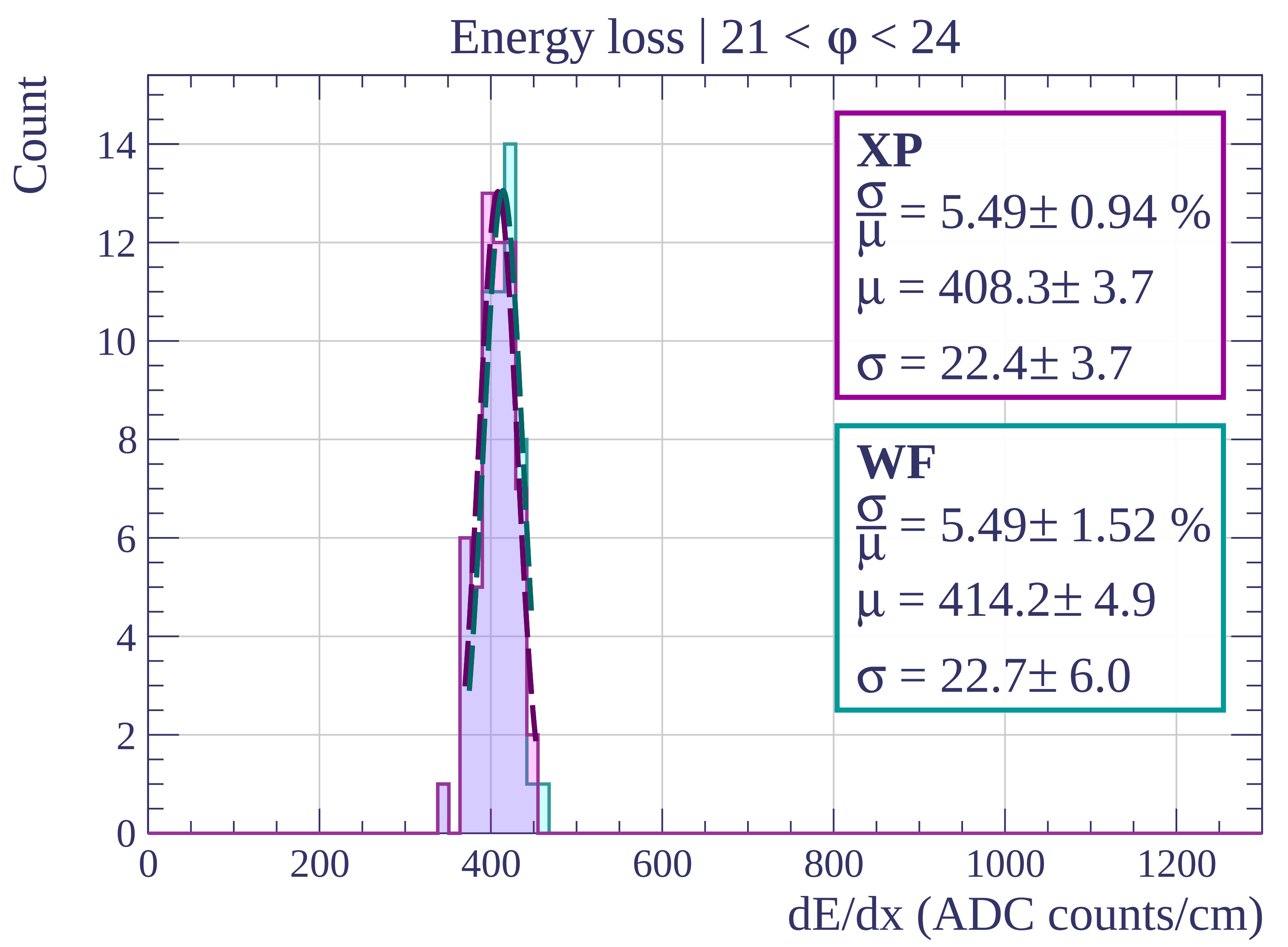

Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.12± 0.36 %
μ = 416.2± 1.5
σ = 25.5± 1.4
WF
$\frac{\sigma}{\mu}$ = 6.36± 0.44 %
μ = 421.3± 1.8
σ = 26.8± 1.7



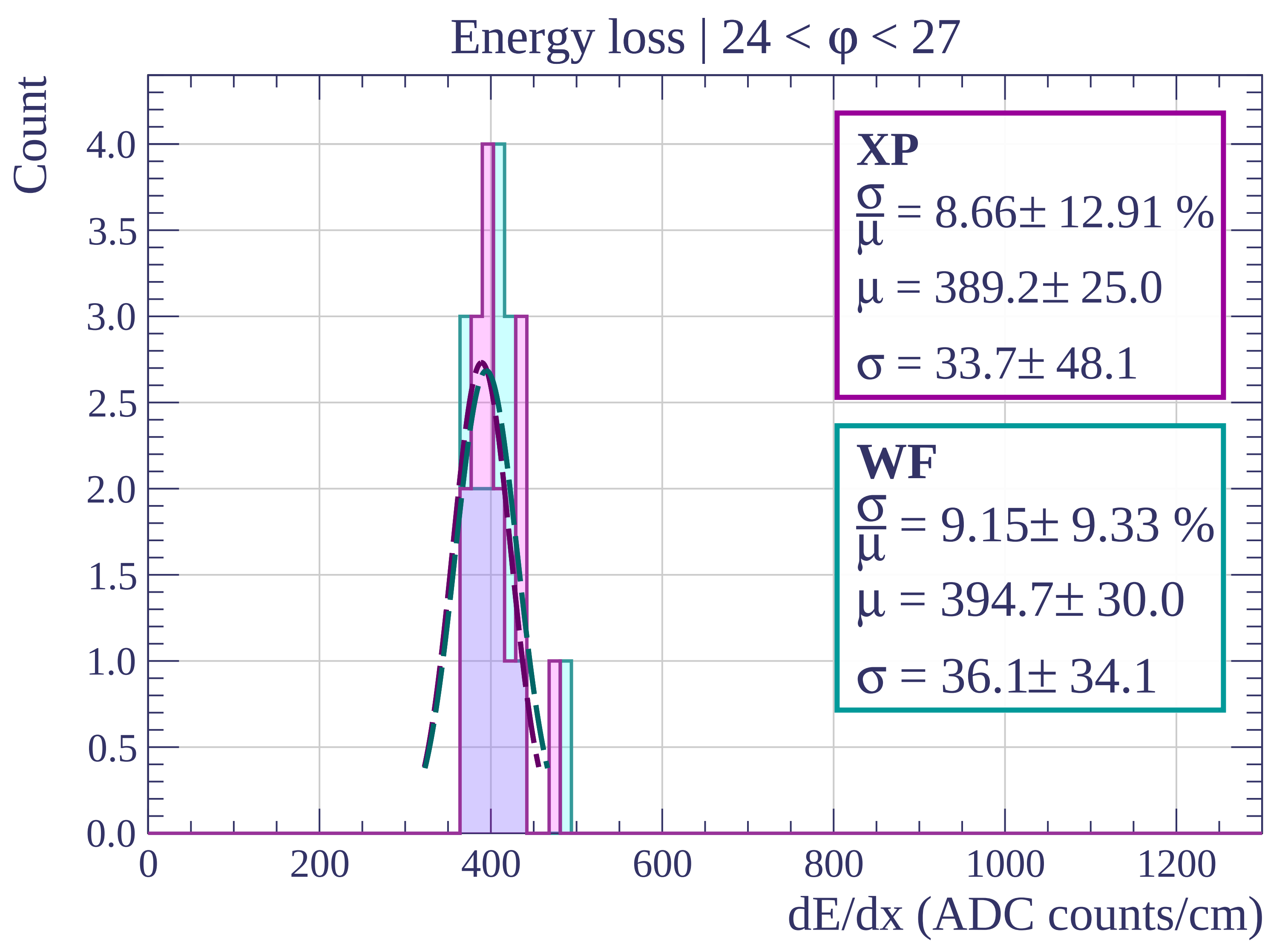
Energy loss | 15 < φ < 18
Count
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.78± 0.34 %
μ = 412.6± 1.6
σ = 23.8± 1.3
WF
$\frac{\sigma}{\mu}$ = 6.13± 0.42 %
μ = 415.3± 1.7
σ = 25.5± 1.6



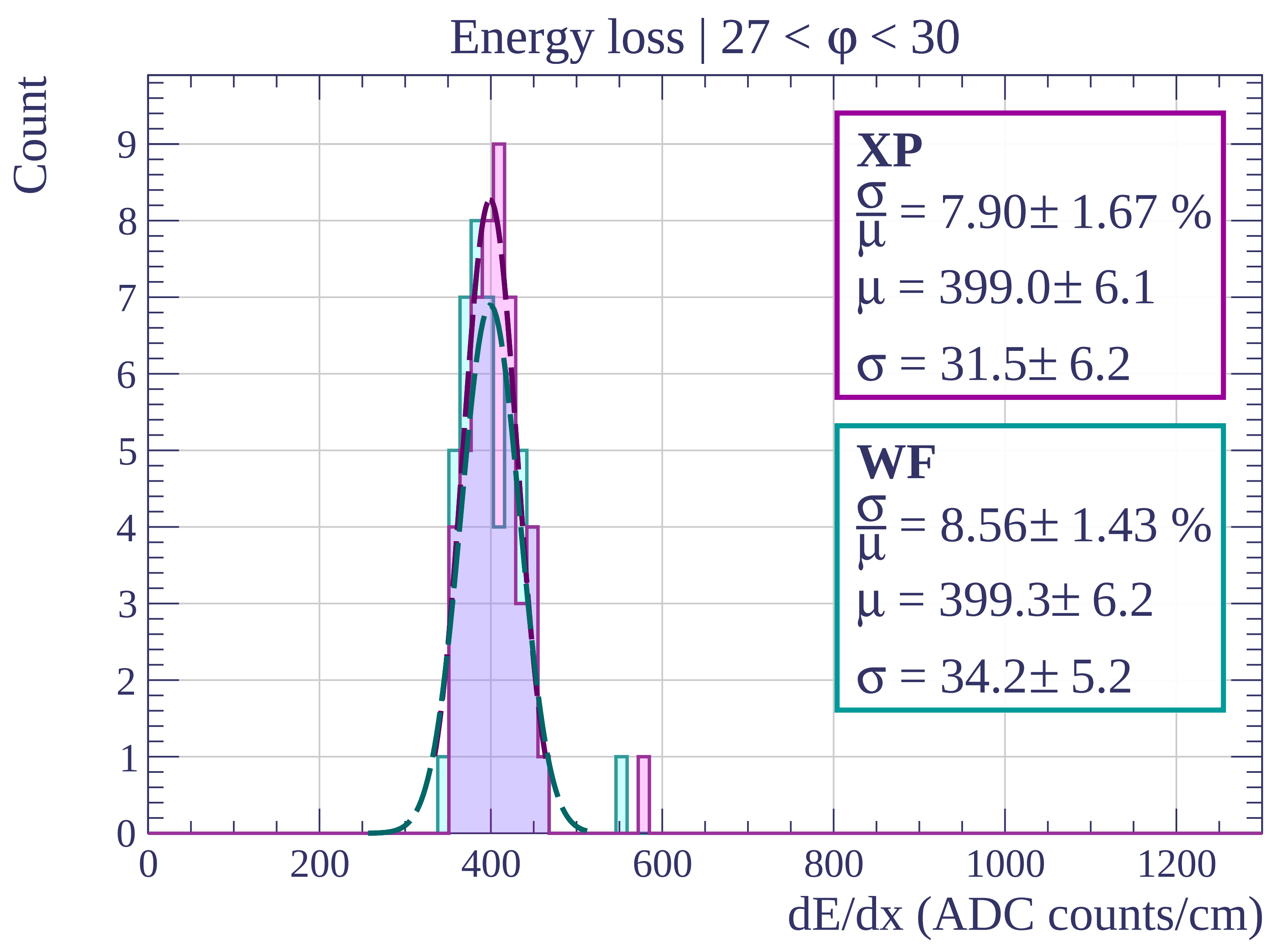
Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 4.31± 0.52 %
μ = 409.1± 1.8
σ = 17.6± 2.0
WF
σ/μ = 4.53± 0.38 %
μ = 413.2± 1.7
σ = 18.7± 1.5



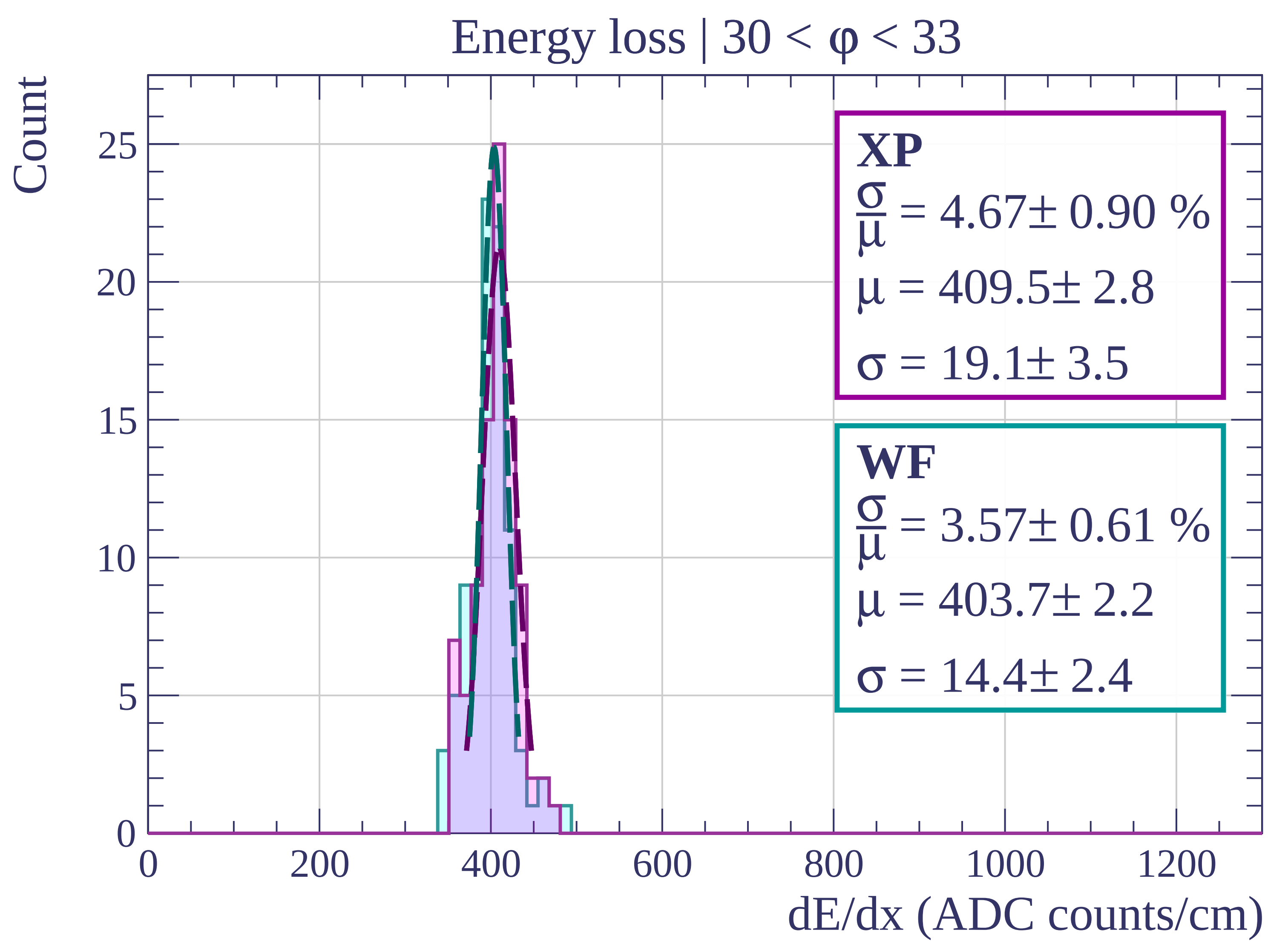
Energy loss | 21 < φ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
XP
σ/μ = 5.49± 0.94 %
μ = 408.3± 3.7
σ = 22.4± 3.7
WF
σ/μ = 5.49± 1.52 %
μ = 414.2± 4.9
σ = 22.7± 6.0



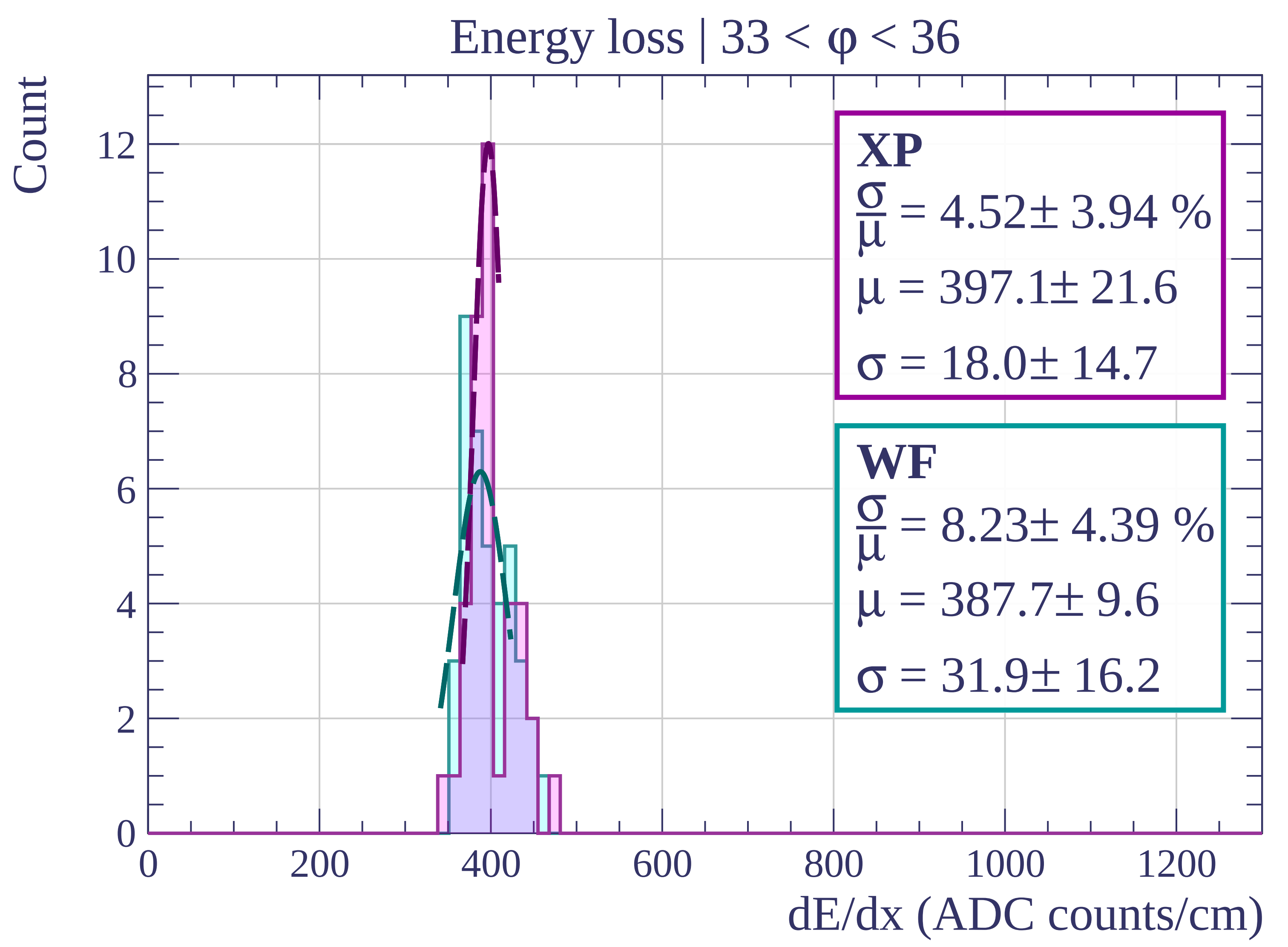
Energy loss | 24 < φ < 27
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
σ/μ = 8.66± 12.91 %
μ = 389.2± 25.0
σ = 33.7± 48.1
WF
σ/μ = 9.15± 9.33 %
μ = 394.7± 30.0
σ = 36.1± 34.1



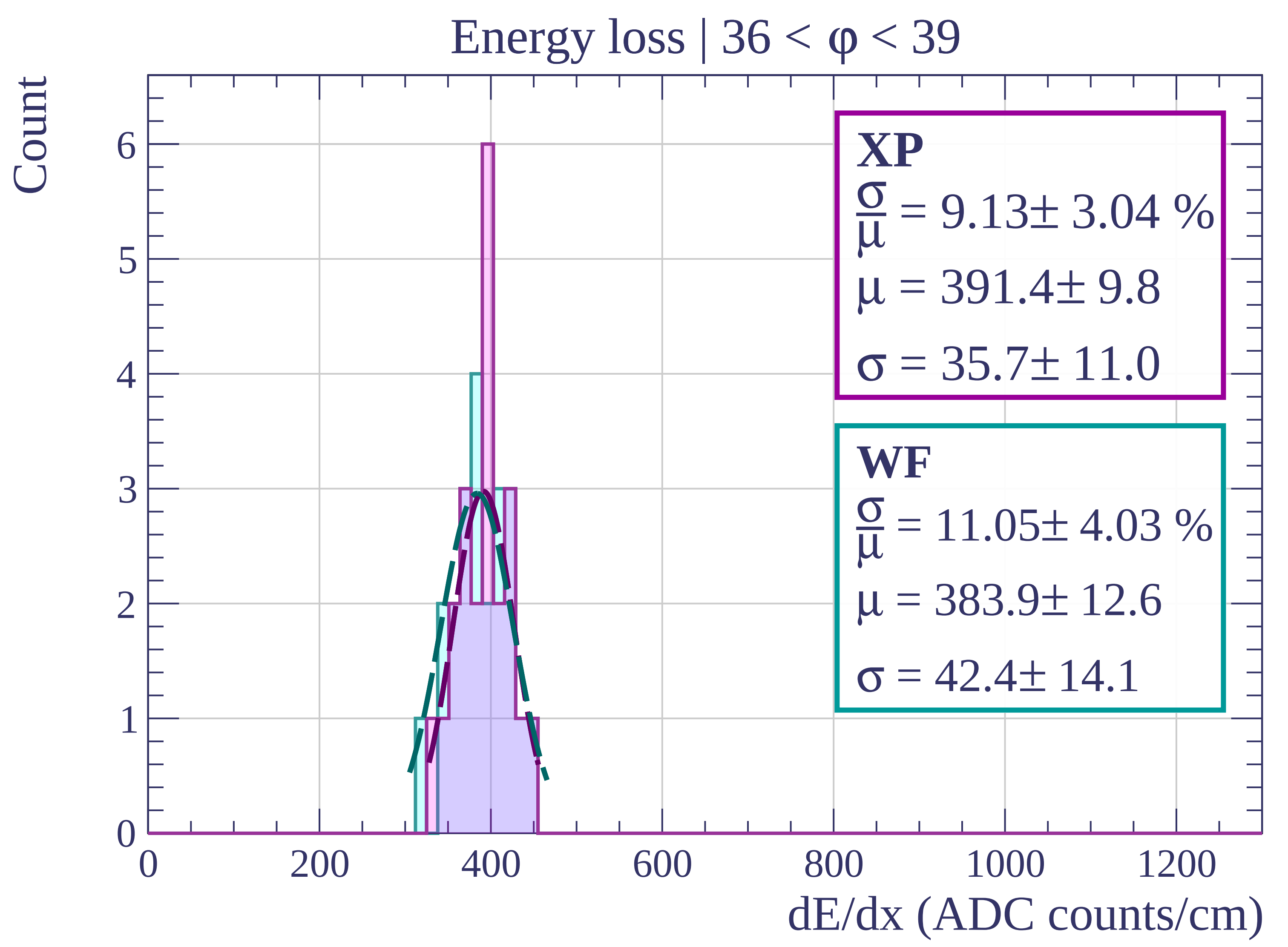
Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 7.90± 1.67 %
μ = 399.0± 6.1
σ = 31.5± 6.2
WF
σ/μ = 8.56± 1.43 %
μ = 399.3± 6.2
σ = 34.2± 5.2

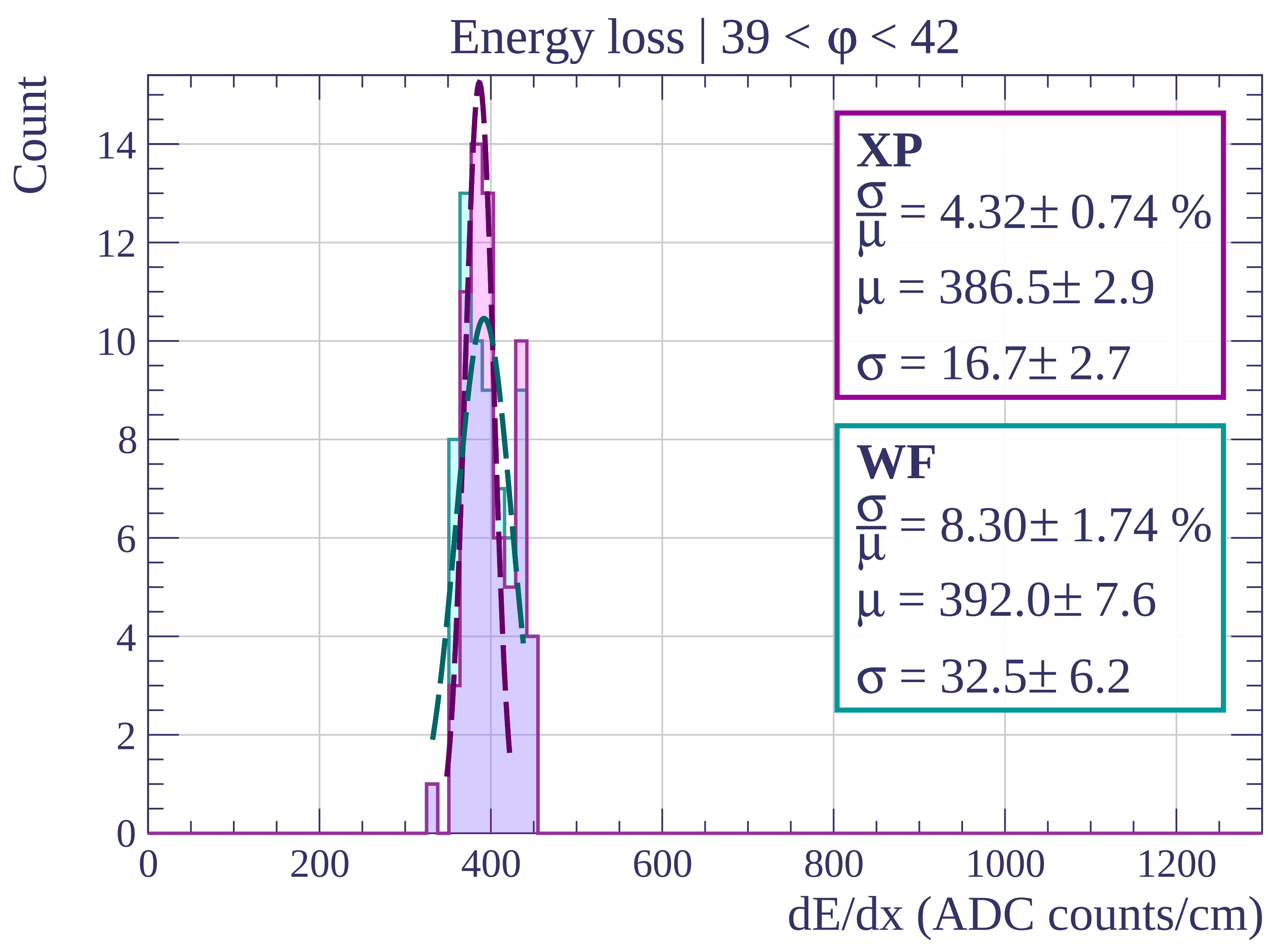


Energy loss | 30 < φ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
σ/μ = 4.67± 0.90 %
μ = 409.5± 2.8
σ = 19.1± 3.5
WF
σ/μ = 3.57± 0.61 %
μ = 403.7± 2.2
σ = 14.4± 2.4

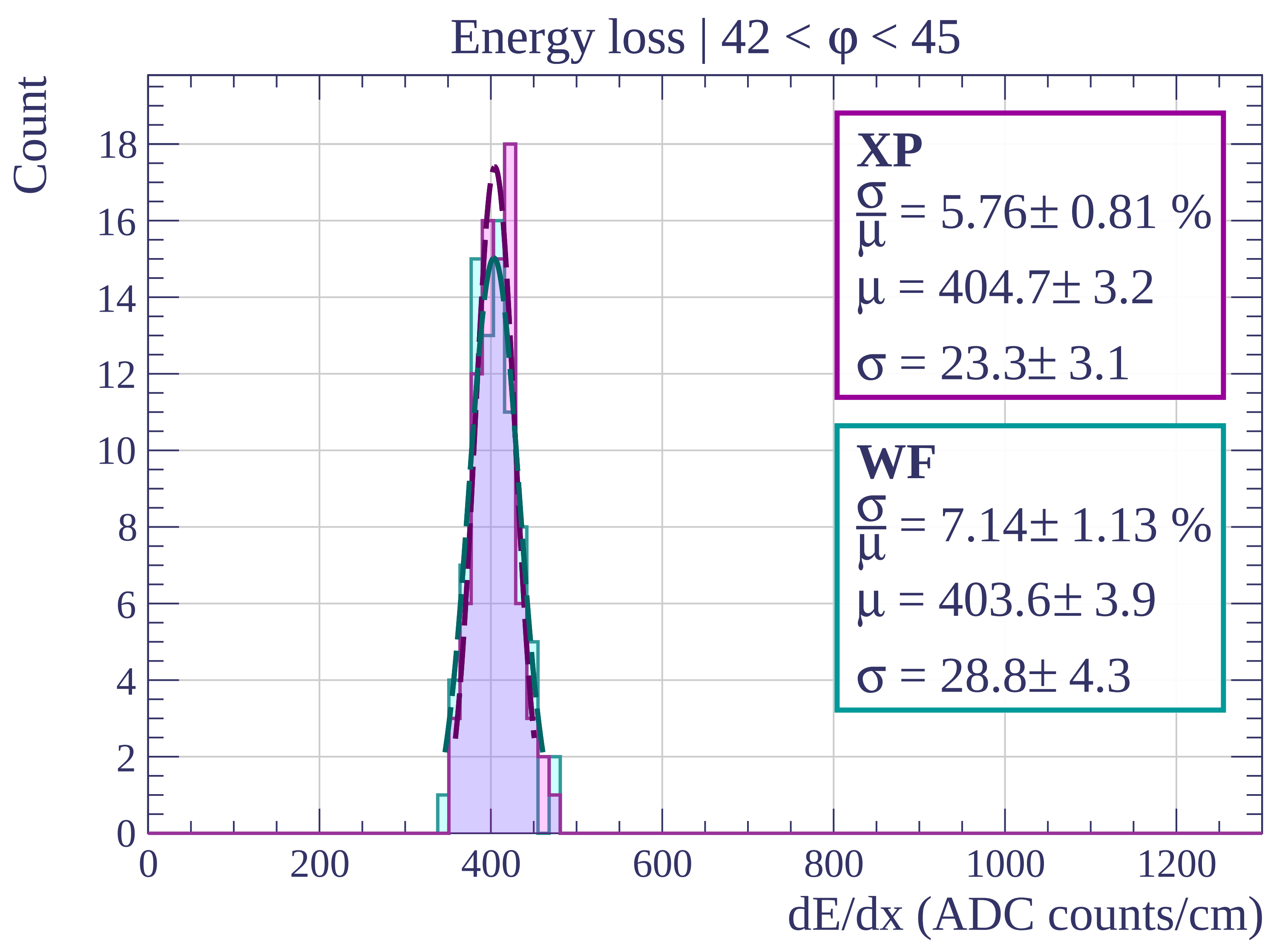


Energy loss | 33 < φ < 36
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.52± 3.94 %
μ = 397.1± 21.6
σ = 18.0± 14.7
WF
$\frac{\sigma}{\mu}$ = 8.23± 4.39 %
μ = 387.7± 9.6
σ = 31.9± 16.2

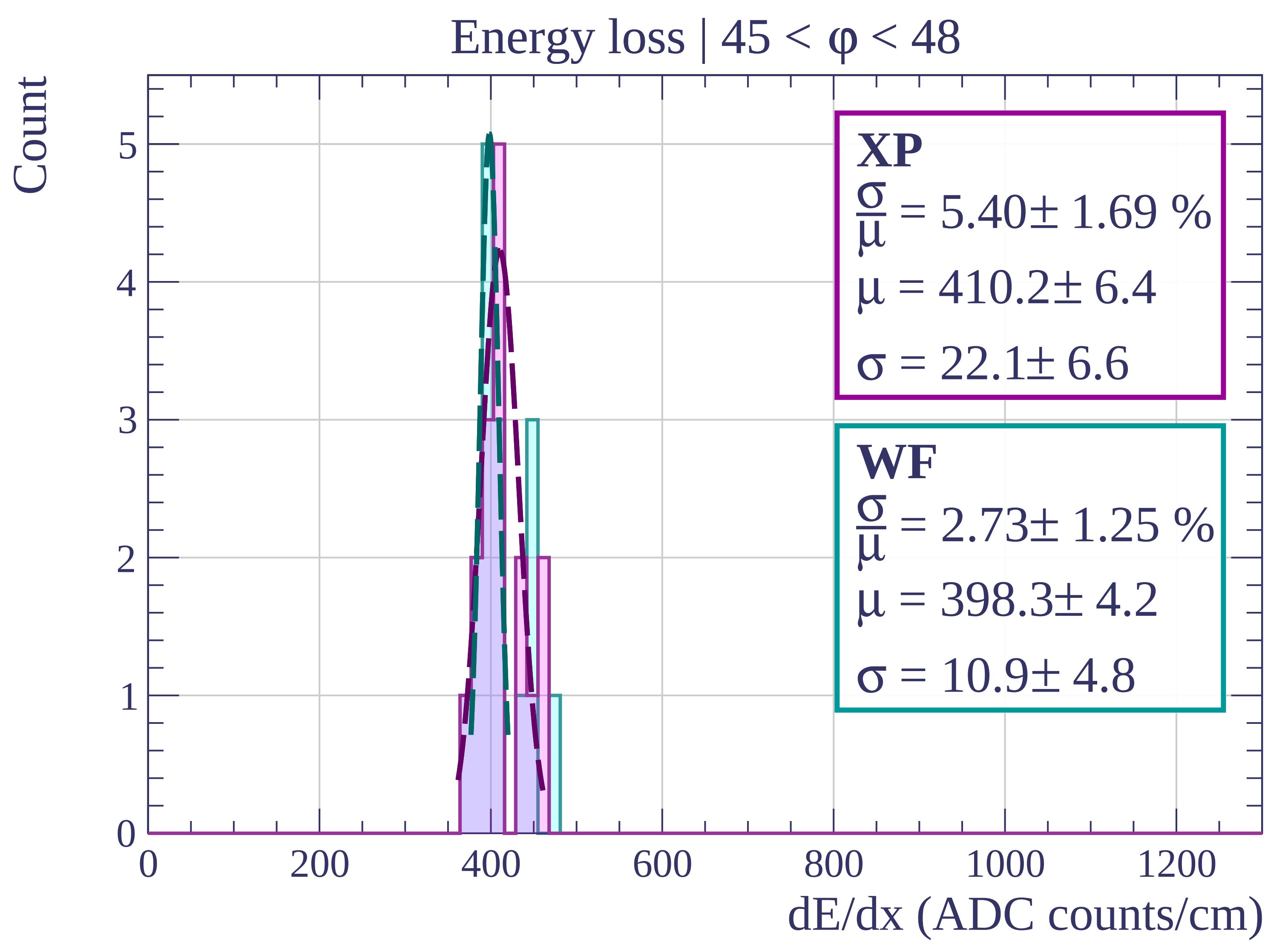

Energy loss | 36 < φ < 39
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
$\frac{\sigma}{\mu}$ = 9.13± 3.04 %
μ = 391.4± 9.8
σ = 35.7± 11.0
WF
$\frac{\sigma}{\mu}$ = 11.05± 4.03 %
μ = 383.9± 12.6
σ = 42.4± 14.1



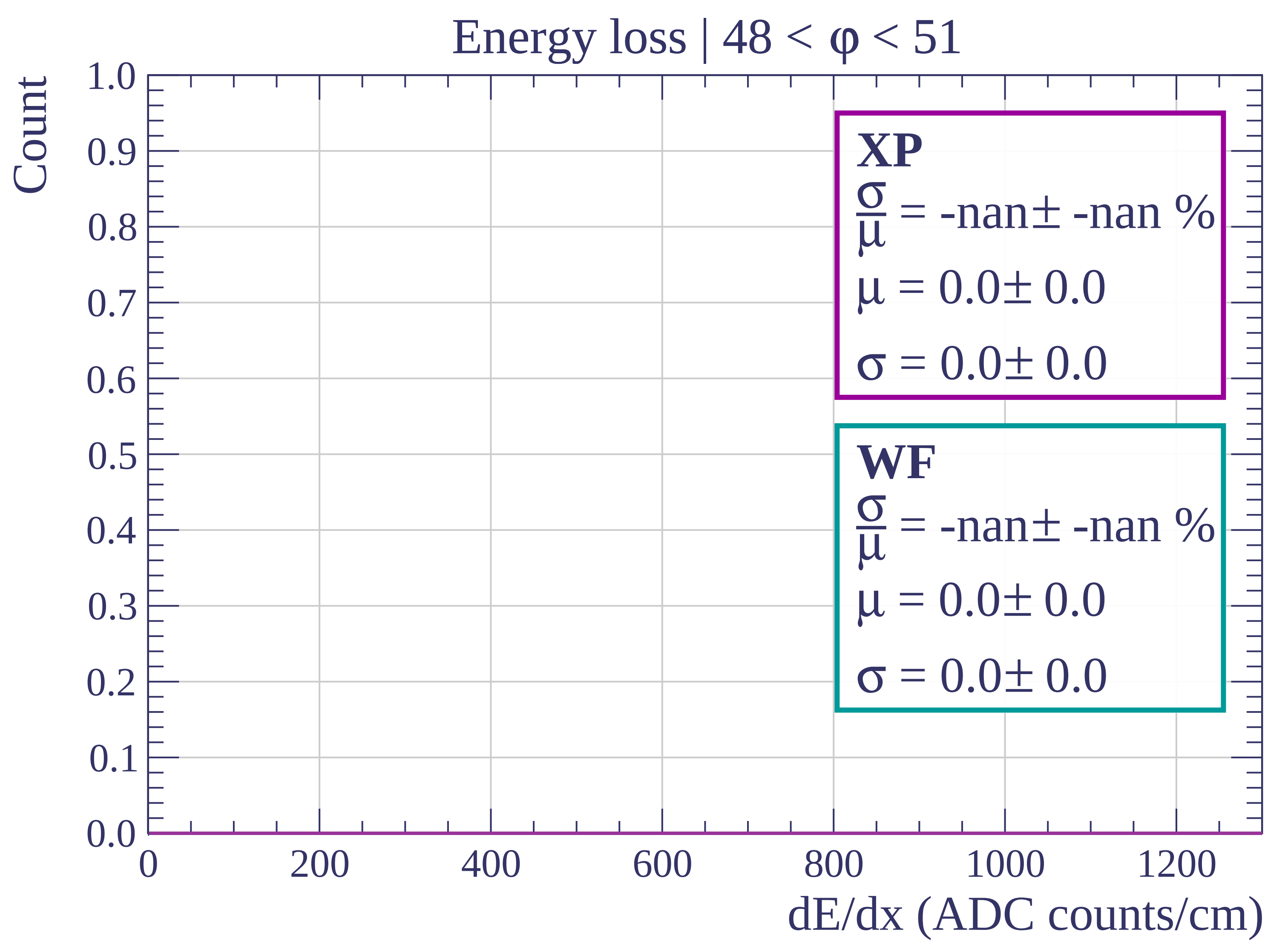
Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
σ/μ = 4.32± 0.74 %
μ = 386.5± 2.9
σ = 16.7± 2.7
WF
σ/μ = 8.30± 1.74 %
μ = 392.0± 7.6
σ = 32.5± 6.2



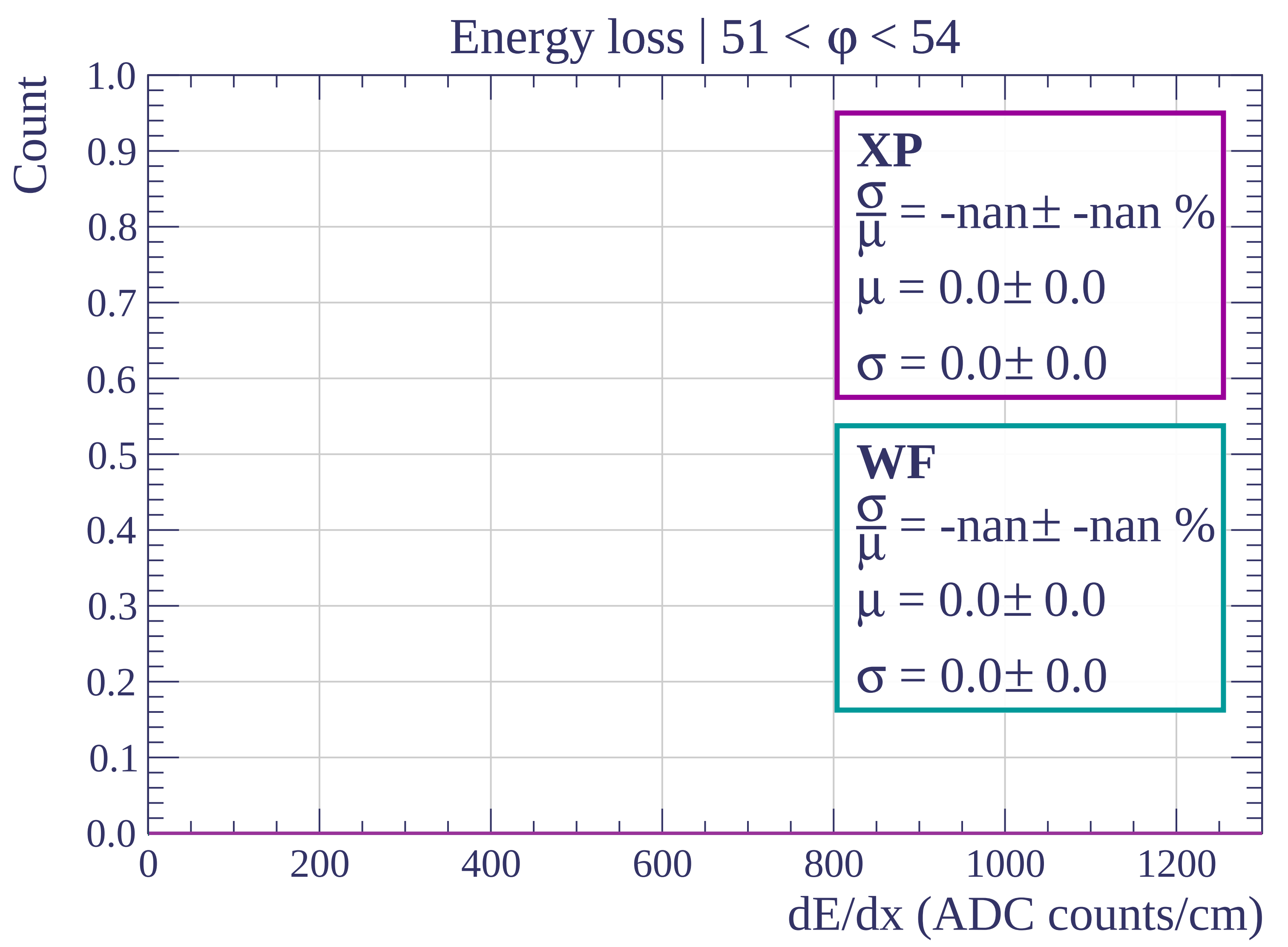
Energy loss | 42 < φ < 45
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 5.76± 0.81 %
μ = 404.7± 3.2
σ = 23.3± 3.1
WF
σ/μ = 7.14± 1.13 %
μ = 403.6± 3.9
σ = 28.8± 4.3



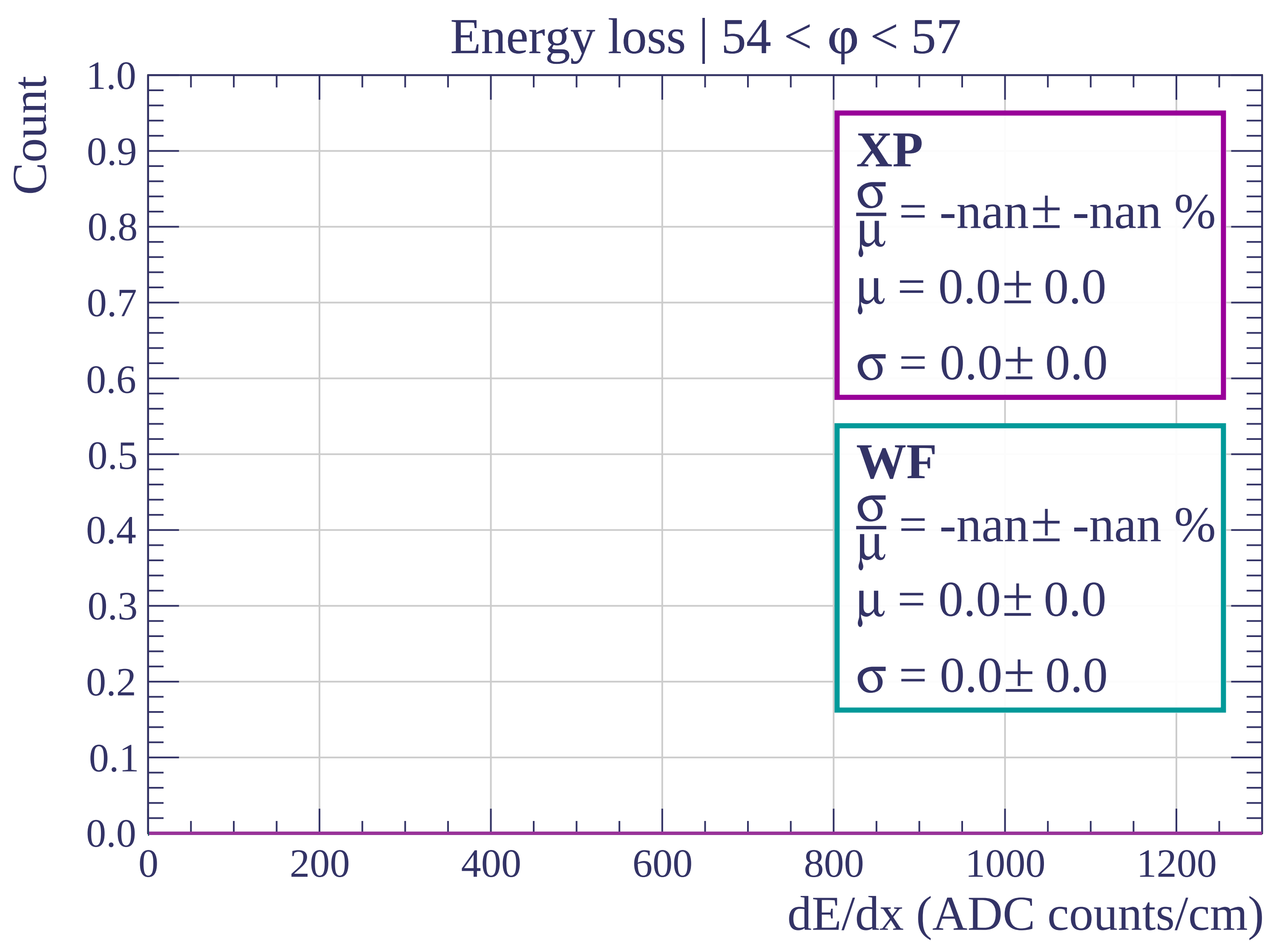
Energy loss | 45 < φ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 5.40± 1.69 %
μ = 410.2± 6.4
σ = 22.1± 6.6
WF
σ/μ = 2.73± 1.25 %
μ = 398.3± 4.2
σ = 10.9± 4.8



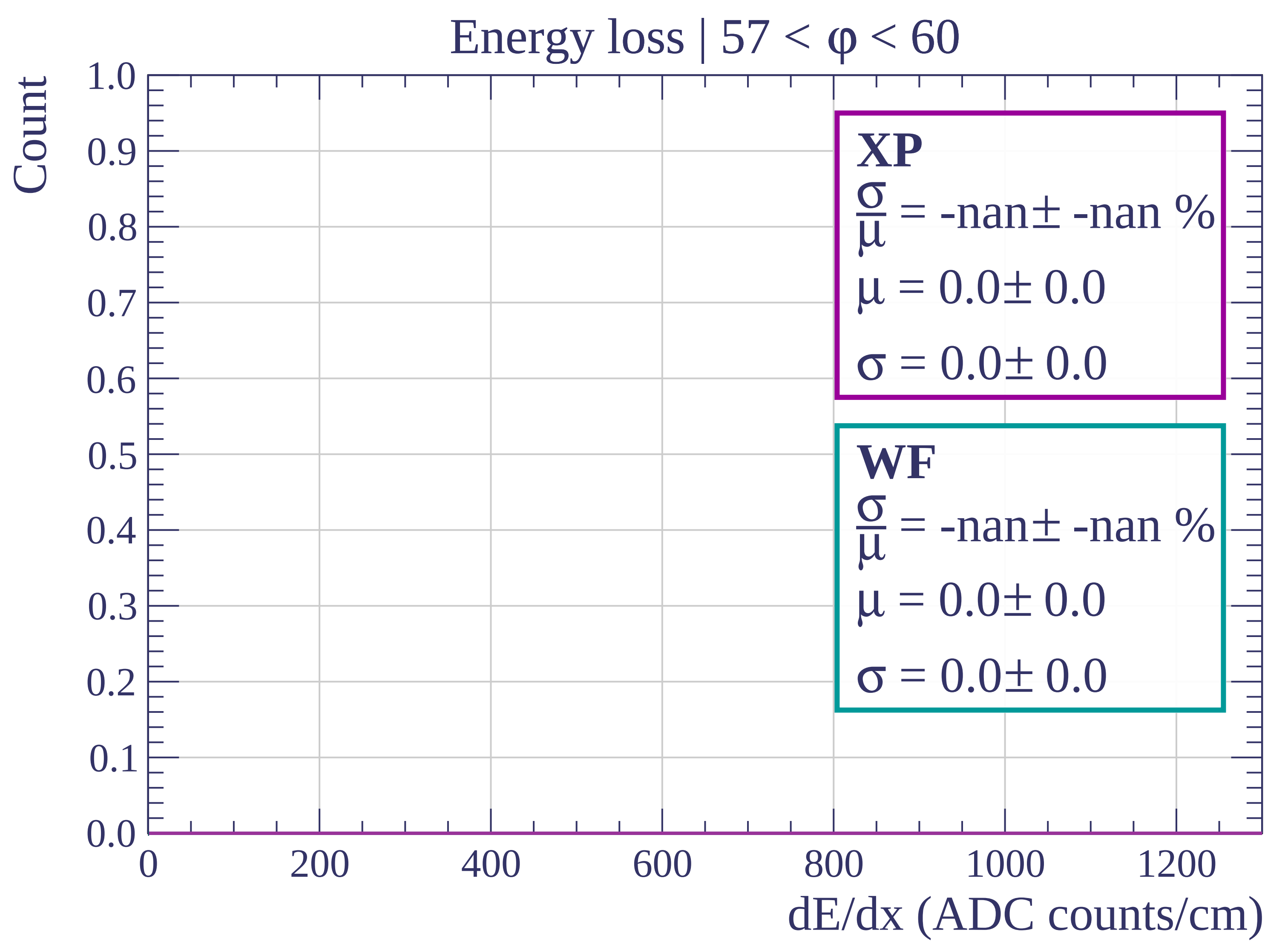
Energy loss | 48 < φ < 51
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



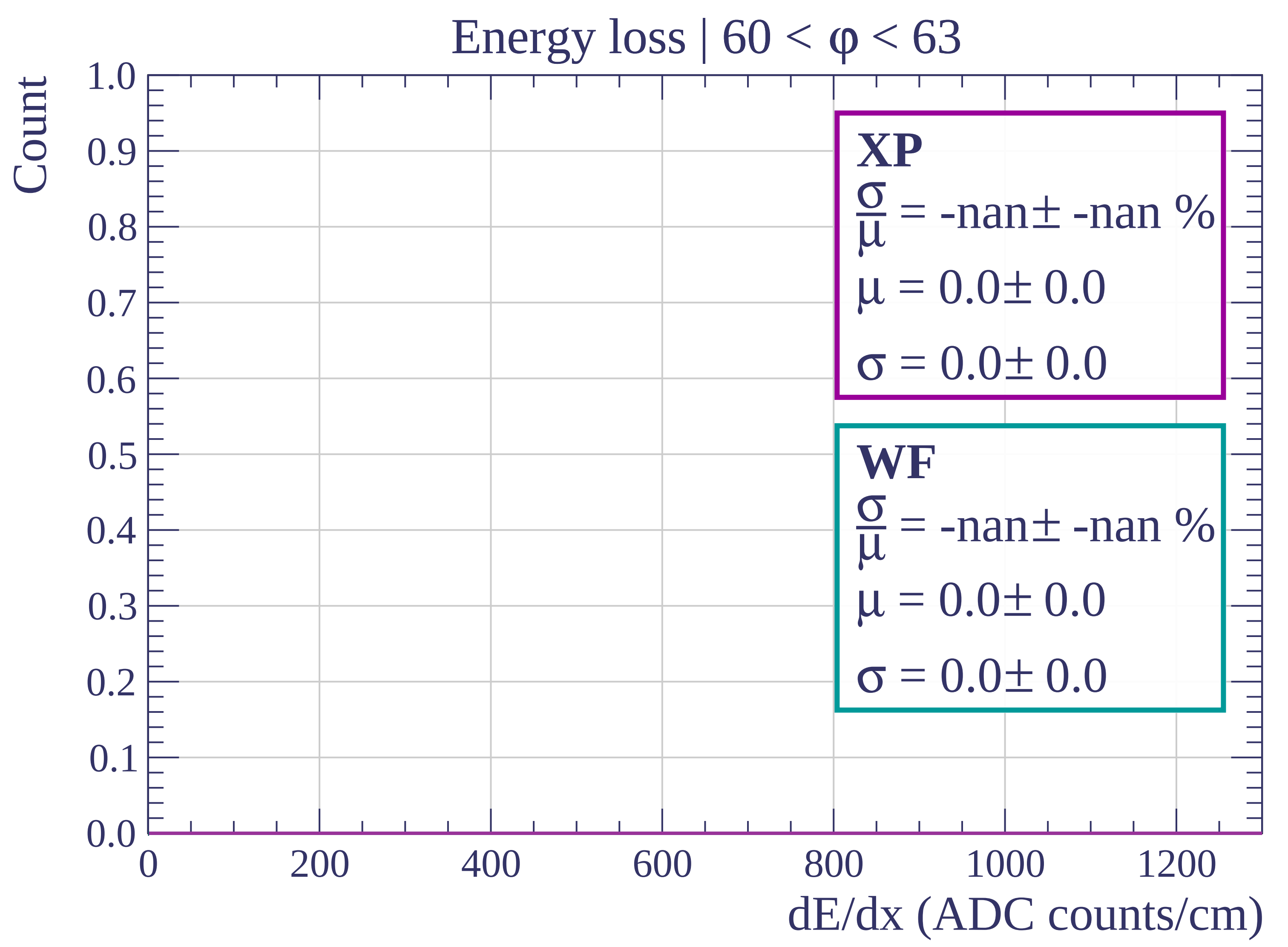
Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



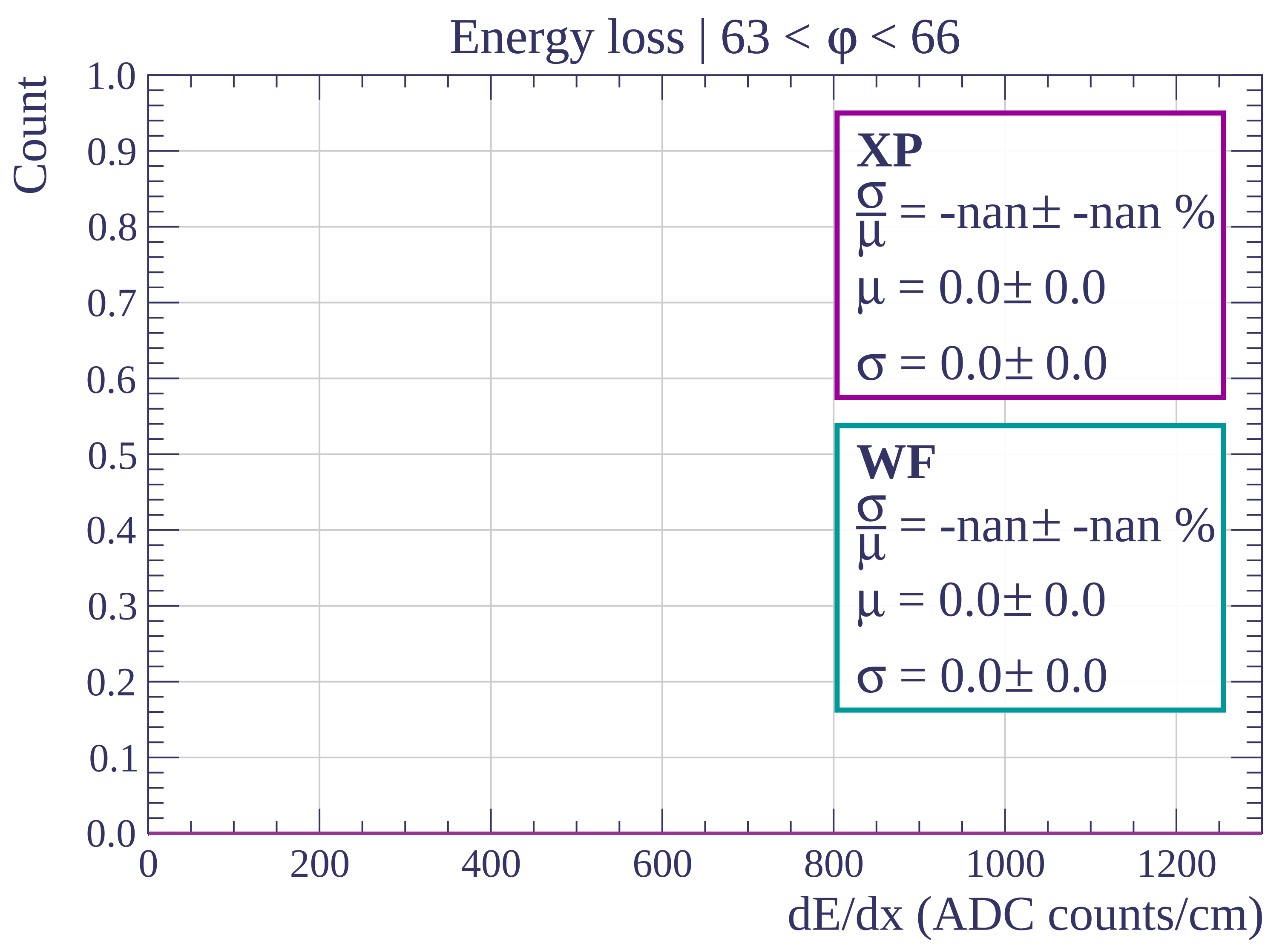
Energy loss | 54 < φ < 57
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



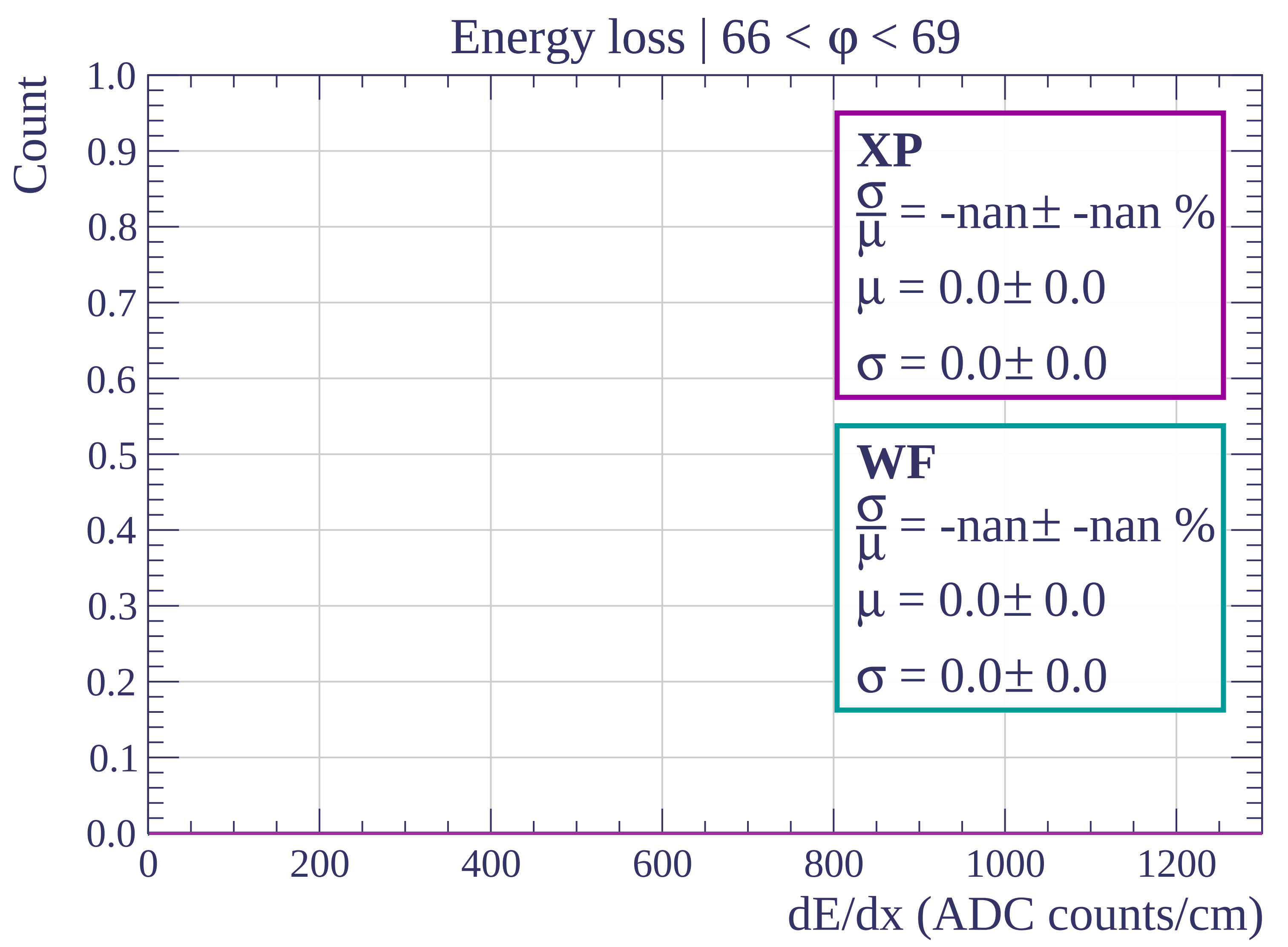
Energy loss | 57 < φ < 60
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



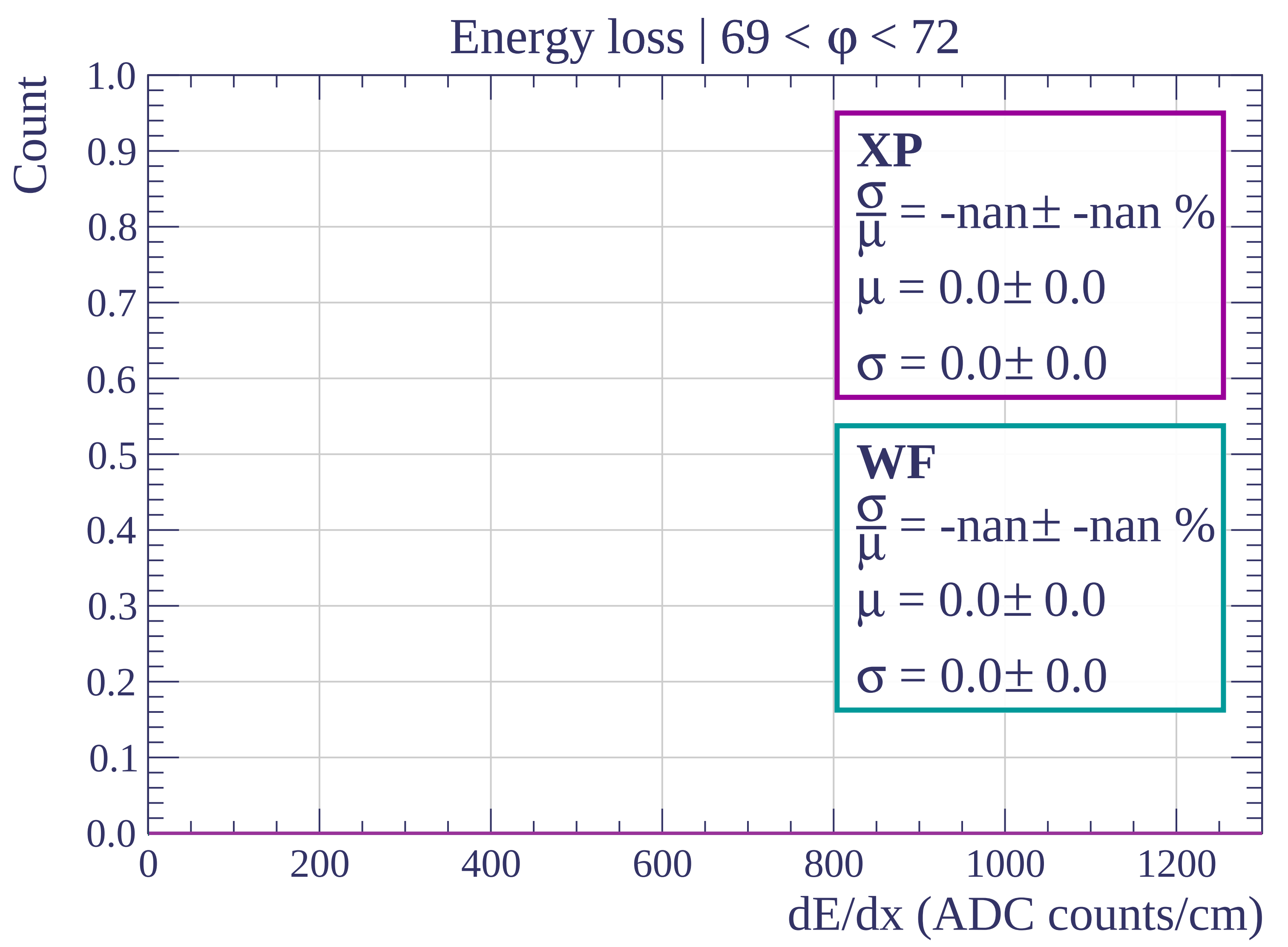
Energy loss | 60 < φ < 63
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



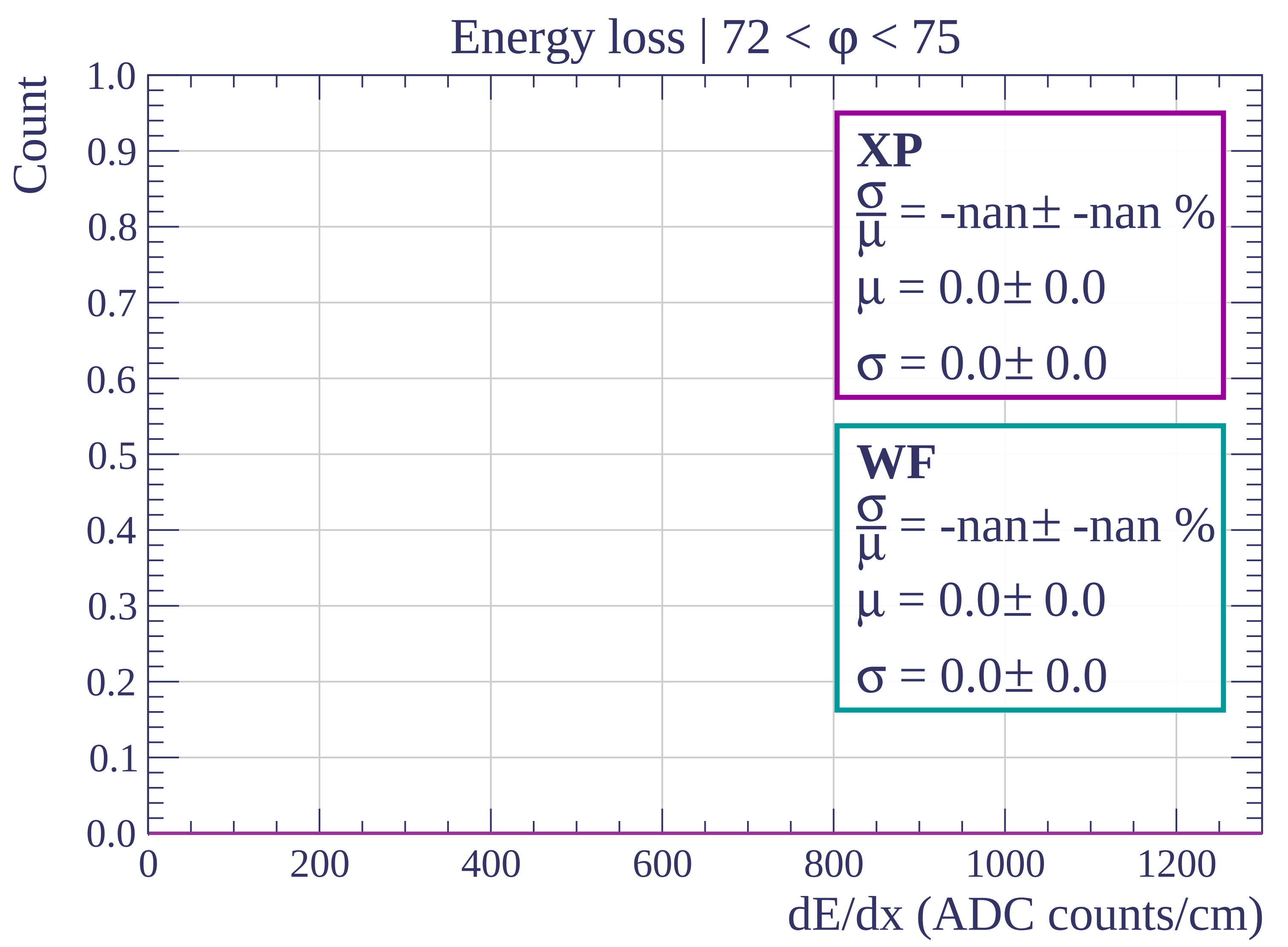
Energy loss | 63 < φ < 66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

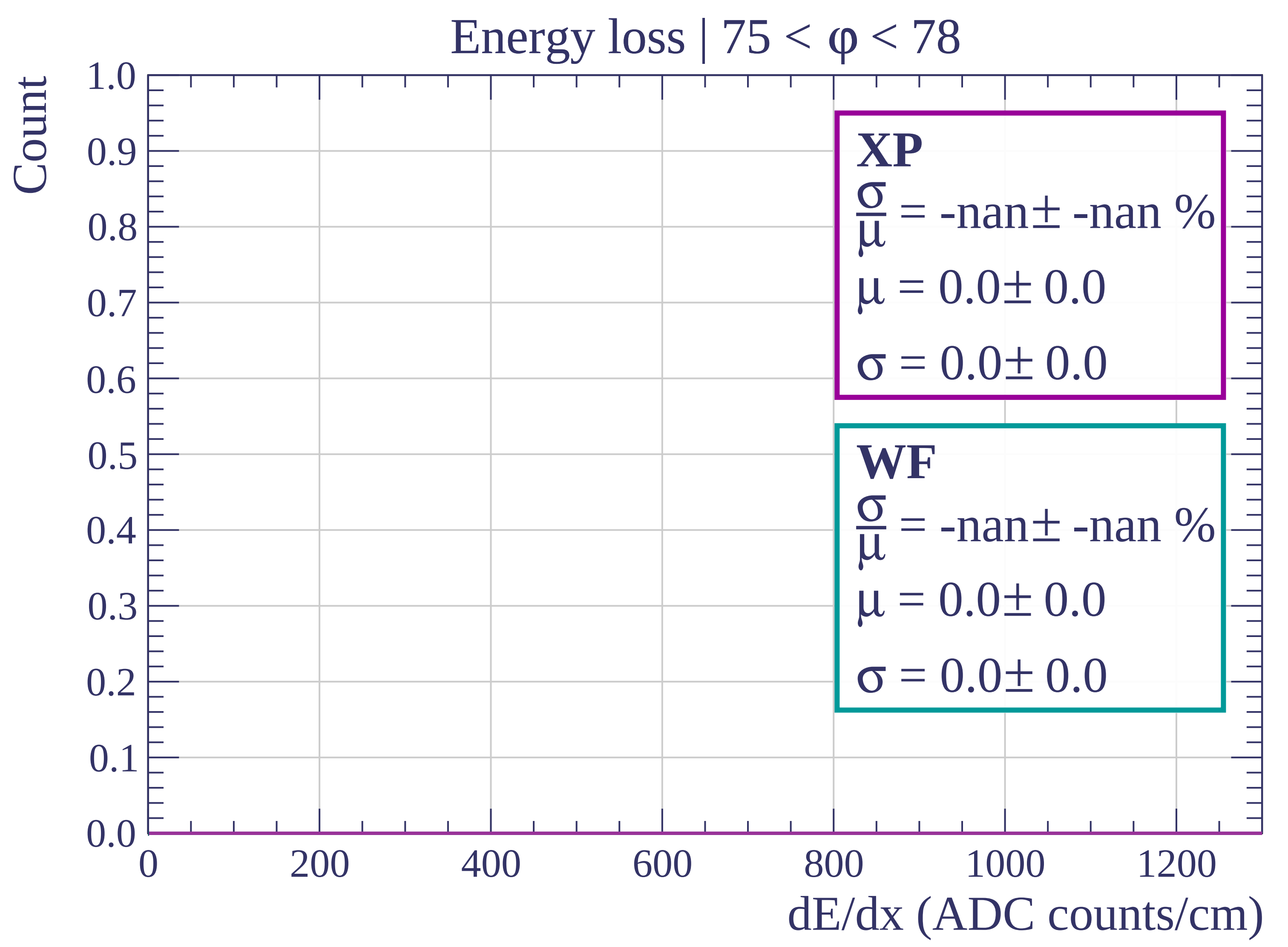


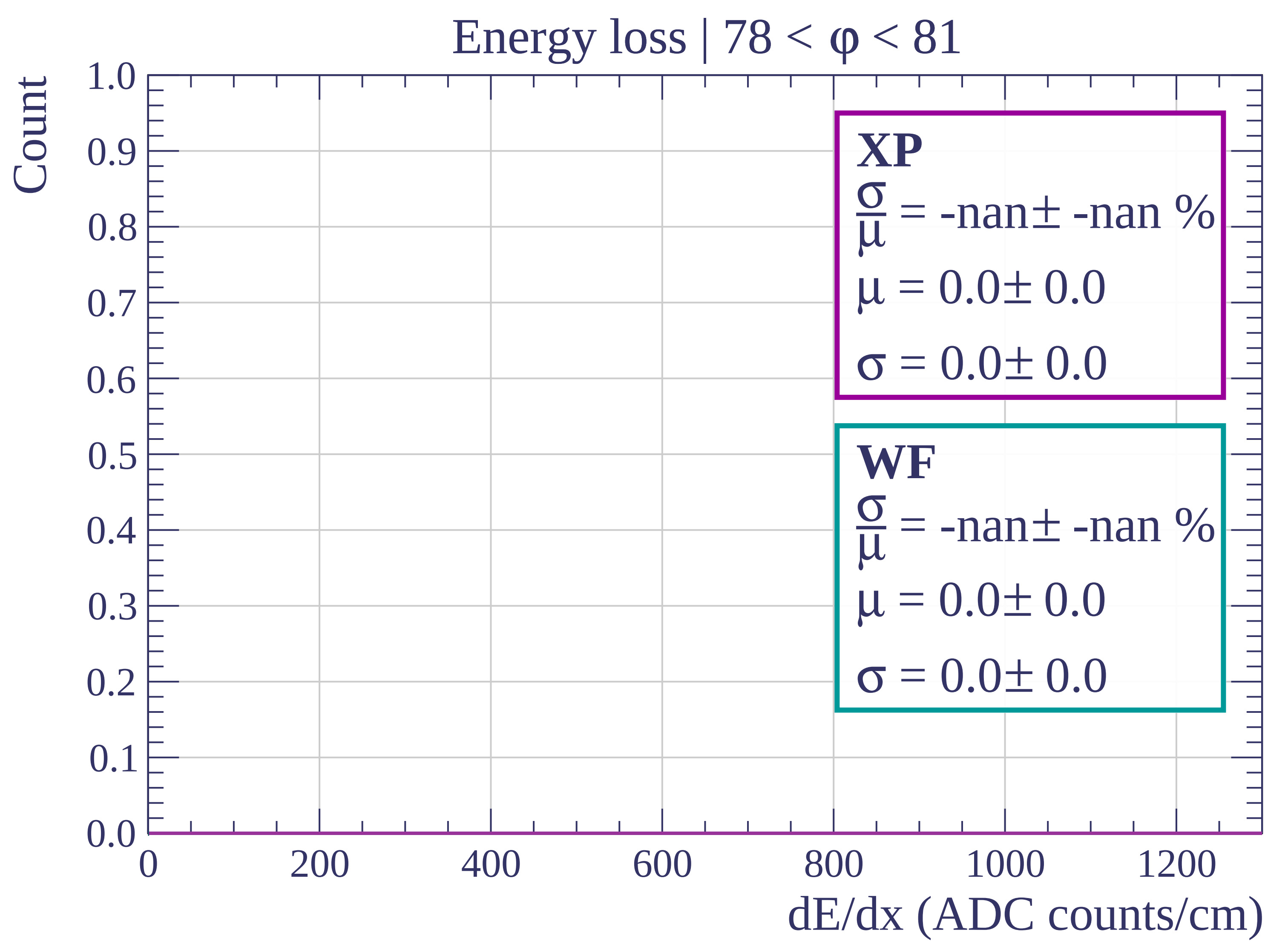
Energy loss | 66 < φ < 69
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



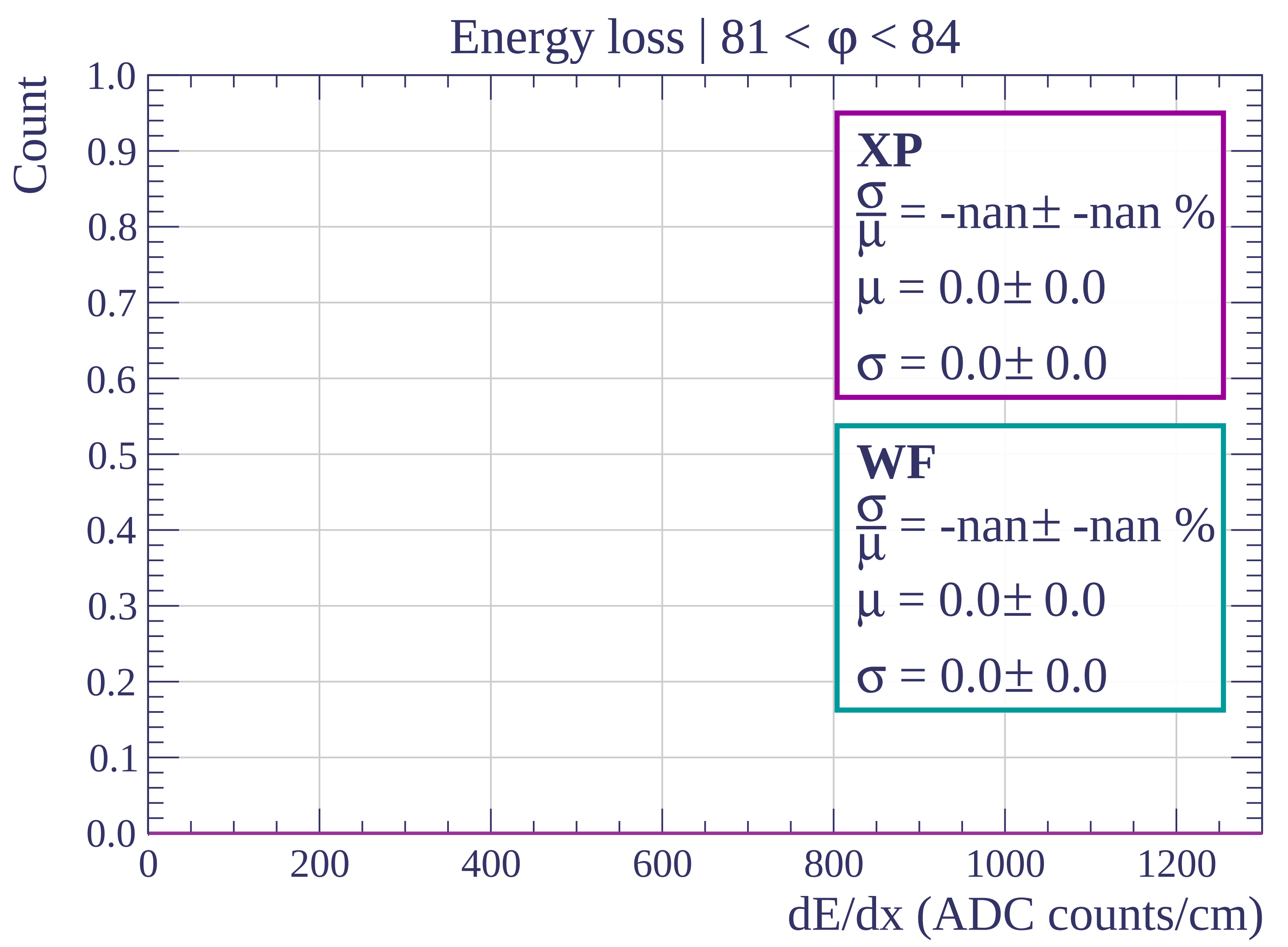
Energy loss | 69 < φ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 72 < φ < 75
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

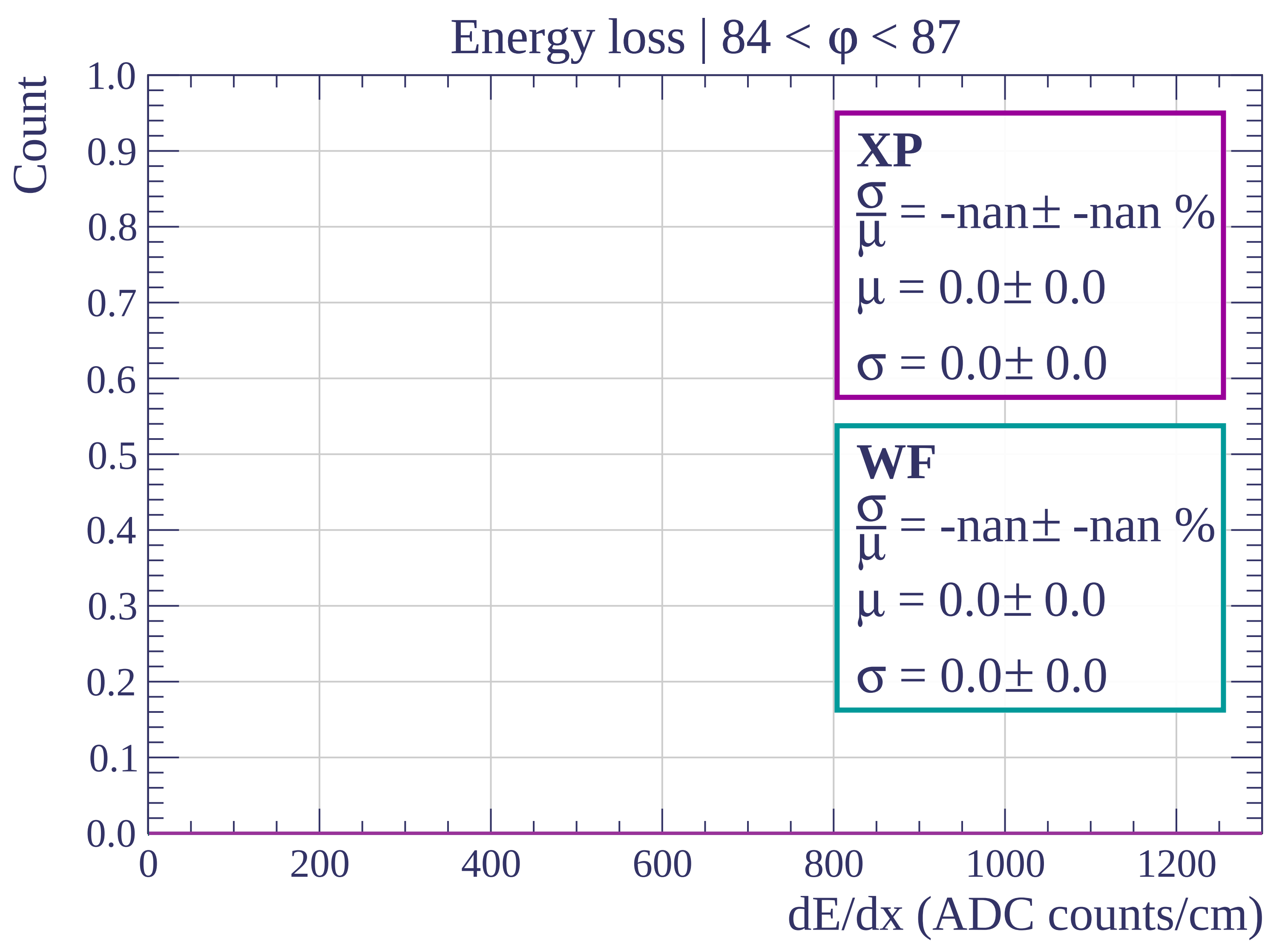

Energy loss | 75 < φ < 78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

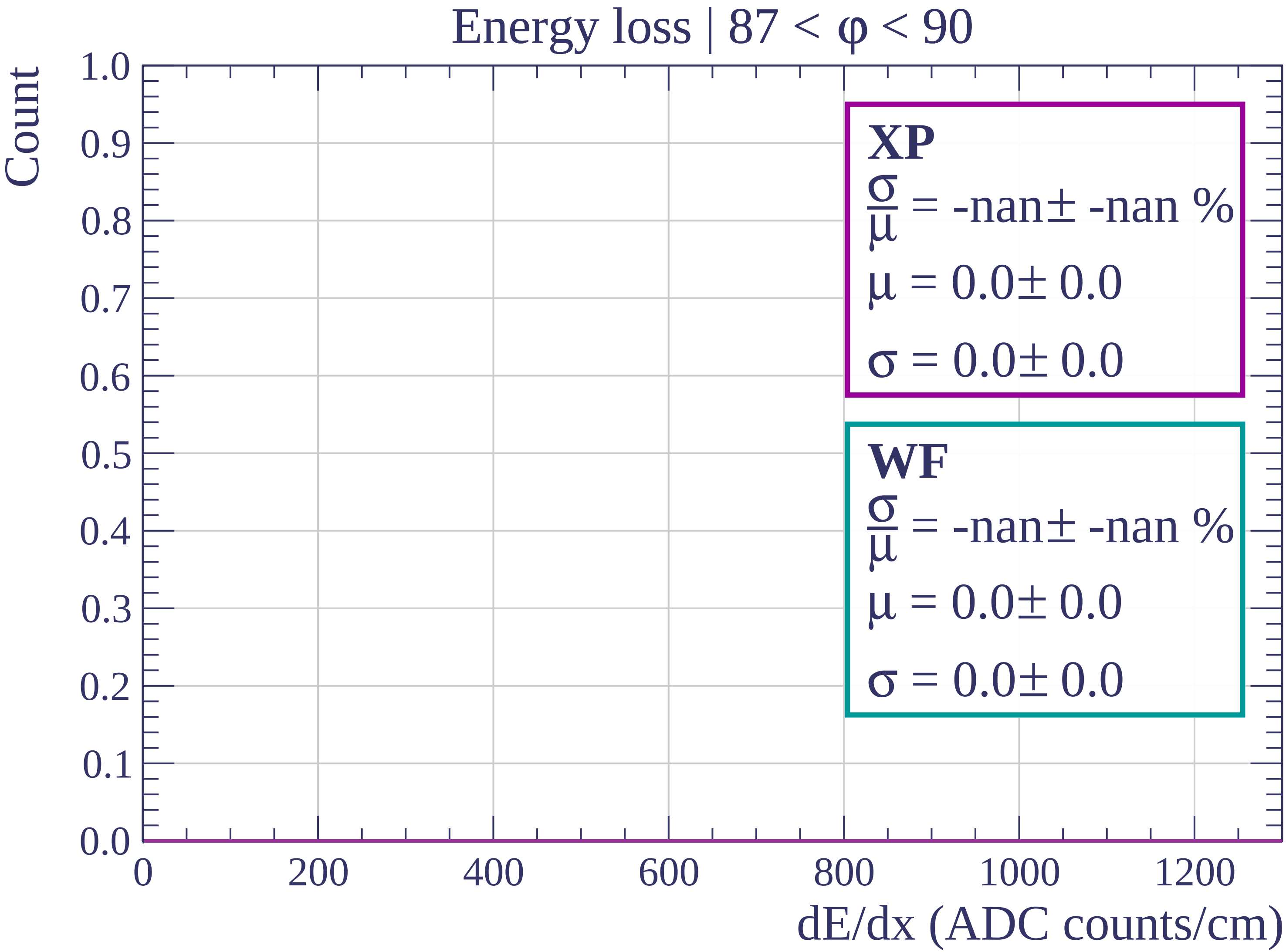


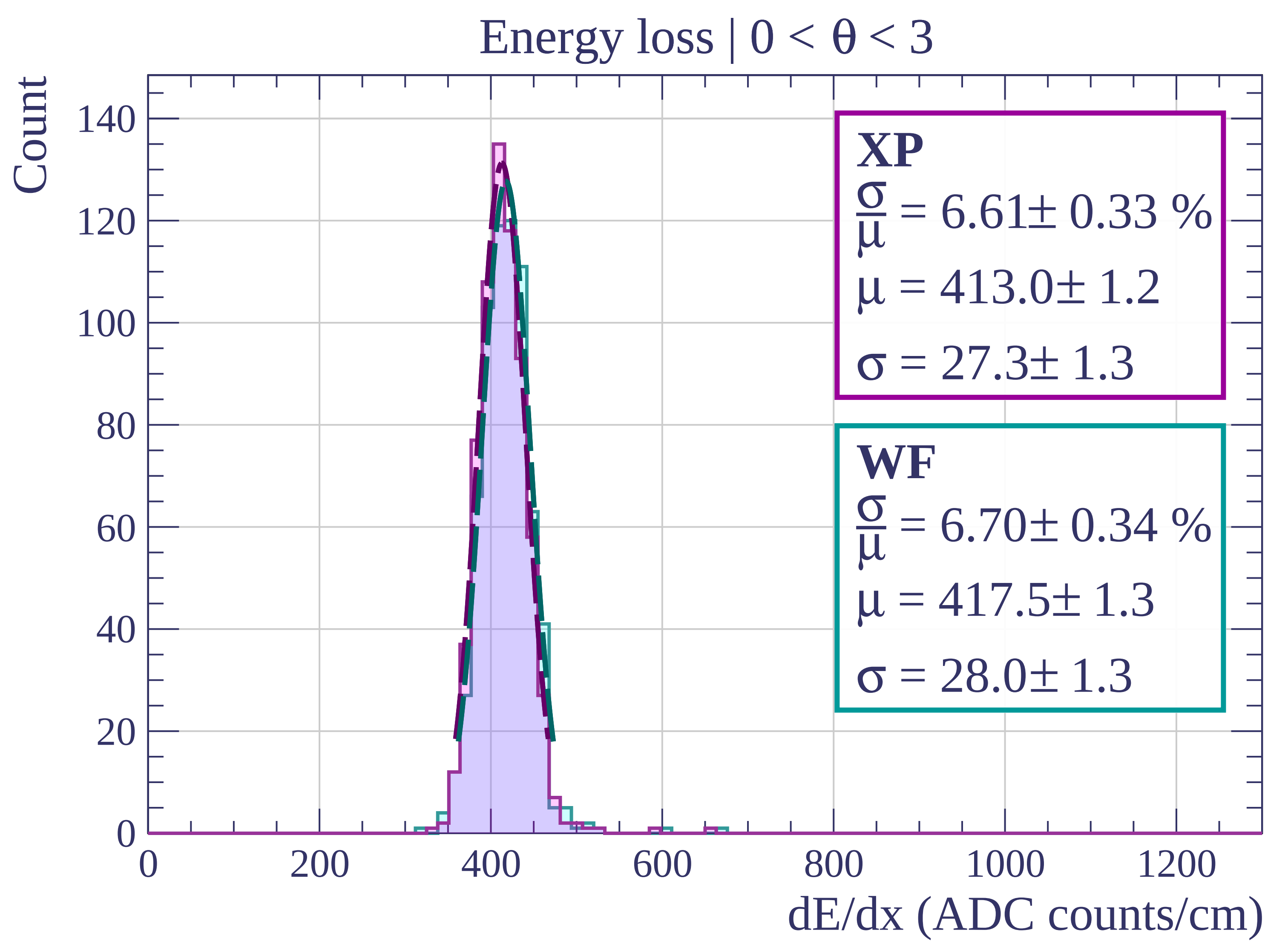
Energy loss | 78 < φ < 81
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



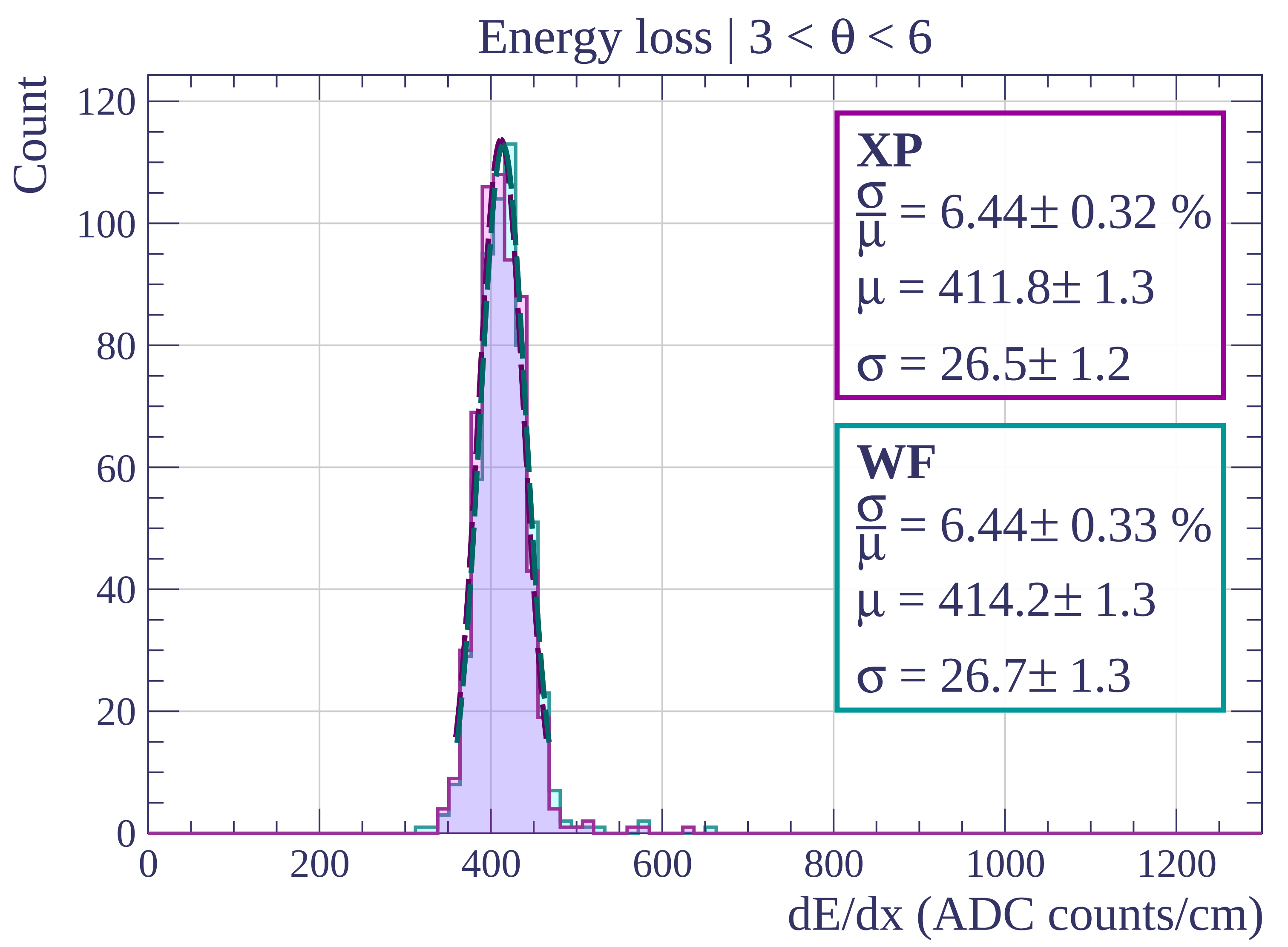
Energy loss | 81 < φ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 84 < φ < 87
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

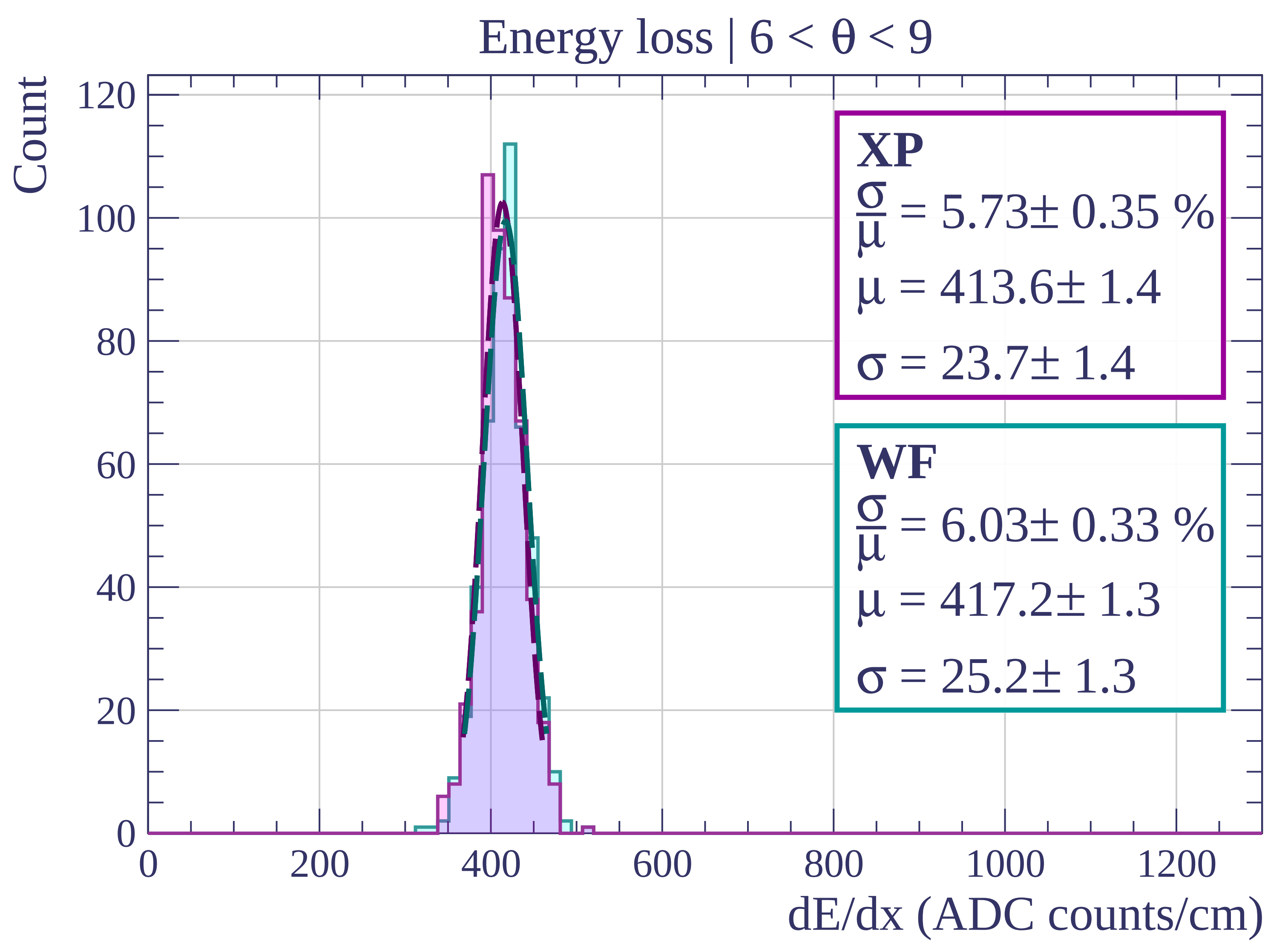

Energy loss | 87 < φ < 90
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



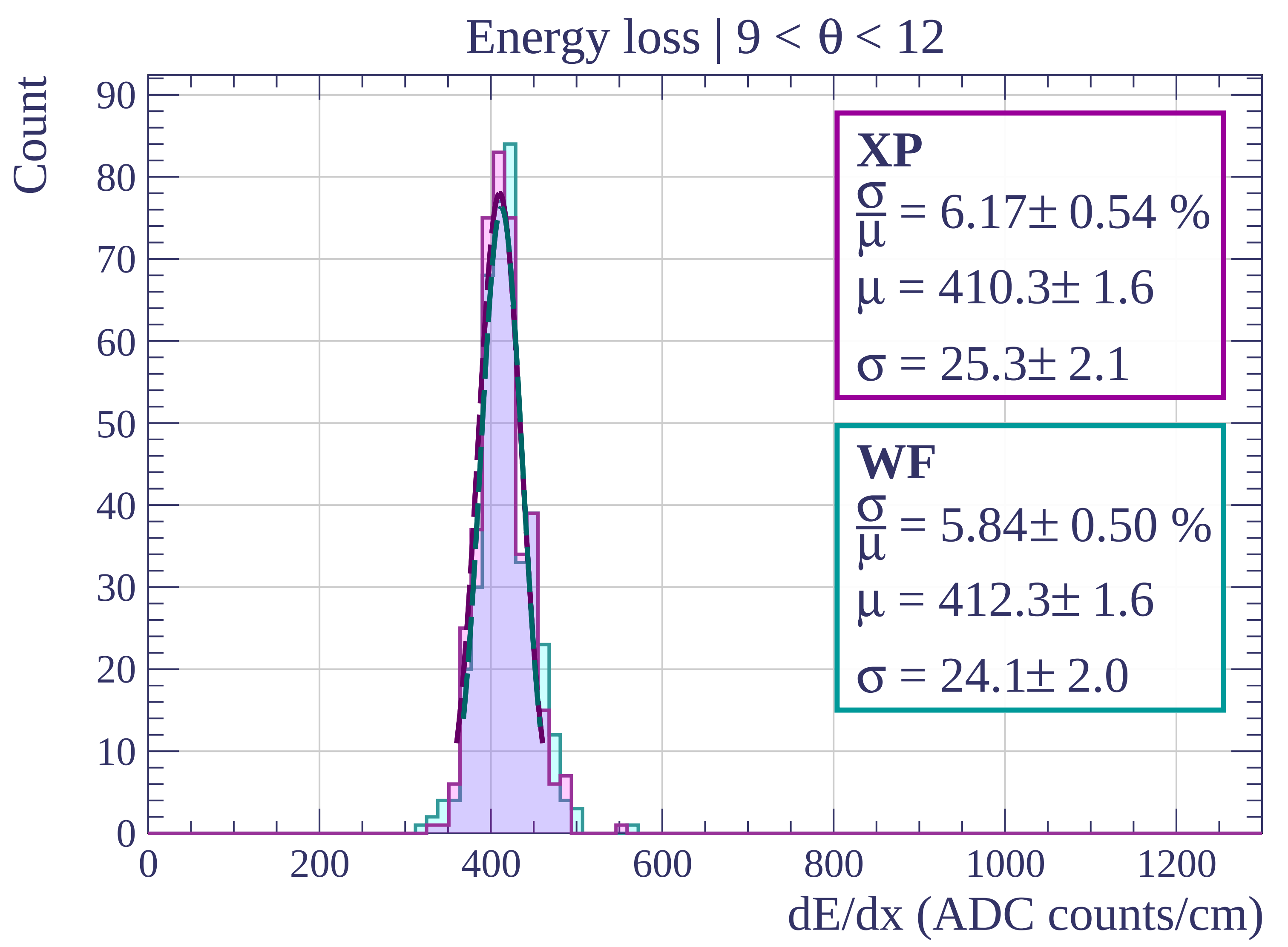
Energy loss | 0 < θ < 3
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 6.61± 0.33 %
μ = 413.0± 1.2
σ = 27.3± 1.3
WF
σ/μ = 6.70± 0.34 %
μ = 417.5± 1.3
σ = 28.0± 1.3



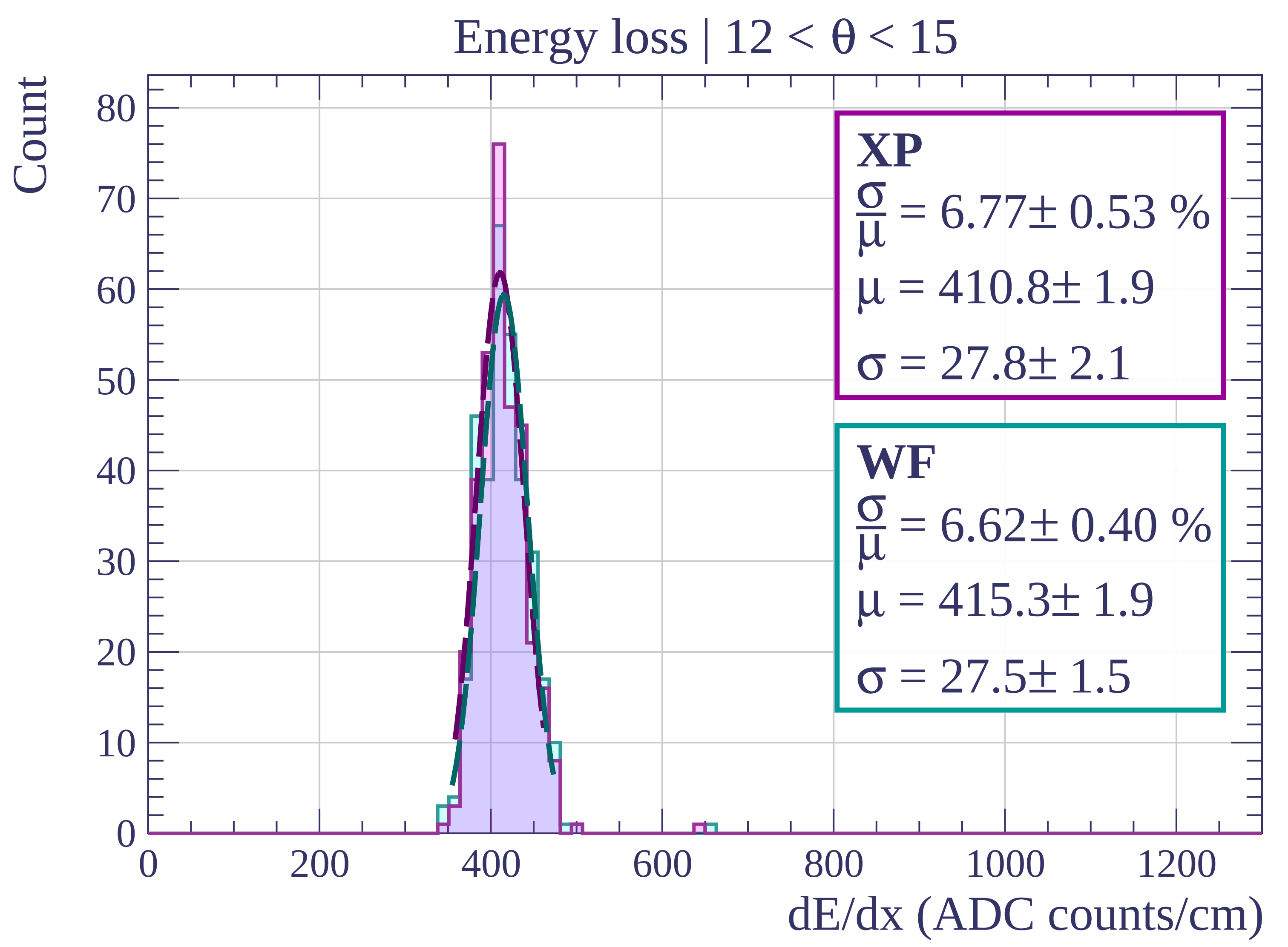
Energy loss | 3 < θ < 6
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 6.44± 0.32 %
μ = 411.8± 1.3
σ = 26.5± 1.2
WF
σ/μ = 6.44± 0.33 %
μ = 414.2± 1.3
σ = 26.7± 1.3



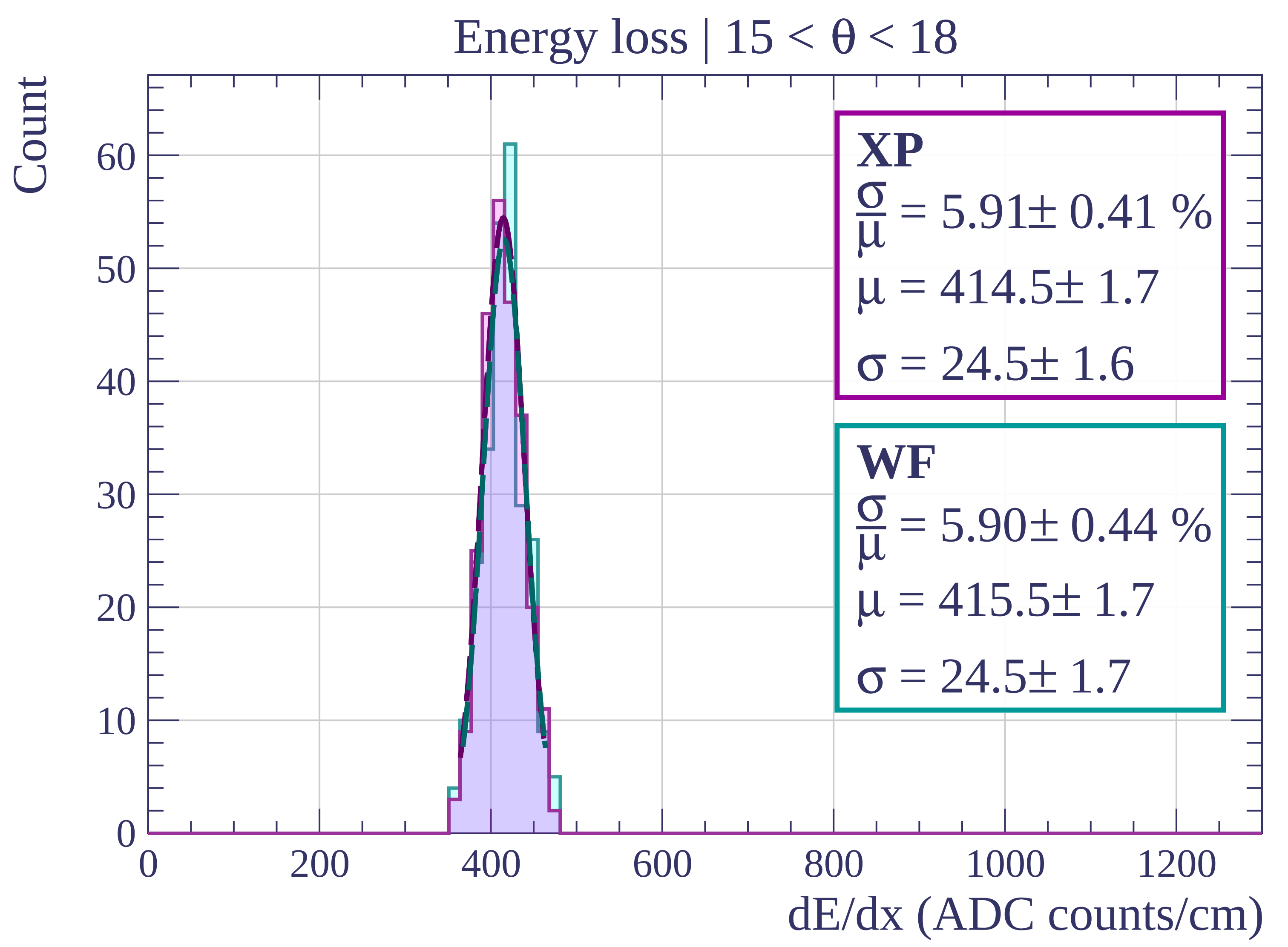
Energy loss | 6 < θ < 9
Count
XP
σ/μ = 5.73± 0.35 %
μ = 413.6± 1.4
σ = 23.7± 1.4
WF
σ/μ = 6.03± 0.33 %
μ = 417.2± 1.3
σ = 25.2± 1.3
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



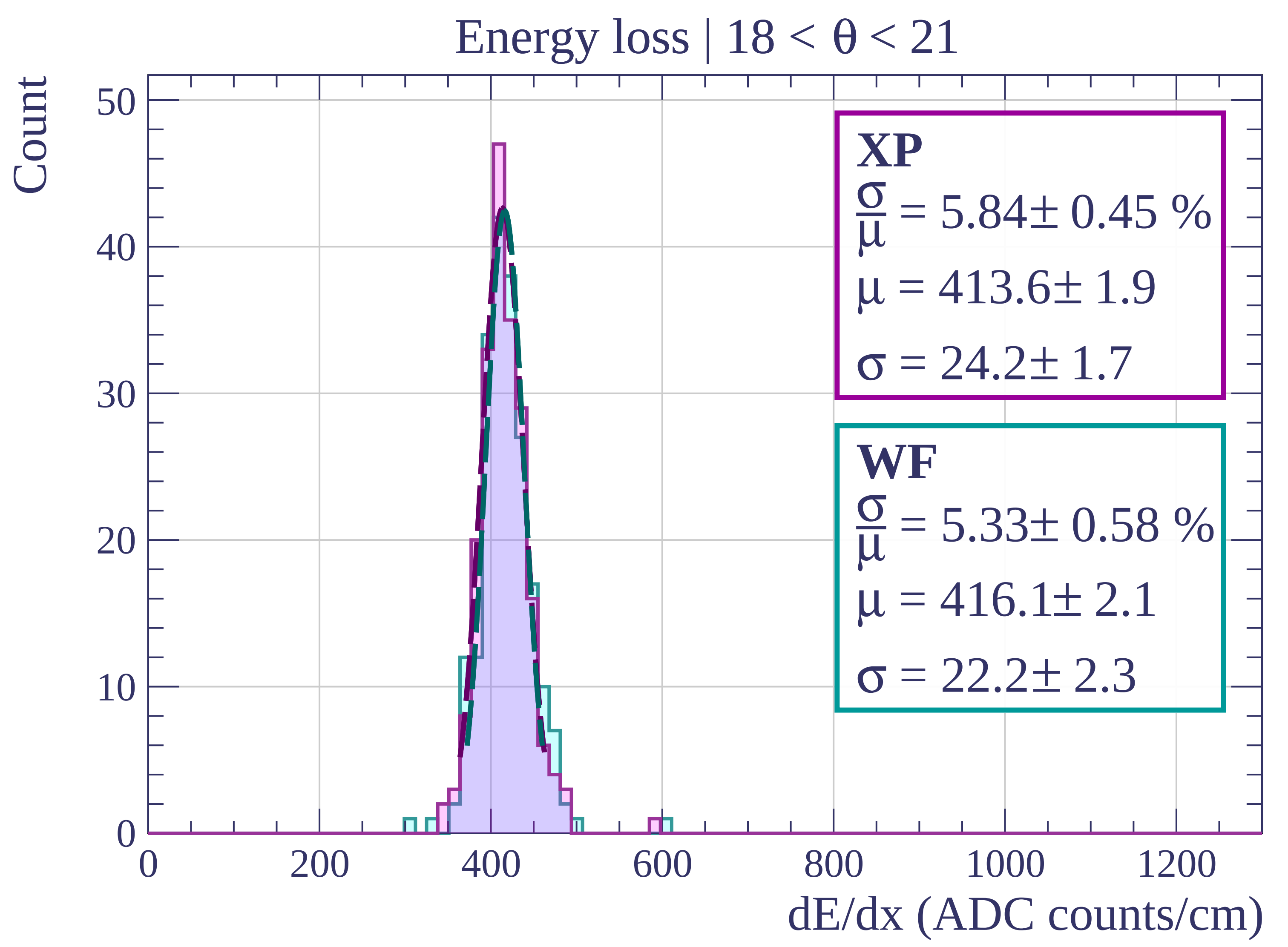
Energy loss | 9 < θ < 12
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 6.17± 0.54 %
μ = 410.3± 1.6
σ = 25.3± 2.1
WF
σ/μ = 5.84± 0.50 %
μ = 412.3± 1.6
σ = 24.1± 2.0



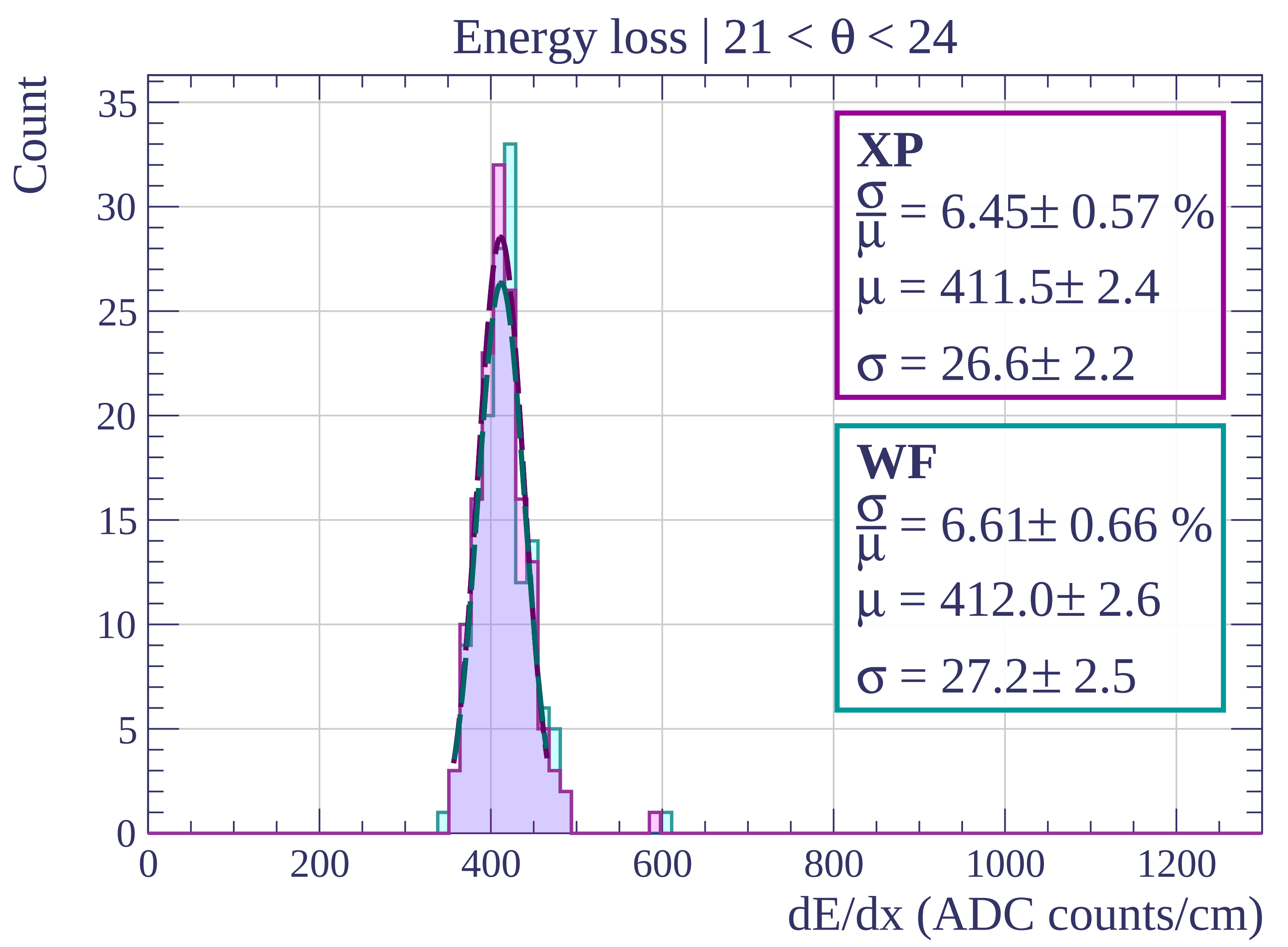
Energy loss | 12 < θ < 15
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.77± 0.53 %
μ = 410.8± 1.9
σ = 27.8± 2.1
WF
σ/μ = 6.62± 0.40 %
μ = 415.3± 1.9
σ = 27.5± 1.5
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200



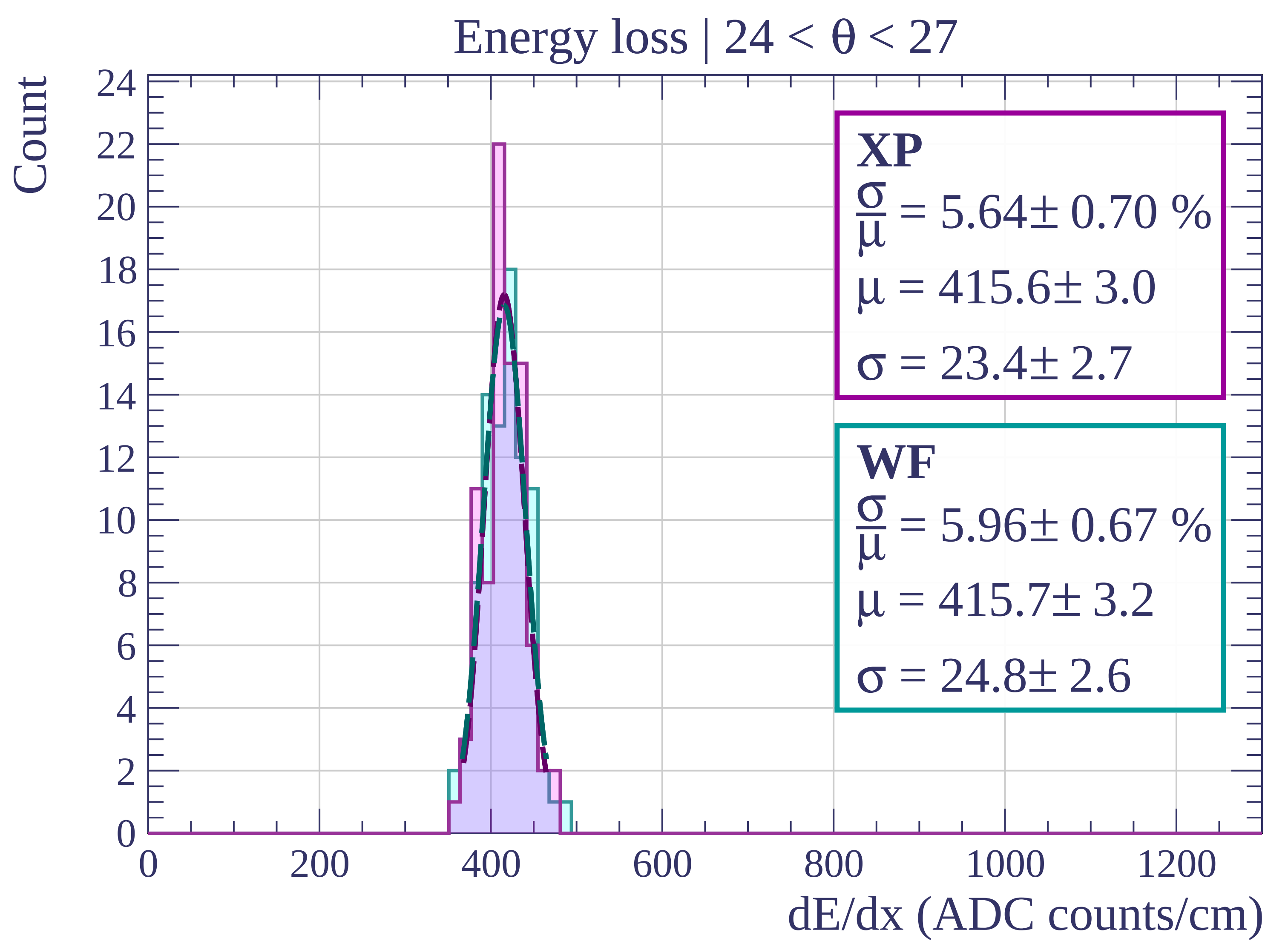
Energy loss | 15 < θ < 18
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.91± 0.41 %
μ = 414.5± 1.7
σ = 24.5± 1.6
WF
$\frac{\sigma}{\mu}$ = 5.90± 0.44 %
μ = 415.5± 1.7
σ = 24.5± 1.7



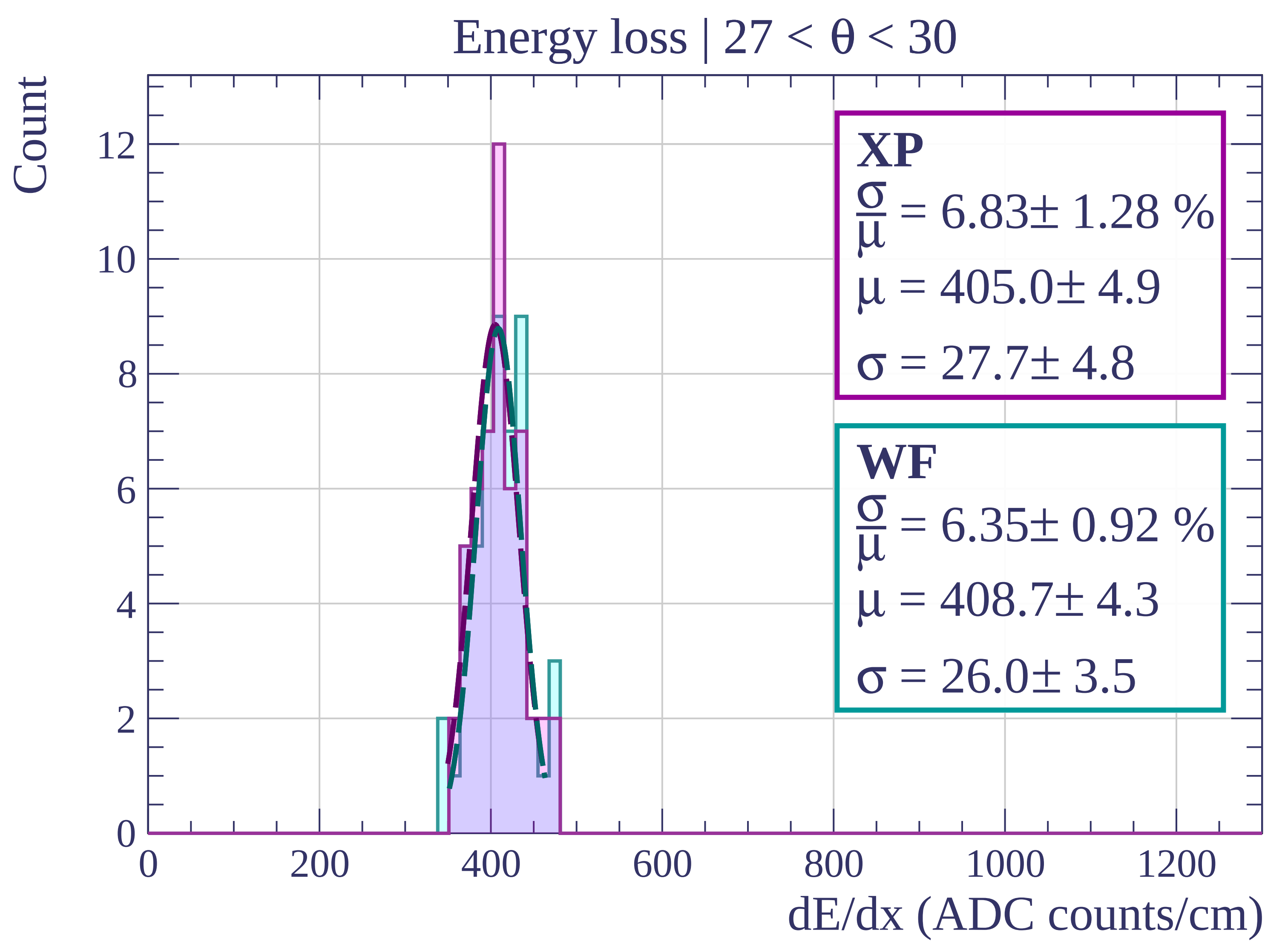
Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 5.84± 0.45 %
μ = 413.6± 1.9
σ = 24.2± 1.7
WF
σ/μ = 5.33± 0.58 %
μ = 416.1± 2.1
σ = 22.2± 2.3



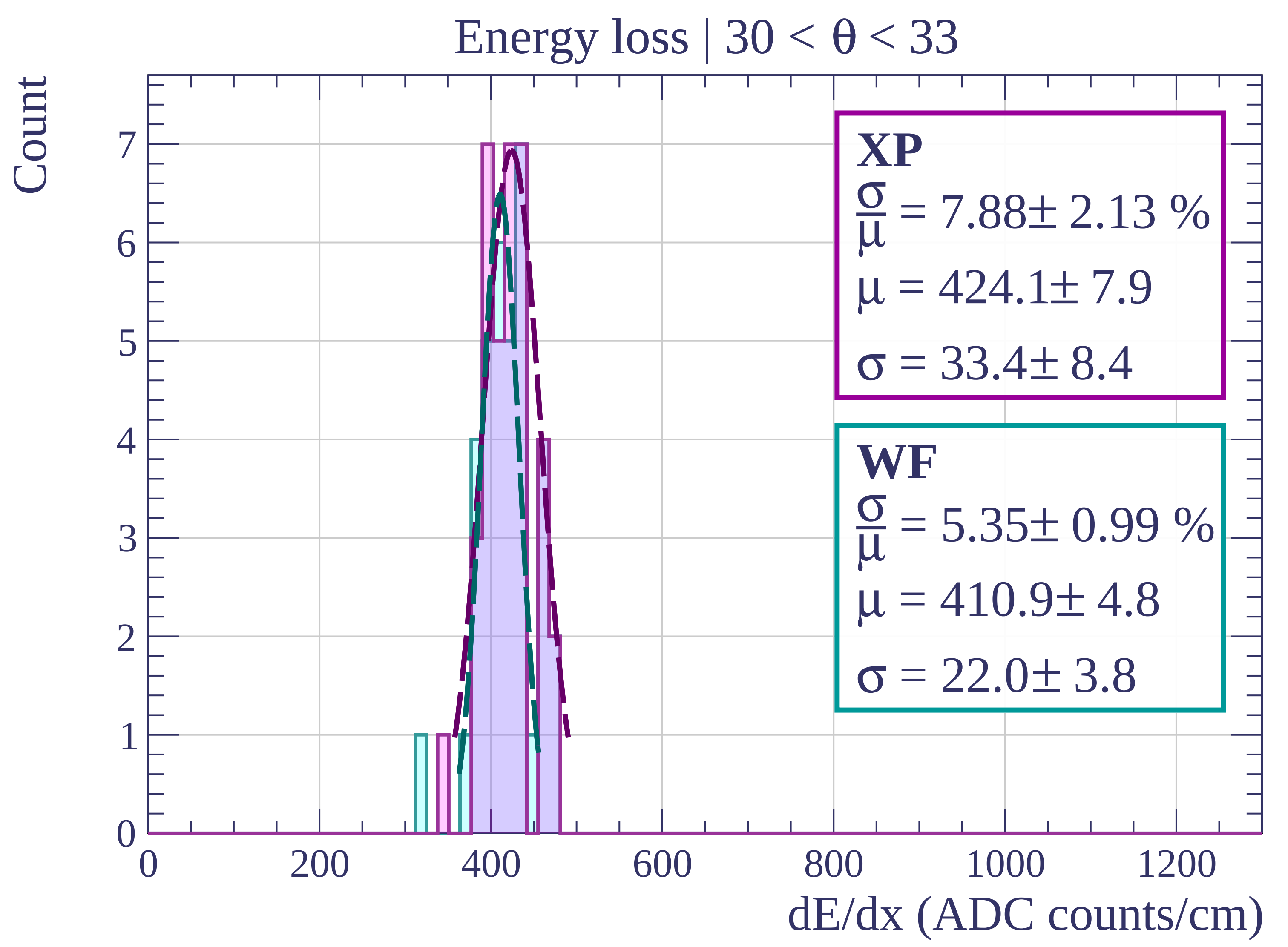
Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP
σ/μ = 6.45± 0.57 %
μ = 411.5± 2.4
σ = 26.6± 2.2
WF
σ/μ = 6.61± 0.66 %
μ = 412.0± 2.6
σ = 27.2± 2.5



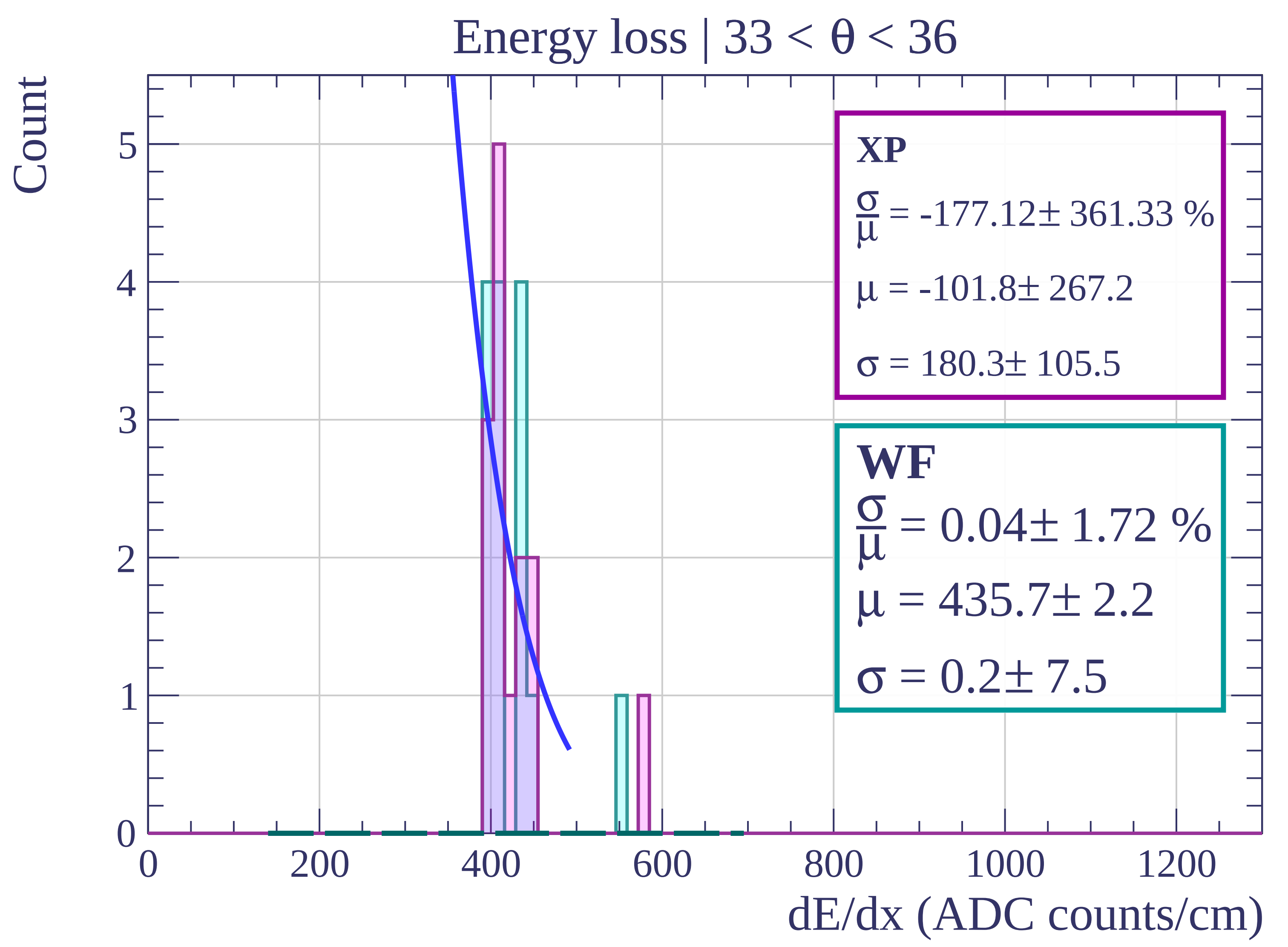
Energy loss | 24 < θ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
24
XP
σ/μ = 5.64± 0.70 %
μ = 415.6± 3.0
σ = 23.4± 2.7
WF
σ/μ = 5.96± 0.67 %
μ = 415.7± 3.2
σ = 24.8± 2.6



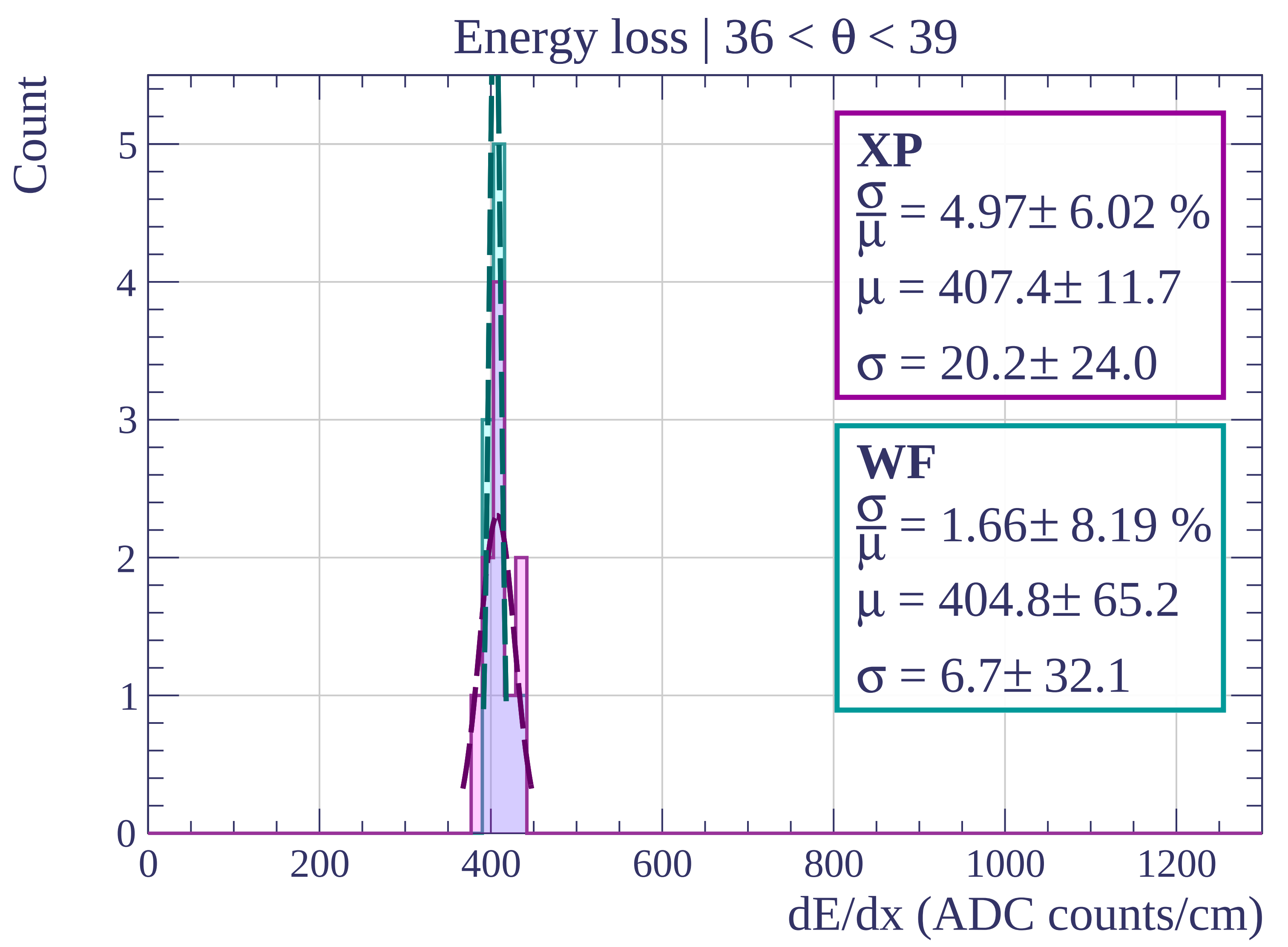
Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.83± 1.28 %
μ = 405.0± 4.9
σ = 27.7± 4.8
WF
$\frac{\sigma}{\mu}$ = 6.35± 0.92 %
μ = 408.7± 4.3
σ = 26.0± 3.5



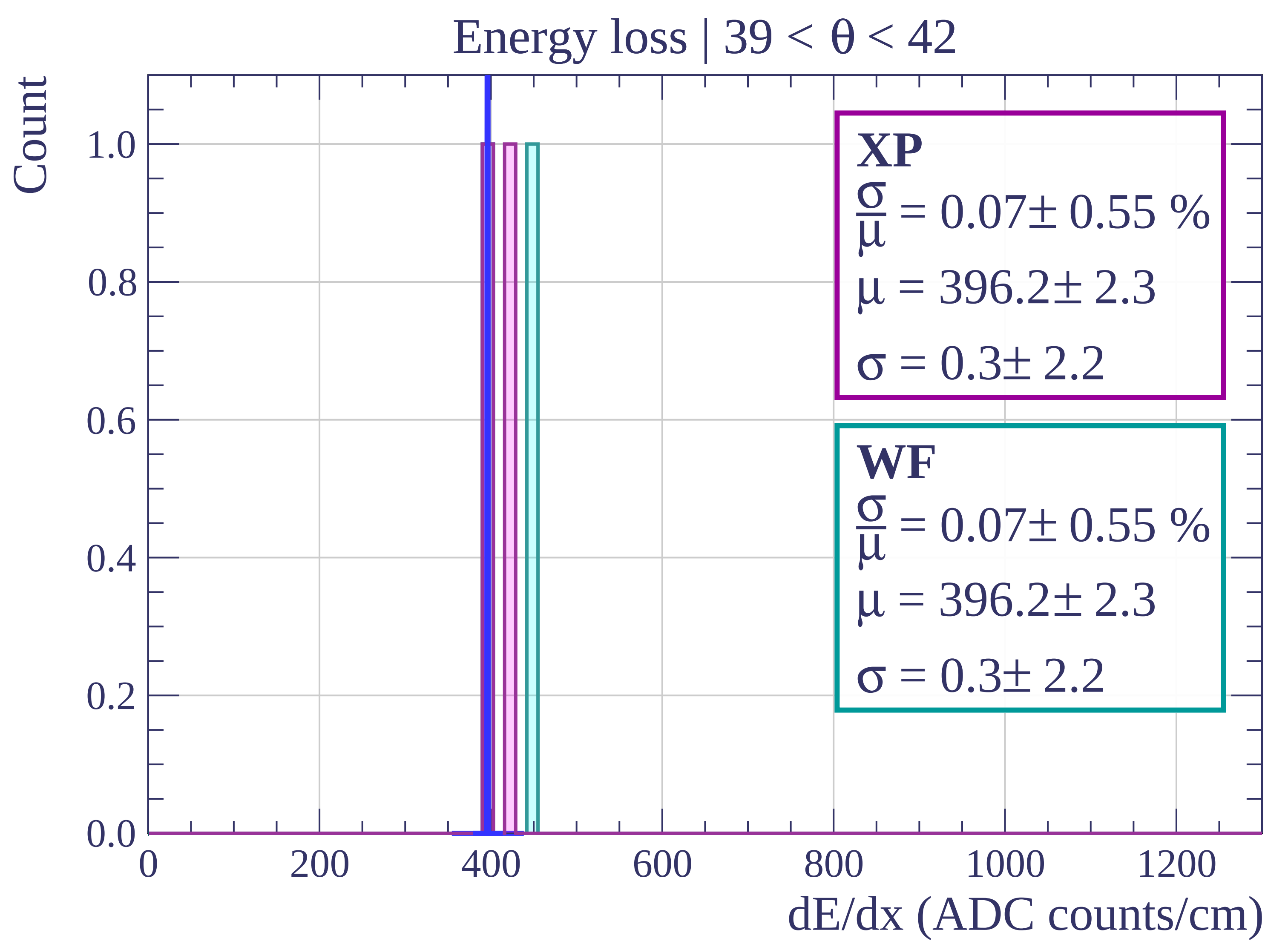
Energy loss | 30 < θ < 33
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.88± 2.13 %
μ = 424.1± 7.9
σ = 33.4± 8.4
WF
σ/μ = 5.35± 0.99 %
μ = 410.9± 4.8
σ = 22.0± 3.8
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200



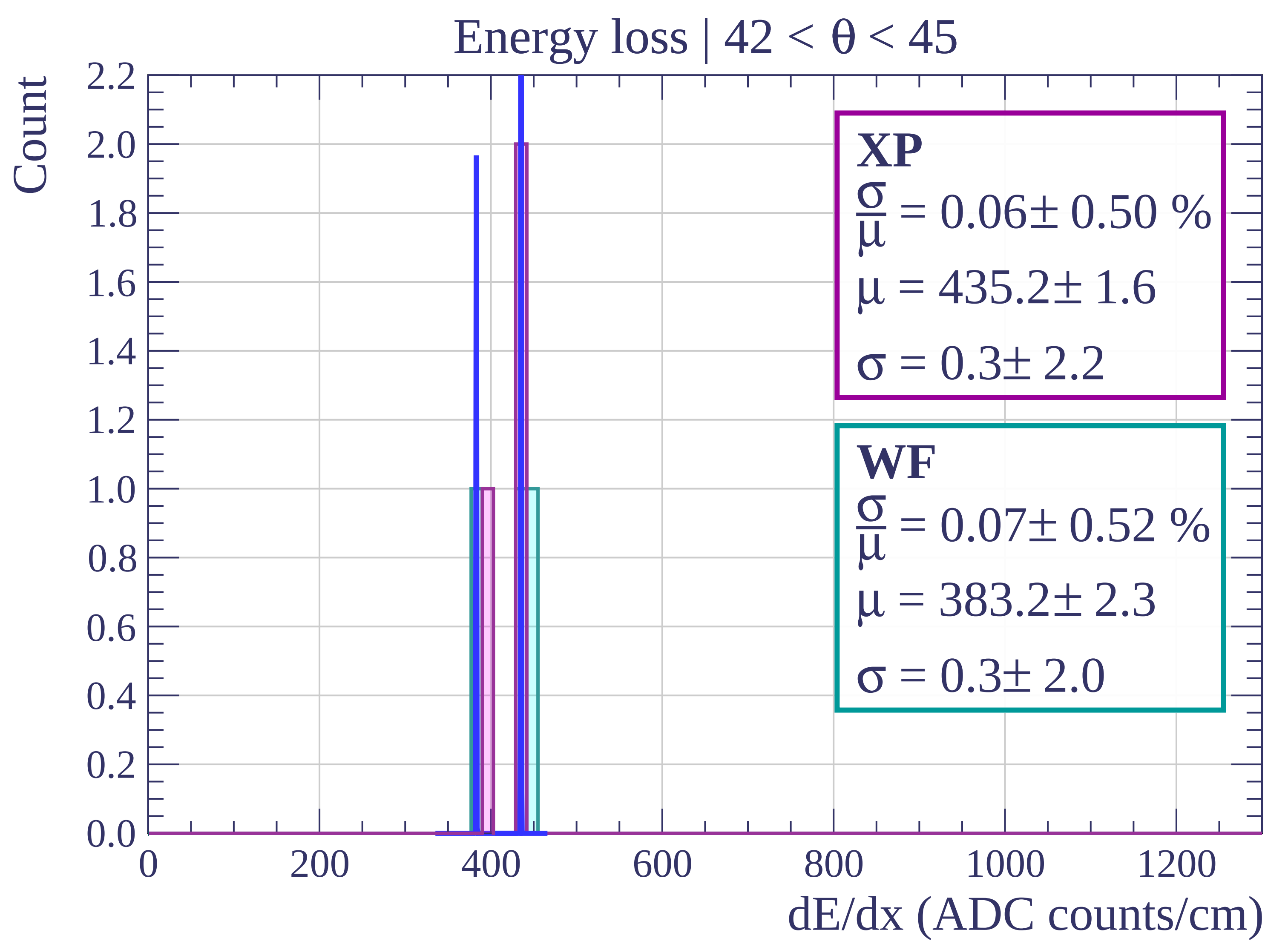
Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = -177.12± 361.33 %
μ = -101.8± 267.2
σ = 180.3± 105.5
WF
σ/μ = 0.04± 1.72 %
μ = 435.7± 2.2
σ = 0.2± 7.5



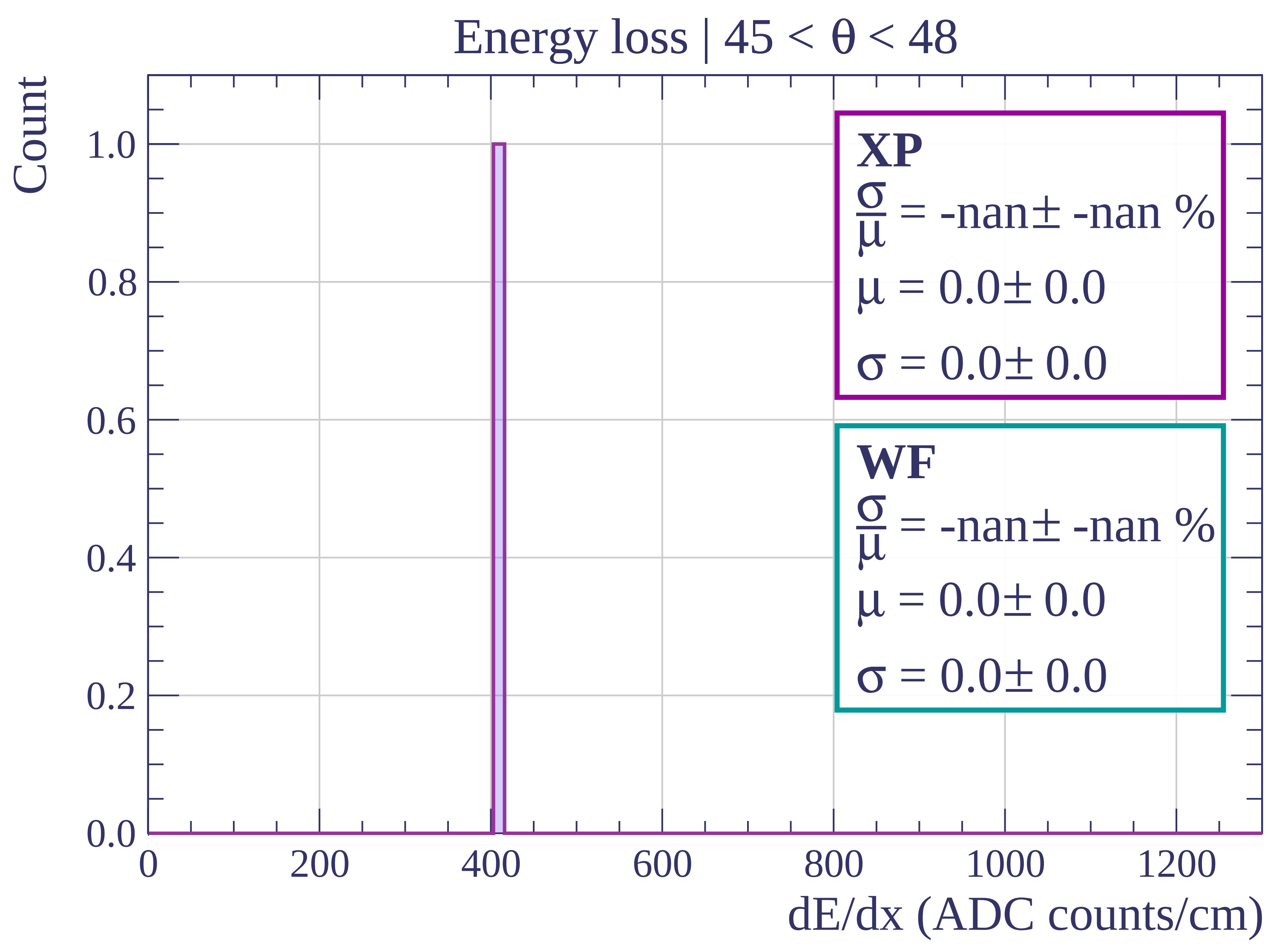
Energy loss | 36 < θ < 39
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 4.97± 6.02 %
μ = 407.4± 11.7
σ = 20.2± 24.0
WF
σ/μ = 1.66± 8.19 %
μ = 404.8± 65.2
σ = 6.7± 32.1



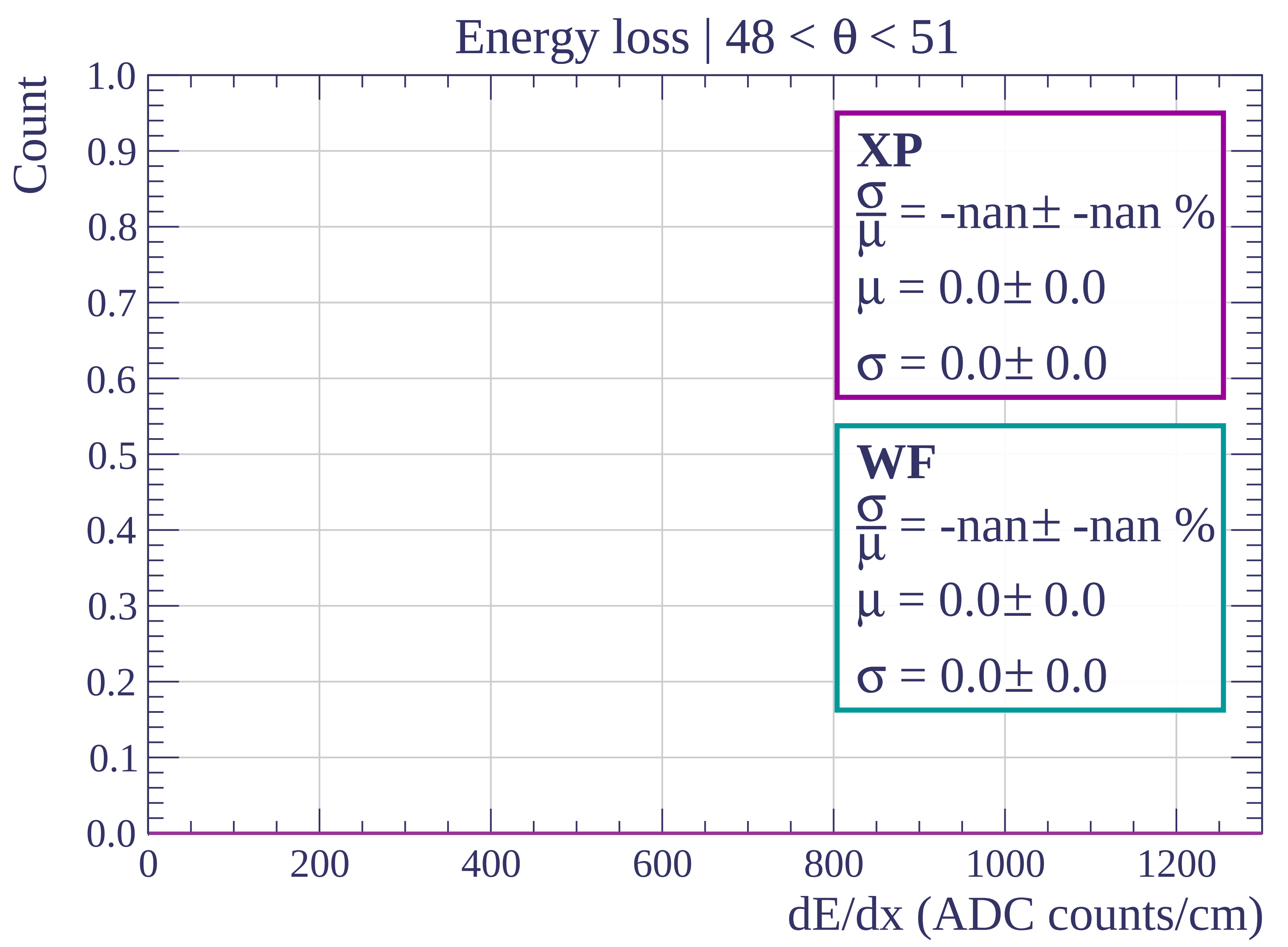
Energy loss | 39 < θ < 42
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 0.07 \pm 0.55$ %
$\mu = 396.2 \pm 2.3$
$\sigma = 0.3 \pm 2.2$
WF
$\frac{\sigma}{\mu} = 0.07 \pm 0.55$ %
$\mu = 396.2 \pm 2.3$
$\sigma = 0.3 \pm 2.2$



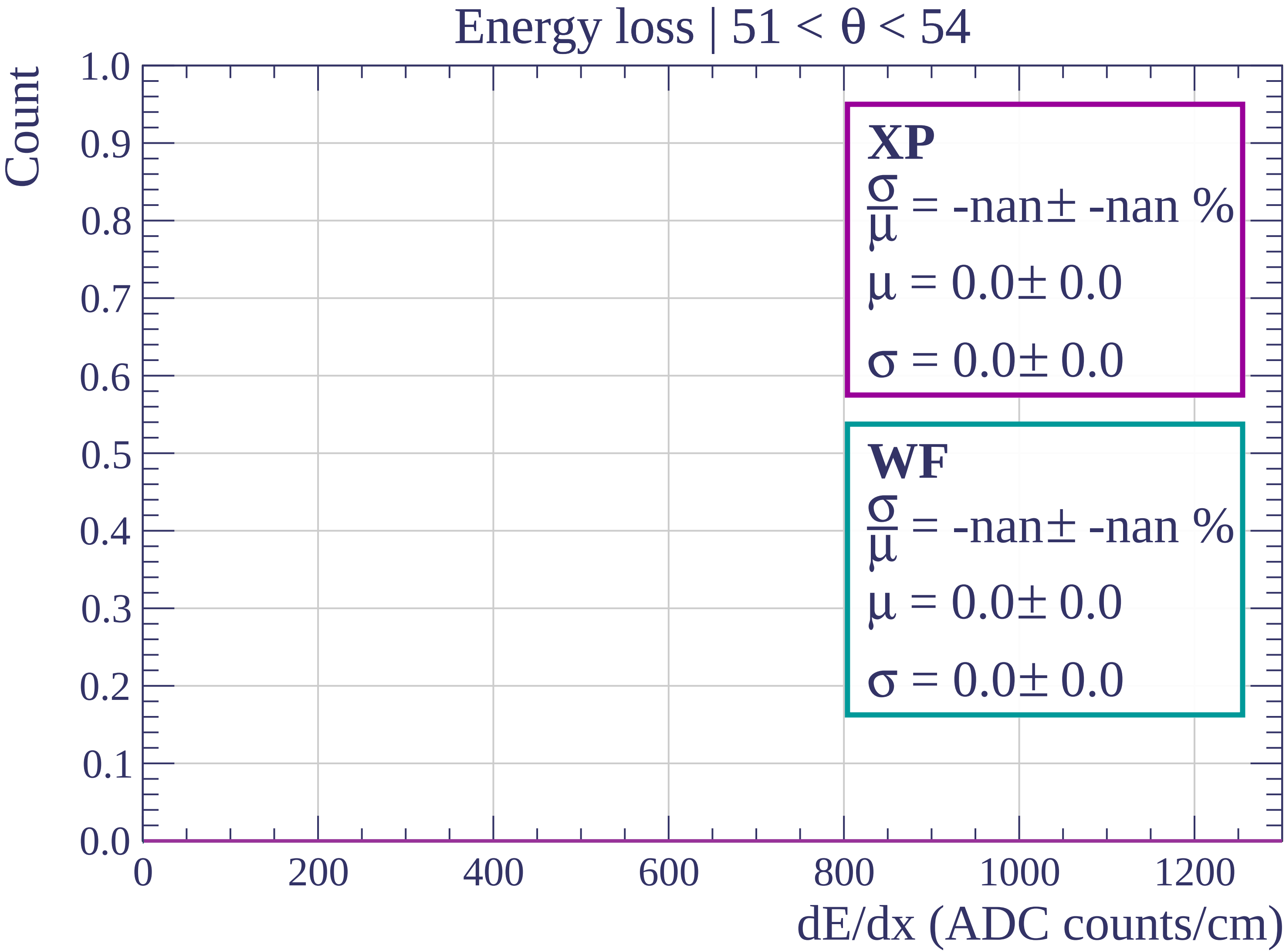
Energy loss | 42 < θ < 45
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.06± 0.50 %
μ = 435.2± 1.6
σ = 0.3± 2.2
WF
$\frac{\sigma}{\mu}$ = 0.07± 0.52 %
μ = 383.2± 2.3
σ = 0.3± 2.0



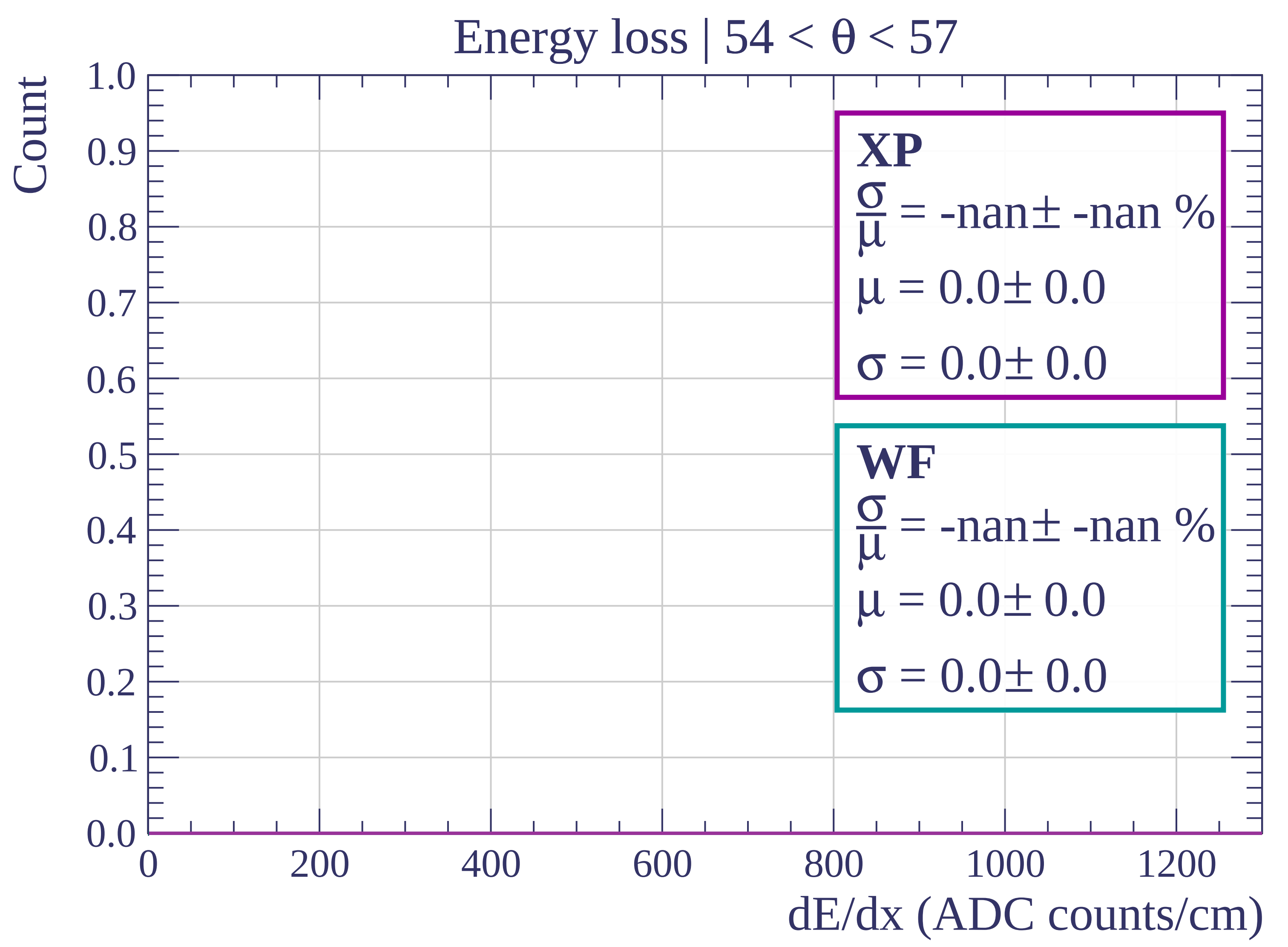
Energy loss | 45 < θ < 48
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



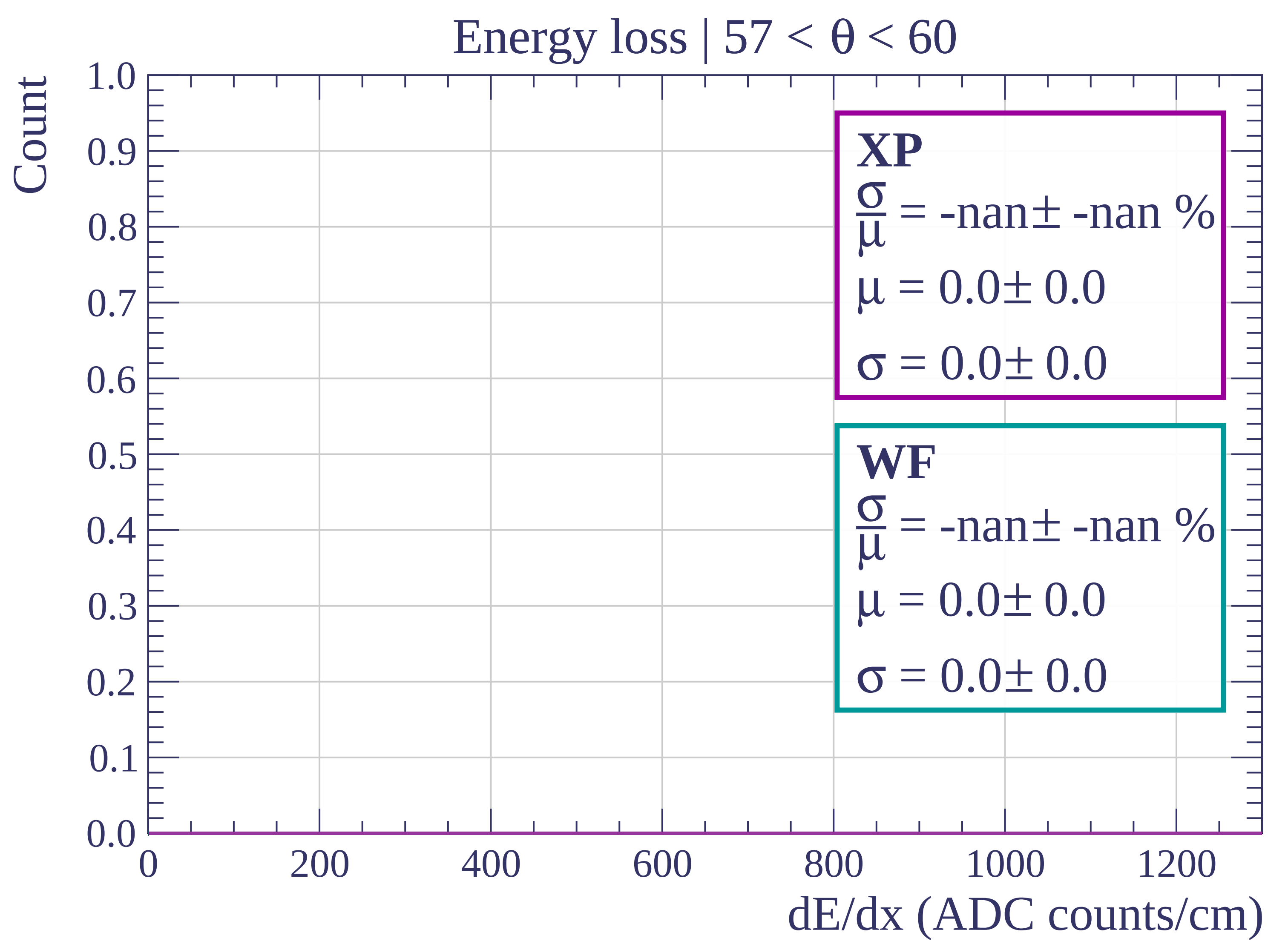
Energy loss | 48 < θ < 51
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



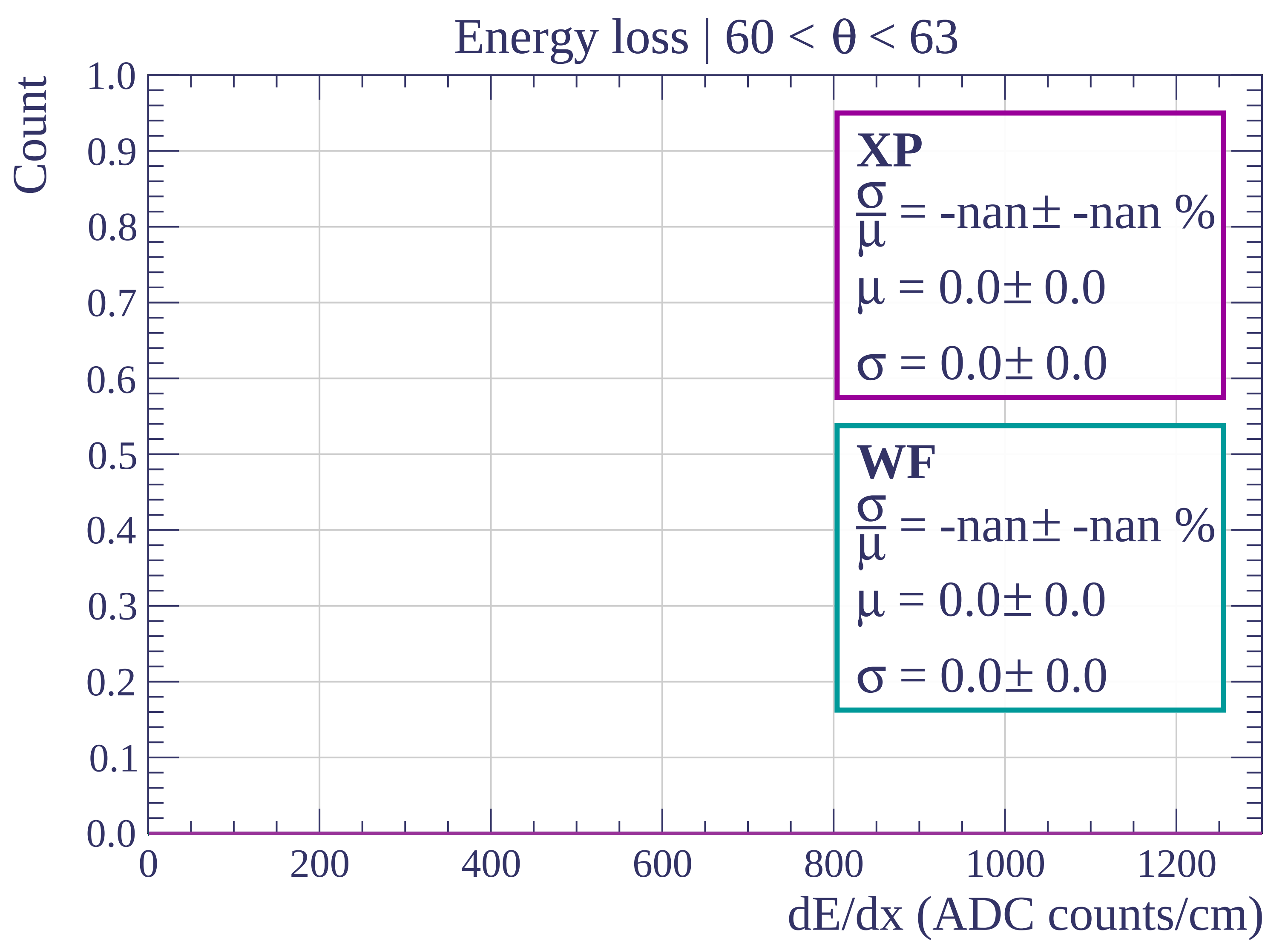
Energy loss | 51 < θ < 54
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



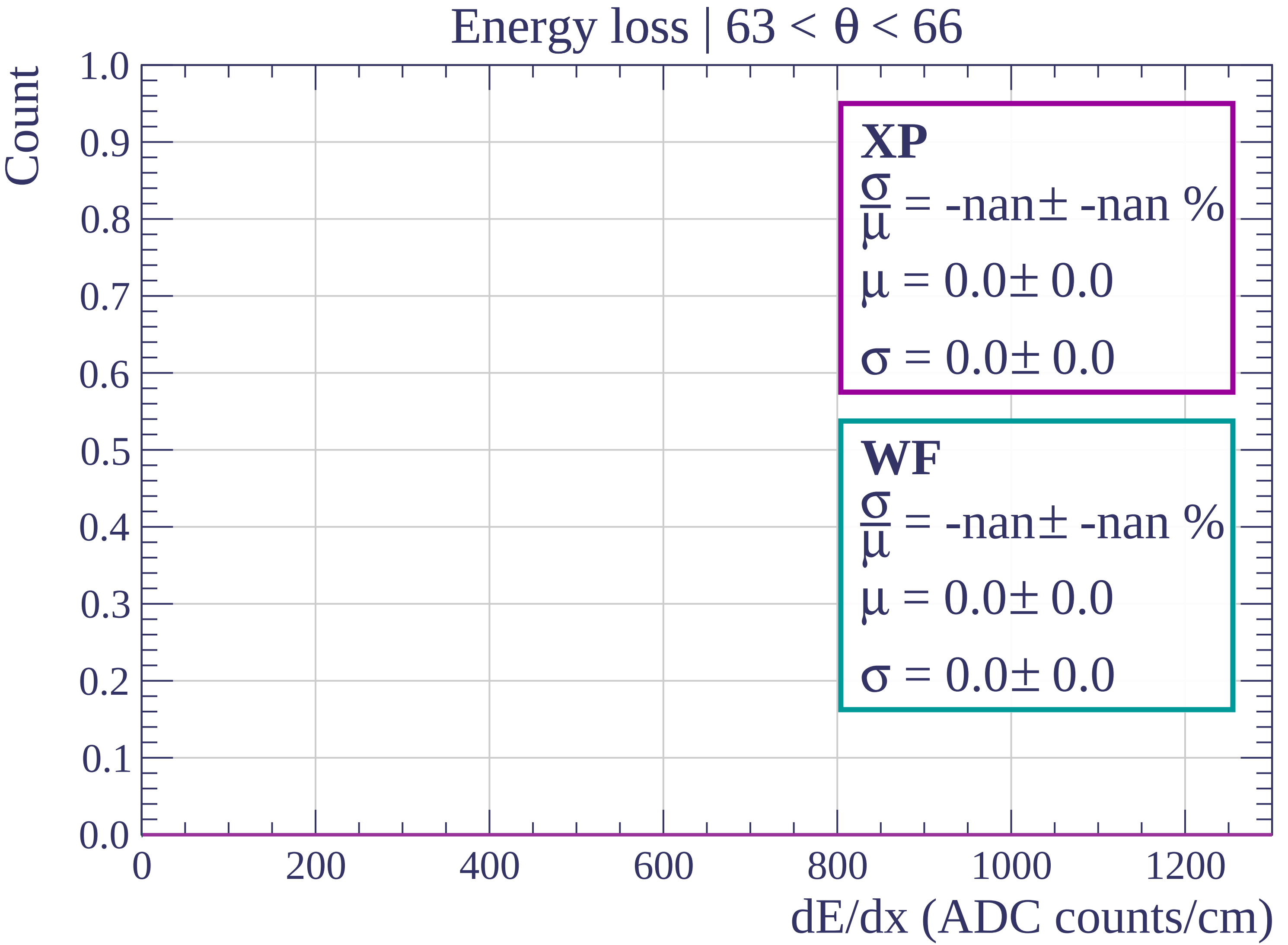
Energy loss | 54 < θ < 57
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



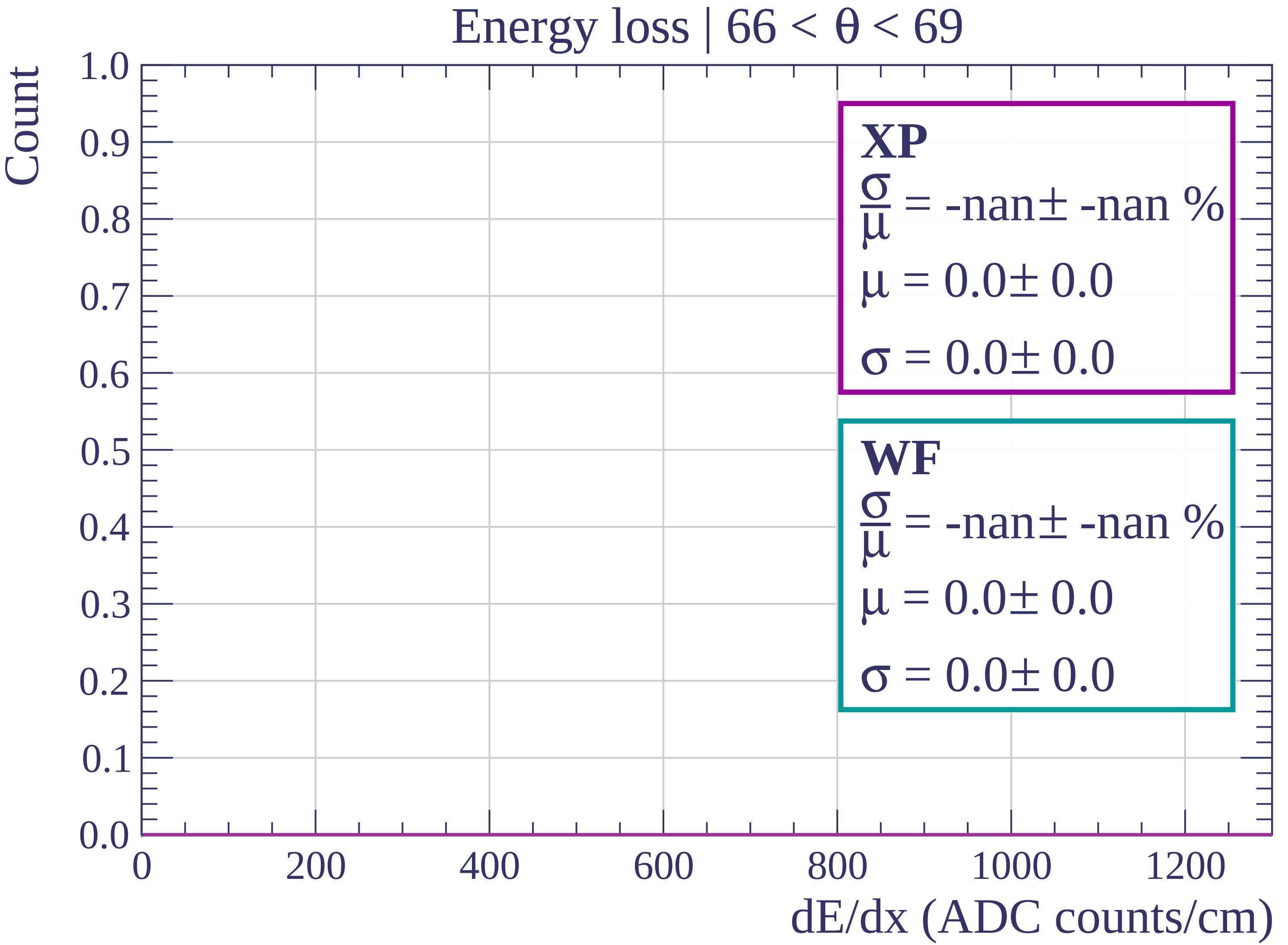
Energy loss | 57 < θ < 60
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



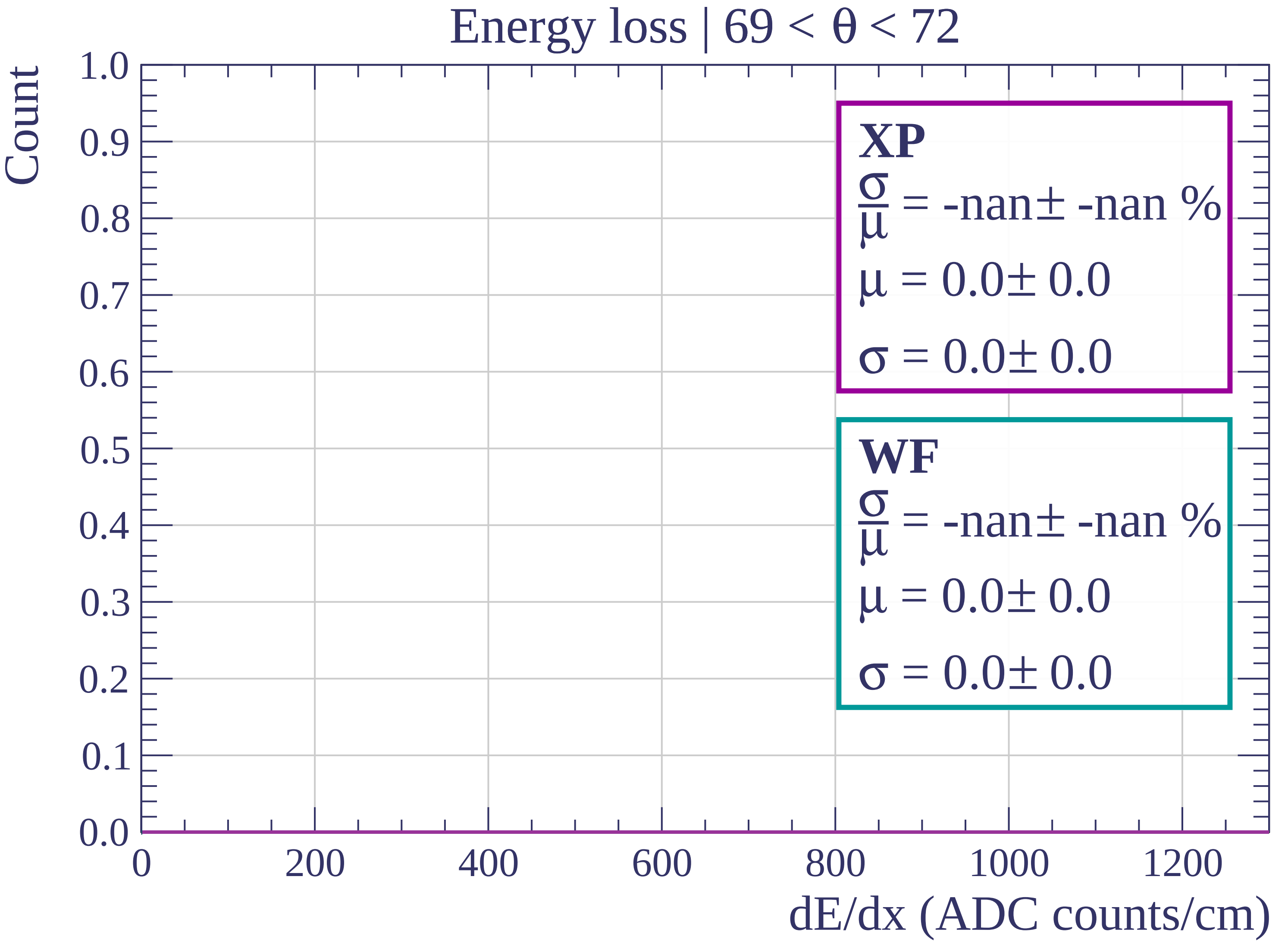
Energy loss | 60 < θ < 63
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



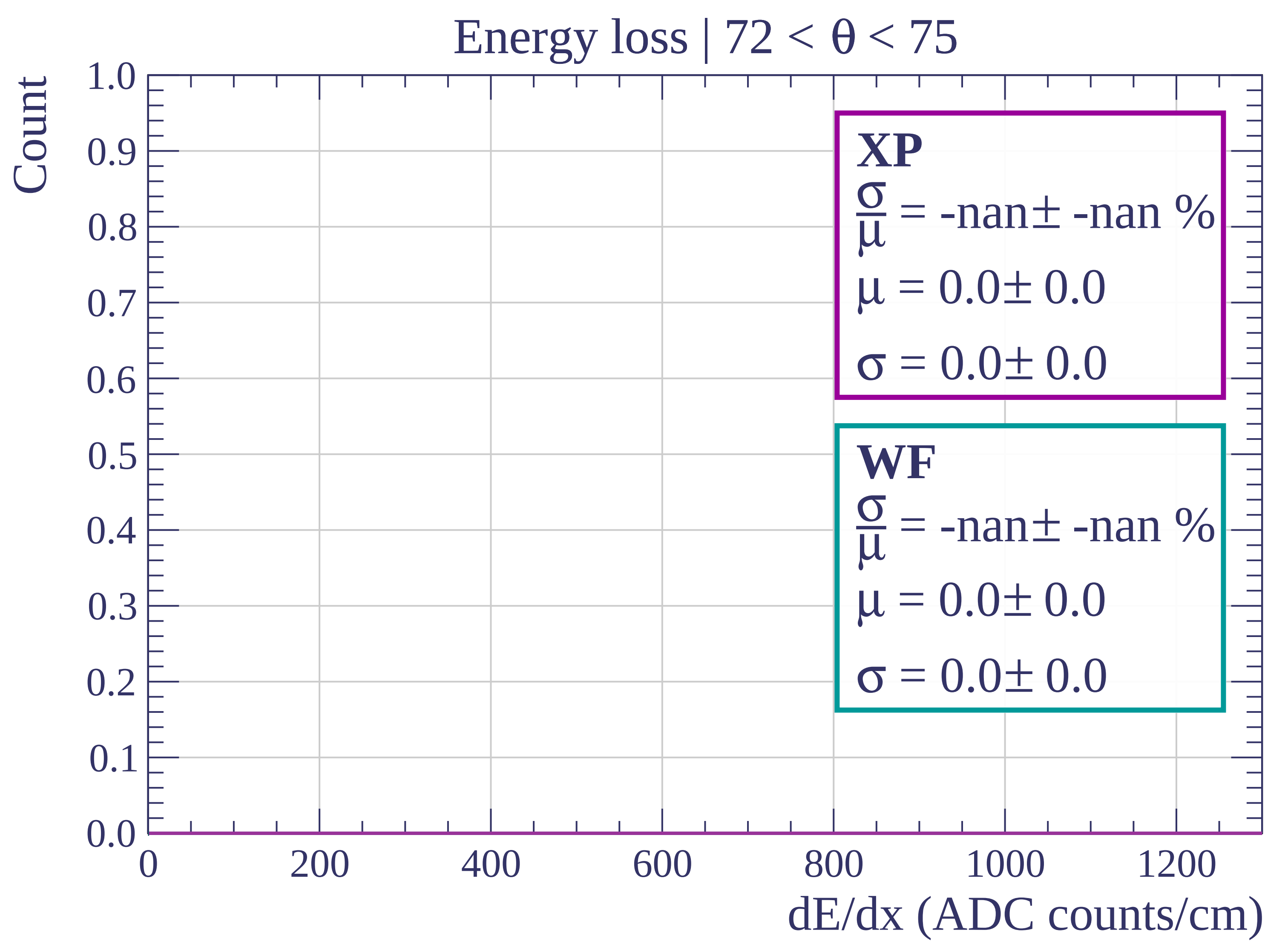
Energy loss | 63 < θ < 66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



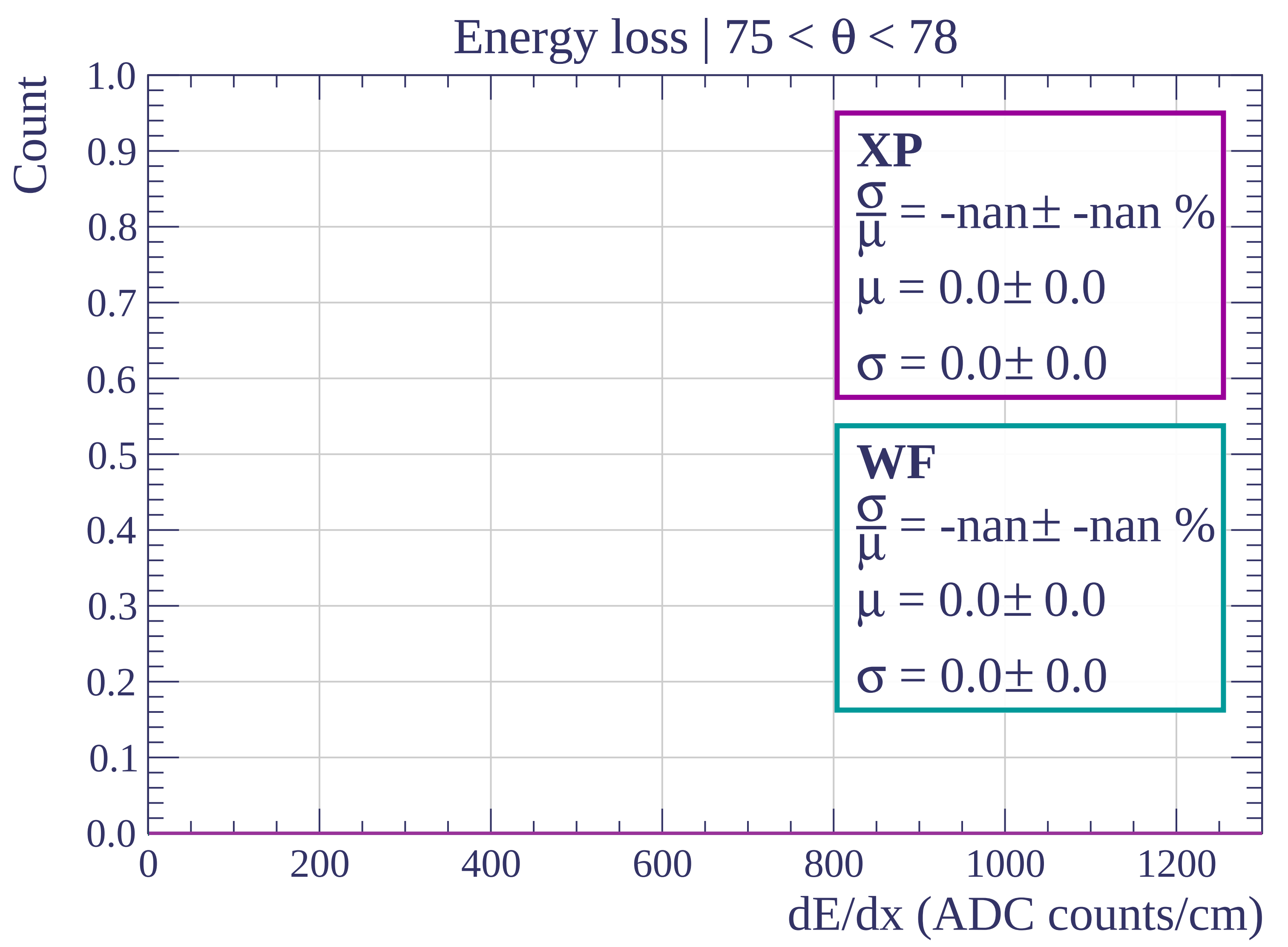
Energy loss | 66 < θ < 69
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



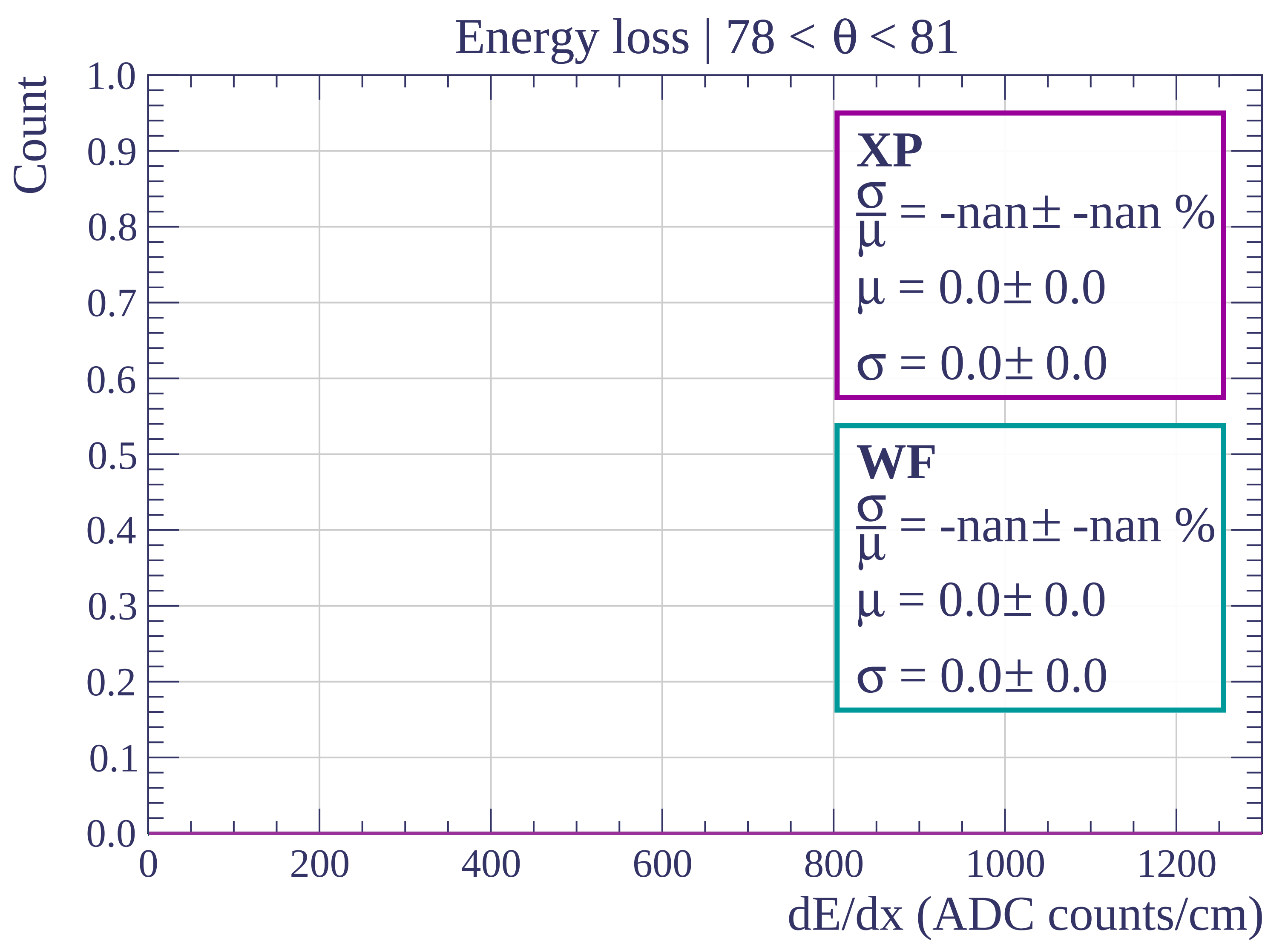
Energy loss | 69 < θ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

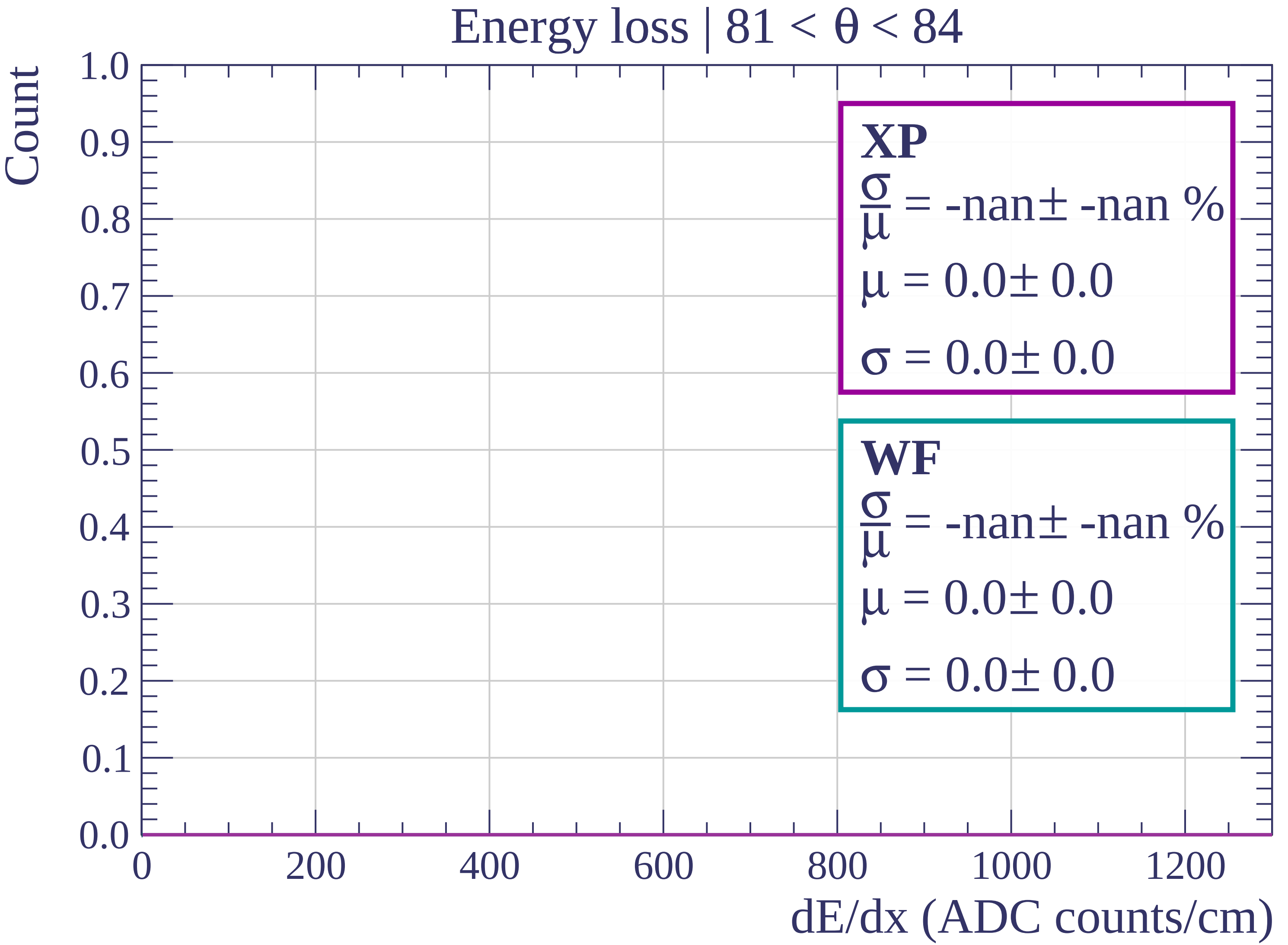


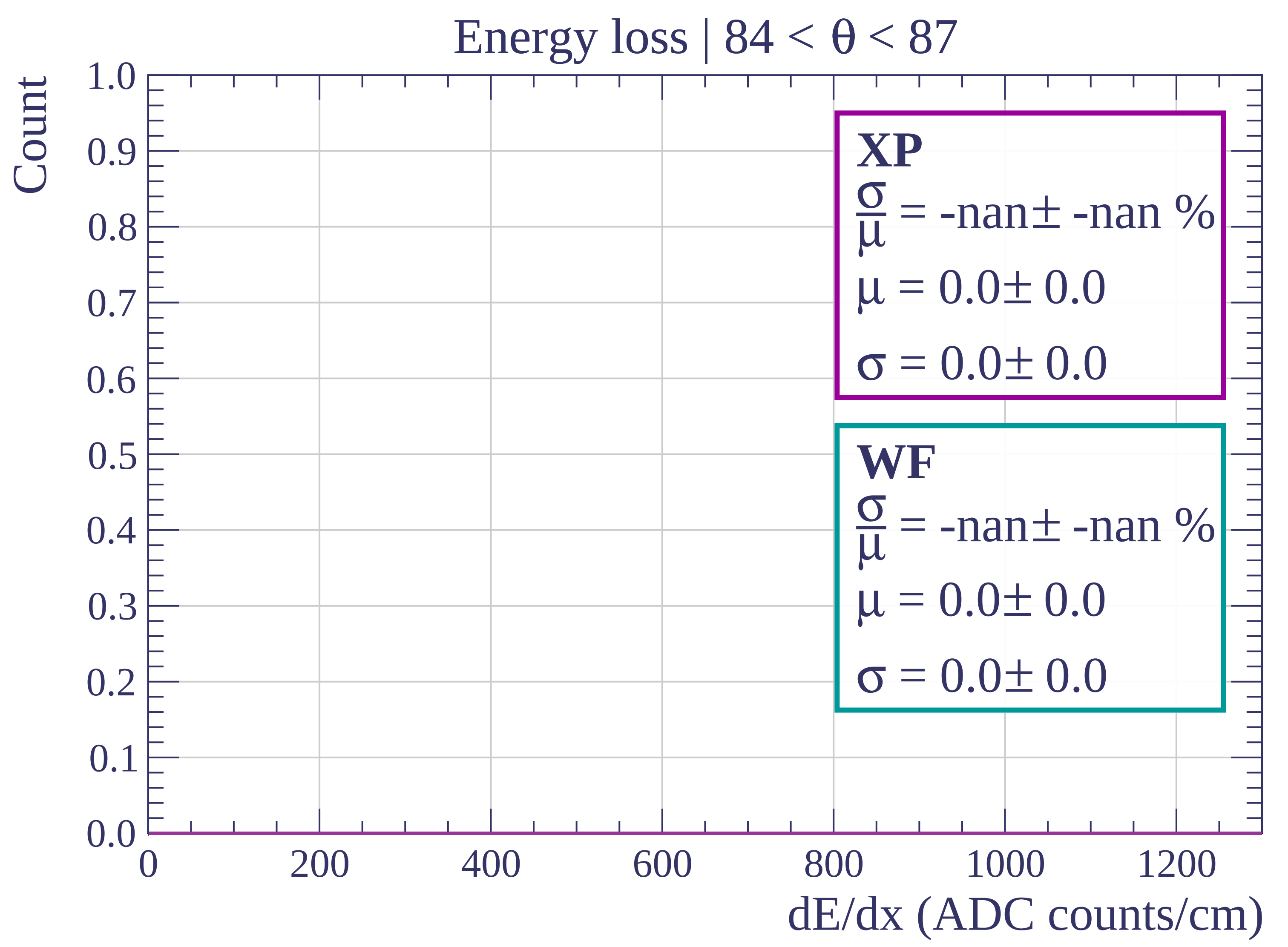
Energy loss | 72 < θ < 75
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



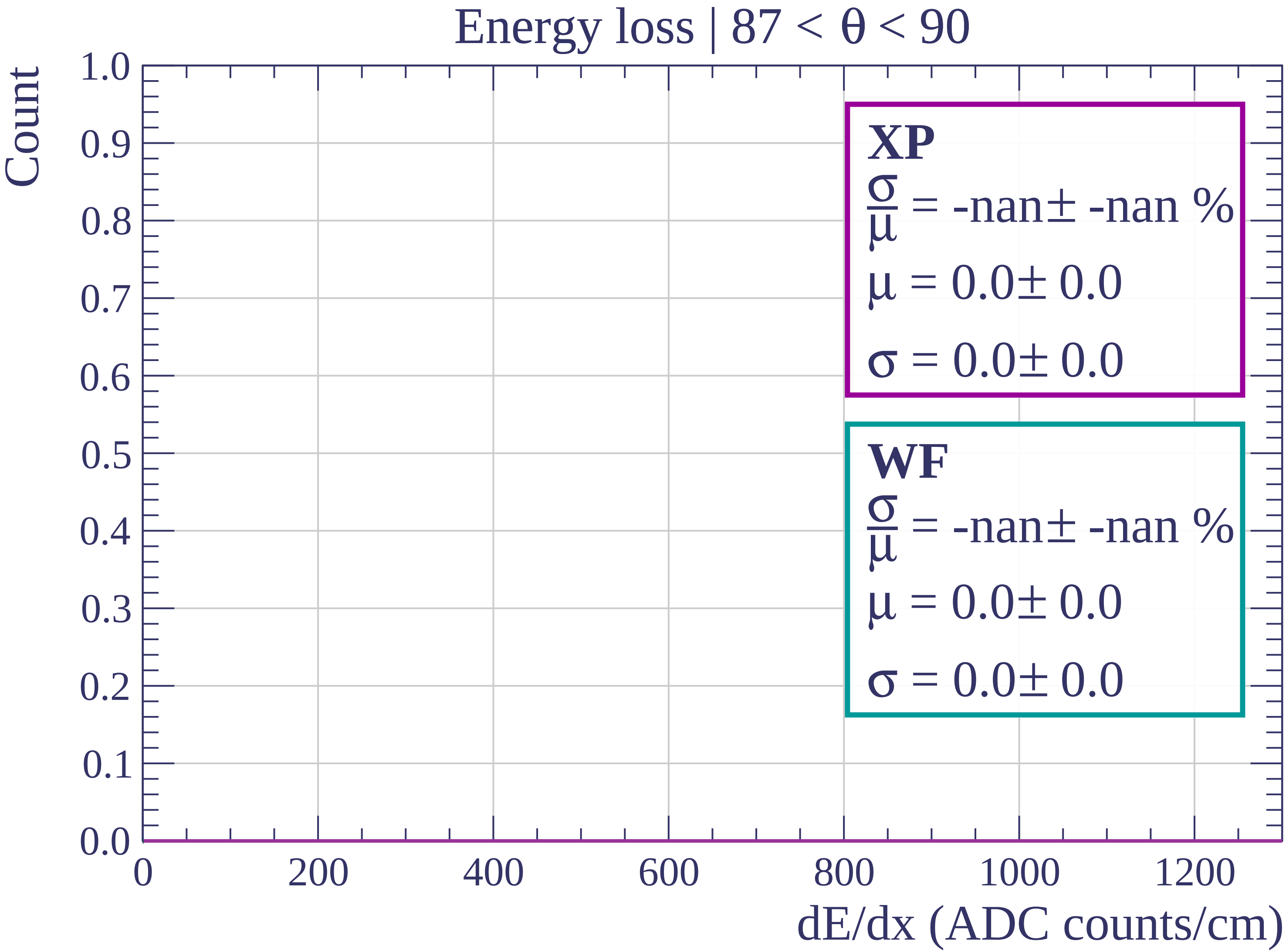
Energy loss | 75 < θ < 78
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 78 < θ < 81
Count
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

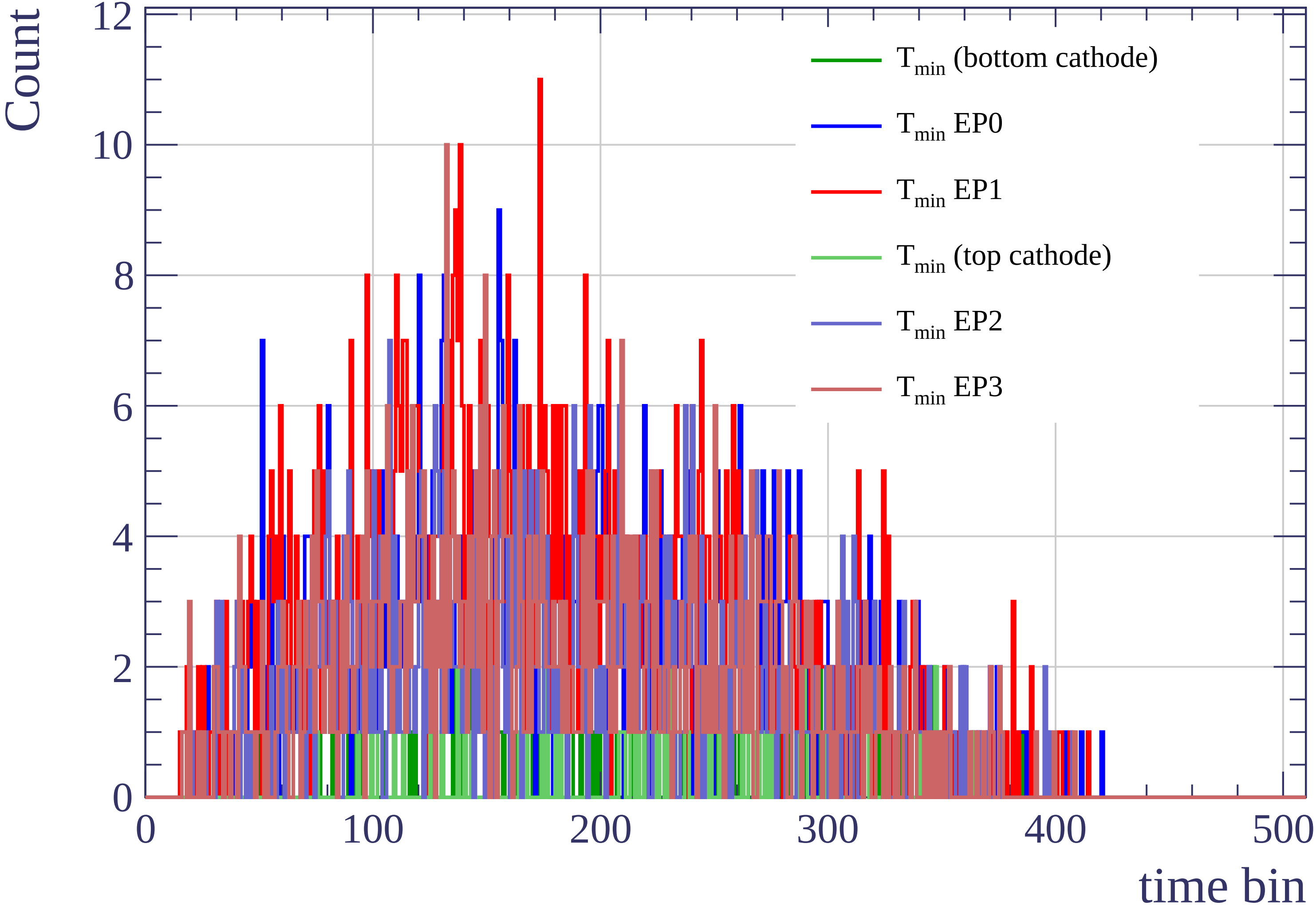

Energy loss | 81 < θ < 84
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



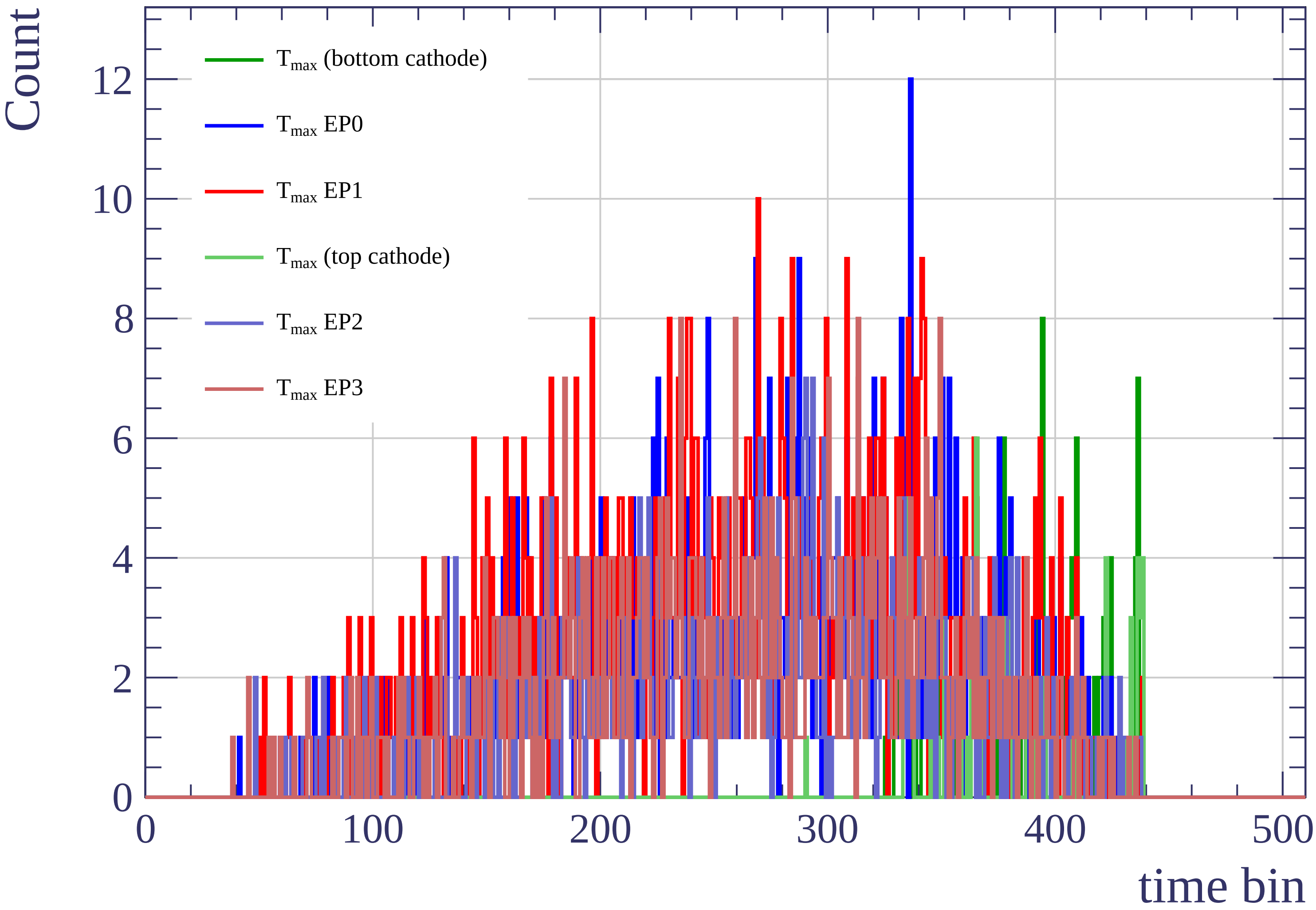
Energy loss | 84 < θ < 87
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 87 < θ < 90
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_max (bottom cathode)
T_max EP0
T_max EP1
T_max (top cathode)
T_max EP2
T_max EP3